

restart

► Solution with two media

▼ Solution with 3 media

restart

$$\begin{aligned} eq1 &:= \frac{(\epsilon_{eff}-1)}{\epsilon_{eff}+2} = \frac{(\epsilon_1-1)}{\epsilon_1+2} \cdot \eta_1 + \frac{(\epsilon_2-1)}{\epsilon_2+2} \cdot \eta_2 + \frac{(\epsilon_3-1)}{\epsilon_3+2} \cdot \eta_3 \\ \frac{\epsilon_{eff}-1}{\epsilon_{eff}+2} &= \frac{(\epsilon_1-1) \eta_1}{\epsilon_1+2} + \frac{(\epsilon_2-1) \eta_2}{\epsilon_2+2} + \frac{(\epsilon_3-1) \eta_3}{\epsilon_3+2} \end{aligned} \quad (2.1)$$

$$\begin{aligned} \epsilon_{eff} &:= \epsilon_{reff} + I \epsilon_{ceff}; \epsilon_1 := \epsilon_{r1} + I \epsilon_{c1}; \epsilon_2 := \epsilon_{r2} + I \epsilon_{c2}; \epsilon_3 := \epsilon_{r3} + I \epsilon_{c3}; \\ &\quad \epsilon_{reff} + I \epsilon_{ceff} \\ &\quad \epsilon_{r1} + I \epsilon_{c1} \\ &\quad \epsilon_{r2} + I \epsilon_{c2} \\ &\quad \epsilon_{r3} + I \epsilon_{c3} \end{aligned} \quad (2.2)$$

eq2 := evalc(eq1)

$$\begin{aligned} &\frac{(\epsilon_{reff}-1) (\epsilon_{reff}+2)}{(\epsilon_{reff}+2)^2 + \epsilon_{ceff}^2} + \frac{\epsilon_{ceff}^2}{(\epsilon_{reff}+2)^2 + \epsilon_{ceff}^2} + I \left(\frac{\epsilon_{ceff} (\epsilon_{reff}+2)}{(\epsilon_{reff}+2)^2 + \epsilon_{ceff}^2} \right. \\ &\quad \left. - \frac{(\epsilon_{reff}-1) \epsilon_{ceff}}{(\epsilon_{reff}+2)^2 + \epsilon_{ceff}^2} \right) = \left(\frac{(\epsilon_{r1}-1) (\epsilon_{r1}+2)}{(\epsilon_{r1}+2)^2 + \epsilon_{c1}^2} + \frac{\epsilon_{c1}^2}{(\epsilon_{r1}+2)^2 + \epsilon_{c1}^2} \right) \eta_1 \\ &\quad + \left(\frac{(\epsilon_{r2}-1) (\epsilon_{r2}+2)}{(\epsilon_{r2}+2)^2 + \epsilon_{c2}^2} + \frac{\epsilon_{c2}^2}{(\epsilon_{r2}+2)^2 + \epsilon_{c2}^2} \right) \eta_2 + \left(\frac{(\epsilon_{r3}-1) (\epsilon_{r3}+2)}{(\epsilon_{r3}+2)^2 + \epsilon_{c3}^2} \right. \\ &\quad \left. + \frac{\epsilon_{c3}^2}{(\epsilon_{r3}+2)^2 + \epsilon_{c3}^2} \right) \eta_3 + I \left(\left(\frac{\epsilon_{c1} (\epsilon_{r1}+2)}{(\epsilon_{r1}+2)^2 + \epsilon_{c1}^2} - \frac{(\epsilon_{r1}-1) \epsilon_{c1}}{(\epsilon_{r1}+2)^2 + \epsilon_{c1}^2} \right) \eta_1 \right. \\ &\quad \left. + \left(\frac{\epsilon_{c2} (\epsilon_{r2}+2)}{(\epsilon_{r2}+2)^2 + \epsilon_{c2}^2} - \frac{(\epsilon_{r2}-1) \epsilon_{c2}}{(\epsilon_{r2}+2)^2 + \epsilon_{c2}^2} \right) \eta_2 + \left(\frac{\epsilon_{c3} (\epsilon_{r3}+2)}{(\epsilon_{r3}+2)^2 + \epsilon_{c3}^2} \right. \right. \\ &\quad \left. \left. - \frac{(\epsilon_{r3}-1) \epsilon_{c3}}{(\epsilon_{r3}+2)^2 + \epsilon_{c3}^2} \right) \eta_3 \right) \end{aligned} \quad (2.3)$$

$$\begin{aligned}
eq2r := & \frac{(\epsilon_{reff}-1)(\epsilon_{reff}+2)}{(\epsilon_{reff}+2)^2 + \epsilon_{ceff}^2} + \frac{\epsilon_{ceff}^2}{(\epsilon_{reff}+2)^2 + \epsilon_{ceff}^2} = \left(\frac{(\epsilon_{r1}-1)(\epsilon_{r1}+2)}{(\epsilon_{r1}+2)^2 + \epsilon_{c1}^2} \right. \\
& + \left. \frac{\epsilon_{c1}^2}{(\epsilon_{r1}+2)^2 + \epsilon_{c1}^2} \right) \eta_1 + \left(\frac{(\epsilon_{r2}-1)(\epsilon_{r2}+2)}{(\epsilon_{r2}+2)^2 + \epsilon_{c2}^2} + \frac{\epsilon_{c2}^2}{(\epsilon_{r2}+2)^2 + \epsilon_{c2}^2} \right) \eta_2 \\
& + \left(\frac{(\epsilon_{r3}-1)(\epsilon_{r3}+2)}{(\epsilon_{r3}+2)^2 + \epsilon_{c3}^2} + \frac{\epsilon_{c3}^2}{(\epsilon_{r3}+2)^2 + \epsilon_{c3}^2} \right) \eta_3 \\
& \frac{(\epsilon_{reff}-1)(\epsilon_{reff}+2)}{(\epsilon_{reff}+2)^2 + \epsilon_{ceff}^2} + \frac{\epsilon_{ceff}^2}{(\epsilon_{reff}+2)^2 + \epsilon_{ceff}^2} = \left(\frac{(\epsilon_{r1}-1)(\epsilon_{r1}+2)}{(\epsilon_{r1}+2)^2 + \epsilon_{c1}^2} \right. \\
& + \left. \frac{\epsilon_{c1}^2}{(\epsilon_{r1}+2)^2 + \epsilon_{c1}^2} \right) \eta_1 + \left(\frac{(\epsilon_{r2}-1)(\epsilon_{r2}+2)}{(\epsilon_{r2}+2)^2 + \epsilon_{c2}^2} + \frac{\epsilon_{c2}^2}{(\epsilon_{r2}+2)^2 + \epsilon_{c2}^2} \right) \eta_2 \\
& + \left(\frac{(\epsilon_{r3}-1)(\epsilon_{r3}+2)}{(\epsilon_{r3}+2)^2 + \epsilon_{c3}^2} + \frac{\epsilon_{c3}^2}{(\epsilon_{r3}+2)^2 + \epsilon_{c3}^2} \right) \eta_3
\end{aligned} \tag{2.4}$$

$$\begin{aligned}
eq2c := & \left(\frac{\epsilon_{ceff}(\epsilon_{reff}+2)}{(\epsilon_{reff}+2)^2 + \epsilon_{ceff}^2} - \frac{(\epsilon_{reff}-1)\epsilon_{ceff}}{(\epsilon_{reff}+2)^2 + \epsilon_{ceff}^2} \right) = \left(\left(\frac{\epsilon_{c1}(\epsilon_{r1}+2)}{(\epsilon_{r1}+2)^2 + \epsilon_{c1}^2} \right. \right. \\
& - \left. \frac{(\epsilon_{r1}-1)\epsilon_{c1}}{(\epsilon_{r1}+2)^2 + \epsilon_{c1}^2} \right) \eta_1 + \left(\frac{\epsilon_{c2}(\epsilon_{r2}+2)}{(\epsilon_{r2}+2)^2 + \epsilon_{c2}^2} - \frac{(\epsilon_{r2}-1)\epsilon_{c2}}{(\epsilon_{r2}+2)^2 + \epsilon_{c2}^2} \right) \eta_2 \\
& + \left(\frac{\epsilon_{c3}(\epsilon_{r3}+2)}{(\epsilon_{r3}+2)^2 + \epsilon_{c3}^2} - \frac{(\epsilon_{r3}-1)\epsilon_{c3}}{(\epsilon_{r3}+2)^2 + \epsilon_{c3}^2} \right) \eta_3 \\
& \frac{\epsilon_{ceff}(\epsilon_{reff}+2)}{(\epsilon_{reff}+2)^2 + \epsilon_{ceff}^2} - \frac{(\epsilon_{reff}-1)\epsilon_{ceff}}{(\epsilon_{reff}+2)^2 + \epsilon_{ceff}^2} = \left(\frac{\epsilon_{c1}(\epsilon_{r1}+2)}{(\epsilon_{r1}+2)^2 + \epsilon_{c1}^2} \right. \\
& - \left. \frac{(\epsilon_{r1}-1)\epsilon_{c1}}{(\epsilon_{r1}+2)^2 + \epsilon_{c1}^2} \right) \eta_1 + \left(\frac{\epsilon_{c2}(\epsilon_{r2}+2)}{(\epsilon_{r2}+2)^2 + \epsilon_{c2}^2} - \frac{(\epsilon_{r2}-1)\epsilon_{c2}}{(\epsilon_{r2}+2)^2 + \epsilon_{c2}^2} \right) \eta_2 \\
& + \left(\frac{\epsilon_{c3}(\epsilon_{r3}+2)}{(\epsilon_{r3}+2)^2 + \epsilon_{c3}^2} - \frac{(\epsilon_{r3}-1)\epsilon_{c3}}{(\epsilon_{r3}+2)^2 + \epsilon_{c3}^2} \right) \eta_3
\end{aligned} \tag{2.5}$$

$$\begin{aligned}
sol := & solve(\{eq2r, eq2c\}, \{\epsilon_{reff}, \epsilon_{ceff}\}) \\
& \left\{ \epsilon_{ceff} = (9 (16 \epsilon_{c1} \eta_1 + 16 \epsilon_{c2} \eta_2 + 16 \epsilon_{c3} \eta_3 + 16 \epsilon_{c1} \eta_1 \epsilon_{r2} + 16 \epsilon_{c1} \eta_1 \epsilon_{r3} + 4 \epsilon_{c1} \eta_1 \right. \tag{2.6}
\end{aligned}$$

$$\begin{aligned}
& \epsilon_{r2}^2 + 4 \epsilon_{c1} \eta_1 \epsilon_{r3}^2 + 4 \epsilon_{c1} \eta_1 \epsilon_{c2}^2 + 4 \epsilon_{c1} \eta_1 \epsilon_{c3}^2 + 16 \epsilon_{c2} \eta_2 \epsilon_{r1} + 16 \epsilon_{c2} \eta_2 \epsilon_{r3} \\
& + 4 \epsilon_{c2} \eta_2 \epsilon_{r1}^2 + 4 \epsilon_{c2} \eta_2 \epsilon_{r3}^2 + 4 \epsilon_{c2} \eta_2 \epsilon_{c1}^2 + 4 \epsilon_{c2} \eta_2 \epsilon_{c3}^2 + 16 \epsilon_{c3} \eta_3 \epsilon_{r1} \\
& + 16 \epsilon_{c3} \eta_3 \epsilon_{r2} + 4 \epsilon_{c3} \eta_3 \epsilon_{r1}^2 + 4 \epsilon_{c3} \eta_3 \epsilon_{r2}^2 + 4 \epsilon_{c3} \eta_3 \epsilon_{c1}^2 + 4 \epsilon_{c3} \eta_3 \epsilon_{c2}^2 \\
& + 4 \epsilon_{c1} \eta_1 \epsilon_{r2} \epsilon_{r3}^2 + 16 \epsilon_{c1} \eta_1 \epsilon_{r2} \epsilon_{r3} + 4 \epsilon_{c1} \eta_1 \epsilon_{r2} \epsilon_{c3}^2 + \epsilon_{c1} \eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 + 4 \epsilon_{c1} \eta_1 \\
& \epsilon_{r2}^2 \epsilon_{r3} + \epsilon_{c1} \eta_1 \epsilon_{r2}^2 \epsilon_{c3}^2 + \epsilon_{c1} \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \epsilon_{c1} \eta_1 \epsilon_{c2}^2 \epsilon_{r3} + \epsilon_{c1} \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 + 4 \epsilon_{c2} \eta_2 \epsilon_{r1} \\
& \epsilon_{r3}^2 + 16 \epsilon_{c2} \eta_2 \epsilon_{r1} \epsilon_{r3} + 4 \epsilon_{c2} \eta_2 \epsilon_{r1} \epsilon_{c3}^2 + \epsilon_{c2} \eta_2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 4 \epsilon_{c2} \eta_2 \epsilon_{r1}^2 \epsilon_{r3} + \epsilon_{c2} \eta_2 \epsilon_{r1}^2 \\
& \epsilon_{c3}^2 + \epsilon_{c2} \eta_2 \epsilon_{c1}^2 \epsilon_{r3}^2 + 4 \epsilon_{c2} \eta_2 \epsilon_{c1}^2 \epsilon_{r3} + \epsilon_{c2} \eta_2 \epsilon_{c1}^2 \epsilon_{c3}^2 + \epsilon_{c3} \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 + 4 \epsilon_{c3} \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \\
& + \epsilon_{c3} \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 + 4 \epsilon_{c3} \eta_3 \epsilon_{r1} \epsilon_{r2}^2 + 16 \epsilon_{c3} \eta_3 \epsilon_{r1} \epsilon_{r2} + 4 \epsilon_{c3} \eta_3 \epsilon_{r1} \epsilon_{c2}^2 + \epsilon_{c3} \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \\
& + 4 \epsilon_{c3} \eta_3 \epsilon_{c1}^2 \epsilon_{r2} + \epsilon_{c3} \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2) \Big/ \left(64 + 4 \epsilon_{c2}^2 \eta_2^2 \epsilon_{c3}^2 + 64 \eta_1 + 64 \eta_2 + 64 \eta_3 \right. \\
& + 64 \epsilon_{r1} + 64 \epsilon_{r2} + 64 \epsilon_{r3} + 16 \epsilon_{c1}^2 + 16 \epsilon_{c2}^2 + 16 \epsilon_{c3}^2 - 8 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r3} - 8 \eta_3^2 \epsilon_{r2}^2 \epsilon_{r3} \\
& + 18 \epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 \epsilon_{r2}^2 + 4 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r3}^2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r2}^2 + 4 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2}^2 - 16 \eta_2 \epsilon_{r2} \eta_1 \\
& + 8 \eta_3 \epsilon_{r1}^2 \eta_1 \epsilon_{r3}^2 - 16 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \eta_2 - 16 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \eta_2 - 4 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \eta_2 - 8 \eta_3^2 \\
& \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} - 2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 4 \eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_2 - 4 \eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_3 - 16 \eta_1 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 \\
& \left. + 8 \eta_3 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 + 16 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1} \epsilon_{r3} + 4 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3} + 16 \epsilon_{c3}^2 \right)
\end{aligned}$$

$$\begin{aligned}
& \eta_3^2 \epsilon_{r1} \epsilon_{r2} + 8 \eta_2 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 + 8 \eta_2 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 - 4 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \eta_3 - 16 \eta_2 \epsilon_{r1} \epsilon_{r3}^2 \eta_3 \\
& - 16 \eta_2 \epsilon_{r1} \epsilon_{r3} \eta_3 - 16 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 \eta_3 + \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{c3}^2 + 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \eta_1 - 4 \eta_2 \epsilon_{r1} \\
& \epsilon_{r3}^2 \eta_1 - 16 \eta_2 \epsilon_{r1} \epsilon_{r3} \eta_1 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{c1}^2 + 8 \eta_3 \epsilon_{c2}^2 \eta_2 \epsilon_{r3}^2 - 4 \eta_2 \epsilon_{c1}^2 \epsilon_{r3}^2 \eta_1 - 16 \eta_2 \\
& \epsilon_{c1}^2 \epsilon_{r3} \eta_1 - 4 \eta_2 \epsilon_{c1}^2 \epsilon_{c3}^2 \eta_1 + 8 \eta_3 \epsilon_{r2}^2 \eta_2 \epsilon_{c3}^2 + 8 \eta_3 \epsilon_{r2}^2 \eta_2 \epsilon_{r3}^2 - 4 \eta_1 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_2 - 4 \eta_1 \\
& \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_3 - 4 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_2 - 4 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_3 - 16 \eta_1 \epsilon_{c2}^2 \epsilon_{r3} \eta_2 - 4 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_2 \\
& - 4 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_3 - 4 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \eta_2 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r3}^2 \eta_1 - 16 \eta_2 \epsilon_{r1}^2 \epsilon_{r3} \eta_1 - 4 \eta_2 \epsilon_{r1}^2 \\
& \epsilon_{c3}^2 \eta_1 + \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 + 8 \eta_3 \eta_1 \epsilon_{c1}^2 \epsilon_{r3}^2 + 8 \eta_3 \eta_1 \epsilon_{c1}^2 \\
& \epsilon_{c3}^2 + 16 \eta_3^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 + 4 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 + \eta_3^2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_2 - 4 \eta_3 \epsilon_{r1}^2 \\
& \epsilon_{c2}^2 \eta_2 - 2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 8 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} + 8 \eta_3 \epsilon_{r1}^2 \eta_1 \epsilon_{c3}^2 + 4 \eta_2^2 \\
& \epsilon_{c3}^2 \epsilon_{r2}^2 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 16 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \eta_1 - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_1 - 4 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \eta_1 \\
& - 16 \eta_3 \epsilon_{r1} \epsilon_{r2} \eta_1 + 72 \epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 + 72 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_1 \\
& + 72 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 - 2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 8 \eta_3 \epsilon_{c2}^2 \eta_2 \epsilon_{c3}^2 + 16 \epsilon_{c1}^2 \eta_1^2 \\
& + 16 \eta_3^2 \epsilon_{r1} + 16 \eta_3^2 \epsilon_{r2} - 32 \eta_3^2 \epsilon_{r3} + 4 \eta_3^2 \epsilon_{r1}^2 + 4 \eta_3^2 \epsilon_{r2}^2 + 4 \eta_3^2 \epsilon_{c1}^2 + 4 \eta_3^2 \epsilon_{c2}^2 \\
& + 16 \eta_2^2 \epsilon_{r1} - 32 \eta_2^2 \epsilon_{r2} + 16 \eta_2^2 \epsilon_{r3} + 4 \eta_2^2 \epsilon_{r1}^2 + 4 \eta_2^2 \epsilon_{r3}^2 + 4 \eta_2^2 \epsilon_{c1}^2 + 4 \eta_2^2 \epsilon_{c3}^2
\end{aligned}$$

$$\begin{aligned}
& + 16 \epsilon_{c3}^2 \eta_3^2 + 4 \epsilon_{r1}^2 \epsilon_{r2}^2 + 16 \epsilon_{r1}^2 \epsilon_{r2} + 4 \epsilon_{r1}^2 \epsilon_{c2}^2 + 16 \epsilon_{r1} \epsilon_{r2}^2 + 64 \epsilon_{r1} \epsilon_{r2} + 16 \epsilon_{r1} \epsilon_{c2}^2 \\
& + 4 \epsilon_{c1}^2 \epsilon_{r2}^2 + 16 \epsilon_{c1}^2 \epsilon_{r2} + 16 \epsilon_{r1}^2 + 16 \epsilon_{r2}^2 + 16 \epsilon_{r3}^2 + 16 \eta_1^2 + 16 \eta_2^2 + 16 \eta_3^2 - 2 \eta_2 \\
& \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 \\
& - 32 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 32 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 32 \eta_2 \epsilon_{r1} \\
& \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 8 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - 8 \eta_2 \\
& \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 - 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} \\
& - 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 2 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_3 \\
& \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \\
& - 8 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 - 32 \eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 + 18 \epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 \epsilon_{c2}^2 + 72 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \epsilon_{r1} \\
& + 18 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \epsilon_{r1}^2 + 18 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \epsilon_{c1}^2 + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 \eta_3 \\
& + 72 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 \epsilon_{r3} + 18 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 \epsilon_{r3}^2 + 18 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 \epsilon_{c3}^2 \\
& + 72 \epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 \epsilon_{r2} + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_1 + 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 \eta_3 - 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 \\
& - 32 \eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 2 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r1} \\
& \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 2 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 32 \eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - 2 \eta_3
\end{aligned}$$

$$\begin{aligned}
& \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 - 2 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} \\
& - 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 2 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 2 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \\
& + \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 4 \eta_2^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 + \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \eta_2^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 \\
& - 8 \eta_2^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2} + \epsilon_{c2}^2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{c3}^2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1} \epsilon_{c2}^2 - 8 \eta_2^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 \\
& + 4 \epsilon_{c1}^2 \epsilon_{c2}^2 + 16 \epsilon_{r1} \epsilon_{r3}^2 + 64 \epsilon_{r1} \epsilon_{r3} + 16 \epsilon_{r1} \epsilon_{c3}^2 + 16 \epsilon_{r2} \epsilon_{r3}^2 + 64 \epsilon_{r2} \epsilon_{r3} + 16 \epsilon_{r2} \\
& \epsilon_{c3}^2 + 4 \epsilon_{r1}^2 \epsilon_{r3}^2 + 16 \epsilon_{r1}^2 \epsilon_{r3} + 4 \epsilon_{r1}^2 \epsilon_{c3}^2 + 4 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \epsilon_{r2}^2 \epsilon_{r3} + 4 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \epsilon_{c1}^2 \epsilon_{r3}^2 \\
& + 16 \epsilon_{c1}^2 \epsilon_{r3} + 4 \epsilon_{c1}^2 \epsilon_{c3}^2 + 4 \epsilon_{c2}^2 \epsilon_{r3}^2 + 16 \epsilon_{c2}^2 \epsilon_{r3} + 4 \epsilon_{c2}^2 \epsilon_{c3}^2 - 32 \eta_1 \epsilon_{r1} + 64 \eta_1 \epsilon_{r2} \\
& + 64 \eta_1 \epsilon_{r3} - 32 \eta_1 \epsilon_{r1}^2 + 16 \eta_1 \epsilon_{r2}^2 + 16 \eta_1 \epsilon_{r3}^2 - 32 \eta_1 \epsilon_{c1}^2 + 16 \eta_1 \epsilon_{c2}^2 + 16 \eta_1 \epsilon_{c3}^2 \\
& + 64 \eta_2 \epsilon_{r1} - 32 \eta_2 \epsilon_{r2} + 64 \eta_2 \epsilon_{r3} + 16 \eta_2 \epsilon_{r1}^2 - 32 \eta_2 \epsilon_{r2}^2 + 16 \eta_2 \epsilon_{r3}^2 + 16 \eta_2 \epsilon_{c1}^2 \\
& - 32 \eta_2 \epsilon_{c2}^2 + 16 \eta_2 \epsilon_{c3}^2 + 64 \eta_3 \epsilon_{r1} + 64 \eta_3 \epsilon_{r2} - 32 \eta_3 \epsilon_{r3} + 16 \eta_3 \epsilon_{r1}^2 + 16 \eta_3 \epsilon_{r2}^2 \\
& - 32 \eta_3 \epsilon_{r3}^2 + 16 \eta_3 \epsilon_{c1}^2 + 16 \eta_3 \epsilon_{c2}^2 - 32 \eta_3 \epsilon_{c3}^2 + 16 \eta_1^2 \epsilon_{r1}^2 + 16 \eta_2^2 \epsilon_{r2}^2 + 16 \epsilon_{c2}^2 \eta_2^2 \\
& + 32 \eta_2 \eta_1 + 32 \eta_3 \eta_1 + 32 \eta_2 \eta_3 + 16 \eta_3^2 \epsilon_{r3}^2 - 32 \eta_2^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} + 4 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \\
& + 4 \eta_2^2 \epsilon_{r2}^2 \epsilon_{r1} \epsilon_{c3}^2 + \epsilon_{c3}^2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{c2}^2 + \epsilon_{c1}^2 \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{c2}^2 + 4 \\
& \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1} \epsilon_{r3}^2 + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2} \epsilon_{c3}^2 + \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} + \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4
\end{aligned}$$

$$\eta_3^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \epsilon_{c2}^2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r3}^2 + 8 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \eta_2 + 8 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \eta_2 + 4 \eta_3^2 \epsilon_{r3}^2 \epsilon_{r1}^2 \epsilon_{r2}^2 +$$

$$\eta_3^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + \eta_3^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \epsilon_{r1}^2 - 4 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \eta_1 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + \eta_1^2$$

$$\epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 16 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + \eta_1^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + \eta_1^2$$

$$\epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} \eta_3 + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_1 + 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_1 + 2 \eta_2 \eta_3 \epsilon_{r1}^2$$

$$\epsilon_{c2}^2 \epsilon_{c3}^2 + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_3 + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_1 + 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_1 + 2 \eta_3 \epsilon_{c1}^2$$

$$\epsilon_{c2}^2 \epsilon_{r3}^2 \eta_1 + 2 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_1 + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_3 + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_1 + 2 \eta_3 \epsilon_{r1}^2$$

$$\epsilon_{r2}^2 \epsilon_{r3}^2 \eta_1 + 8 \eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} \eta_1 + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_2 + 2 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_2 + 2 \eta_3 \epsilon_{r1}^2$$

$$\epsilon_{c2}^2 \epsilon_{r3}^2 \eta_2 + 8 \eta_1 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 \eta_3 + 2 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_2 + 2 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_3 + 8 \eta_1 \epsilon_{r1}^2$$

$$\epsilon_{c2}^2 \epsilon_{r3}^2 \eta_2 + 8 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_2 + 8 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_3 + 2 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_2 + 2 \eta_1 \epsilon_{r1} \epsilon_{r2}^2$$

$$\epsilon_{r3}^2 \eta_3 + 2 \eta_2 \epsilon_{r3}^2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 - 8 \eta_1^2 \epsilon_{r1} \epsilon_{r2}^2 - 32 \eta_1^2 \epsilon_{r1} \epsilon_{r2} + 16 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_1^2 \epsilon_{r1}$$

$$\epsilon_{c2}^2 - 8 \eta_1^2 \epsilon_{r1} \epsilon_{r3}^2 - 32 \eta_1^2 \epsilon_{r1} \epsilon_{r3} - 8 \eta_1^2 \epsilon_{r1} \epsilon_{c3}^2 + 8 \eta_2 \epsilon_{r3}^2 \eta_1 + 16 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1} + 8 \eta_3$$

$$\epsilon_{r2}^2 \eta_1 - 16 \eta_2 \epsilon_{r1}^2 \eta_1 + 4 \epsilon_{c2}^2 \eta_2^2 \epsilon_{c1}^2 - 16 \eta_3 \epsilon_{r1}^2 \eta_1 + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r3}^2 + 4 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 + 4$$

$$\epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 + \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + 16 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 +$$

$$\epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 64 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} + \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} + \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}$$

$$\begin{aligned}
& + 4 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 4 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 + 4 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 4 \epsilon_{r1} \\
& \epsilon_{c2}^2 \epsilon_{c3}^2 + \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} + 4 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 + \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \epsilon_{c1}^2 \\
& \epsilon_{c2}^2 \epsilon_{r3}^2 + \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 - 32 \eta_1 \epsilon_{r1}^2 \epsilon_{r2} - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 - 8 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \\
& - 32 \eta_1 \epsilon_{r1} \epsilon_{r2} - 8 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 - 32 \eta_1 \epsilon_{c1}^2 \epsilon_{r2} - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 - 8 \eta_1 \epsilon_{r1} \\
& \epsilon_{r3}^2 + 2 \eta_2 \epsilon_{r3}^2 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 + 2 \eta_2 \epsilon_{r3}^2 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 + 2 \eta_2 \epsilon_{c3}^2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 + 2 \eta_2 \epsilon_{c3}^2 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 \\
& + 2 \eta_2 \epsilon_{c3}^2 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 + 2 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_2 + 2 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_3 + 8 \eta_2 \epsilon_{r3} \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \\
& + 8 \eta_2 \epsilon_{r3} \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 + 8 \eta_2 \epsilon_{r3} \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 + 2 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_2 + 2 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_3 \\
& + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} \eta_1 + 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} \eta_1 + 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 \eta_1 + 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 \eta_1 \\
& + 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} \eta_1 + 2 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 \eta_1 + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 \eta_1 + 2 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 \eta_1 \\
& + 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} \eta_1 + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 \eta_1 + 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} \eta_1 + 8 \eta_3 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 \\
& + 2 \eta_3 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{r2} \epsilon_{c3}^2 \eta_1 - 32 \eta_1^2 \epsilon_{r1} + 16 \eta_1^2 \epsilon_{r2} + 16 \eta_1^2 \epsilon_{r3} + 4 \eta_1^2 \epsilon_{r2}^2 \\
& + 4 \eta_1^2 \epsilon_{r3}^2 + 4 \eta_1^2 \epsilon_{c2}^2 + 4 \eta_1^2 \epsilon_{c3}^2 + 4 \eta_2^2 \epsilon_{r3}^2 \epsilon_{r2}^2 - 32 \eta_1 \epsilon_{r1} \epsilon_{r3} - 8 \eta_1 \epsilon_{r1} \epsilon_{c3}^2 \\
& + 16 \eta_1 \epsilon_{r2} \epsilon_{r3}^2 + 64 \eta_1 \epsilon_{r2} \epsilon_{r3} + 16 \eta_1 \epsilon_{r2} \epsilon_{c3}^2 - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{r3}^2 - 32 \eta_1 \epsilon_{r1}^2 \epsilon_{r3} - 8 \eta_1 \\
& \epsilon_{r1}^2 \epsilon_{c3}^2 + 4 \eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \eta_1 \epsilon_{r2}^2 \epsilon_{r3} + 4 \eta_1 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{r3}^2 - 32 \eta_1 \epsilon_{c1}^2 \epsilon_{r3}
\end{aligned}$$

$$-8 \eta_1 \epsilon_{cl}^2 \epsilon_{c3}^2 + 4 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 + 16 \eta_1 \epsilon_{c2}^2 \epsilon_{r3} + 4 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 - 8 \eta_2$$

$$\epsilon_{r1}^2 \epsilon_{r2} - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 - 32 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 - 32 \eta_2 \epsilon_{r1} \epsilon_{r2} - 32 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 - 8 \eta_2 \epsilon_{cl}^2 \epsilon_{r2}^2$$

$$-8 \eta_2 \epsilon_{cl}^2 \epsilon_{r2} - 8 \eta_2 \epsilon_{cl}^2 \epsilon_{c2}^2 + 16 \eta_2 \epsilon_{r1} \epsilon_{r3}^2 + 64 \eta_2 \epsilon_{r1} \epsilon_{r3} + 16 \eta_2 \epsilon_{r1} \epsilon_{c3}^2$$

$$-8 \eta_2 \epsilon_{r2} \epsilon_{r3}^2 - 32 \eta_2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_2 \epsilon_{r2} \epsilon_{c3}^2 + 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 16 \eta_2 \epsilon_{r1}^2 \epsilon_{r3} + 4 \eta_2 \epsilon_{r1}^2$$

$$\epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 32 \eta_2 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \eta_2 \epsilon_{cl}^2 \epsilon_{r3}^2 + 16 \eta_2 \epsilon_{cl}^2 \epsilon_{r3} + 4 \eta_2$$

$$\epsilon_{cl}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 32 \eta_2 \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_2 \epsilon_{c2}^2 \epsilon_{c3}^2 + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 + 16 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}$$

$$+ 4 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 + 16 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 + 64 \eta_3 \epsilon_{r1} \epsilon_{r2} + 16 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 + 4 \eta_3 \epsilon_{cl}^2 \epsilon_{r2}^2 + 16 \eta_3$$

$$\epsilon_{cl}^2 \epsilon_{r2} + 4 \eta_3 \epsilon_{cl}^2 \epsilon_{c2}^2 - 32 \eta_3 \epsilon_{r1} \epsilon_{r3}^2 - 32 \eta_3 \epsilon_{r1} \epsilon_{r3} - 32 \eta_3 \epsilon_{r1} \epsilon_{c3}^2 - 32 \eta_3 \epsilon_{r2} \epsilon_{r3}^2$$

$$- 32 \eta_3 \epsilon_{r2} \epsilon_{r3} - 32 \eta_3 \epsilon_{r2} \epsilon_{c3}^2 - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r3} - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{c3}^2 - 8 \eta_3 \epsilon_{r2}^2$$

$$\epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_3 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_3 \epsilon_{cl}^2 \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{cl}^2 \epsilon_{r3} - 8 \eta_3 \epsilon_{cl}^2 \epsilon_{c3}^2 - 8 \eta_3 \epsilon_{c2}^2$$

$$\epsilon_{r3}^2 - 8 \eta_3 \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_3 \epsilon_{c2}^2 \epsilon_{c3}^2 + 4 \eta_3^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 4 \eta_3^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 4 \eta_3^2 \epsilon_{cl}^2$$

$$\epsilon_{r3}^2 + 16 \eta_3^2 \epsilon_{r1} \epsilon_{r3}^2 + 16 \eta_3^2 \epsilon_{r2} \epsilon_{r3}^2 - 16 \eta_3 \epsilon_{r3} \eta_1 + 8 \eta_3 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 + 2 \eta_3 \eta_2 \epsilon_{r1}^2$$

$$\epsilon_{c2}^2 \epsilon_{r3}^2 + 2 \eta_3 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 2 \eta_3 \eta_2 \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 8 \eta_3 \epsilon_{r2} \eta_1 \epsilon_{cl}^2 \epsilon_{c3}^2 + 2 \eta_3 \epsilon_{r2}^2 \eta_1 \epsilon_{cl}^2$$

$$\epsilon_{c3}^2 + 2 \eta_3 \epsilon_{cl}^2 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 + 2 \eta_3 \epsilon_{cl}^2 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 + 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 \eta_3 + 2 \eta_2 \epsilon_{cl}^2 \epsilon_{r2} \epsilon_{r3}^2 \eta_3$$

$$\begin{aligned}
& + 2 \eta_2 \epsilon_{cl}^2 \epsilon_{r2} \epsilon_{r3} \eta_3 + 2 \eta_2 \epsilon_{cl}^2 \epsilon_{r2} \epsilon_{c3}^2 \eta_3 + 8 \eta_3 \epsilon_{r1}^2 \eta_1 \epsilon_{r2} \epsilon_{r3}^2 + 8 \eta_3 \epsilon_{cl}^2 \eta_1 \epsilon_{r2} \epsilon_{r3}^2 \\
& + 8 \eta_3 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 + 2 \eta_3 \eta_2 \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 2 \eta_3 \eta_2 \epsilon_{cl}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 2 \eta_3 \eta_2 \epsilon_{cl}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 \\
& + 8 \eta_3 \epsilon_{r2}^2 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_1 \epsilon_{r3}^2 + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_1 \epsilon_{c3}^2 + 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \eta_1 \epsilon_{c3}^2 \\
& + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_1 \epsilon_{r3}^2 + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_1 \epsilon_{c3}^2 + 2 \eta_3 \epsilon_{cl}^2 \epsilon_{r2}^2 \eta_1 \epsilon_{r3}^2 + 8 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} \eta_2 \\
& + 8 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 2 \eta_3 \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 - 16 \eta_3 \epsilon_{r1} \eta_1 - 16 \eta_2 \epsilon_{cl}^2 \eta_1 + 8 \eta_2 \epsilon_{c3}^2 \eta_1 \\
& - 16 \eta_3 \epsilon_{cl}^2 \eta_1 + 8 \eta_3 \epsilon_{c2}^2 \eta_1 - 16 \eta_2 \epsilon_{r1} \eta_1 + 4 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 + 4 \epsilon_{cl}^2 \eta_1^2 \epsilon_{c2}^2 + 16 \epsilon_{c3}^2 \\
& \eta_3^2 \epsilon_{r1} + 4 \epsilon_{cl}^2 \eta_1^2 \epsilon_{c3}^2 + \eta_1^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3} + \eta_1^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3} \\
& + \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \eta_1^2 \epsilon_{r2} \epsilon_{r3} + 4 \eta_1^2 \epsilon_{r2} \epsilon_{c3}^2 + 4 \eta_1^2 \epsilon_{r2} \epsilon_{r3}^2 + 16 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r3} + 16 \epsilon_{cl}^2 \\
& \eta_1^2 \epsilon_{r3} - 16 \eta_2 \epsilon_{r2} \eta_3 - 16 \eta_2 \epsilon_{r3} \eta_3 + 8 \eta_2 \epsilon_{r1}^2 \eta_3 - 16 \eta_2 \epsilon_{r3}^2 \eta_3 + 16 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2} + 4 \\
& \eta_1^2 \epsilon_{r1}^2 \epsilon_{c2}^2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{c2}^2 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 - 16 \eta_1 \epsilon_{r2}^2 \eta_2 - 16 \eta_1 \epsilon_{r3}^2 \eta_3 - 16 \eta_1 \epsilon_{c2}^2 \eta_2 \\
& - 16 \eta_1 \epsilon_{c3}^2 \eta_3 + 16 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r2} + 32 \eta_2 \epsilon_{r3} \eta_1 + 4 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 + 4 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2} + \eta_3^2 \epsilon_{r1}^2 \\
& \epsilon_{c2}^2 + 4 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 - 4 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 \eta_1 - 4 \eta_2 \epsilon_{r2} \epsilon_{r3}^2 \eta_1 - 16 \eta_2 \epsilon_{r2} \epsilon_{r3} \eta_1 - 32 \\
& \eta_3^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - 8 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_3^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} - 4 \eta_2 \epsilon_{cl}^2 \epsilon_{r3} \eta_3 - 4 \eta_2 \epsilon_{cl}^2 \epsilon_{c3}^2 \eta_3 \\
& + 4 \epsilon_{cl}^2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3} - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r3}^2 \eta_3 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r3} \eta_3 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{c3}^2 \eta_3 - 4 \eta_2 \epsilon_{cl}^2
\end{aligned}$$

$$\epsilon_{r3}^2 \eta_3 + 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \eta_1 + 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \eta_1 + 8 \eta_2 \epsilon_{r2} \epsilon_{r3}^2 \eta_3 + 8 \eta_2 \epsilon_{r2} \epsilon_{r3} \eta_3 + 8 \eta_2 \epsilon_{r2}$$

$$\epsilon_{c3}^2 \eta_3 - 8 \eta_1^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - 2 \eta_1^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 + 16 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2} \epsilon_{r3} - 2 \eta_1^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 32$$

$$\eta_1^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - 8 \eta_1^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_1^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_1^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 2 \eta_1^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2$$

$$- 8 \eta_1^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 - 4 \eta_3 \epsilon_{r2}^2 \epsilon_{r3} \eta_1 + 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r3} \eta_1 - 4 \eta_3 \epsilon_{c2}^2 \epsilon_{r3} \eta_1 - 16 \eta_3$$

$$\epsilon_{c1}^2 \epsilon_{r2} \eta_1 - 4 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \eta_1 + 8 \eta_3 \epsilon_{r1} \epsilon_{r3} \eta_1 - 16 \eta_3 \epsilon_{r2} \epsilon_{r3} \eta_1 + 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r3} \eta_1$$

$$- 4 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \eta_1 + \epsilon_{c2}^2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r3}^2 - 2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 8 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2} \epsilon_{r3}^2$$

$$- 16 \eta_1 \epsilon_{r2} \epsilon_{r3}^2 \eta_3 - 16 \eta_1 \epsilon_{r2} \epsilon_{c3}^2 \eta_3 + 8 \eta_3 \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 8 \eta_1 \epsilon_{r1} \epsilon_{r3}^2 \eta_3 + 8 \eta_1 \epsilon_{r1}$$

$$\epsilon_{c3}^2 \eta_3 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2} + 8 \eta_2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 + 8 \eta_2 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 + \epsilon_{c3}^2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2}^2 + 4 \epsilon_{c2}^2$$

$$\eta_2^2 \epsilon_{r1} \epsilon_{c3}^2 - 2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 8 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} - 2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3} + \epsilon_{c1}^2$$

$$\eta_1^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \eta_3 - 16 \eta_2 \epsilon_{r1} \epsilon_{r2} \eta_3 - 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 32 \eta_1 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}$$

$$+ 16 \eta_3 \epsilon_{r1} \epsilon_{r2} + 4 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 + \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2}^2 + 4 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2} + \eta_3^2 \epsilon_{c1}^2 \epsilon_{c2}^2 - 32 \eta_3^2 \epsilon_{r1} \epsilon_{r3}$$

$$- 32 \eta_3^2 \epsilon_{r2} \epsilon_{r3} + \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 4 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3} + \eta_2^2 \epsilon_{r1}^2 \epsilon_{c3}^2 + \eta_2^2 \epsilon_{c1}^2 \epsilon_{r3}^2 + 4 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r3} +$$

$$\eta_2^2 \epsilon_{c1}^2 \epsilon_{c3}^2 + \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 - 8 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2} - 32 \eta_2^2 \epsilon_{r1} \epsilon_{r2} - 8 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2} + 4 \eta_2^2 \epsilon_{r1} \epsilon_{r3}^2 + 16$$

$$\eta_2^2 \epsilon_{r1} \epsilon_{r3} + 4 \eta_2^2 \epsilon_{r1} \epsilon_{c3}^2 - 8 \eta_2^2 \epsilon_{r2} \epsilon_{r3}^2 - 32 \eta_2^2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_2^2 \epsilon_{r2} \epsilon_{c3}^2 + 16 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2}$$

$$\begin{aligned}
& -16 \eta_3 \epsilon_{r2}^2 \eta_2 - 16 \eta_3 \epsilon_{c2}^2 \eta_2 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{c3}^2 + 16 \eta_1^2 \\
& \epsilon_{r1}^2 \epsilon_{r3}^2 + 16 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2}^2 + 16 \eta_2^2 \epsilon_{r3}^2 \epsilon_{r2}^2 + 32 \eta_2 \epsilon_{r1} \eta_3 - 8 \eta_3^2 \epsilon_{c2}^2 \epsilon_{r3}^2 \\
& + 8 \eta_2 \epsilon_{c1}^2 \eta_3 - 16 \eta_2 \epsilon_{c3}^2 \eta_3 + 32 \eta_3 \epsilon_{r2} \eta_1 - 2 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \\
& - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 2 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 2 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_1 \epsilon_{r1} \epsilon_{r2} \\
& \epsilon_{c3}^2 - 2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 32 \eta_1 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - 2 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 8 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 2 \eta_1 \epsilon_{c1}^2 \\
& \epsilon_{r2}^2 \epsilon_{r3}^2 - 8 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 2 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \\
& - 8 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 2 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 - 2 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{r2} \\
& \epsilon_{r3}^2 - 32 \eta_1 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 2 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_1 \epsilon_{c1}^2 \\
& \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 8 \eta_3^2 \\
& \epsilon_{r1}^2 \epsilon_{r3}^2), \epsilon_{\text{reff}} = - \left(-64 + 8 \epsilon_{c2}^2 \eta_2^2 \epsilon_{c3}^2 + 32 \eta_1 + 32 \eta_2 + 32 \eta_3 - 64 \epsilon_{r1} \right. \\
& - 64 \epsilon_{r2} - 64 \epsilon_{r3} - 16 \epsilon_{c1}^2 - 16 \epsilon_{c2}^2 - 16 \epsilon_{c3}^2 - 16 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r3}^2 - 16 \eta_3^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \\
& + 36 \epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 \epsilon_{r2}^2 + 8 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r3}^2 + 8 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r2}^2 + 8 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2}^2 - 32 \eta_2 \epsilon_{r2} \eta_1 \\
& + 16 \eta_3 \epsilon_{r1}^2 \eta_1 \epsilon_{r3}^2 - 32 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \eta_2 - 32 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \eta_2 - 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \eta_2 - 16 \eta_3^2 \\
& \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} - 4 \eta_3^2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 8 \eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_2 - 8 \eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_3 - 32 \eta_1 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 \\
& + 16 \eta_3 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 + 8 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 + 32 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1} \epsilon_{r3} + 8 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3} + 32 \epsilon_{c3}^2 \\
& \eta_3^2 \epsilon_{r1} \epsilon_{r2} + 16 \eta_2 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 + 16 \eta_2 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \eta_3 - 32 \eta_2 \epsilon_{r1} \epsilon_{r3}^2 \eta_3 \\
& - 32 \eta_2 \epsilon_{r1} \epsilon_{r3} \eta_3 - 32 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 \eta_3 + 2 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{c3}^2 + 16 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \eta_1 - 8 \eta_2 \epsilon_{r1} \\
& \epsilon_{r3}^2 \eta_1 - 32 \eta_2 \epsilon_{r1} \epsilon_{r3} \eta_1 + 8 \epsilon_{c3}^2 \eta_3^2 \epsilon_{c1}^2 + 16 \eta_3 \epsilon_{c2}^2 \eta_2 \epsilon_{r3}^2 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r3}^2 \eta_1 - 32 \eta_2 \\
& \epsilon_{c1}^2 \epsilon_{r3} \eta_1 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{c3}^2 \eta_1 + 16 \eta_3 \epsilon_{r2}^2 \eta_2 \epsilon_{c3}^2 + 16 \eta_3 \epsilon_{r2}^2 \eta_2 \epsilon_{r3}^2 - 8 \eta_1 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_2
\end{aligned}$$

$$\begin{aligned}
& -8\eta_1\epsilon_{r2}^2\epsilon_{c3}^2\eta_3-8\eta_1\epsilon_{c2}^2\epsilon_{r3}^2\eta_2-8\eta_1\epsilon_{c2}^2\epsilon_{r3}^2\eta_3-32\eta_1\epsilon_{c2}^2\epsilon_{r3}\eta_2-8\eta_1\epsilon_{c2}^2\epsilon_{c3}^2\eta_2-8\eta_1\epsilon_{c2}^2\epsilon_{c3}^2\eta_3-8\eta_3\epsilon_{c1}^2\epsilon_{c2}^2\eta_2-8\eta_2\epsilon_{r1}^2\epsilon_{r3}^2\eta_1-32\eta_2\epsilon_{r1}^2\epsilon_{r3}\eta_1-8\eta_2\epsilon_{r1}^2\epsilon_{c3}^2\eta_1+2\epsilon_{c1}^2\eta_1^2\epsilon_{r2}^2\epsilon_{r3}^2+2\epsilon_{c2}^2\eta_2^2\epsilon_{r1}^2\epsilon_{r3}^2+2\epsilon_{c3}^2\eta_3^2\epsilon_{r1}^2\epsilon_{r2}^2+16\eta_3\eta_1\epsilon_{c1}^2\epsilon_{r3}^2\\
& +16\eta_3\eta_1\epsilon_{c1}^2\epsilon_{c3}^2+32\eta_3^2\epsilon_{r1}\epsilon_{r2}\epsilon_{r3}^2+8\eta_3^2\epsilon_{c1}^2\epsilon_{r2}\epsilon_{r3}^2+2\eta_3^2\epsilon_{c1}^2\epsilon_{c2}^2\epsilon_{r3}^2-8\eta_3\epsilon_{r1}^2\epsilon_{r2}^2\eta_2-8\eta_3\epsilon_{r1}^2\epsilon_{c2}^2\eta_2-4\eta_3^2\epsilon_{c1}^2\epsilon_{r2}^2\epsilon_{r3}-4\eta_3^2\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{r3}-16\eta_3^2\epsilon_{r1}^2\epsilon_{r2}\epsilon_{r3}\\
& +16\eta_3\epsilon_{r1}^2\eta_1\epsilon_{c3}^2+8\eta_2^2\epsilon_{c3}^2\epsilon_{r2}^2+8\eta_1^2\epsilon_{r1}^2\epsilon_{r2}\epsilon_{c3}^2-32\eta_3\epsilon_{r1}^2\epsilon_{r2}\eta_1-8\eta_3\epsilon_{r1}^2\epsilon_{c2}^2\eta_1-8\eta_3\epsilon_{r1}\epsilon_{r2}^2\eta_1-32\eta_3\epsilon_{r1}\epsilon_{r2}\eta_1+144\epsilon_{c1}\eta_1\epsilon_{c3}\eta_3+144\epsilon_{c2}\eta_2\epsilon_{c3}\eta_3\\
& -8\eta_3\epsilon_{r1}^2\epsilon_{r2}^2\eta_1+144\epsilon_{c1}\eta_1\epsilon_{c2}\eta_2-4\eta_3^2\epsilon_{r1}^2\epsilon_{c2}^2\epsilon_{r3}+8\eta_3^2\epsilon_{r1}\epsilon_{r2}^2\epsilon_{r3}^2+16\eta_3\epsilon_{c2}^2\eta_2\epsilon_{c3}^2+32\epsilon_{c1}^2\eta_1^2+32\eta_3^2\epsilon_{r1}+32\eta_3^2\epsilon_{r2}-64\eta_3^2\epsilon_{r3}+8\eta_3^2\epsilon_{r1}^2+8\eta_3^2\epsilon_{r2}^2+8\eta_3^2\epsilon_{c1}^2+8\eta_3^2\epsilon_{c2}^2+32\eta_2^2\epsilon_{r1}-64\eta_2^2\epsilon_{r2}+32\eta_2^2\epsilon_{r3}+8\eta_2^2\epsilon_{r1}^2+8\eta_2^2\epsilon_{r3}^2+8\eta_2^2\epsilon_{c1}^2\\
& +8\eta_2^2\epsilon_{c3}^2+32\epsilon_{c3}^2\eta_3^2-4\epsilon_{r1}^2\epsilon_{r2}^2-16\epsilon_{r1}^2\epsilon_{r2}-4\epsilon_{r1}^2\epsilon_{c2}^2-16\epsilon_{r1}\epsilon_{r2}^2-64\epsilon_{r1}\epsilon_{r2}-16\epsilon_{r1}\epsilon_{c2}^2-4\epsilon_{c1}^2\epsilon_{r2}^2-16\epsilon_{c1}^2\epsilon_{r2}-16\epsilon_{r1}^2-16\epsilon_{r2}^2-16\epsilon_{r3}^2+32\eta_1^2+32\eta_2^2\\
& +32\eta_3^2-\eta_2\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{c3}^2-4\eta_2\epsilon_{r1}\epsilon_{r2}^2\epsilon_{r3}^2-\eta_2\epsilon_{r1}^2\epsilon_{c2}^2\epsilon_{c3}^2-4\eta_2\epsilon_{r1}\epsilon_{r2}\epsilon_{c3}^2-\eta_2\epsilon_{r1}^2\epsilon_{c2}^2\epsilon_{r3}^2-16\eta_2\epsilon_{r1}\epsilon_{r2}\epsilon_{r3}-\eta_2\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{r3}^2-16\eta_2\epsilon_{r1}\epsilon_{r2}^2\epsilon_{r3}-\eta_2\epsilon_{c1}^2\epsilon_{r2}^2\epsilon_{r3}^2\\
& -16\eta_2\epsilon_{r1}\epsilon_{c2}^2\epsilon_{r3}-\eta_2\epsilon_{r1}^2\epsilon_{r2}\epsilon_{r3}^2-4\eta_2\epsilon_{r1}\epsilon_{r2}^2\epsilon_{c3}^2-4\eta_2\epsilon_{c1}^2\epsilon_{r2}^2\epsilon_{r3}-4\eta_2\epsilon_{r1}\epsilon_{r2}\epsilon_{r3}^2-4\eta_2\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{r3}-4\eta_2\epsilon_{r1}\epsilon_{c2}^2\epsilon_{c3}^2-4\eta_2\epsilon_{c1}^2\epsilon_{r2}\epsilon_{c3}^2-4\eta_2\epsilon_{c1}^2\epsilon_{r2}\epsilon_{c3}^2-4\eta_3\epsilon_{r1}^2\epsilon_{r2}\epsilon_{r3}-\eta_2\epsilon_{c1}^2\epsilon_{r2}\epsilon_{c3}^2-4\eta_3\epsilon_{r1}\epsilon_{c2}^2\epsilon_{r3}^2-4\eta_3\epsilon_{r1}^2\epsilon_{r2}\epsilon_{c3}^2-4\eta_3\epsilon_{r1}^2\epsilon_{c2}^2\epsilon_{r3}-\eta_3\epsilon_{r1}^2\epsilon_{c2}^2\epsilon_{r3}-\eta_3\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{c3}^2\\
& -4\eta_3\epsilon_{r1}\epsilon_{r2}^2\epsilon_{r3}^2-16\eta_3\epsilon_{r1}\epsilon_{r2}\epsilon_{c3}^2+36\epsilon_{c1}\eta_1\epsilon_{c3}\eta_3\epsilon_{c2}^2+144\epsilon_{c2}\eta_2\epsilon_{c3}\eta_3\epsilon_{r1}^2+36\epsilon_{c2}\eta_2\epsilon_{c3}\eta_3\epsilon_{r1}^2+36\epsilon_{c2}\eta_2\epsilon_{c3}\eta_3\epsilon_{c1}^2+4\eta_2\epsilon_{r1}^2\epsilon_{r2}\epsilon_{r3}^2\\
& +144\epsilon_{c1}\eta_1\epsilon_{c2}\eta_2\epsilon_{r3}+36\epsilon_{c1}\eta_1\epsilon_{c2}\eta_2\epsilon_{r3}^2+36\epsilon_{c1}\eta_1\epsilon_{c2}\eta_2\epsilon_{c3}^2+144\epsilon_{c1}\eta_1\epsilon_{c3}\eta_3\epsilon_{r2}+4\eta_2\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{r3}^2\eta_1+16\eta_2\epsilon_{r1}\epsilon_{r2}\epsilon_{c3}^2\eta_3-\eta_3\epsilon_{r1}^2\epsilon_{c2}^2\epsilon_{r3}^2\\
& -16\eta_3\epsilon_{r1}\epsilon_{r2}\epsilon_{r3}-\eta_3\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{r3}^2-4\eta_3\epsilon_{r1}\epsilon_{r2}^2\epsilon_{r3}-\eta_3\epsilon_{c1}^2\epsilon_{r2}^2\epsilon_{r3}^2-4\eta_3\epsilon_{r1}\epsilon_{c2}^2\epsilon_{r3}^2-4\eta_3\epsilon_{r1}^2\epsilon_{r2}\epsilon_{c3}^2-4\eta_3\epsilon_{r1}\epsilon_{r2}^2\epsilon_{c3}^2-4\eta_3\epsilon_{c1}^2\epsilon_{r2}^2\epsilon_{r3}-16\eta_3\epsilon_{r1}\epsilon_{r2}\epsilon_{r3}^2-\eta_3\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{r3}^2\\
& -4\eta_3\epsilon_{r1}\epsilon_{c2}^2\epsilon_{c3}^2-\eta_3\epsilon_{c1}^2\epsilon_{r2}^2\epsilon_{c3}^2-4\eta_3\epsilon_{c1}^2\epsilon_{r2}\epsilon_{r3}^2-4\eta_3\epsilon_{c1}^2\epsilon_{r2}\epsilon_{r3}-4\eta_3\epsilon_{c1}^2\epsilon_{r2}\epsilon_{r3}-4\eta_3\epsilon_{c1}^2\epsilon_{r2}\epsilon_{r3}^2-4\eta_3\epsilon_{c1}^2\epsilon_{r2}\epsilon_{c3}^2-4\eta_3\epsilon_{c1}^2\epsilon_{r2}\epsilon_{c3}^2+2\eta_2^2\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{c3}^2+2\eta_2^2\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{r3}^2\\
& +8\eta_2^2\epsilon_{r1}\epsilon_{r2}^2\epsilon_{r3}^2+2\eta_2^2\epsilon_{c1}^2\epsilon_{r2}^2\epsilon_{r3}^2+32\eta_2^2\epsilon_{r1}\epsilon_{r2}^2\epsilon_{r3}-4\eta_2^2\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{r3}-16
\end{aligned}$$

$$\begin{aligned}
& \eta_2^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 + 8 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2} + 2 \epsilon_{c2}^2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{c3}^2 + 8 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1} \epsilon_{c2}^2 - 16 \eta_2^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 \\
& - 4 \epsilon_{c1}^2 \epsilon_{c2}^2 - 16 \epsilon_{r1} \epsilon_{r3}^2 - 64 \epsilon_{r1} \epsilon_{r3} - 16 \epsilon_{r1} \epsilon_{c3}^2 - 16 \epsilon_{r2} \epsilon_{r3}^2 - 64 \epsilon_{r2} \epsilon_{r3} - 16 \epsilon_{r2} \\
& \epsilon_{c3}^2 - 4 \epsilon_{r1}^2 \epsilon_{r3}^2 - 16 \epsilon_{r1}^2 \epsilon_{r3} - 4 \epsilon_{r1}^2 \epsilon_{c3}^2 - 4 \epsilon_{r2}^2 \epsilon_{r3}^2 - 16 \epsilon_{r2}^2 \epsilon_{r3} - 4 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \epsilon_{c1}^2 \epsilon_{r3}^2 \\
& - 16 \epsilon_{c1}^2 \epsilon_{r3} - 4 \epsilon_{c1}^2 \epsilon_{c3}^2 - 4 \epsilon_{c2}^2 \epsilon_{r3}^2 - 16 \epsilon_{c2}^2 \epsilon_{r3} - 4 \epsilon_{c2}^2 \epsilon_{c3}^2 - 16 \eta_1 \epsilon_{r1} + 32 \eta_1 \epsilon_{r2} \\
& + 32 \eta_1 \epsilon_{r3} - 16 \eta_1 \epsilon_{r1}^2 + 8 \eta_1 \epsilon_{r2}^2 + 8 \eta_1 \epsilon_{r3}^2 - 16 \eta_1 \epsilon_{c1}^2 + 8 \eta_1 \epsilon_{c2}^2 + 8 \eta_1 \epsilon_{c3}^2 \\
& + 32 \eta_2 \epsilon_{r1} - 16 \eta_2 \epsilon_{r2} + 32 \eta_2 \epsilon_{r3} + 8 \eta_2 \epsilon_{r1}^2 - 16 \eta_2 \epsilon_{r2}^2 + 8 \eta_2 \epsilon_{r3}^2 + 8 \eta_2 \epsilon_{c1}^2 \\
& - 16 \eta_2 \epsilon_{c2}^2 + 8 \eta_2 \epsilon_{c3}^2 + 32 \eta_3 \epsilon_{r1} + 32 \eta_3 \epsilon_{r2} - 16 \eta_3 \epsilon_{r3} + 8 \eta_3 \epsilon_{r1}^2 + 8 \eta_3 \epsilon_{r2}^2 \\
& - 16 \eta_3 \epsilon_{r3}^2 + 8 \eta_3 \epsilon_{c1}^2 + 8 \eta_3 \epsilon_{c2}^2 - 16 \eta_3 \epsilon_{c3}^2 + 32 \eta_1^2 \epsilon_{r1}^2 + 32 \eta_2^2 \epsilon_{r2}^2 + 32 \epsilon_{c2}^2 \eta_2^2 \\
& + 64 \eta_2 \eta_1 + 64 \eta_3 \eta_1 + 64 \eta_2 \eta_3 + 32 \eta_3^2 \epsilon_{r3}^2 - 64 \eta_2^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} + 8 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \\
& + 8 \eta_2^2 \epsilon_{r2}^2 \epsilon_{r1} \epsilon_{c3}^2 + 2 \epsilon_{c3}^2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{c2}^2 + 2 \epsilon_{c1}^2 \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 2 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 \\
& \epsilon_{c2}^2 + 8 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1} \epsilon_{r3}^2 + 8 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2} \epsilon_{c3}^2 + 2 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 8 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} + 2 \eta_2^2 \epsilon_{c1}^2 \\
& \epsilon_{r2}^2 \epsilon_{c3}^2 + 8 \eta_3^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 + 8 \epsilon_{c2}^2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r3} + 16 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \eta_2 + 16 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \eta_2 + 8 \\
& \eta_3^2 \epsilon_{r3}^2 \epsilon_{r1} \epsilon_{r2} + 2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 2 \eta_3^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \epsilon_{r1}^2 - 8 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \eta_1 + 8 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \\
& + 2 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 2 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 32 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} + 8 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 2 \eta_1^2 \epsilon_{r1}^2 \\
& \epsilon_{c2}^2 \epsilon_{c3}^2 + 8 \eta_1^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 2 \eta_1^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 16 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} \eta_3 + 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_1 \\
& + 16 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_1 + 4 \eta_2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_3 + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_1 \\
& + 16 \eta_3 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} \eta_1 + 4 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} \eta_1 + 4 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_1 + 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_3 \\
& + 4 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} \eta_1 + 4 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} \eta_1 + 16 \eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} \eta_1 + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 \\
& + 4 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 16 \eta_1 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 \eta_3 + 4 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_2 \\
& + 4 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_3 + 16 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 16 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} \eta_2 + 16 \eta_1 \epsilon_{r1} \epsilon_{r2} \\
& \epsilon_{c3}^2 \eta_3 + 4 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_2 + 4 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_3 + 4 \eta_2 \epsilon_{r3}^2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 - 16 \eta_1^2 \epsilon_{r1} \epsilon_{r2}^2 \\
& - 64 \eta_1^2 \epsilon_{r1} \epsilon_{r2} - 16 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} - 16 \eta_1^2 \epsilon_{r1} \epsilon_{c2}^2 - 16 \eta_1^2 \epsilon_{r1} \epsilon_{r3}^2 - 64 \eta_1^2 \epsilon_{r1} \epsilon_{r3} - 16 \\
& \eta_1^2 \epsilon_{r1} \epsilon_{c3}^2 + 16 \eta_2 \epsilon_{r3}^2 \eta_1 + 32 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1} + 16 \eta_3 \epsilon_{r2}^2 \eta_1 - 32 \eta_2 \epsilon_{r1}^2 \eta_1 + 8 \epsilon_{c2}^2 \eta_2^2 \epsilon_{c1}^2 \\
& - 32 \eta_3 \epsilon_{r1}^2 \eta_1 + 8 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r3}^2 - 4 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 4 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} - \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \\
& - 4 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 - \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 16 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 - \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 64 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \\
& - 16 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 16 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} - 4 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 4 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \epsilon_{c1}^2 \\
& \epsilon_{r2}^2 \epsilon_{r3} - 16 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - 4 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 4 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 - \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 16
\end{aligned}$$

$$\begin{aligned}
& \epsilon_{cl}^2 \epsilon_{r2} \epsilon_{r3} - 4 \epsilon_{cl}^2 \epsilon_{r2} \epsilon_{c3}^2 - \epsilon_{cl}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 4 \epsilon_{cl}^2 \epsilon_{c2}^2 \epsilon_{r3} - \epsilon_{cl}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 4 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \\
& - 16 \eta_1 \epsilon_{r1}^2 \epsilon_{r2} - 4 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 - 4 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 - 16 \eta_1 \epsilon_{r1} \epsilon_{r2} - 4 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 - 4 \eta_1 \epsilon_{c1}^2 \\
& \epsilon_{r2}^2 - 16 \eta_1 \epsilon_{c1}^2 \epsilon_{r2} - 4 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 - 4 \eta_1 \epsilon_{r1} \epsilon_{r3}^2 + 4 \eta_2 \epsilon_{r3}^2 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 + 4 \eta_2 \epsilon_{r3}^2 \eta_1 \epsilon_{c1}^2 \\
& \epsilon_{c2}^2 + 4 \eta_2 \epsilon_{c3}^2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 + 4 \eta_2 \epsilon_{c3}^2 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 + 4 \eta_2 \epsilon_{c3}^2 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 + 4 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_2 \\
& + 4 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_3 + 16 \eta_2 \epsilon_{r3} \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 + 16 \eta_2 \epsilon_{r3} \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 + 16 \eta_2 \epsilon_{r3} \eta_1 \epsilon_{c1}^2 \\
& \epsilon_{c2}^2 + 4 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_2 + 4 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_3 + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} \eta_1 + 16 \eta_3 \\
& \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} \eta_1 + 4 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 \eta_1 + 4 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 \eta_1 + 16 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} \eta_1 \\
& + 4 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 \eta_1 + 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 \eta_1 + 4 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 \eta_1 + 16 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} \eta_1 \\
& + 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 \eta_1 + 16 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} \eta_1 + 16 \eta_3 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \eta_3 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \\
& - 8 \eta_2 \epsilon_{r2} \epsilon_{c3}^2 \eta_1 - 64 \eta_1^2 \epsilon_{r1} + 32 \eta_1^2 \epsilon_{r2} + 32 \eta_1^2 \epsilon_{r3} + 8 \eta_1^2 \epsilon_{r2}^2 + 8 \eta_1^2 \epsilon_{r3}^2 + 8 \eta_1^2 \\
& \epsilon_{c2}^2 + 8 \eta_1^2 \epsilon_{c3}^2 + 8 \eta_2^2 \epsilon_{r3}^2 \epsilon_{r2}^2 - 16 \eta_1 \epsilon_{r1} \epsilon_{r3} - 4 \eta_1 \epsilon_{r1} \epsilon_{c3}^2 + 8 \eta_1 \epsilon_{r2} \epsilon_{r3}^2 \\
& + 32 \eta_1 \epsilon_{r2} \epsilon_{r3} + 8 \eta_1 \epsilon_{r2} \epsilon_{c3}^2 - 4 \eta_1 \epsilon_{r1}^2 \epsilon_{r3}^2 - 16 \eta_1 \epsilon_{r1}^2 \epsilon_{r3} - 4 \eta_1 \epsilon_{r1}^2 \epsilon_{c3}^2 + 2 \eta_1 \epsilon_{r2}^2 \\
& \epsilon_{r3}^2 + 8 \eta_1 \epsilon_{r2}^2 \epsilon_{r3} + 2 \eta_1 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \eta_1 \epsilon_{c1}^2 \epsilon_{r3}^2 - 16 \eta_1 \epsilon_{c1}^2 \epsilon_{r3} - 4 \eta_1 \epsilon_{c1}^2 \epsilon_{c3}^2 + 2 \eta_1 \\
& \epsilon_{c2}^2 \epsilon_{r3}^2 + 8 \eta_1 \epsilon_{c2}^2 \epsilon_{r3} + 2 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \\
& - 16 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 - 16 \eta_2 \epsilon_{r1} \epsilon_{r2} - 16 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 - 4 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 - 4 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} - 4 \eta_2 \\
& \epsilon_{c1}^2 \epsilon_{c2}^2 + 8 \eta_2 \epsilon_{r1} \epsilon_{r3}^2 + 32 \eta_2 \epsilon_{r1} \epsilon_{r3} + 8 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{r2} \epsilon_{r3}^2 - 16 \eta_2 \epsilon_{r2} \epsilon_{r3} \\
& - 4 \eta_2 \epsilon_{r2} \epsilon_{c3}^2 + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r3} + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 16 \eta_2 \\
& \epsilon_{r2}^2 \epsilon_{r3} - 4 \eta_2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r3}^2 + 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r3} + 2 \eta_2 \epsilon_{c1}^2 \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{c2}^2 \epsilon_{r3}^2 \\
& - 16 \eta_2 \epsilon_{c2}^2 \epsilon_{r3} - 4 \eta_2 \epsilon_{c2}^2 \epsilon_{c3}^2 + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 + 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 + 8 \eta_3 \epsilon_{r1} \\
& \epsilon_{r2}^2 + 32 \eta_3 \epsilon_{r1} \epsilon_{r2} + 8 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 + 2 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 + 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r2} + 2 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \\
& - 16 \eta_3 \epsilon_{r1} \epsilon_{r3}^2 - 16 \eta_3 \epsilon_{r1} \epsilon_{r3} - 16 \eta_3 \epsilon_{r1} \epsilon_{c3}^2 - 16 \eta_3 \epsilon_{r2} \epsilon_{r3}^2 - 16 \eta_3 \epsilon_{r2} \epsilon_{r3} \\
& - 16 \eta_3 \epsilon_{r2} \epsilon_{c3}^2 - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r3}^2 - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r3} - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{c3}^2 - 4 \eta_3 \epsilon_{r2}^2 \epsilon_{r3}^2 - 4 \eta_3 \\
& \epsilon_{r2}^2 \epsilon_{r3} - 4 \eta_3 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \eta_3 \epsilon_{c1}^2 \epsilon_{r3}^2 - 4 \eta_3 \epsilon_{c1}^2 \epsilon_{r3} - 4 \eta_3 \epsilon_{c1}^2 \epsilon_{c3}^2 - 4 \eta_3 \epsilon_{c2}^2 \epsilon_{r3}^2 - 4 \eta_3 \\
& \epsilon_{c2}^2 \epsilon_{r3} - 4 \eta_3 \epsilon_{c2}^2 \epsilon_{c3}^2 + 8 \eta_3^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 8 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 8 \eta_3^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 8 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r3}^2 + 32 \\
& \eta_3^2 \epsilon_{r1} \epsilon_{r3}^2 + 32 \eta_3^2 \epsilon_{r2} \epsilon_{r3}^2 - 32 \eta_3 \epsilon_{r3} \eta_1 + 16 \eta_3 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 + 4 \eta_3 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 \\
& + 4 \eta_3 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 4 \eta_3 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \eta_3 \epsilon_{r2} \eta_1 \epsilon_{c1}^2 \epsilon_{c3}^2 + 4 \eta_3 \epsilon_{r2}^2 \eta_1 \epsilon_{c1}^2 \epsilon_{c3}^2 \\
& + 4 \eta_3 \epsilon_{c1}^2 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \eta_3 \epsilon_{c1}^2 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 + 16 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 \eta_3 + 4 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 \eta_3
\end{aligned}$$

$$\begin{aligned}
& + 4 \eta_2 \epsilon_{cl}^2 \epsilon_{r2} \epsilon_{r3} \eta_3 + 4 \eta_2 \epsilon_{cl}^2 \epsilon_{r2} \epsilon_{c3}^2 \eta_3 + 16 \eta_3 \epsilon_{rl}^2 \eta_1 \epsilon_{r2} \epsilon_{r3}^2 + 16 \eta_3 \epsilon_{cl}^2 \eta_1 \epsilon_{r2} \epsilon_{r3}^2 \\
& + 16 \eta_3 \eta_2 \epsilon_{rl} \epsilon_{c2}^2 \epsilon_{c3}^2 + 4 \eta_3 \eta_2 \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \eta_3 \eta_2 \epsilon_{cl}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \eta_3 \eta_2 \epsilon_{cl}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 \\
& + 16 \eta_3 \epsilon_{r2}^2 \eta_2 \epsilon_{rl} \epsilon_{c3}^2 + 4 \eta_3 \epsilon_{rl}^2 \epsilon_{r2}^2 \eta_1 \epsilon_{r3}^2 + 4 \eta_3 \epsilon_{rl}^2 \epsilon_{r2}^2 \eta_1 \epsilon_{c3}^2 + 16 \eta_3 \epsilon_{rl}^2 \epsilon_{r2} \eta_1 \epsilon_{c3}^2 \\
& + 4 \eta_3 \epsilon_{rl}^2 \epsilon_{c2}^2 \eta_1 \epsilon_{r3}^2 + 4 \eta_3 \epsilon_{rl}^2 \epsilon_{c2}^2 \eta_1 \epsilon_{c3}^2 + 4 \eta_3 \epsilon_{cl}^2 \epsilon_{r2}^2 \eta_1 \epsilon_{r3}^2 + 16 \eta_3 \epsilon_{rl} \epsilon_{r2}^2 \epsilon_{r3} \eta_2 \\
& + 16 \eta_3 \epsilon_{rl} \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 4 \eta_3 \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 - 32 \eta_3 \epsilon_{rl} \eta_1 - 32 \eta_2 \epsilon_{cl}^2 \eta_1 + 16 \eta_2 \\
& \epsilon_{c3}^2 \eta_1 - 32 \eta_3 \epsilon_{cl}^2 \eta_1 + 16 \eta_3 \epsilon_{c2}^2 \eta_1 - 32 \eta_2 \epsilon_{rl} \eta_1 + 8 \epsilon_{c2}^2 \eta_2^2 \epsilon_{rl}^2 + 8 \epsilon_{cl}^2 \eta_1^2 \epsilon_{c2}^2 \\
& + 32 \epsilon_{c3}^2 \eta_3^2 \epsilon_{rl} + 8 \epsilon_{cl}^2 \eta_1^2 \epsilon_{c3}^2 + 2 \eta_1^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + 2 \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 8 \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3} + 2 \eta_1^2 \epsilon_{r2}^2 \\
& \epsilon_{c3}^2 + 8 \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3} + 2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 32 \eta_1^2 \epsilon_{r2} \epsilon_{r3} + 8 \eta_1^2 \epsilon_{r2} \epsilon_{c3}^2 + 8 \eta_1^2 \epsilon_{r2} \epsilon_{r3}^2 + 32 \epsilon_{c2}^2 \\
& \eta_2^2 \epsilon_{r3} + 32 \epsilon_{cl}^2 \eta_1^2 \epsilon_{r3} - 32 \eta_2 \epsilon_{r2} \eta_3 - 32 \eta_2 \epsilon_{r3} \eta_3 + 16 \eta_2 \epsilon_{rl}^2 \eta_3 - 32 \eta_2 \epsilon_{r3}^2 \eta_3 \\
& + 32 \eta_1^2 \epsilon_{rl}^2 \epsilon_{r2} + 8 \eta_1^2 \epsilon_{rl}^2 \epsilon_{c2}^2 + 8 \epsilon_{c3}^2 \eta_3^2 \epsilon_{c2}^2 + 8 \eta_1^2 \epsilon_{rl}^2 \epsilon_{r2}^2 - 32 \eta_1 \epsilon_{r2}^2 \eta_2 - 32 \eta_1 \\
& \epsilon_{r3}^2 \eta_3 - 32 \eta_1 \epsilon_{c2}^2 \eta_2 - 32 \eta_1 \epsilon_{c3}^2 \eta_3 + 32 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r2} + 64 \eta_2 \epsilon_{r3} \eta_1 + 8 \eta_2^2 \epsilon_{rl}^2 \epsilon_{r2}^2 \\
& + 8 \eta_3^2 \epsilon_{rl}^2 \epsilon_{r2} + 2 \eta_3^2 \epsilon_{rl}^2 \epsilon_{c2}^2 + 8 \eta_3^2 \epsilon_{rl} \epsilon_{r2}^2 - 8 \eta_2 \epsilon_{rl} \epsilon_{c3}^2 \eta_1 - 8 \eta_2 \epsilon_{r2} \epsilon_{r3}^2 \eta_1 \\
& - 32 \eta_2 \epsilon_{r2} \epsilon_{r3} \eta_1 - 64 \eta_3^2 \epsilon_{rl} \epsilon_{r2} \epsilon_{r3} - 16 \eta_3^2 \epsilon_{rl} \epsilon_{r2}^2 \epsilon_{r3} - 16 \eta_3^2 \epsilon_{rl} \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_2 \\
& \epsilon_{cl}^2 \epsilon_{r3} \eta_3 - 8 \eta_2 \epsilon_{cl}^2 \epsilon_{c3}^2 \eta_3 + 8 \epsilon_{cl}^2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_2 \epsilon_{rl}^2 \epsilon_{r3}^2 \eta_3 - 8 \eta_2 \epsilon_{rl}^2 \epsilon_{r3} \eta_3 \\
& - 8 \eta_2 \epsilon_{rl}^2 \epsilon_{c3}^2 \eta_3 - 8 \eta_2 \epsilon_{cl}^2 \epsilon_{r3}^2 \eta_3 + 16 \eta_2 \epsilon_{rl}^2 \epsilon_{r2} \eta_1 + 16 \eta_2 \epsilon_{rl} \epsilon_{r2} \eta_1 + 16 \eta_2 \epsilon_{r2} \\
& \epsilon_{r3}^2 \eta_3 + 16 \eta_2 \epsilon_{r2} \epsilon_{r3} \eta_3 + 16 \eta_2 \epsilon_{r2} \epsilon_{c3}^2 \eta_3 - 16 \eta_1^2 \epsilon_{rl} \epsilon_{r2} \epsilon_{r3}^2 - 4 \eta_1^2 \epsilon_{rl} \epsilon_{c2}^2 \epsilon_{c3}^2 + 32 \\
& \epsilon_{cl}^2 \eta_1^2 \epsilon_{r2} \epsilon_{r3} - 4 \eta_1^2 \epsilon_{rl} \epsilon_{r2}^2 \epsilon_{c3}^2 - 64 \eta_1^2 \epsilon_{rl} \epsilon_{r2} \epsilon_{r3} - 16 \eta_1^2 \epsilon_{rl} \epsilon_{r2}^2 \epsilon_{r3} - 16 \eta_1^2 \epsilon_{rl} \\
& \epsilon_{c2}^2 \epsilon_{r3} - 4 \eta_1^2 \epsilon_{rl} \epsilon_{c2}^2 \epsilon_{r3}^2 - 4 \eta_1^2 \epsilon_{rl} \epsilon_{r2}^2 \epsilon_{r3}^2 - 16 \eta_1^2 \epsilon_{rl} \epsilon_{r2} \epsilon_{c3}^2 - 8 \eta_3 \epsilon_{r2}^2 \epsilon_{r3} \eta_1 \\
& + 16 \eta_3 \epsilon_{cl}^2 \epsilon_{r3} \eta_1 - 8 \eta_3 \epsilon_{c2}^2 \epsilon_{r3} \eta_1 - 32 \eta_3 \epsilon_{cl}^2 \epsilon_{r2} \eta_1 - 8 \eta_3 \epsilon_{cl}^2 \epsilon_{c2}^2 \eta_1 \\
& + 16 \eta_3 \epsilon_{rl} \epsilon_{r3} \eta_1 - 32 \eta_3 \epsilon_{r2} \epsilon_{r3} \eta_1 + 16 \eta_3 \epsilon_{rl}^2 \epsilon_{r3} \eta_1 - 8 \eta_3 \epsilon_{cl}^2 \epsilon_{r2}^2 \eta_1 + 2 \epsilon_{c2}^2 \eta_2^2 \\
& \epsilon_{cl}^2 \epsilon_{r3}^2 - 4 \eta_2^2 \epsilon_{rl}^2 \epsilon_{r2} \epsilon_{c3}^2 - 16 \eta_2^2 \epsilon_{rl}^2 \epsilon_{r2} \epsilon_{r3} + 8 \epsilon_{cl}^2 \eta_1^2 \epsilon_{r2} \epsilon_{r3}^2 - 32 \eta_1 \epsilon_{r2} \epsilon_{r3}^2 \eta_3 \\
& - 32 \eta_1 \epsilon_{r2} \epsilon_{c3}^2 \eta_3 + 16 \eta_3 \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 16 \eta_1 \epsilon_{rl} \epsilon_{r3}^2 \eta_3 + 16 \eta_1 \epsilon_{rl} \epsilon_{c3}^2 \eta_3 + 8 \epsilon_{c3}^2 \eta_3^2 \\
& \epsilon_{cl}^2 \epsilon_{r2} + 16 \eta_2 \eta_1 \epsilon_{rl}^2 \epsilon_{c2}^2 + 16 \eta_2 \eta_1 \epsilon_{rl}^2 \epsilon_{r2}^2 + 2 \epsilon_{c3}^2 \eta_3^2 \epsilon_{cl}^2 \epsilon_{r2}^2 + 8 \epsilon_{c2}^2 \eta_2^2 \epsilon_{rl} \epsilon_{c3}^2 - 4 \\
& \eta_2^2 \epsilon_{cl}^2 \epsilon_{r2} \epsilon_{r3}^2 - 16 \eta_2^2 \epsilon_{cl}^2 \epsilon_{r2} \epsilon_{r3} - 4 \eta_2^2 \epsilon_{cl}^2 \epsilon_{r2} \epsilon_{c3}^2 + 8 \epsilon_{cl}^2 \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3} + 2 \epsilon_{cl}^2 \eta_1^2 \epsilon_{c2}^2 \epsilon_{c3}^2 \\
& - 8 \eta_2 \epsilon_{rl}^2 \epsilon_{r2} \eta_3 - 32 \eta_2 \epsilon_{rl} \epsilon_{r2} \eta_3 - \eta_3 \epsilon_{rl}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 16 \eta_1 \epsilon_{rl}^2 \epsilon_{r2} \epsilon_{r3} + 32 \eta_3^2 \epsilon_{rl} \epsilon_{r2} \\
& + 8 \eta_3^2 \epsilon_{rl} \epsilon_{c2}^2 + 2 \eta_3^2 \epsilon_{cl}^2 \epsilon_{r2}^2 + 8 \eta_3^2 \epsilon_{cl}^2 \epsilon_{r2} + 2 \eta_3^2 \epsilon_{cl}^2 \epsilon_{c2}^2 - 64 \eta_3^2 \epsilon_{rl} \epsilon_{r3} - 64 \\
& \eta_3^2 \epsilon_{r2} \epsilon_{r3} + 2 \eta_2^2 \epsilon_{rl}^2 \epsilon_{r3}^2 + 8 \eta_2^2 \epsilon_{rl}^2 \epsilon_{r3} + 2 \eta_2^2 \epsilon_{rl}^2 \epsilon_{c3}^2 + 2 \eta_2^2 \epsilon_{cl}^2 \epsilon_{r3}^2 + 8 \eta_2^2 \epsilon_{cl}^2 \epsilon_{r3} + 2
\end{aligned}$$

$$\begin{aligned}
& \eta_2^2 \epsilon_{cl}^2 \epsilon_{c3}^2 + 2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 - 16 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2} - 64 \eta_2^2 \epsilon_{r1} \epsilon_{r2} - 16 \eta_2^2 \epsilon_{cl}^2 \epsilon_{r2} + 8 \eta_2^2 \epsilon_{r1} \epsilon_{r3}^2 \\
& + 32 \eta_2^2 \epsilon_{r1} \epsilon_{r3} + 8 \eta_2^2 \epsilon_{r1} \epsilon_{c3}^2 - 16 \eta_2^2 \epsilon_{r2} \epsilon_{r3}^2 - 64 \eta_2^2 \epsilon_{r2} \epsilon_{r3} - 16 \eta_2^2 \epsilon_{r2} \epsilon_{c3}^2 + 32 \\
& \epsilon_{cl}^2 \eta_1^2 \epsilon_{r2} - 32 \eta_3 \epsilon_{r2}^2 \eta_2 - 32 \eta_3 \epsilon_{c2}^2 \eta_2 + 8 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 8 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 + 8 \eta_1^2 \epsilon_{r1}^2 \epsilon_{c3}^2 \\
& + 32 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r3} + 32 \eta_2^2 \epsilon_{r1} \epsilon_{r2}^2 + 8 \epsilon_{cl}^2 \eta_1^2 \epsilon_{r2}^2 + 32 \eta_2^2 \epsilon_{r3} \epsilon_{r2}^2 + 64 \eta_2 \epsilon_{r1} \eta_3 - 16 \eta_3^2 \\
& \epsilon_{c2}^2 \epsilon_{r3} + 16 \eta_2 \epsilon_{cl}^2 \eta_3 - 32 \eta_2 \epsilon_{c3}^2 \eta_3 + 64 \eta_3 \epsilon_{r2} \eta_1 - \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 4 \eta_1 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 \\
& - 4 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} - \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 4 \eta_1 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 \\
& - \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 16 \eta_1 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 4 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - \eta_1 \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \\
& - 4 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} - 4 \eta_1 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 - \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \eta_1 \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{r3} - 4 \eta_1 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 \\
& - 4 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 - \eta_1 \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \eta_1 \epsilon_{cl}^2 \epsilon_{r2} \epsilon_{r3}^2 - 16 \eta_1 \epsilon_{cl}^2 \epsilon_{r2} \epsilon_{r3} \\
& - 4 \eta_1 \epsilon_{cl}^2 \epsilon_{r2} \epsilon_{c3}^2 - \eta_1 \epsilon_{cl}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 4 \eta_1 \epsilon_{cl}^2 \epsilon_{c2}^2 \epsilon_{r3} - \eta_1 \epsilon_{cl}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} \\
& - 4 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - \eta_2 \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 4 \eta_2 \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{r3} - 16 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} \\
& - 4 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 16 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r3}^2) / (64 + 4 \epsilon_{c2}^2 \eta_2^2 \\
& \epsilon_{c3}^2 + 64 \eta_1 + 64 \eta_2 + 64 \eta_3 + 64 \epsilon_{r1} + 64 \epsilon_{r2} + 64 \epsilon_{r3} + 16 \epsilon_{cl}^2 + 16 \epsilon_{c2}^2 + 16 \epsilon_{c3}^2 \\
& - 8 \eta_3^2 \epsilon_{cl}^2 \epsilon_{r3} - 8 \eta_3^2 \epsilon_{r2}^2 \epsilon_{r3} + 18 \epsilon_{cl} \eta_1 \epsilon_{c3} \eta_3 \epsilon_{r2}^2 + 4 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r3}^2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r2}^2 + 4 \\
& \eta_2^2 \epsilon_{cl}^2 \epsilon_{r2}^2 - 16 \eta_2 \epsilon_{r2} \eta_1 + 8 \eta_3 \epsilon_{r1}^2 \eta_1 \epsilon_{r3}^2 - 16 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \eta_2 - 16 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \eta_2 \\
& - 4 \eta_3 \epsilon_{cl}^2 \epsilon_{r2}^2 \eta_2 - 8 \eta_3^2 \epsilon_{cl}^2 \epsilon_{r2} \epsilon_{r3} - 2 \eta_3^2 \epsilon_{cl}^2 \epsilon_{c2}^2 \epsilon_{r3} - 4 \eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_2 - 4 \eta_1 \epsilon_{r2}^2 \\
& \epsilon_{r3}^2 \eta_3 - 16 \eta_1 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 + 8 \eta_3 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 + 16 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1} \epsilon_{r3} + 4 \\
& \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3} + 16 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1} \epsilon_{r2} + 8 \eta_2 \eta_1 \epsilon_{cl}^2 \epsilon_{r2}^2 + 8 \eta_2 \eta_1 \epsilon_{cl}^2 \epsilon_{c2}^2 - 4 \eta_2 \epsilon_{cl}^2 \epsilon_{r2} \eta_3 \\
& - 16 \eta_2 \epsilon_{r1} \epsilon_{r3}^2 \eta_3 - 16 \eta_2 \epsilon_{r1} \epsilon_{r3} \eta_3 - 16 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 \eta_3 + \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{c3}^2 + 8 \eta_2 \\
& \epsilon_{cl}^2 \epsilon_{r2} \eta_1 - 4 \eta_2 \epsilon_{r1} \epsilon_{r3}^2 \eta_1 - 16 \eta_2 \epsilon_{r1} \epsilon_{r3} \eta_1 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{cl}^2 + 8 \eta_3 \epsilon_{c2}^2 \eta_2 \epsilon_{r3}^2 - 4 \eta_2 \\
& \epsilon_{cl}^2 \epsilon_{r3}^2 \eta_1 - 16 \eta_2 \epsilon_{cl}^2 \epsilon_{r3} \eta_1 - 4 \eta_2 \epsilon_{cl}^2 \epsilon_{c3}^2 \eta_1 + 8 \eta_3 \epsilon_{r2}^2 \eta_2 \epsilon_{c3}^2 + 8 \eta_3 \epsilon_{r2}^2 \eta_2 \epsilon_{r3}^2 \\
& - 4 \eta_1 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_2 - 4 \eta_1 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_3 - 4 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_2 - 4 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_3 - 16 \eta_1 \\
& \epsilon_{c2}^2 \epsilon_{r3} \eta_2 - 4 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_2 - 4 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_3 - 4 \eta_3 \epsilon_{cl}^2 \epsilon_{c2}^2 \eta_2 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r3}^2 \eta_1 \\
& - 16 \eta_2 \epsilon_{r1}^2 \epsilon_{r3} \eta_1 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{c3}^2 \eta_1 + \epsilon_{cl}^2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \\
& + 8 \eta_3 \eta_1 \epsilon_{cl}^2 \epsilon_{r3}^2 + 8 \eta_3 \eta_1 \epsilon_{cl}^2 \epsilon_{c3}^2 + 16 \eta_3^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 + 4 \eta_3^2 \epsilon_{cl}^2 \epsilon_{r2} \epsilon_{r3}^2 + \eta_3^2 \epsilon_{cl}^2 \epsilon_{c2}^2 \\
& \epsilon_{r3}^2 - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_2 - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_2 - 2 \eta_3^2 \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{r3} - 2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_3^2 \\
& \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} + 8 \eta_3 \epsilon_{r1}^2 \eta_1 \epsilon_{c3}^2 + 4 \eta_2^2 \epsilon_{c3}^2 \epsilon_{r2}^2 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 16 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \eta_1 - 4 \eta_3 \\
& \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_1 - 4 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \eta_1 - 16 \eta_3 \epsilon_{r1} \epsilon_{r2} \eta_1 + 72 \epsilon_{cl} \eta_1 \epsilon_{c3} \eta_3 + 72 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3
\end{aligned}$$

$$\begin{aligned}
& -4\eta_3\epsilon_{r1}^2\epsilon_{r2}^2\eta_1 + 72\epsilon_{c1}\eta_1\epsilon_{c2}\eta_2 - 2\eta_3^2\epsilon_{r1}^2\epsilon_{c2}^2\epsilon_{r3} + 4\eta_3^2\epsilon_{r1}\epsilon_{r2}^2\epsilon_{r3}^2 + 8\eta_3\epsilon_{c2}^2\eta_2 \\
& \epsilon_{c3}^2 + 16\epsilon_{c1}^2\eta_1^2 + 16\eta_3^2\epsilon_{r1} + 16\eta_3^2\epsilon_{r2} - 32\eta_3^2\epsilon_{r3} + 4\eta_3^2\epsilon_{r1}^2 + 4\eta_3^2\epsilon_{r2}^2 + 4\eta_3^2\epsilon_{c1}^2 \\
& + 4\eta_3^2\epsilon_{c2}^2 + 16\eta_2^2\epsilon_{r1} - 32\eta_2^2\epsilon_{r2} + 16\eta_2^2\epsilon_{r3} + 4\eta_2^2\epsilon_{r1}^2 + 4\eta_2^2\epsilon_{r2}^2 + 4\eta_2^2\epsilon_{c1}^2 + 4 \\
& \eta_2^2\epsilon_{c3}^2 + 16\epsilon_{c3}^2\eta_3^2 + 4\epsilon_{r1}^2\epsilon_{r2}^2 + 16\epsilon_{r1}^2\epsilon_{r2} + 4\epsilon_{r1}^2\epsilon_{c2}^2 + 16\epsilon_{r1}\epsilon_{r2}^2 + 64\epsilon_{r1}\epsilon_{r2} \\
& + 16\epsilon_{r1}\epsilon_{c2}^2 + 4\epsilon_{c1}^2\epsilon_{r2}^2 + 16\epsilon_{c1}^2\epsilon_{r2} + 16\epsilon_{r1}^2 + 16\epsilon_{r2}^2 + 16\epsilon_{r3}^2 + 16\eta_1^2 + 16\eta_2^2 \\
& + 16\eta_3^2 - 2\eta_2\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{c3}^2 - 8\eta_2\epsilon_{r1}\epsilon_{r2}^2\epsilon_{r3}^2 - 2\eta_2\epsilon_{r1}^2\epsilon_{c2}^2\epsilon_{c3}^2 - 8\eta_2\epsilon_{r1}\epsilon_{r2}\epsilon_{c3}^2 \\
& - 2\eta_2\epsilon_{r1}^2\epsilon_{c2}^2\epsilon_{r3}^2 - 32\eta_2\epsilon_{r1}\epsilon_{r2}\epsilon_{r3} - 2\eta_2\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{r3}^2 - 32\eta_2\epsilon_{r1}\epsilon_{r2}^2\epsilon_{r3} - 2\eta_2\epsilon_{c1}^2 \\
& \epsilon_{r2}^2\epsilon_{r3}^2 - 32\eta_2\epsilon_{r1}\epsilon_{c2}^2\epsilon_{r3} - 2\eta_2\epsilon_{r1}^2\epsilon_{r2}\epsilon_{r3}^2 - 8\eta_2\epsilon_{r1}\epsilon_{r2}^2\epsilon_{c3}^2 - 8\eta_2\epsilon_{c1}^2\epsilon_{r2}^2\epsilon_{r3} \\
& - 8\eta_2\epsilon_{r1}\epsilon_{r2}\epsilon_{r3}^2 - 8\eta_2\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{r3} - 8\eta_2\epsilon_{r1}\epsilon_{c2}^2\epsilon_{c3}^2 - 2\eta_2\epsilon_{c1}^2\epsilon_{r2}^2\epsilon_{c3}^2 - 2\eta_2\epsilon_{c1}^2\epsilon_{r2} \\
& \epsilon_{r3}^2 - 8\eta_2\epsilon_{c1}^2\epsilon_{r2}\epsilon_{r3} - 2\eta_2\epsilon_{c1}^2\epsilon_{r2}\epsilon_{c3}^2 - 2\eta_2\epsilon_{c1}^2\epsilon_{c2}^2\epsilon_{r3}^2 - 8\eta_2\epsilon_{c1}^2\epsilon_{c2}^2\epsilon_{r3} - 2\eta_2\epsilon_{c1}^2 \\
& \epsilon_{c2}^2\epsilon_{c3}^2 - 8\eta_3\epsilon_{r1}^2\epsilon_{r2}\epsilon_{r3} - 8\eta_3\epsilon_{r1}\epsilon_{c2}^2\epsilon_{r3}^2 - 8\eta_3\epsilon_{r1}^2\epsilon_{r2}\epsilon_{c3}^2 - 2\eta_3\epsilon_{r1}^2\epsilon_{c2}^2\epsilon_{r3} - 2\eta_3 \\
& \epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{c3}^2 - 8\eta_3\epsilon_{r1}\epsilon_{r2}^2\epsilon_{r3}^2 - 32\eta_3\epsilon_{r1}\epsilon_{r2}\epsilon_{c3}^2 + 18\epsilon_{c1}\eta_1\epsilon_{c3}\eta_3\epsilon_{c2}^2 \\
& + 72\epsilon_{c2}\eta_2\epsilon_{c3}\eta_3\epsilon_{r1} + 18\epsilon_{c2}\eta_2\epsilon_{c3}\eta_3\epsilon_{r1}^2 + 18\epsilon_{c2}\eta_2\epsilon_{c3}\eta_3\epsilon_{c1}^2 + 2\eta_2\epsilon_{r1}^2\epsilon_{r2} \\
& \epsilon_{r3}^2\eta_3 + 72\epsilon_{c1}\eta_1\epsilon_{c2}\eta_2\epsilon_{r3} + 18\epsilon_{c1}\eta_1\epsilon_{c2}\eta_2\epsilon_{r3}^2 + 18\epsilon_{c1}\eta_1\epsilon_{c2}\eta_2\epsilon_{c3}^2 \\
& + 72\epsilon_{c1}\eta_1\epsilon_{c3}\eta_3\epsilon_{r2} + 2\eta_2\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{r3}^2\eta_1 + 8\eta_2\epsilon_{r1}\epsilon_{r2}\epsilon_{c3}^2\eta_3 - 2\eta_3\epsilon_{r1}^2\epsilon_{c2}^2\epsilon_{r3}^2 \\
& - 32\eta_3\epsilon_{r1}\epsilon_{r2}\epsilon_{r3} - 2\eta_3\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{r3}^2 - 8\eta_3\epsilon_{r1}\epsilon_{r2}^2\epsilon_{r3} - 2\eta_3\epsilon_{c1}^2\epsilon_{r2}^2\epsilon_{r3}^2 - 8\eta_3\epsilon_{r1} \\
& \epsilon_{c2}^2\epsilon_{r3} - 8\eta_3\epsilon_{r1}^2\epsilon_{r2}\epsilon_{r3}^2 - 8\eta_3\epsilon_{r1}\epsilon_{r2}^2\epsilon_{c3}^2 - 2\eta_3\epsilon_{c1}^2\epsilon_{r2}^2\epsilon_{r3} - 32\eta_3\epsilon_{r1}\epsilon_{r2}\epsilon_{r3}^2 - 2\eta_3 \\
& \epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{r3} - 8\eta_3\epsilon_{r1}\epsilon_{c2}^2\epsilon_{c3}^2 - 2\eta_3\epsilon_{c1}^2\epsilon_{r2}^2\epsilon_{c3}^2 - 8\eta_3\epsilon_{c1}^2\epsilon_{r2}\epsilon_{r3}^2 - 8\eta_3\epsilon_{c1}^2\epsilon_{r2}\epsilon_{r3} \\
& - 8\eta_3\epsilon_{c1}^2\epsilon_{r2}\epsilon_{c3}^2 - 2\eta_3\epsilon_{c1}^2\epsilon_{c2}^2\epsilon_{r3}^2 - 2\eta_3\epsilon_{c1}^2\epsilon_{c2}^2\epsilon_{r3} - 2\eta_3\epsilon_{c1}^2\epsilon_{c2}^2\epsilon_{c3}^2 + \eta_2^2\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{c3}^2 \\
& + \eta_2^2\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{r3}^2 + 4\eta_2^2\epsilon_{r1}\epsilon_{r2}^2\epsilon_{r3}^2 + \eta_2^2\epsilon_{c1}^2\epsilon_{r2}^2\epsilon_{r3}^2 + 16\eta_2^2\epsilon_{r1}\epsilon_{r2}^2\epsilon_{r3} - 2\eta_2^2\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{r3}^2 \\
& - 8\eta_2^2\epsilon_{r1}\epsilon_{r2}\epsilon_{r3}^2 + 4\epsilon_{c3}^2\eta_3^2\epsilon_{r1}^2\epsilon_{r2} + \epsilon_{c2}^2\eta_2^2\epsilon_{c1}^2\epsilon_{c3}^2 + 4\epsilon_{c3}^2\eta_3^2\epsilon_{r1}\epsilon_{c2}^2 - 8\eta_2^2\epsilon_{r1}\epsilon_{r2}\epsilon_{c3}^2 \\
& + 4\epsilon_{c1}^2\epsilon_{c2}^2 + 16\epsilon_{r1}\epsilon_{r3}^2 + 64\epsilon_{r1}\epsilon_{r3} + 16\epsilon_{r1}\epsilon_{c3}^2 + 16\epsilon_{r2}\epsilon_{r3}^2 + 64\epsilon_{r2}\epsilon_{r3} + 16\epsilon_{r2} \\
& \epsilon_{c3}^2 + 4\epsilon_{r1}^2\epsilon_{r3}^2 + 16\epsilon_{r1}^2\epsilon_{r3} + 4\epsilon_{r1}^2\epsilon_{c3}^2 + 4\epsilon_{r2}^2\epsilon_{r3}^2 + 16\epsilon_{r2}^2\epsilon_{r3} + 4\epsilon_{r2}^2\epsilon_{c3}^2 + 4\epsilon_{c1}^2\epsilon_{r3}^2 \\
& + 16\epsilon_{c1}^2\epsilon_{r3} + 4\epsilon_{c1}^2\epsilon_{c3}^2 + 4\epsilon_{c2}^2\epsilon_{r3}^2 + 16\epsilon_{c2}^2\epsilon_{r3} + 4\epsilon_{c2}^2\epsilon_{c3}^2 - 32\eta_1\epsilon_{r1} + 64\eta_1\epsilon_{r2} \\
& + 64\eta_1\epsilon_{r3} - 32\eta_1\epsilon_{r1}^2 + 16\eta_1\epsilon_{r2}^2 + 16\eta_1\epsilon_{r3}^2 - 32\eta_1\epsilon_{c1}^2 + 16\eta_1\epsilon_{c2}^2 + 16\eta_1\epsilon_{c3}^2 \\
& + 64\eta_2\epsilon_{r1} - 32\eta_2\epsilon_{r2} + 64\eta_2\epsilon_{r3} + 16\eta_2\epsilon_{r1}^2 - 32\eta_2\epsilon_{r2}^2 + 16\eta_2\epsilon_{r3}^2 + 16\eta_2\epsilon_{c1}^2 \\
& - 32\eta_2\epsilon_{c2}^2 + 16\eta_2\epsilon_{c3}^2 + 64\eta_3\epsilon_{r1} + 64\eta_3\epsilon_{r2} - 32\eta_3\epsilon_{r3} + 16\eta_3\epsilon_{r1}^2 + 16\eta_3\epsilon_{r2}^2
\end{aligned}$$

$$\begin{aligned}
& -32\eta_3\epsilon_{r3}^2 + 16\eta_3\epsilon_{cl}^2 + 16\eta_3\epsilon_{c2}^2 - 32\eta_3\epsilon_{c3}^2 + 16\eta_1^2\epsilon_{r1}^2 + 16\eta_2^2\epsilon_{r2}^2 + 16\epsilon_{c2}^2\eta_2^2 \\
& + 32\eta_2\eta_1 + 32\eta_3\eta_1 + 32\eta_2\eta_3 + 16\eta_3^2\epsilon_{r3}^2 - 32\eta_2^2\epsilon_{r1}\epsilon_{r2}\epsilon_{r3} + 4\eta_2^2\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{r3} \\
& + 4\eta_2^2\epsilon_{r2}^2\epsilon_{r1}\epsilon_{c3}^2 + \epsilon_{c3}^2\eta_3^2\epsilon_{cl}^2\epsilon_{c2}^2 + \epsilon_{cl}^2\eta_1^2\epsilon_{c2}^2\epsilon_{r3}^2 + \eta_3^2\epsilon_{cl}^2\epsilon_{r2}^2\epsilon_{r3}^2 + \epsilon_{c3}^2\eta_3^2\epsilon_{r1}^2\epsilon_{c2}^2 + 4 \\
& \epsilon_{c2}^2\eta_2^2\epsilon_{r1}\epsilon_{r3}^2 + 4\epsilon_{cl}^2\eta_1^2\epsilon_{r2}\epsilon_{c3}^2 + \epsilon_{cl}^2\eta_1^2\epsilon_{r2}^2\epsilon_{c3}^2 + 4\eta_2^2\epsilon_{cl}^2\epsilon_{r2}^2\epsilon_{r3}^2 + \eta_2^2\epsilon_{cl}^2\epsilon_{r2}^2\epsilon_{c3}^2 + 4 \\
& \eta_3^2\epsilon_{r1}\epsilon_{c2}^2\epsilon_{r3}^2 + 4\epsilon_{c2}^2\eta_2^2\epsilon_{cl}^2\epsilon_{r3}^2 + 8\eta_1\epsilon_{r1}\epsilon_{r2}^2\eta_2 + 8\eta_1\epsilon_{r1}\epsilon_{c2}^2\eta_2 + 4\eta_3^2\epsilon_{r3}^2\epsilon_{r1}^2\epsilon_{r2}^2 + \\
& \eta_3^2\epsilon_{r1}^2\epsilon_{c2}^2\epsilon_{r3}^2 + \eta_3^2\epsilon_{r2}^2\epsilon_{r3}^2\epsilon_{r1}^2 - 4\eta_3\epsilon_{r1}\epsilon_{c2}^2\eta_1 + 4\eta_1^2\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{r3}^2 + \eta_1^2\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{r3}^2 + \eta_1^2 \\
& \epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{c3}^2 + 16\eta_1^2\epsilon_{r1}^2\epsilon_{r2}\epsilon_{r3}^2 + 4\eta_1^2\epsilon_{r1}^2\epsilon_{r2}\epsilon_{r3}^2 + \eta_1^2\epsilon_{r1}^2\epsilon_{c2}^2\epsilon_{c3}^2 + 4\eta_1^2\epsilon_{r1}^2\epsilon_{c2}^2\epsilon_{r3}^2 + \eta_1^2 \\
& \epsilon_{r1}^2\epsilon_{c2}^2\epsilon_{r3}^2 + 8\eta_2\epsilon_{r1}\epsilon_{r2}\epsilon_{r3}\eta_3 + 2\eta_2\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{c3}^2\eta_1 + 8\eta_2\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{r3}\eta_1 + 2\eta_2\eta_3\epsilon_{r1}^2 \\
& \epsilon_{c2}^2\epsilon_{c3}^2 + 2\eta_2\epsilon_{r1}^2\epsilon_{r2}\epsilon_{c3}^2\eta_3 + 2\eta_3\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{r3}\eta_1 + 8\eta_3\epsilon_{cl}^2\epsilon_{r2}\epsilon_{r3}\eta_1 + 2\eta_3\epsilon_{cl}^2 \\
& \epsilon_{c2}^2\epsilon_{r3}\eta_1 + 2\eta_3\epsilon_{cl}^2\epsilon_{r2}^2\epsilon_{r3}\eta_1 + 2\eta_2\epsilon_{r1}^2\epsilon_{r2}\epsilon_{r3}\eta_3 + 2\eta_3\epsilon_{r1}\epsilon_{c2}^2\epsilon_{r3}\eta_1 + 2\eta_3\epsilon_{r1} \\
& \epsilon_{r2}^2\epsilon_{r3}\eta_1 + 8\eta_3\epsilon_{r1}\epsilon_{r2}\epsilon_{r3}\eta_1 + 2\eta_3\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{r3}\eta_2 + 2\eta_3\epsilon_{cl}^2\epsilon_{c2}^2\epsilon_{r3}\eta_2 + 2\eta_3\epsilon_{r1}^2 \\
& \epsilon_{c2}^2\epsilon_{r3}\eta_2 + 8\eta_1\epsilon_{r1}\epsilon_{r2}\epsilon_{r3}^2\eta_3 + 2\eta_1\epsilon_{r1}\epsilon_{r2}^2\epsilon_{c3}^2\eta_2 + 2\eta_1\epsilon_{r1}\epsilon_{r2}^2\epsilon_{c3}^2\eta_3 + 8\eta_1\epsilon_{r1} \\
& \epsilon_{c2}^2\epsilon_{r3}\eta_2 + 8\eta_1\epsilon_{r1}\epsilon_{r2}^2\epsilon_{r3}\eta_2 + 8\eta_1\epsilon_{r1}\epsilon_{r2}\epsilon_{c3}^2\eta_3 + 2\eta_1\epsilon_{r1}\epsilon_{r2}^2\epsilon_{r3}\eta_2 + 2\eta_1\epsilon_{r1}\epsilon_{r2}^2 \\
& \epsilon_{r3}^2\eta_3 + 2\eta_2\epsilon_{r3}^2\eta_1\epsilon_{r1}^2\epsilon_{c2}^2 - 8\eta_1^2\epsilon_{r1}\epsilon_{r2}^2 - 32\eta_1^2\epsilon_{r1}\epsilon_{r2} + 16\epsilon_{r1}^2\epsilon_{r2}\epsilon_{r3} - 8\eta_1^2\epsilon_{r1} \\
& \epsilon_{c2}^2 - 8\eta_1^2\epsilon_{r1}\epsilon_{r3}^2 - 32\eta_1^2\epsilon_{r1}\epsilon_{r3} - 8\eta_1^2\epsilon_{r1}\epsilon_{c3}^2 + 8\eta_2\epsilon_{r3}^2\eta_1 + 16\epsilon_{c2}^2\eta_2^2\epsilon_{r1} + 8\eta_3 \\
& \epsilon_{r2}^2\eta_1 - 16\eta_2\epsilon_{r1}^2\eta_1 + 4\epsilon_{c2}^2\eta_2^2\epsilon_{cl}^2 - 16\eta_3\epsilon_{r1}^2\eta_1 + 4\epsilon_{cl}^2\eta_1^2\epsilon_{r3}^2 + 4\epsilon_{r1}\epsilon_{c2}^2\epsilon_{r3}^2 + 4 \\
& \epsilon_{r1}^2\epsilon_{r2}\epsilon_{c3}^2 + 4\epsilon_{r1}^2\epsilon_{c2}^2\epsilon_{r3}^2 + \epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{c3}^2 + 4\epsilon_{r1}\epsilon_{r2}^2\epsilon_{r3}^2 + \epsilon_{r1}^2\epsilon_{c2}^2\epsilon_{c3}^2 + 16\epsilon_{r1}\epsilon_{r2}\epsilon_{c3}^2 + \\
& \epsilon_{r1}^2\epsilon_{c2}^2\epsilon_{r3}^2 + 64\epsilon_{r1}\epsilon_{r2}\epsilon_{r3} + \epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{r3}^2 + 16\epsilon_{r1}\epsilon_{r2}^2\epsilon_{r3} + \epsilon_{cl}^2\epsilon_{r2}^2\epsilon_{r3}^2 + 16\epsilon_{r1}\epsilon_{c2}^2\epsilon_{r3} \\
& + 4\epsilon_{r1}^2\epsilon_{r2}\epsilon_{r3}^2 + 4\epsilon_{r1}\epsilon_{r2}^2\epsilon_{c3}^2 + 4\epsilon_{cl}^2\epsilon_{r2}^2\epsilon_{r3} + 16\epsilon_{r1}\epsilon_{r2}\epsilon_{r3}^2 + 4\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{r3} + 4\epsilon_{r1} \\
& \epsilon_{c2}^2\epsilon_{c3}^2 + \epsilon_{cl}^2\epsilon_{r2}^2\epsilon_{c3}^2 + 4\epsilon_{cl}^2\epsilon_{r2}\epsilon_{r3}^2 + 16\epsilon_{cl}^2\epsilon_{r2}\epsilon_{r3} + 4\epsilon_{cl}^2\epsilon_{r2}\epsilon_{c3}^2 + \epsilon_{cl}^2\epsilon_{c2}^2\epsilon_{r3}^2 + 4\epsilon_{cl}^2 \\
& \epsilon_{c2}^2\epsilon_{r3} + \epsilon_{cl}^2\epsilon_{c2}^2\epsilon_{c3}^2 - 8\eta_1\epsilon_{r1}^2\epsilon_{r2}^2 - 32\eta_1\epsilon_{r1}^2\epsilon_{r2} - 8\eta_1\epsilon_{r1}^2\epsilon_{c2}^2 - 8\eta_1\epsilon_{r1}\epsilon_{r2}^2 \\
& - 32\eta_1\epsilon_{r1}\epsilon_{r2} - 8\eta_1\epsilon_{r1}\epsilon_{c2}^2 - 8\eta_1\epsilon_{cl}^2\epsilon_{r2}^2 - 32\eta_1\epsilon_{cl}^2\epsilon_{r2} - 8\eta_1\epsilon_{cl}^2\epsilon_{c2}^2 - 8\eta_1\epsilon_{r1} \\
& \epsilon_{r3}^2 + 2\eta_2\epsilon_{r3}^2\eta_1\epsilon_{cl}^2\epsilon_{r2}^2 + 2\eta_2\epsilon_{r3}^2\eta_1\epsilon_{cl}^2\epsilon_{c2}^2 + 2\eta_2\epsilon_{c3}^2\eta_1\epsilon_{r1}^2\epsilon_{c2}^2 + 2\eta_2\epsilon_{c3}^2\eta_1\epsilon_{cl}^2\epsilon_{r2}^2 \\
& + 2\eta_2\epsilon_{c3}^2\eta_1\epsilon_{cl}^2\epsilon_{c2}^2 + 2\eta_1\epsilon_{r1}\epsilon_{c2}^2\epsilon_{r3}^2\eta_2 + 2\eta_1\epsilon_{r1}\epsilon_{c2}^2\epsilon_{r3}^2\eta_3 + 8\eta_2\epsilon_{r3}\eta_1\epsilon_{r1}^2\epsilon_{c2}^2 \\
& + 8\eta_2\epsilon_{r3}\eta_1\epsilon_{cl}^2\epsilon_{r2}^2 + 8\eta_2\epsilon_{r3}\eta_1\epsilon_{cl}^2\epsilon_{c2}^2 + 2\eta_1\epsilon_{r1}\epsilon_{c2}^2\epsilon_{c3}^2\eta_2 + 2\eta_1\epsilon_{r1}\epsilon_{c2}^2\epsilon_{c3}^2\eta_3 \\
& + 2\eta_3\epsilon_{r1}^2\epsilon_{c2}^2\epsilon_{r3}\eta_1 + 8\eta_3\epsilon_{r1}^2\epsilon_{r2}\epsilon_{r3}\eta_1 + 2\eta_2\epsilon_{cl}^2\epsilon_{r2}^2\epsilon_{c3}^2\eta_1 + 2\eta_2\epsilon_{cl}^2\epsilon_{r2}^2\epsilon_{r3}^2\eta_1 \\
& + 8\eta_2\epsilon_{cl}^2\epsilon_{r2}\epsilon_{r3}\eta_1 + 2\eta_2\epsilon_{r1}\epsilon_{r2}^2\epsilon_{r3}^2\eta_1 + 2\eta_2\epsilon_{r1}^2\epsilon_{r2}\epsilon_{c3}^2\eta_1 + 2\eta_2\epsilon_{r1}\epsilon_{r2}\epsilon_{c3}^2\eta_1
\end{aligned}$$

[illegible]

$$\begin{aligned}
& \eta_1^2 \varepsilon_{r3} - 16 \eta_2 \varepsilon_{r2} \eta_3 - 16 \eta_2 \varepsilon_{r3} \eta_3 + 8 \eta_2 \varepsilon_{r1}^2 \eta_3 - 16 \eta_2 \varepsilon_{r3}^2 \eta_3 + 16 \eta_1^2 \varepsilon_{r1}^2 \varepsilon_{r2} + 4 \\
& \eta_1^2 \varepsilon_{r1}^2 \varepsilon_{c2}^2 + 4 \varepsilon_{c3}^2 \eta_3^2 \varepsilon_{c2}^2 + 4 \eta_1^2 \varepsilon_{r1}^2 \varepsilon_{r2}^2 - 16 \eta_1 \varepsilon_{r2}^2 \eta_2 - 16 \eta_1 \varepsilon_{r3}^2 \eta_3 - 16 \eta_1 \varepsilon_{c2}^2 \eta_2 \\
& - 16 \eta_1 \varepsilon_{c3}^2 \eta_3 + 16 \varepsilon_{c3}^2 \eta_3^2 \varepsilon_{r2} + 32 \eta_2 \varepsilon_{r3} \eta_1 + 4 \eta_2^2 \varepsilon_{r1}^2 \varepsilon_{r2}^2 + 4 \eta_3^2 \varepsilon_{r1}^2 \varepsilon_{r2} + \eta_3^2 \varepsilon_{r1}^2 \\
& \varepsilon_{c2}^2 + 4 \eta_3^2 \varepsilon_{r1} \varepsilon_{r2}^2 - 4 \eta_2 \varepsilon_{r1} \varepsilon_{c3}^2 \eta_1 - 4 \eta_2 \varepsilon_{r2} \varepsilon_{r3}^2 \eta_1 - 16 \eta_2 \varepsilon_{r2} \varepsilon_{r3} \eta_1 - 32 \\
& \eta_3^2 \varepsilon_{r1} \varepsilon_{r2} \varepsilon_{r3} - 8 \eta_3^2 \varepsilon_{r1} \varepsilon_{r2}^2 \varepsilon_{r3} - 8 \eta_3^2 \varepsilon_{r1} \varepsilon_{c2}^2 \varepsilon_{r3} - 4 \eta_2 \varepsilon_{c1}^2 \varepsilon_{r3} \eta_3 - 4 \eta_2 \varepsilon_{c1}^2 \varepsilon_{c3}^2 \eta_3 \\
& + 4 \varepsilon_{c1}^2 \eta_1^2 \varepsilon_{r2}^2 \varepsilon_{r3} - 4 \eta_2 \varepsilon_{r1}^2 \varepsilon_{r3}^2 \eta_3 - 4 \eta_2 \varepsilon_{r1}^2 \varepsilon_{r3} \eta_3 - 4 \eta_2 \varepsilon_{r1}^2 \varepsilon_{c3}^2 \eta_3 - 4 \eta_2 \varepsilon_{c1}^2 \\
& \varepsilon_{r3}^2 \eta_3 + 8 \eta_2 \varepsilon_{r1}^2 \varepsilon_{r2} \eta_1 + 8 \eta_2 \varepsilon_{r1} \varepsilon_{r2} \eta_1 + 8 \eta_2 \varepsilon_{r2} \varepsilon_{r3}^2 \eta_3 + 8 \eta_2 \varepsilon_{r2} \varepsilon_{r3} \eta_3 + 8 \eta_2 \varepsilon_{r2} \\
& \varepsilon_{c3}^2 \eta_3 - 8 \eta_1^2 \varepsilon_{r1} \varepsilon_{r2} \varepsilon_{r3}^2 - 2 \eta_1^2 \varepsilon_{r1} \varepsilon_{c2}^2 \varepsilon_{c3}^2 + 16 \varepsilon_{c1}^2 \eta_1^2 \varepsilon_{r2} \varepsilon_{r3} - 2 \eta_1^2 \varepsilon_{r1} \varepsilon_{r2}^2 \varepsilon_{c3}^2 - 32 \\
& \eta_1^2 \varepsilon_{r1} \varepsilon_{r2} \varepsilon_{r3} - 8 \eta_1^2 \varepsilon_{r1} \varepsilon_{r2}^2 \varepsilon_{r3} - 8 \eta_1^2 \varepsilon_{r1} \varepsilon_{c2}^2 \varepsilon_{r3} - 2 \eta_1^2 \varepsilon_{r1} \varepsilon_{c2}^2 \varepsilon_{r3}^2 - 2 \eta_1^2 \varepsilon_{r1} \varepsilon_{r2}^2 \varepsilon_{r3}^2 \\
& - 8 \eta_1^2 \varepsilon_{r1} \varepsilon_{r2} \varepsilon_{c3}^2 - 4 \eta_3 \varepsilon_{r2}^2 \varepsilon_{r3} \eta_1 + 8 \eta_3 \varepsilon_{c1}^2 \varepsilon_{r3} \eta_1 - 4 \eta_3 \varepsilon_{c2}^2 \varepsilon_{r3} \eta_1 - 16 \eta_3 \\
& \varepsilon_{c1}^2 \varepsilon_{r2} \eta_1 - 4 \eta_3 \varepsilon_{c1}^2 \varepsilon_{c2}^2 \eta_1 + 8 \eta_3 \varepsilon_{r1} \varepsilon_{r3} \eta_1 - 16 \eta_3 \varepsilon_{r2} \varepsilon_{r3} \eta_1 + 8 \eta_3 \varepsilon_{r1}^2 \varepsilon_{r3} \eta_1 \\
& - 4 \eta_3 \varepsilon_{c1}^2 \varepsilon_{r2}^2 \eta_1 + \varepsilon_{c2}^2 \eta_2^2 \varepsilon_{c1}^2 \varepsilon_{r3}^2 - 2 \eta_2^2 \varepsilon_{r1}^2 \varepsilon_{r2}^2 \varepsilon_{c3}^2 - 8 \eta_2^2 \varepsilon_{r1}^2 \varepsilon_{r2} \varepsilon_{r3} + 4 \varepsilon_{c1}^2 \eta_1^2 \varepsilon_{r2} \varepsilon_{r3}^2 \\
& - 16 \eta_1 \varepsilon_{r2} \varepsilon_{r3}^2 \eta_3 - 16 \eta_1 \varepsilon_{r2} \varepsilon_{c3}^2 \eta_3 + 8 \eta_3 \varepsilon_{c2}^2 \varepsilon_{r3} \eta_2 + 8 \eta_1 \varepsilon_{r1} \varepsilon_{r3}^2 \eta_3 + 8 \eta_1 \varepsilon_{r1} \\
& \varepsilon_{c3}^2 \eta_3 + 4 \varepsilon_{c3}^2 \eta_3^2 \varepsilon_{c1}^2 \varepsilon_{r2} + 8 \eta_2 \eta_1 \varepsilon_{r1}^2 \varepsilon_{c2}^2 + 8 \eta_2 \eta_1 \varepsilon_{r1}^2 \varepsilon_{r2}^2 + \varepsilon_{c3}^2 \eta_3^2 \varepsilon_{c1}^2 \varepsilon_{r2}^2 + 4 \varepsilon_{c2}^2 \\
& \eta_2^2 \varepsilon_{r1} \varepsilon_{c3}^2 - 2 \eta_2^2 \varepsilon_{c1}^2 \varepsilon_{r2} \varepsilon_{r3}^2 - 8 \eta_2^2 \varepsilon_{c1}^2 \varepsilon_{r2} \varepsilon_{r3} - 2 \eta_2^2 \varepsilon_{c1}^2 \varepsilon_{r2} \varepsilon_{c3}^2 + 4 \varepsilon_{c1}^2 \eta_1^2 \varepsilon_{c2}^2 \varepsilon_{r3} + \varepsilon_{c1}^2 \\
& \eta_1^2 \varepsilon_{c2}^2 \varepsilon_{c3}^2 - 4 \eta_2 \varepsilon_{r1}^2 \varepsilon_{r2} \eta_3 - 16 \eta_2 \varepsilon_{r1} \varepsilon_{r2} \eta_3 - 2 \eta_3 \varepsilon_{r1}^2 \varepsilon_{c2}^2 \varepsilon_{c3}^2 - 32 \eta_1 \varepsilon_{r1}^2 \varepsilon_{r2} \varepsilon_{r3} \\
& + 16 \eta_3^2 \varepsilon_{r1} \varepsilon_{r2} + 4 \eta_3^2 \varepsilon_{r1} \varepsilon_{c2}^2 + \eta_3^2 \varepsilon_{c1}^2 \varepsilon_{r2}^2 + 4 \eta_3^2 \varepsilon_{c1}^2 \varepsilon_{r2} + \eta_3^2 \varepsilon_{c1}^2 \varepsilon_{c2}^2 - 32 \eta_3^2 \varepsilon_{r1} \varepsilon_{r3} \\
& - 32 \eta_3^2 \varepsilon_{r2} \varepsilon_{r3} + \eta_2^2 \varepsilon_{r1}^2 \varepsilon_{r3}^2 + 4 \eta_2^2 \varepsilon_{r1}^2 \varepsilon_{r3} + \eta_2^2 \varepsilon_{r1}^2 \varepsilon_{c3}^2 + \eta_2^2 \varepsilon_{c1}^2 \varepsilon_{r3}^2 + 4 \eta_2^2 \varepsilon_{c1}^2 \varepsilon_{r3} + \\
& \eta_2^2 \varepsilon_{c1}^2 \varepsilon_{c3}^2 + \eta_3^2 \varepsilon_{r1}^2 \varepsilon_{r2}^2 - 8 \eta_2^2 \varepsilon_{r1}^2 \varepsilon_{r2} - 32 \eta_2^2 \varepsilon_{r1} \varepsilon_{r2} - 8 \eta_2^2 \varepsilon_{c1}^2 \varepsilon_{r2} + 4 \eta_2^2 \varepsilon_{r1} \varepsilon_{r3}^2 + 16 \\
& \eta_2^2 \varepsilon_{r1} \varepsilon_{r3} + 4 \eta_2^2 \varepsilon_{r1} \varepsilon_{c3}^2 - 8 \eta_2^2 \varepsilon_{r2} \varepsilon_{r3}^2 - 32 \eta_2^2 \varepsilon_{r2} \varepsilon_{r3} - 8 \eta_2^2 \varepsilon_{r2} \varepsilon_{c3}^2 + 16 \varepsilon_{c1}^2 \eta_1^2 \varepsilon_{r2} \\
& - 16 \eta_3 \varepsilon_{r2}^2 \eta_2 - 16 \eta_3 \varepsilon_{c2}^2 \eta_2 + 4 \eta_1^2 \varepsilon_{r1}^2 \varepsilon_{r3}^2 + 4 \varepsilon_{c3}^2 \eta_3^2 \varepsilon_{r1}^2 + 4 \eta_1^2 \varepsilon_{r1} \varepsilon_{c3}^2 + 16 \eta_1^2 \\
& \varepsilon_{r1} \varepsilon_{r3} + 16 \eta_2^2 \varepsilon_{r1} \varepsilon_{r2}^2 + 4 \varepsilon_{c1}^2 \eta_1^2 \varepsilon_{r2}^2 + 16 \eta_2^2 \varepsilon_{r3} \varepsilon_{r2}^2 + 32 \eta_2 \varepsilon_{r1} \eta_3 - 8 \eta_3^2 \varepsilon_{c2}^2 \varepsilon_{r3} \\
& + 8 \eta_2 \varepsilon_{c1}^2 \eta_3 - 16 \eta_2 \varepsilon_{c3}^2 \eta_3 + 32 \eta_3 \varepsilon_{r2} \eta_1 - 2 \eta_1 \varepsilon_{r1} \varepsilon_{c2}^2 \varepsilon_{r3}^2 - 8 \eta_1 \varepsilon_{r1}^2 \varepsilon_{r2} \varepsilon_{c3}^2 \\
& - 8 \eta_1 \varepsilon_{r1}^2 \varepsilon_{c2}^2 \varepsilon_{r3} - 2 \eta_1 \varepsilon_{r1}^2 \varepsilon_{r2}^2 \varepsilon_{c3}^2 - 2 \eta_1 \varepsilon_{r1} \varepsilon_{r2}^2 \varepsilon_{r3}^2 - 2 \eta_1 \varepsilon_{r1}^2 \varepsilon_{c2}^2 \varepsilon_{c3}^2 - 8 \eta_1 \varepsilon_{r1} \varepsilon_{r2} \\
& \varepsilon_{c3}^2 - 2 \eta_1 \varepsilon_{r1}^2 \varepsilon_{c2}^2 \varepsilon_{r3}^2 - 32 \eta_1 \varepsilon_{r1} \varepsilon_{r2} \varepsilon_{r3} - 2 \eta_1 \varepsilon_{r1}^2 \varepsilon_{r2}^2 \varepsilon_{r3}^2 - 8 \eta_1 \varepsilon_{r1} \varepsilon_{r2}^2 \varepsilon_{r3} - 2 \eta_1 \varepsilon_{c1}^2 \\
& \varepsilon_{r2}^2 \varepsilon_{r3}^2 - 8 \eta_1 \varepsilon_{r1} \varepsilon_{c2}^2 \varepsilon_{r3} - 8 \eta_1 \varepsilon_{r1}^2 \varepsilon_{r2} \varepsilon_{r3}^2 - 2 \eta_1 \varepsilon_{r1} \varepsilon_{r2}^2 \varepsilon_{c3}^2 - 8 \eta_1 \varepsilon_{c1}^2 \varepsilon_{r2}^2 \varepsilon_{r3} \\
& - 8 \eta_1 \varepsilon_{r1} \varepsilon_{r2} \varepsilon_{r3}^2 - 8 \eta_1 \varepsilon_{r1}^2 \varepsilon_{r2}^2 \varepsilon_{r3} - 2 \eta_1 \varepsilon_{r1} \varepsilon_{c2}^2 \varepsilon_{c3}^2 - 2 \eta_1 \varepsilon_{c1}^2 \varepsilon_{r2}^2 \varepsilon_{c3}^2 - 8 \eta_1 \varepsilon_{c1}^2 \varepsilon_{r2}
\end{aligned}$$

$$\begin{aligned} & \epsilon_{r3}^2 - 32 \eta_1 \epsilon_{cl}^2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_1 \epsilon_{cl}^2 \epsilon_{r2} \epsilon_{c3}^2 - 2 \eta_1 \epsilon_{cl}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 8 \eta_1 \epsilon_{cl}^2 \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_1 \epsilon_{cl}^2 \\ & \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_3^2 \\ & \epsilon_{r1}^2 \epsilon_{r3}^2 \} \end{aligned}$$

$$eps_c := rhs(sol[1])$$

$$\begin{aligned} & (9 (16 \epsilon_{cl} \eta_1 + 16 \epsilon_{c2} \eta_2 + 16 \epsilon_{c3} \eta_3 + 16 \epsilon_{cl} \eta_1 \epsilon_{r2} + 16 \epsilon_{cl} \eta_1 \epsilon_{r3} + 4 \epsilon_{cl} \eta_1 \epsilon_{r2}^2 \\ & + 4 \epsilon_{cl} \eta_1 \epsilon_{r3}^2 + 4 \epsilon_{cl} \eta_1 \epsilon_{c2}^2 + 4 \epsilon_{cl} \eta_1 \epsilon_{c3}^2 + 16 \epsilon_{c2} \eta_2 \epsilon_{r1} + 16 \epsilon_{c2} \eta_2 \epsilon_{r3} + 4 \epsilon_{c2} \eta_2 \\ & \epsilon_{r1}^2 + 4 \epsilon_{c2} \eta_2 \epsilon_{r3}^2 + 4 \epsilon_{c2} \eta_2 \epsilon_{cl}^2 + 4 \epsilon_{c2} \eta_2 \epsilon_{c3}^2 + 16 \epsilon_{c3} \eta_3 \epsilon_{r1} + 16 \epsilon_{c3} \eta_3 \epsilon_{r2} \\ & + 4 \epsilon_{c3} \eta_3 \epsilon_{r1}^2 + 4 \epsilon_{c3} \eta_3 \epsilon_{r2}^2 + 4 \epsilon_{c3} \eta_3 \epsilon_{cl}^2 + 4 \epsilon_{c3} \eta_3 \epsilon_{c2}^2 + 4 \epsilon_{cl} \eta_1 \epsilon_{r2} \epsilon_{r3}^2 \\ & + 16 \epsilon_{cl} \eta_1 \epsilon_{r2} \epsilon_{r3} + 4 \epsilon_{cl} \eta_1 \epsilon_{r2} \epsilon_{c3}^2 + \epsilon_{cl} \eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 + 4 \epsilon_{cl} \eta_1 \epsilon_{r2}^2 \epsilon_{r3} + \epsilon_{cl} \eta_1 \epsilon_{r2}^2 \epsilon_{c3}^2 \\ & + \epsilon_{cl} \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \epsilon_{cl} \eta_1 \epsilon_{c2}^2 \epsilon_{r3} + \epsilon_{cl} \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 + 4 \epsilon_{c2} \eta_2 \epsilon_{r1} \epsilon_{r3}^2 + 16 \epsilon_{c2} \eta_2 \epsilon_{r1} \epsilon_{r3} \\ & + 4 \epsilon_{c2} \eta_2 \epsilon_{r1} \epsilon_{c3}^2 + \epsilon_{c2} \eta_2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 4 \epsilon_{c2} \eta_2 \epsilon_{r1}^2 \epsilon_{r3} + \epsilon_{c2} \eta_2 \epsilon_{r1}^2 \epsilon_{c3}^2 + \epsilon_{c2} \eta_2 \epsilon_{cl}^2 \epsilon_{r3}^2 \\ & + 4 \epsilon_{c2} \eta_2 \epsilon_{cl}^2 \epsilon_{r3} + \epsilon_{c2} \eta_2 \epsilon_{cl}^2 \epsilon_{c3}^2 + \epsilon_{c3} \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 + 4 \epsilon_{c3} \eta_3 \epsilon_{r1}^2 \epsilon_{r2} + \epsilon_{c3} \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \\ & + 4 \epsilon_{c3} \eta_3 \epsilon_{r1} \epsilon_{r2}^2 + 16 \epsilon_{c3} \eta_3 \epsilon_{r1} \epsilon_{r2} + 4 \epsilon_{c3} \eta_3 \epsilon_{r1} \epsilon_{c2}^2 + \epsilon_{c3} \eta_3 \epsilon_{cl}^2 \epsilon_{r2}^2 + 4 \epsilon_{c3} \eta_3 \\ & \epsilon_{cl}^2 \epsilon_{r2} + \epsilon_{c3} \eta_3 \epsilon_{cl}^2 \epsilon_{c2}^2)) / (64 + 4 \epsilon_{c2}^2 \eta_2^2 \epsilon_{c3}^2 + 64 \eta_1 + 64 \eta_2 + 64 \eta_3 + 64 \epsilon_{r1} \\ & + 64 \epsilon_{r2} + 64 \epsilon_{r3} + 16 \epsilon_{cl}^2 + 16 \epsilon_{c2}^2 + 16 \epsilon_{c3}^2 - 8 \eta_3^2 \epsilon_{cl}^2 \epsilon_{r3} - 8 \eta_3^2 \epsilon_{r2}^2 \epsilon_{r3} \\ & + 18 \epsilon_{cl} \eta_1 \epsilon_{c3} \eta_3 \epsilon_{r2}^2 + 4 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r3}^2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r2}^2 + 4 \eta_2^2 \epsilon_{cl}^2 \epsilon_{r2}^2 - 16 \eta_2 \epsilon_{r2} \eta_1 \\ & + 8 \eta_3 \epsilon_{r1}^2 \eta_1 \epsilon_{r3}^2 - 16 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \eta_2 - 16 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \eta_2 - 4 \eta_3 \epsilon_{cl}^2 \epsilon_{r2}^2 \eta_2 - 8 \eta_3^2 \\ & \epsilon_{cl}^2 \epsilon_{r2} \epsilon_{r3} - 2 \eta_3^2 \epsilon_{cl}^2 \epsilon_{c2}^2 \epsilon_{r3} - 4 \eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_2 - 4 \eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_3 - 16 \eta_1 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 \\ & + 8 \eta_3 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 + 16 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1} \epsilon_{r3} + 4 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3} + 16 \epsilon_{c3}^2 \\ & \eta_3^2 \epsilon_{r1} \epsilon_{r2} + 8 \eta_2 \eta_1 \epsilon_{cl}^2 \epsilon_{r2}^2 + 8 \eta_2 \eta_1 \epsilon_{cl}^2 \epsilon_{c2}^2 - 4 \eta_2 \epsilon_{cl}^2 \epsilon_{r2} \eta_3 - 16 \eta_2 \epsilon_{r1} \epsilon_{r3}^2 \eta_3 \\ & - 16 \eta_2 \epsilon_{r1} \epsilon_{r3} \eta_3 - 16 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 \eta_3 + \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{c3}^2 + 8 \eta_2 \epsilon_{cl}^2 \epsilon_{r2} \eta_1 - 4 \eta_2 \epsilon_{r1} \\ & \epsilon_{r3}^2 \eta_1 - 16 \eta_2 \epsilon_{r1} \epsilon_{r3} \eta_1 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{cl}^2 + 8 \eta_3 \epsilon_{c2}^2 \eta_2 \epsilon_{r3}^2 - 4 \eta_2 \epsilon_{cl}^2 \epsilon_{r3}^2 \eta_1 - 16 \eta_2 \\ & \epsilon_{cl}^2 \epsilon_{r3} \eta_1 - 4 \eta_2 \epsilon_{cl}^2 \epsilon_{c3}^2 \eta_1 + 8 \eta_3 \epsilon_{r2}^2 \eta_2 \epsilon_{c3}^2 + 8 \eta_3 \epsilon_{r2}^2 \eta_2 \epsilon_{r3}^2 - 4 \eta_1 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_2 - 4 \eta_1 \\ & \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_3 - 4 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_2 - 4 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_3 - 16 \eta_1 \epsilon_{c2}^2 \epsilon_{r3} \eta_2 - 4 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_2 \\ & - 4 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_3 - 4 \eta_3 \epsilon_{cl}^2 \epsilon_{c2}^2 \eta_2 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r3}^2 \eta_1 - 16 \eta_2 \epsilon_{r1}^2 \epsilon_{r3} \eta_1 - 4 \eta_2 \epsilon_{r1}^2 \\ & \epsilon_{c3}^2 \eta_1 + \epsilon_{cl}^2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 + 8 \eta_3 \eta_1 \epsilon_{cl}^2 \epsilon_{r3}^2 + 8 \eta_3 \eta_1 \epsilon_{cl}^2 \\ & \epsilon_{c3}^2 + 16 \eta_3^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 + 4 \eta_3^2 \epsilon_{cl}^2 \epsilon_{r2} \epsilon_{r3}^2 + \eta_3^2 \epsilon_{cl}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_2 - 4 \eta_3 \epsilon_{r1}^2 \end{aligned} \quad (2.7)$$

$$\begin{aligned}
& \epsilon_{c2}^2 \eta_2 - 2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} + 8 \eta_3 \epsilon_{r1}^2 \eta_1 \epsilon_{c3}^2 + 4 \eta_2^2 \\
& \epsilon_{c3}^2 \epsilon_{r2}^2 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 16 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_1 - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_1 - 4 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \eta_1 \\
& - 16 \eta_3 \epsilon_{r1} \epsilon_{r2} \eta_1 + 72 \epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 + 72 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_1 \\
& + 72 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 - 2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} + 4 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 + 8 \eta_3 \epsilon_{c2}^2 \eta_2 \epsilon_{c3}^2 + 16 \epsilon_{c1}^2 \eta_1^2 \\
& + 16 \eta_3^2 \epsilon_{r1} + 16 \eta_3^2 \epsilon_{r2} - 32 \eta_3^2 \epsilon_{r3} + 4 \eta_3^2 \epsilon_{r1}^2 + 4 \eta_3^2 \epsilon_{r2}^2 + 4 \eta_3^2 \epsilon_{c1}^2 + 4 \eta_3^2 \epsilon_{c2}^2 \\
& + 16 \eta_2^2 \epsilon_{r1} - 32 \eta_2^2 \epsilon_{r2} + 16 \eta_2^2 \epsilon_{r3} + 4 \eta_2^2 \epsilon_{r1}^2 + 4 \eta_2^2 \epsilon_{r3}^2 + 4 \eta_2^2 \epsilon_{c1}^2 + 4 \eta_2^2 \epsilon_{c3}^2 \\
& + 16 \epsilon_{c3}^2 \eta_3^2 + 4 \epsilon_{r1}^2 \epsilon_{r2}^2 + 16 \epsilon_{r1}^2 \epsilon_{r2} + 4 \epsilon_{r1}^2 \epsilon_{c2}^2 + 16 \epsilon_{r1} \epsilon_{r2}^2 + 64 \epsilon_{r1} \epsilon_{r2} + 16 \epsilon_{r1} \epsilon_{c2}^2 \\
& + 4 \epsilon_{c1}^2 \epsilon_{r2}^2 + 16 \epsilon_{c1}^2 \epsilon_{r2} + 16 \epsilon_{r1}^2 + 16 \epsilon_{r2}^2 + 16 \epsilon_{r3}^2 + 16 \eta_1^2 + 16 \eta_2^2 + 16 \eta_3^2 - 2 \eta_2 \\
& \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 \\
& - 32 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 32 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 32 \eta_2 \epsilon_{r1} \\
& \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 8 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - 8 \eta_2 \\
& \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 - 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} \\
& - 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 2 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_3 \\
& \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \\
& - 8 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 - 32 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 + 18 \epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 \epsilon_{c2}^2 + 72 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \epsilon_{r1} \\
& + 18 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \epsilon_{r1}^2 + 18 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \epsilon_{c1}^2 + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_3 \\
& + 72 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 \epsilon_{r3} + 18 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 \epsilon_{r3}^2 + 18 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 \epsilon_{c3}^2 \\
& + 72 \epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 \epsilon_{r2} + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_1 + 8 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_3 - 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 \\
& - 32 \eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 2 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r1} \\
& \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 2 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 32 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 - 2 \eta_3 \\
& \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 - 2 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} \\
& - 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 2 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 2 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \\
& + \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 4 \eta_2^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 + \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \eta_2^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \\
& - 8 \eta_2^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 + \epsilon_{c2}^2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{c3}^2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1} \epsilon_{c2}^2 - 8 \eta_2^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 \\
& + 4 \epsilon_{c1}^2 \epsilon_{c2}^2 + 16 \epsilon_{r1} \epsilon_{r3}^2 + 64 \epsilon_{r1} \epsilon_{r3} + 16 \epsilon_{r1} \epsilon_{c3}^2 + 16 \epsilon_{r2} \epsilon_{r3}^2 + 64 \epsilon_{r2} \epsilon_{r3} + 16 \epsilon_{r2} \\
& \epsilon_{c3}^2 + 4 \epsilon_{r1}^2 \epsilon_{r3}^2 + 16 \epsilon_{r1}^2 \epsilon_{r3} + 4 \epsilon_{r1}^2 \epsilon_{c3}^2 + 4 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \epsilon_{r2}^2 \epsilon_{r3} + 4 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \epsilon_{c1}^2 \epsilon_{r3}^2 \\
& + 16 \epsilon_{c1}^2 \epsilon_{r3} + 4 \epsilon_{c1}^2 \epsilon_{c3}^2 + 4 \epsilon_{c2}^2 \epsilon_{r3}^2 + 16 \epsilon_{c2}^2 \epsilon_{r3} + 4 \epsilon_{c2}^2 \epsilon_{c3}^2 - 32 \eta_1 \epsilon_{r1} + 64 \eta_1 \epsilon_{r2}
\end{aligned}$$

$$\begin{aligned}
& + 64 \eta_1 \epsilon_{r3} - 32 \eta_1 \epsilon_{r1}^2 + 16 \eta_1 \epsilon_{r2}^2 + 16 \eta_1 \epsilon_{r3}^2 - 32 \eta_1 \epsilon_{c1}^2 + 16 \eta_1 \epsilon_{c2}^2 + 16 \eta_1 \epsilon_{c3}^2 \\
& + 64 \eta_2 \epsilon_{r1} - 32 \eta_2 \epsilon_{r2} + 64 \eta_2 \epsilon_{r3} + 16 \eta_2 \epsilon_{r1}^2 - 32 \eta_2 \epsilon_{r2}^2 + 16 \eta_2 \epsilon_{r3}^2 + 16 \eta_2 \epsilon_{c1}^2 \\
& - 32 \eta_2 \epsilon_{c2}^2 + 16 \eta_2 \epsilon_{c3}^2 + 64 \eta_3 \epsilon_{r1} + 64 \eta_3 \epsilon_{r2} - 32 \eta_3 \epsilon_{r3} + 16 \eta_3 \epsilon_{r1}^2 + 16 \eta_3 \epsilon_{r2}^2 \\
& - 32 \eta_3 \epsilon_{r3}^2 + 16 \eta_3 \epsilon_{c1}^2 + 16 \eta_3 \epsilon_{c2}^2 - 32 \eta_3 \epsilon_{c3}^2 + 16 \eta_1^2 \epsilon_{r1}^2 + 16 \eta_2^2 \epsilon_{r2}^2 + 16 \epsilon_{c2}^2 \eta_2^2 \\
& + 32 \eta_2 \eta_1 + 32 \eta_3 \eta_1 + 32 \eta_2 \eta_3 + 16 \eta_3^2 \epsilon_{r3}^2 - 32 \eta_2^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} + 4 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \\
& + 4 \eta_2^2 \epsilon_{r2}^2 \epsilon_{r1} \epsilon_{c3}^2 + \epsilon_{c3}^2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{c2}^2 + \epsilon_{c1}^2 \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{c2}^2 + 4 \\
& \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1} \epsilon_{r3}^2 + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2} \epsilon_{c3}^2 + \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \\
& \eta_3^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \epsilon_{c2}^2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r3}^2 + 8 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \eta_2 + 8 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \eta_2 + 4 \eta_3^2 \epsilon_{r3}^2 \epsilon_{r1} \epsilon_{r2} + \\
& \eta_3^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + \eta_3^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \epsilon_{r1}^2 - 4 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \eta_1 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + \eta_1^2 \\
& \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 16 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + \eta_1^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + \eta_1^2 \\
& \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} \eta_3 + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_1 + 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_1 + 2 \eta_2 \eta_3 \epsilon_{r1}^2 \\
& \epsilon_{c2}^2 \epsilon_{c3}^2 + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_3 + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_1 + 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} \eta_1 + 2 \eta_3 \epsilon_{c1}^2 \\
& \epsilon_{c2}^2 \epsilon_{r3} \eta_1 + 2 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_1 + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} \eta_3 + 2 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} \eta_1 + 2 \eta_3 \epsilon_{r1} \\
& \epsilon_{r2}^2 \epsilon_{r3} \eta_1 + 8 \eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} \eta_1 + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 + 2 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 2 \eta_3 \epsilon_{r1}^2 \\
& \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 8 \eta_1 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 \eta_3 + 2 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_2 + 2 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_3 + 8 \eta_1 \epsilon_{r1} \\
& \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 8 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} \eta_2 + 8 \eta_1 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 \eta_3 + 2 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_2 + 2 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \\
& \epsilon_{r3}^2 \eta_3 + 2 \eta_2 \epsilon_{r3}^2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 - 8 \eta_1^2 \epsilon_{r1} \epsilon_{r2}^2 - 32 \eta_1^2 \epsilon_{r1} \epsilon_{r2} + 16 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_1^2 \epsilon_{r1} \\
& \epsilon_{c2}^2 - 8 \eta_1^2 \epsilon_{r1} \epsilon_{r3}^2 - 32 \eta_1^2 \epsilon_{r1} \epsilon_{r3} - 8 \eta_1^2 \epsilon_{r1} \epsilon_{c3}^2 + 8 \eta_2 \epsilon_{r3}^2 \eta_1 + 16 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1} + 8 \eta_3 \\
& \epsilon_{r2}^2 \eta_1 - 16 \eta_2 \epsilon_{r1}^2 \eta_1 + 4 \epsilon_{c2}^2 \eta_2^2 \epsilon_{c1}^2 - 16 \eta_3 \epsilon_{r1}^2 \eta_1 + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r3}^2 + 4 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \\
& \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 + \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + 16 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 + \\
& \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 64 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} + \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} + \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} \\
& + 4 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 4 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 + 4 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} + 4 \epsilon_{r1} \\
& \epsilon_{c2}^2 \epsilon_{c3}^2 + \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 + 16 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} + 4 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 + \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \epsilon_{c1}^2 \\
& \epsilon_{c2}^2 \epsilon_{r3} + \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 - 32 \eta_1 \epsilon_{r1}^2 \epsilon_{r2} - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 - 8 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \\
& - 32 \eta_1 \epsilon_{r1} \epsilon_{r2} - 8 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 - 32 \eta_1 \epsilon_{c1}^2 \epsilon_{r2} - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 - 8 \eta_1 \epsilon_{r1} \\
& \epsilon_{r3}^2 + 2 \eta_2 \epsilon_{r3}^2 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 + 2 \eta_2 \epsilon_{r3}^2 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 + 2 \eta_2 \epsilon_{c3}^2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 + 2 \eta_2 \epsilon_{c3}^2 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 \\
& + 2 \eta_2 \epsilon_{c3}^2 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 + 2 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_2 + 2 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_3 + 8 \eta_2 \epsilon_{r3} \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2
\end{aligned}$$

[illegible]

$$\begin{aligned}
& -16\eta_3\epsilon_{cl}^2\eta_1+8\eta_3\epsilon_{c2}^2\eta_1-16\eta_2\epsilon_{r1}\eta_1+4\epsilon_{c2}^2\eta_2^2\epsilon_{r1}^2+4\epsilon_{cl}^2\eta_1^2\epsilon_{c2}^2+16\epsilon_{c3}^2 \\
& \eta_3^2\epsilon_{r1}+4\epsilon_{cl}^2\eta_1^2\epsilon_{c3}^2+\eta_1^2\epsilon_{c2}^2\epsilon_{c3}^2+\eta_1^2\epsilon_{c2}^2\epsilon_{r3}^2+4\eta_1^2\epsilon_{c2}^2\epsilon_{r3}+\eta_1^2\epsilon_{r2}^2\epsilon_{c3}^2+4\eta_1^2\epsilon_{r2}^2\epsilon_{r3} \\
& +\eta_1^2\epsilon_{r2}^2\epsilon_{r3}^2+16\eta_1^2\epsilon_{r2}\epsilon_{r3}+4\eta_1^2\epsilon_{r2}\epsilon_{c3}^2+4\eta_1^2\epsilon_{r2}\epsilon_{r3}^2+16\epsilon_{c2}^2\eta_2^2\epsilon_{r3}+16\epsilon_{cl}^2 \\
& \eta_1^2\epsilon_{r3}-16\eta_2\epsilon_{r2}\eta_3-16\eta_2\epsilon_{r3}\eta_3+8\eta_2\epsilon_{r1}^2\eta_3-16\eta_2\epsilon_{r3}^2\eta_3+16\eta_1^2\epsilon_{r1}^2\epsilon_{r2}+4 \\
& \eta_1^2\epsilon_{r1}^2\epsilon_{c2}^2+4\epsilon_{c3}^2\eta_3^2\epsilon_{c2}^2+4\eta_1^2\epsilon_{r1}^2\epsilon_{r2}^2-16\eta_1\epsilon_{r2}^2\eta_2-16\eta_1\epsilon_{r3}^2\eta_3-16\eta_1\epsilon_{c2}^2\eta_2 \\
& -16\eta_1\epsilon_{c3}^2\eta_3+16\epsilon_{c3}^2\eta_3^2\epsilon_{r2}+32\eta_2\epsilon_{r3}\eta_1+4\eta_2^2\epsilon_{r1}^2\epsilon_{r2}^2+4\eta_3^2\epsilon_{r1}^2\epsilon_{r2}+\eta_3^2\epsilon_{r1}^2 \\
& \epsilon_{c2}^2+4\eta_3^2\epsilon_{r1}\epsilon_{r2}^2-4\eta_2\epsilon_{r1}\epsilon_{c3}^2\eta_1-4\eta_2\epsilon_{r2}\epsilon_{r3}^2\eta_1-16\eta_2\epsilon_{r2}\epsilon_{r3}\eta_1-32 \\
& \eta_3^2\epsilon_{r1}\epsilon_{r2}\epsilon_{r3}-8\eta_3^2\epsilon_{r1}\epsilon_{r2}^2\epsilon_{r3}-8\eta_3^2\epsilon_{r1}\epsilon_{c2}^2\epsilon_{r3}-4\eta_2\epsilon_{cl}^2\epsilon_{r3}\eta_3-4\eta_2\epsilon_{cl}^2\epsilon_{c3}^2\eta_3 \\
& +4\epsilon_{cl}^2\eta_1^2\epsilon_{r2}^2\epsilon_{r3}-4\eta_2\epsilon_{r1}^2\epsilon_{r3}^2\eta_3-4\eta_2\epsilon_{r1}^2\epsilon_{r3}\eta_3-4\eta_2\epsilon_{r1}^2\epsilon_{c3}^2\eta_3-4\eta_2\epsilon_{cl}^2 \\
& \epsilon_{r3}^2\eta_3+8\eta_2\epsilon_{r1}^2\epsilon_{r2}\eta_1+8\eta_2\epsilon_{r1}\epsilon_{r2}\eta_1+8\eta_2\epsilon_{r2}\epsilon_{r3}^2\eta_3+8\eta_2\epsilon_{r2}\epsilon_{r3}\eta_3+8\eta_2\epsilon_{r2} \\
& \epsilon_{c3}^2\eta_3-8\eta_1^2\epsilon_{r1}\epsilon_{r2}\epsilon_{r3}^2-2\eta_1^2\epsilon_{r1}\epsilon_{c2}^2\epsilon_{c3}^2+16\epsilon_{cl}^2\eta_1^2\epsilon_{r2}\epsilon_{r3}-2\eta_1^2\epsilon_{r1}\epsilon_{r2}^2\epsilon_{c3}^2-32 \\
& \eta_1^2\epsilon_{r1}\epsilon_{r2}\epsilon_{r3}-8\eta_1^2\epsilon_{r1}\epsilon_{r2}^2\epsilon_{r3}-8\eta_1^2\epsilon_{r1}\epsilon_{c2}^2\epsilon_{r3}-2\eta_1^2\epsilon_{r1}\epsilon_{c2}^2\epsilon_{r3}^2-2\eta_1^2\epsilon_{r1}\epsilon_{r2}^2\epsilon_{r3}^2 \\
& -8\eta_1^2\epsilon_{r1}\epsilon_{r2}\epsilon_{c3}^2-4\eta_3\epsilon_{r2}^2\epsilon_{r3}\eta_1+8\eta_3\epsilon_{cl}^2\epsilon_{r3}\eta_1-4\eta_3\epsilon_{c2}^2\epsilon_{r3}\eta_1-16\eta_3 \\
& \epsilon_{cl}^2\epsilon_{r2}\eta_1-4\eta_3\epsilon_{cl}^2\epsilon_{c2}^2\eta_1+8\eta_3\epsilon_{r1}\epsilon_{r3}\eta_1-16\eta_3\epsilon_{r2}\epsilon_{r3}\eta_1+8\eta_3\epsilon_{r1}^2\epsilon_{r3}\eta_1 \\
& -4\eta_3\epsilon_{cl}^2\epsilon_{r2}^2\eta_1+\epsilon_{c2}^2\eta_2^2\epsilon_{cl}^2\epsilon_{r3}^2-2\eta_2^2\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{c3}^2-8\eta_2^2\epsilon_{r1}^2\epsilon_{r2}\epsilon_{r3}+4\epsilon_{cl}^2\eta_1^2\epsilon_{r2}\epsilon_{r3}^2 \\
& -16\eta_1\epsilon_{r2}\epsilon_{r3}^2\eta_3-16\eta_1\epsilon_{r2}\epsilon_{c3}^2\eta_3+8\eta_3\epsilon_{c2}^2\epsilon_{r3}\eta_2+8\eta_1\epsilon_{r1}\epsilon_{r3}^2\eta_3+8\eta_1\epsilon_{r1} \\
& \epsilon_{c3}^2\eta_3+4\epsilon_{c3}^2\eta_3^2\epsilon_{cl}^2\epsilon_{r2}+8\eta_2\eta_1\epsilon_{r1}^2\epsilon_{c2}^2+8\eta_2\eta_1\epsilon_{r1}^2\epsilon_{r2}^2+\epsilon_{c3}^2\eta_3^2\epsilon_{cl}^2\epsilon_{r2}^2+4\epsilon_{c2}^2 \\
& \eta_2^2\epsilon_{r1}\epsilon_{c3}^2-2\eta_2^2\epsilon_{cl}^2\epsilon_{r2}\epsilon_{r3}^2-8\eta_2^2\epsilon_{cl}^2\epsilon_{r2}\epsilon_{r3}-2\eta_2^2\epsilon_{cl}^2\epsilon_{r2}\epsilon_{c3}^2+4\epsilon_{cl}^2\eta_1^2\epsilon_{c2}^2\epsilon_{r3}+\epsilon_{cl}^2 \\
& \eta_1^2\epsilon_{c2}^2\epsilon_{c3}^2-4\eta_2\epsilon_{r1}^2\epsilon_{r2}\eta_3-16\eta_2\epsilon_{r1}\epsilon_{r2}\eta_3-2\eta_3\epsilon_{r1}^2\epsilon_{c2}^2\epsilon_{c3}^2-32\eta_1\epsilon_{r1}^2\epsilon_{r2}\epsilon_{r3} \\
& +16\eta_3^2\epsilon_{r1}\epsilon_{r2}+4\eta_3^2\epsilon_{r1}\epsilon_{c2}^2+\eta_3^2\epsilon_{cl}^2\epsilon_{r2}^2+4\eta_3^2\epsilon_{cl}^2\epsilon_{r2}+\eta_3^2\epsilon_{cl}^2\epsilon_{c2}^2-32\eta_3^2\epsilon_{r1}\epsilon_{r3} \\
& -32\eta_3^2\epsilon_{r2}\epsilon_{r3}+\eta_2^2\epsilon_{r1}^2\epsilon_{r3}^2+4\eta_2^2\epsilon_{r1}^2\epsilon_{r3}+\eta_2^2\epsilon_{r1}^2\epsilon_{c3}^2+\eta_2^2\epsilon_{cl}^2\epsilon_{r3}^2+4\eta_2^2\epsilon_{cl}^2\epsilon_{r3}+ \\
& \eta_2^2\epsilon_{cl}^2\epsilon_{c3}^2+\eta_3^2\epsilon_{r1}^2\epsilon_{r2}^2-8\eta_2^2\epsilon_{r1}^2\epsilon_{r2}-32\eta_2^2\epsilon_{r1}\epsilon_{r2}-8\eta_2^2\epsilon_{cl}^2\epsilon_{r2}+4\eta_2^2\epsilon_{r1}\epsilon_{r3}^2+16 \\
& \eta_2^2\epsilon_{r1}\epsilon_{r3}+4\eta_2^2\epsilon_{r1}\epsilon_{c3}^2-8\eta_2^2\epsilon_{r2}\epsilon_{r3}^2-32\eta_2^2\epsilon_{r2}\epsilon_{r3}-8\eta_2^2\epsilon_{r2}\epsilon_{c3}^2+16\epsilon_{cl}^2\eta_1^2\epsilon_{r2} \\
& -16\eta_3\epsilon_{r2}^2\eta_2-16\eta_3\epsilon_{c2}^2\eta_2+4\eta_1^2\epsilon_{r1}^2\epsilon_{r3}^2+4\epsilon_{c3}^2\eta_3^2\epsilon_{r1}^2+4\eta_1^2\epsilon_{r1}^2\epsilon_{c3}^2+16\eta_1^2 \\
& \epsilon_{r1}^2\epsilon_{r3}+16\eta_2^2\epsilon_{r1}\epsilon_{r2}^2+4\epsilon_{cl}^2\eta_1^2\epsilon_{r2}^2+16\eta_2^2\epsilon_{r3}\epsilon_{r2}^2+32\eta_2\epsilon_{r1}\eta_3-8\eta_3^2\epsilon_{c2}^2\epsilon_{r3} \\
& +8\eta_2\epsilon_{cl}^2\eta_3-16\eta_2\epsilon_{c3}^2\eta_3+32\eta_3\epsilon_{r2}\eta_1-2\eta_1\epsilon_{r1}\epsilon_{c2}^2\epsilon_{r3}^2-8\eta_1\epsilon_{r1}^2\epsilon_{r2}\epsilon_{c3}^2 \\
& -8\eta_1\epsilon_{r1}^2\epsilon_{c2}^2\epsilon_{r3}-2\eta_1\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{c3}^2-2\eta_1\epsilon_{r1}\epsilon_{r2}^2\epsilon_{r3}^2-2\eta_1\epsilon_{r1}^2\epsilon_{c2}^2\epsilon_{c3}^2-8\eta_1\epsilon_{r1}\epsilon_{r2}
\end{aligned}$$

$$\begin{aligned}
& \epsilon_{c3}^2 - 2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 32 \eta_1 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - 2 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 8 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 2 \eta_1 \epsilon_{c1}^2 \\
& \epsilon_{r2}^2 \epsilon_{r3}^2 - 8 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 2 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} \\
& - 8 \eta_1 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 2 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 - 2 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{r2} \\
& \epsilon_{r3}^2 - 32 \eta_1 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 2 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_1 \epsilon_{c1}^2 \\
& \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_3^2 \\
& \epsilon_{r1}^2 \epsilon_{r3}^2)
\end{aligned}$$

$eps_r := rhs(sol[2])$

$$\begin{aligned}
& - \left(-64 + 8 \epsilon_{c2}^2 \eta_2^2 \epsilon_{c3}^2 + 32 \eta_1 + 32 \eta_2 + 32 \eta_3 - 64 \epsilon_{r1} - 64 \epsilon_{r2} - 64 \epsilon_{r3} - 16 \epsilon_{c1}^2 \right. \\
& - 16 \epsilon_{c2}^2 - 16 \epsilon_{c3}^2 - 16 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r3} - 16 \eta_3^2 \epsilon_{r2}^2 \epsilon_{r3} + 36 \epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 \epsilon_{r2}^2 + 8 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r3}^2 \\
& + 8 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r2}^2 + 8 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2}^2 - 32 \eta_2 \epsilon_{r2} \eta_1 + 16 \eta_3 \epsilon_{r1}^2 \eta_1 \epsilon_{r3}^2 - 32 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \eta_2 \\
& - 32 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \eta_2 - 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \eta_2 - 16 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} - 4 \eta_3^2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_1 \epsilon_{r2}^2 \\
& \epsilon_{r3}^2 \eta_2 - 8 \eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_3 - 32 \eta_1 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 + 16 \eta_3 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 + 8 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 + 32 \\
& \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1} \epsilon_{r3} + 8 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3} + 32 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1} \epsilon_{r2} + 16 \eta_2 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 + 16 \eta_2 \eta_1 \epsilon_{c1}^2 \\
& \epsilon_{c2}^2 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \eta_3 - 32 \eta_2 \epsilon_{r1} \epsilon_{r3}^2 \eta_3 - 32 \eta_2 \epsilon_{r1} \epsilon_{r3} \eta_3 - 32 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 \eta_3 + 2 \epsilon_{c2}^2 \\
& \eta_2^2 \epsilon_{r1}^2 \epsilon_{c3}^2 + 16 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \eta_1 - 8 \eta_2 \epsilon_{r1} \epsilon_{r3}^2 \eta_1 - 32 \eta_2 \epsilon_{r1} \epsilon_{r3} \eta_1 + 8 \epsilon_{c3}^2 \eta_3^2 \epsilon_{c1}^2 \\
& + 16 \eta_3 \epsilon_{c2}^2 \eta_2 \epsilon_{r3}^2 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r3}^2 \eta_1 - 32 \eta_2 \epsilon_{c1}^2 \epsilon_{r3} \eta_1 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{c3}^2 \eta_1 + 16 \eta_3 \\
& \epsilon_{r2}^2 \eta_2 \epsilon_{c3}^2 + 16 \eta_3 \epsilon_{r2}^2 \eta_2 \epsilon_{r3}^2 - 8 \eta_1 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_2 - 8 \eta_1 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_3 - 8 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_2 \\
& - 8 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_3 - 32 \eta_1 \epsilon_{c2}^2 \epsilon_{r3} \eta_2 - 8 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_2 - 8 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_3 - 8 \eta_3 \epsilon_{c1}^2 \\
& \epsilon_{c2}^2 \eta_2 - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r3}^2 \eta_1 - 32 \eta_2 \epsilon_{r1}^2 \epsilon_{r3} \eta_1 - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{c3}^2 \eta_1 + 2 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 2 \epsilon_{c2}^2 \\
& \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 2 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 + 16 \eta_3 \eta_1 \epsilon_{c1}^2 \epsilon_{r3}^2 + 16 \eta_3 \eta_1 \epsilon_{c1}^2 \epsilon_{c3}^2 + 32 \eta_3^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 \\
& + 8 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 + 2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_2 - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_2 - 4 \eta_3^2 \epsilon_{c1}^2 \\
& \epsilon_{r2}^2 \epsilon_{r3}^2 - 4 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 16 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} + 16 \eta_3 \epsilon_{r1}^2 \eta_1 \epsilon_{c3}^2 + 8 \eta_2^2 \epsilon_{c3}^2 \epsilon_{r2}^2 + 8 \eta_1^2 \\
& \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 32 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \eta_1 - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_1 - 8 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \eta_1 - 32 \eta_3 \epsilon_{r1} \epsilon_{r2} \eta_1 \\
& + 144 \epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 + 144 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_1 + 144 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 - 4 \eta_3^2 \\
& \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 8 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \eta_3 \epsilon_{c2}^2 \eta_2 \epsilon_{c3}^2 + 32 \epsilon_{c1}^2 \eta_1^2 + 32 \eta_3^2 \epsilon_{r1} + 32 \eta_3^2 \epsilon_{r2} \\
& - 64 \eta_3^2 \epsilon_{r3} + 8 \eta_3^2 \epsilon_{r1}^2 + 8 \eta_3^2 \epsilon_{r2}^2 + 8 \eta_3^2 \epsilon_{c1}^2 + 8 \eta_3^2 \epsilon_{c2}^2 + 32 \eta_2^2 \epsilon_{r1} - 64 \eta_2^2 \epsilon_{r2} \\
& + 32 \eta_2^2 \epsilon_{r3} + 8 \eta_2^2 \epsilon_{r1}^2 + 8 \eta_2^2 \epsilon_{r3}^2 + 8 \eta_2^2 \epsilon_{c1}^2 + 8 \eta_2^2 \epsilon_{c3}^2 + 32 \epsilon_{c3}^2 \eta_3^2 - 4 \epsilon_{r1}^2 \epsilon_{r2}^2 - 16
\end{aligned} \tag{2.8}$$

$$\begin{aligned}
& \epsilon_{r1}^2 \epsilon_{r2} - 4 \epsilon_{r1}^2 \epsilon_{c2}^2 - 16 \epsilon_{r1} \epsilon_{r2}^2 - 64 \epsilon_{r1} \epsilon_{r2} - 16 \epsilon_{r1} \epsilon_{c2}^2 - 4 \epsilon_{c1}^2 \epsilon_{r2}^2 - 16 \epsilon_{c1}^2 \epsilon_{r2} - 16 \\
& \epsilon_{r1}^2 - 16 \epsilon_{r2}^2 - 16 \epsilon_{r3}^2 + 32 \eta_1^2 + 32 \eta_2^2 + 32 \eta_3^2 - \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 \\
& - \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 - \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 16 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \\
& - 16 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 16 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} - \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 4 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \\
& \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 4 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 4 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 - \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \\
& \epsilon_{c3}^2 - \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 4 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} - \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 - \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 4 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} \\
& - \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} - 4 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 - \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} \\
& - \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 - 16 \eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 + 36 \epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 \epsilon_{c2}^2 \\
& + 144 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \epsilon_{r1} + 36 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \epsilon_{r1}^2 + 36 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \epsilon_{c1}^2 + 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \\
& \epsilon_{r3}^2 \eta_3 + 144 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 \epsilon_{r3} + 36 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 \epsilon_{r3}^2 + 36 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 \epsilon_{c3}^2 \\
& + 144 \epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 \epsilon_{r2} + 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_1 + 16 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 \eta_3 - \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 \\
& - 16 \eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 4 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 4 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} \\
& - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 4 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 16 \eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \\
& - 4 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 - \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \eta_3 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 4 \eta_3 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} - 4 \eta_3 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 \\
& - \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} - \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + 2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \\
& + 8 \eta_2^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 + 2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 32 \eta_2^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 4 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 16 \\
& \eta_2^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 + 8 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2} + 2 \epsilon_{c2}^2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{c3}^2 + 8 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1} \epsilon_{c2}^2 - 16 \eta_2^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 \\
& - 4 \epsilon_{c1}^2 \epsilon_{c2}^2 - 16 \epsilon_{r1} \epsilon_{r3}^2 - 64 \epsilon_{r1} \epsilon_{r3} - 16 \epsilon_{r1} \epsilon_{c3}^2 - 16 \epsilon_{r2} \epsilon_{r3}^2 - 64 \epsilon_{r2} \epsilon_{r3} - 16 \epsilon_{r2} \\
& \epsilon_{c3}^2 - 4 \epsilon_{r1}^2 \epsilon_{r3}^2 - 16 \epsilon_{r1}^2 \epsilon_{r3} - 4 \epsilon_{r1}^2 \epsilon_{c3}^2 - 4 \epsilon_{r2}^2 \epsilon_{r3}^2 - 16 \epsilon_{r2}^2 \epsilon_{r3} - 4 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \epsilon_{c1}^2 \epsilon_{r3}^2 \\
& - 16 \epsilon_{c1}^2 \epsilon_{r3} - 4 \epsilon_{c1}^2 \epsilon_{c3}^2 - 4 \epsilon_{c2}^2 \epsilon_{r3}^2 - 16 \epsilon_{c2}^2 \epsilon_{r3} - 4 \epsilon_{c2}^2 \epsilon_{c3}^2 - 16 \eta_1 \epsilon_{r1} + 32 \eta_1 \epsilon_{r2} \\
& + 32 \eta_1 \epsilon_{r3} - 16 \eta_1 \epsilon_{r1}^2 + 8 \eta_1 \epsilon_{r2}^2 + 8 \eta_1 \epsilon_{r3}^2 - 16 \eta_1 \epsilon_{c1}^2 + 8 \eta_1 \epsilon_{c2}^2 + 8 \eta_1 \epsilon_{c3}^2 \\
& + 32 \eta_2 \epsilon_{r1} - 16 \eta_2 \epsilon_{r2} + 32 \eta_2 \epsilon_{r3} + 8 \eta_2 \epsilon_{r1}^2 - 16 \eta_2 \epsilon_{r2}^2 + 8 \eta_2 \epsilon_{r3}^2 + 8 \eta_2 \epsilon_{c1}^2 \\
& - 16 \eta_2 \epsilon_{c2}^2 + 8 \eta_2 \epsilon_{c3}^2 + 32 \eta_3 \epsilon_{r1} + 32 \eta_3 \epsilon_{r2} - 16 \eta_3 \epsilon_{r3} + 8 \eta_3 \epsilon_{r1}^2 + 8 \eta_3 \epsilon_{r2}^2 \\
& - 16 \eta_3 \epsilon_{r3}^2 + 8 \eta_3 \epsilon_{c1}^2 + 8 \eta_3 \epsilon_{c2}^2 - 16 \eta_3 \epsilon_{c3}^2 + 32 \eta_1^2 \epsilon_{r1}^2 + 32 \eta_2^2 \epsilon_{r2}^2 + 32 \epsilon_{c2}^2 \eta_2^2 \\
& + 64 \eta_2 \eta_1 + 64 \eta_3 \eta_1 + 64 \eta_2 \eta_3 + 32 \eta_3^2 \epsilon_{r3}^2 - 64 \eta_2^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} + 8 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \\
& + 8 \eta_2^2 \epsilon_{r2}^2 \epsilon_{r1} \epsilon_{c3}^2 + 2 \epsilon_{c3}^2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{c2}^2 + 2 \epsilon_{c1}^2 \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 2 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 \\
& \epsilon_{c2}^2 + 8 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1} \epsilon_{r3}^2 + 8 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2} \epsilon_{c3}^2 + 2 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 8 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 2 \eta_2^2 \epsilon_{c1}^2
\end{aligned}$$

$$\begin{aligned}
& \epsilon_{r2}^2 \epsilon_{c3}^2 + 8 \eta_3^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 + 8 \epsilon_{c2}^2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r3} + 16 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \eta_2 + 16 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \eta_2 + 8 \\
& \eta_3^2 \epsilon_{r3}^2 \epsilon_{r1}^2 \epsilon_{r2} + 2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 2 \eta_3^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \epsilon_{r1}^2 - 8 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \eta_1 + 8 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \\
& + 2 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 2 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 32 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} + 8 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 + 2 \eta_1^2 \epsilon_{r1}^2 \\
& \epsilon_{c2}^2 \epsilon_{c3}^2 + 8 \eta_1^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} + 2 \eta_1^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 16 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} \eta_3 + 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_1 \\
& + 16 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_1 + 4 \eta_2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_3 + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_1 \\
& + 16 \eta_3 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} \eta_1 + 4 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} \eta_1 + 4 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_1 + 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} \eta_3 \\
& + 4 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} \eta_1 + 4 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} \eta_1 + 16 \eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} \eta_1 + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 \\
& + 4 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 16 \eta_1 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 \eta_3 + 4 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_2 \\
& + 4 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_3 + 16 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 16 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} \eta_2 + 16 \eta_1 \epsilon_{r1} \epsilon_{r2} \\
& \epsilon_{c3}^2 \eta_3 + 4 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_2 + 4 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_3 + 4 \eta_2 \epsilon_{r3}^2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 - 16 \eta_1^2 \epsilon_{r1} \epsilon_{r2}^2 \\
& - 64 \eta_1^2 \epsilon_{r1} \epsilon_{r2} - 16 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} - 16 \eta_1^2 \epsilon_{r1} \epsilon_{c2}^2 - 16 \eta_1^2 \epsilon_{r1} \epsilon_{r3}^2 - 64 \eta_1^2 \epsilon_{r1} \epsilon_{r3} - 16 \\
& \eta_1^2 \epsilon_{r1} \epsilon_{c3}^2 + 16 \eta_2 \epsilon_{r3}^2 \eta_1 + 32 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1} + 16 \eta_3 \epsilon_{r2}^2 \eta_1 - 32 \eta_2 \epsilon_{r1}^2 \eta_1 + 8 \epsilon_{c2}^2 \eta_2^2 \epsilon_{c1}^2 \\
& - 32 \eta_3 \epsilon_{r1}^2 \eta_1 + 8 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r3}^2 - 4 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 4 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 4 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} - \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \\
& - 4 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 - \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 16 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 - \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 64 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \\
& - 16 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 16 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} - 4 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 4 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \epsilon_{c1}^2 \\
& \epsilon_{r2}^2 \epsilon_{r3} - 16 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - 4 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 4 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 - \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 16 \\
& \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} - 4 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 - \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 4 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} - \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 4 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \\
& - 16 \eta_1 \epsilon_{r1}^2 \epsilon_{r2} - 4 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 - 4 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 - 16 \eta_1 \epsilon_{r1} \epsilon_{r2} - 4 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 - 4 \eta_1 \epsilon_{c1}^2 \\
& \epsilon_{r2}^2 - 16 \eta_1 \epsilon_{c1}^2 \epsilon_{r2} - 4 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 - 4 \eta_1 \epsilon_{r1} \epsilon_{r3}^2 + 4 \eta_2 \epsilon_{r3}^2 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 + 4 \eta_2 \epsilon_{r3}^2 \eta_1 \epsilon_{c1}^2 \\
& \epsilon_{c2}^2 + 4 \eta_2 \epsilon_{c3}^2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 + 4 \eta_2 \epsilon_{c3}^2 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 + 4 \eta_2 \epsilon_{c3}^2 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 + 4 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_2 \\
& + 4 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_3 + 16 \eta_2 \epsilon_{r3} \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 + 16 \eta_2 \epsilon_{r3} \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 + 16 \eta_2 \epsilon_{r3} \eta_1 \epsilon_{c1}^2 \\
& \epsilon_{c2}^2 + 4 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_2 + 4 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_3 + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} \eta_1 + 16 \eta_3 \\
& \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} \eta_1 + 4 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 \eta_1 + 4 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 \eta_1 + 16 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} \eta_1 \\
& + 4 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 \eta_1 + 16 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} \eta_1 + 16 \eta_3 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \eta_3 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \\
& - 8 \eta_2 \epsilon_{r2} \epsilon_{c3}^2 \eta_1 - 64 \eta_1^2 \epsilon_{r1} + 32 \eta_1^2 \epsilon_{r2} + 32 \eta_1^2 \epsilon_{r3} + 8 \eta_1^2 \epsilon_{r2}^2 + 8 \eta_1^2 \epsilon_{r3}^2 + 8 \eta_1^2 \\
& \epsilon_{c2}^2 + 8 \eta_1^2 \epsilon_{c3}^2 + 8 \eta_2^2 \epsilon_{r3}^2 \epsilon_{r2}^2 - 16 \eta_1 \epsilon_{r1} \epsilon_{r3} - 4 \eta_1 \epsilon_{r1} \epsilon_{c3}^2 + 8 \eta_1 \epsilon_{r2} \epsilon_{r3}^2
\end{aligned}$$

$$\begin{aligned}
& + 32 \eta_1 \epsilon_{r2} \epsilon_{r3} + 8 \eta_1 \epsilon_{r2} \epsilon_{c3}^2 - 4 \eta_1 \epsilon_{r1}^2 \epsilon_{r3}^2 - 16 \eta_1 \epsilon_{r1}^2 \epsilon_{r3} - 4 \eta_1 \epsilon_{r1}^2 \epsilon_{c3}^2 + 2 \eta_1 \epsilon_{r2}^2 \\
& \epsilon_{r3}^2 + 8 \eta_1 \epsilon_{r2}^2 \epsilon_{r3} + 2 \eta_1 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \eta_1 \epsilon_{c1}^2 \epsilon_{r3}^2 - 16 \eta_1 \epsilon_{c1}^2 \epsilon_{r3} - 4 \eta_1 \epsilon_{c1}^2 \epsilon_{c3}^2 + 2 \eta_1 \\
& \epsilon_{c2}^2 \epsilon_{r3}^2 + 8 \eta_1 \epsilon_{c2}^2 \epsilon_{r3} + 2 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \\
& - 16 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 - 16 \eta_2 \epsilon_{r1} \epsilon_{r2} - 16 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 - 4 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 - 4 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} - 4 \eta_2 \\
& \epsilon_{c1}^2 \epsilon_{c2}^2 + 8 \eta_2 \epsilon_{r1} \epsilon_{r3}^2 + 32 \eta_2 \epsilon_{r1} \epsilon_{r3} + 8 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{r2} \epsilon_{r3}^2 - 16 \eta_2 \epsilon_{r2} \epsilon_{r3} \\
& - 4 \eta_2 \epsilon_{r2} \epsilon_{c3}^2 + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r3} + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 16 \eta_2 \\
& \epsilon_{r2}^2 \epsilon_{r3} - 4 \eta_2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r3}^2 + 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r3} + 2 \eta_2 \epsilon_{c1}^2 \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{c2}^2 \epsilon_{r3}^2 \\
& - 16 \eta_2 \epsilon_{c2}^2 \epsilon_{r3} - 4 \eta_2 \epsilon_{c2}^2 \epsilon_{c3}^2 + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 + 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 + 8 \eta_3 \epsilon_{r1} \\
& \epsilon_{r2}^2 + 32 \eta_3 \epsilon_{r1} \epsilon_{r2} + 8 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 + 2 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 + 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r2} + 2 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \\
& - 16 \eta_3 \epsilon_{r1} \epsilon_{r3}^2 - 16 \eta_3 \epsilon_{r1} \epsilon_{r3} - 16 \eta_3 \epsilon_{r1} \epsilon_{c3}^2 - 16 \eta_3 \epsilon_{r2} \epsilon_{r3}^2 - 16 \eta_3 \epsilon_{r2} \epsilon_{r3} \\
& - 16 \eta_3 \epsilon_{r2} \epsilon_{c3}^2 - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r3}^2 - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r3} - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{c3}^2 - 4 \eta_3 \epsilon_{r2}^2 \epsilon_{r3}^2 - 4 \eta_3 \\
& \epsilon_{r2}^2 \epsilon_{r3} - 4 \eta_3 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \eta_3 \epsilon_{c1}^2 \epsilon_{r3}^2 - 4 \eta_3 \epsilon_{c1}^2 \epsilon_{r3} - 4 \eta_3 \epsilon_{c1}^2 \epsilon_{c3}^2 - 4 \eta_3 \epsilon_{c2}^2 \epsilon_{r3}^2 - 4 \eta_3 \\
& \epsilon_{c2}^2 \epsilon_{r3} - 4 \eta_3 \epsilon_{c2}^2 \epsilon_{c3}^2 + 8 \eta_3^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 8 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 8 \eta_3^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 8 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r3}^2 + 32 \\
& \eta_3^2 \epsilon_{r1} \epsilon_{r3}^2 + 32 \eta_3^2 \epsilon_{r2} \epsilon_{r3}^2 - 32 \eta_3 \epsilon_{r3} \eta_1 + 16 \eta_3 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 + 4 \eta_3 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 \\
& + 4 \eta_3 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 4 \eta_3 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \eta_3 \epsilon_{r2} \eta_1 \epsilon_{c1}^2 \epsilon_{c3}^2 + 4 \eta_3 \epsilon_{r2}^2 \eta_1 \epsilon_{c1}^2 \epsilon_{c3}^2 \\
& + 4 \eta_3 \epsilon_{c1}^2 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \eta_3 \epsilon_{c1}^2 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 + 16 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 \eta_3 + 4 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 \eta_3 \\
& + 4 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} \eta_3 + 4 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 \eta_3 + 16 \eta_3 \epsilon_{r1}^2 \eta_1 \epsilon_{r2} \epsilon_{r3}^2 + 16 \eta_3 \epsilon_{c1}^2 \eta_1 \epsilon_{r2} \epsilon_{r3}^2 \\
& + 16 \eta_3 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 + 4 \eta_3 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \eta_3 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \eta_3 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 \\
& + 16 \eta_3 \epsilon_{r2}^2 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_1 \epsilon_{r3}^2 + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_1 \epsilon_{c3}^2 + 16 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \eta_1 \epsilon_{c3}^2 \\
& + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_1 \epsilon_{r3}^2 + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_1 \epsilon_{c3}^2 + 4 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \eta_1 \epsilon_{r3}^2 + 16 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} \eta_2 \\
& + 16 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 4 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 - 32 \eta_3 \epsilon_{r1} \eta_1 - 32 \eta_2 \epsilon_{c1}^2 \eta_1 + 16 \eta_2 \\
& \epsilon_{c3}^2 \eta_1 - 32 \eta_3 \epsilon_{c1}^2 \eta_1 + 16 \eta_3 \epsilon_{c2}^2 \eta_1 - 32 \eta_2 \epsilon_{r1} \eta_1 + 8 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 + 8 \epsilon_{c1}^2 \eta_1^2 \epsilon_{c2}^2 \\
& + 32 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1} + 8 \epsilon_{c1}^2 \eta_1^2 \epsilon_{c3}^2 + 2 \eta_1^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + 2 \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 8 \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3} + 2 \eta_1^2 \epsilon_{r2}^2 \\
& \epsilon_{c3}^2 + 8 \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3} + 2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 32 \eta_1^2 \epsilon_{r2} \epsilon_{r3} + 8 \eta_1^2 \epsilon_{r2} \epsilon_{c3}^2 + 8 \eta_1^2 \epsilon_{r2} \epsilon_{r3}^2 + 32 \epsilon_{c2}^2 \\
& \eta_2^2 \epsilon_{r3} + 32 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r3} - 32 \eta_2 \epsilon_{r2} \eta_3 - 32 \eta_2 \epsilon_{r3} \eta_3 + 16 \eta_2 \epsilon_{r1}^2 \eta_3 - 32 \eta_2 \epsilon_{r3}^2 \eta_3 \\
& + 32 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2} + 8 \eta_1^2 \epsilon_{r1}^2 \epsilon_{c2}^2 + 8 \epsilon_{c3}^2 \eta_3^2 \epsilon_{c2}^2 + 8 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 - 32 \eta_1 \epsilon_{r2}^2 \eta_2 - 32 \eta_1 \\
& \epsilon_{r3}^2 \eta_3 - 32 \eta_1 \epsilon_{c2}^2 \eta_2 - 32 \eta_1 \epsilon_{c3}^2 \eta_3 + 32 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r2} + 64 \eta_2 \epsilon_{r3} \eta_1 + 8 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2
\end{aligned}$$

$$\begin{aligned}
& + 8 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2} + 2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{c2}^2 + 8 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 - 8 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 \eta_1 - 8 \eta_2 \epsilon_{r2} \epsilon_{r3}^2 \eta_1 \\
& - 32 \eta_2 \epsilon_{r2} \epsilon_{r3} \eta_1 - 64 \eta_3^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - 16 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 16 \eta_3^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_2 \\
& \epsilon_{c1}^2 \epsilon_{r3} \eta_3 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{c3}^2 \eta_3 + 8 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r3}^2 \eta_3 - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r3} \eta_3 \\
& - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{c3}^2 \eta_3 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r3}^2 \eta_3 + 16 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \eta_1 + 16 \eta_2 \epsilon_{r1} \epsilon_{r2} \eta_1 + 16 \eta_2 \epsilon_{r2} \\
& \epsilon_{r3}^2 \eta_3 + 16 \eta_2 \epsilon_{r2} \epsilon_{r3} \eta_3 + 16 \eta_2 \epsilon_{r2} \epsilon_{c3}^2 \eta_3 - 16 \eta_1^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - 4 \eta_1^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 + 32 \\
& \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2} \epsilon_{r3} - 4 \eta_1^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 64 \eta_1^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - 16 \eta_1^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 16 \eta_1^2 \epsilon_{r1} \\
& \epsilon_{c2}^2 \epsilon_{r3} - 4 \eta_1^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 4 \eta_1^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 - 16 \eta_1^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 - 8 \eta_3 \epsilon_{r2}^2 \epsilon_{r3} \eta_1 \\
& + 16 \eta_3 \epsilon_{c1}^2 \epsilon_{r3} \eta_1 - 8 \eta_3 \epsilon_{c2}^2 \epsilon_{r3} \eta_1 - 32 \eta_3 \epsilon_{c1}^2 \epsilon_{r2} \eta_1 - 8 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \eta_1 \\
& + 16 \eta_3 \epsilon_{r1} \epsilon_{r3} \eta_1 - 32 \eta_3 \epsilon_{r2} \epsilon_{r3} \eta_1 + 16 \eta_3 \epsilon_{r1}^2 \epsilon_{r3} \eta_1 - 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \eta_1 + 2 \epsilon_{c2}^2 \eta_2^2 \\
& \epsilon_{c1}^2 \epsilon_{r3}^2 - 4 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 16 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} + 8 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2} \epsilon_{r3}^2 - 32 \eta_1 \epsilon_{r2} \epsilon_{r3}^2 \eta_3 \\
& - 32 \eta_1 \epsilon_{r2} \epsilon_{c3}^2 \eta_3 + 16 \eta_3 \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 16 \eta_1 \epsilon_{r1} \epsilon_{r3}^2 \eta_3 + 16 \eta_1 \epsilon_{r1} \epsilon_{c3}^2 \eta_3 + 8 \epsilon_{c3}^2 \eta_3^2 \\
& \epsilon_{c1}^2 \epsilon_{r2} + 16 \eta_2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 + 16 \eta_2 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 + 2 \epsilon_{c3}^2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2}^2 + 8 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1} \epsilon_{c3}^2 - 4 \\
& \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 16 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} - 4 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 + 8 \epsilon_{c1}^2 \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3} + 2 \epsilon_{c1}^2 \eta_1^2 \epsilon_{c2}^2 \epsilon_{c3}^2 \\
& - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \eta_3 - 32 \eta_2 \epsilon_{r1} \epsilon_{r2} \eta_3 - \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 16 \eta_1 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} + 32 \eta_3^2 \epsilon_{r1} \epsilon_{r2} \\
& + 8 \eta_3^2 \epsilon_{r1} \epsilon_{c2}^2 + 2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2}^2 + 8 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2} + 2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{c2}^2 - 64 \eta_3^2 \epsilon_{r1} \epsilon_{r3} - 64 \\
& \eta_3^2 \epsilon_{r2} \epsilon_{r3} + 2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 8 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3} + 2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{c3}^2 + 2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r3}^2 + 8 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r3} + 2 \\
& \eta_2^2 \epsilon_{c1}^2 \epsilon_{c3}^2 + 2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 - 16 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2} - 64 \eta_2^2 \epsilon_{r1} \epsilon_{r2} - 16 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2} + 8 \eta_2^2 \epsilon_{r1} \epsilon_{r3}^2 \\
& + 32 \eta_2^2 \epsilon_{r1} \epsilon_{r3} + 8 \eta_2^2 \epsilon_{r1} \epsilon_{c3}^2 - 16 \eta_2^2 \epsilon_{r2} \epsilon_{r3}^2 - 64 \eta_2^2 \epsilon_{r2} \epsilon_{r3} - 16 \eta_2^2 \epsilon_{r2} \epsilon_{c3}^2 + 32 \\
& \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2} - 32 \eta_3 \epsilon_{r2}^2 \eta_2 - 32 \eta_3 \epsilon_{c2}^2 \eta_2 + 8 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 8 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 + 8 \eta_1^2 \epsilon_{r1}^2 \epsilon_{c3}^2 \\
& + 32 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r3} + 32 \eta_2^2 \epsilon_{r1} \epsilon_{r2}^2 + 8 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2}^2 + 32 \eta_2^2 \epsilon_{r3} \epsilon_{r2}^2 + 64 \eta_2 \epsilon_{r1} \eta_3 - 16 \eta_3^2 \\
& \epsilon_{c2}^2 \epsilon_{r3} + 16 \eta_2 \epsilon_{c1}^2 \eta_3 - 32 \eta_2 \epsilon_{c3}^2 \eta_3 + 64 \eta_3 \epsilon_{r2} \eta_1 - \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 4 \eta_1 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 \\
& - 4 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} - \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 - \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 4 \eta_1 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 \\
& - \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 16 \eta_1 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 4 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \\
& - 4 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} - 4 \eta_1 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 4 \eta_1 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 \\
& - 4 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 - \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \eta_1 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 16 \eta_1 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} \\
& - 4 \eta_1 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 - \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 4 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} - \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} \\
& - 4 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 16 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r3}^2) / (64 + 4 \epsilon_{c2}^2 \eta_2^2
\end{aligned}$$

$$\begin{aligned}
& \epsilon_{c3}^2 + 64 \eta_1 + 64 \eta_2 + 64 \eta_3 + 64 \epsilon_{r1} + 64 \epsilon_{r2} + 64 \epsilon_{r3} + 16 \epsilon_{c1}^2 + 16 \epsilon_{c2}^2 + 16 \epsilon_{c3}^2 \\
& - 8 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r3} - 8 \eta_3^2 \epsilon_{r2}^2 \epsilon_{r3} + 18 \epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 \epsilon_{r2}^2 + 4 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r3}^2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r2}^2 + 4 \\
& \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2}^2 - 16 \eta_2 \epsilon_{r2} \eta_1 + 8 \eta_3 \epsilon_{r1}^2 \eta_1 \epsilon_{r3}^2 - 16 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \eta_2 - 16 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \eta_2 \\
& - 4 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \eta_2 - 8 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} - 2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 4 \eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_2 - 4 \eta_1 \epsilon_{r2}^2 \\
& \epsilon_{r3}^2 \eta_3 - 16 \eta_1 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 + 8 \eta_3 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 + 16 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1} \epsilon_{r3} + 4 \\
& \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3} + 16 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1} \epsilon_{r2} + 8 \eta_2 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 + 8 \eta_2 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 - 4 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \eta_3 \\
& - 16 \eta_2 \epsilon_{r1} \epsilon_{r3}^2 \eta_3 - 16 \eta_2 \epsilon_{r1} \epsilon_{r3} \eta_3 - 16 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 \eta_3 + \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{c3}^2 + 8 \eta_2 \\
& \epsilon_{c1}^2 \epsilon_{r2} \eta_1 - 4 \eta_2 \epsilon_{r1} \epsilon_{r3}^2 \eta_1 - 16 \eta_2 \epsilon_{r1} \epsilon_{r3} \eta_1 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{c1}^2 + 8 \eta_3 \epsilon_{c2}^2 \eta_2 \epsilon_{r3}^2 - 4 \eta_2 \\
& \epsilon_{c1}^2 \epsilon_{r3}^2 \eta_1 - 16 \eta_2 \epsilon_{c1}^2 \epsilon_{r3} \eta_1 - 4 \eta_2 \epsilon_{c1}^2 \epsilon_{c3}^2 \eta_1 + 8 \eta_3 \epsilon_{r2}^2 \eta_2 \epsilon_{c3}^2 + 8 \eta_3 \epsilon_{r2}^2 \eta_2 \epsilon_{r3}^2 \\
& - 4 \eta_1 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_2 - 4 \eta_1 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_3 - 4 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_2 - 4 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_3 - 16 \eta_1 \\
& \epsilon_{c2}^2 \epsilon_{r3} \eta_2 - 4 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_2 - 4 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_3 - 4 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \eta_2 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r3}^2 \eta_1 \\
& - 16 \eta_2 \epsilon_{r1}^2 \epsilon_{r3} \eta_1 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{c3}^2 \eta_1 + \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \\
& + 8 \eta_3 \eta_1 \epsilon_{c1}^2 \epsilon_{r3}^2 + 8 \eta_3 \eta_1 \epsilon_{c1}^2 \epsilon_{c3}^2 + 16 \eta_3^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 + 4 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 + \eta_3^2 \epsilon_{c1}^2 \epsilon_{c2}^2 \\
& \epsilon_{r3}^2 - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_2 - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_2 - 2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_3^2 \\
& \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} + 8 \eta_3 \epsilon_{r1}^2 \eta_1 \epsilon_{c3}^2 + 4 \eta_2^2 \epsilon_{c3}^2 \epsilon_{r2}^2 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 16 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \eta_1 - 4 \eta_3 \\
& \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_1 - 4 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \eta_1 - 16 \eta_3 \epsilon_{r1} \epsilon_{r2} \eta_1 + 72 \epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 + 72 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \\
& - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_1 + 72 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 - 2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} + 4 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 + 8 \eta_3 \epsilon_{c2}^2 \eta_2 \\
& \epsilon_{c3}^2 + 16 \epsilon_{c1}^2 \eta_1^2 + 16 \eta_3^2 \epsilon_{r1} + 16 \eta_3^2 \epsilon_{r2} - 32 \eta_3^2 \epsilon_{r3} + 4 \eta_3^2 \epsilon_{r1}^2 + 4 \eta_3^2 \epsilon_{r2}^2 + 4 \eta_3^2 \epsilon_{c1}^2 \\
& + 4 \eta_3^2 \epsilon_{c2}^2 + 16 \eta_2^2 \epsilon_{r1} - 32 \eta_2^2 \epsilon_{r2} + 16 \eta_2^2 \epsilon_{r3} + 4 \eta_2^2 \epsilon_{r1}^2 + 4 \eta_2^2 \epsilon_{r3}^2 + 4 \eta_2^2 \epsilon_{c1}^2 + 4 \\
& \eta_2^2 \epsilon_{c3}^2 + 16 \epsilon_{c3}^2 \eta_3^2 + 4 \epsilon_{r1}^2 \epsilon_{r2}^2 + 16 \epsilon_{r1}^2 \epsilon_{r2} + 4 \epsilon_{r1}^2 \epsilon_{c2}^2 + 16 \epsilon_{r1} \epsilon_{r2}^2 + 64 \epsilon_{r1} \epsilon_{r2} \\
& + 16 \epsilon_{r1} \epsilon_{c2}^2 + 4 \epsilon_{c1}^2 \epsilon_{r2}^2 + 16 \epsilon_{c1}^2 \epsilon_{r2} + 16 \epsilon_{r1}^2 + 16 \epsilon_{r2}^2 + 16 \epsilon_{r3}^2 + 16 \eta_1^2 + 16 \eta_2^2 \\
& + 16 \eta_3^2 - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 \\
& - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 32 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 32 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 2 \eta_2 \epsilon_{c1}^2 \\
& \epsilon_{r2}^2 \epsilon_{r3}^2 - 32 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 8 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} \\
& - 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 - 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \\
& \epsilon_{r3}^2 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} - 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 2 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_2 \epsilon_{c1}^2 \\
& \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_3
\end{aligned}$$

$$\begin{aligned}
& \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 - 32 \eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 + 18 \epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 \epsilon_{c2}^2 \\
& + 72 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \epsilon_{r1} + 18 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \epsilon_{r1}^2 + 18 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \epsilon_{c1}^2 + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \\
& \epsilon_{r3}^2 \eta_3 + 72 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 \epsilon_{r3} + 18 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 \epsilon_{r3}^2 + 18 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 \epsilon_{c3}^2 \\
& + 72 \epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 \epsilon_{r2} + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_1 + 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 \eta_3 - 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 \\
& - 32 \eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 2 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r1} \\
& \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 2 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 32 \eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - 2 \eta_3 \\
& \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 - 2 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} \\
& - 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 2 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 2 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \\
& + \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 4 \eta_2^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 + \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \eta_2^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \\
& - 8 \eta_2^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1} \epsilon_{r2} + \epsilon_{c2}^2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{c3}^2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1} \epsilon_{c2}^2 - 8 \eta_2^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 \\
& + 4 \epsilon_{c1}^2 \epsilon_{c2}^2 + 16 \epsilon_{r1} \epsilon_{r3}^2 + 64 \epsilon_{r1} \epsilon_{r3} + 16 \epsilon_{r1} \epsilon_{c3}^2 + 16 \epsilon_{r2} \epsilon_{r3}^2 + 64 \epsilon_{r2} \epsilon_{r3} + 16 \epsilon_{r2} \\
& \epsilon_{c3}^2 + 4 \epsilon_{r1}^2 \epsilon_{r3}^2 + 16 \epsilon_{r1}^2 \epsilon_{r3} + 4 \epsilon_{r1}^2 \epsilon_{c3}^2 + 4 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \epsilon_{r2}^2 \epsilon_{r3} + 4 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \epsilon_{c1}^2 \epsilon_{r3}^2 \\
& + 16 \epsilon_{c1}^2 \epsilon_{r3} + 4 \epsilon_{c1}^2 \epsilon_{c3}^2 + 4 \epsilon_{c2}^2 \epsilon_{r3}^2 + 16 \epsilon_{c2}^2 \epsilon_{r3} + 4 \epsilon_{c2}^2 \epsilon_{c3}^2 - 32 \eta_1 \epsilon_{r1} + 64 \eta_1 \epsilon_{r2} \\
& + 64 \eta_1 \epsilon_{r3} - 32 \eta_1 \epsilon_{r1}^2 + 16 \eta_1 \epsilon_{r2}^2 + 16 \eta_1 \epsilon_{r3}^2 - 32 \eta_1 \epsilon_{c1}^2 + 16 \eta_1 \epsilon_{c2}^2 + 16 \eta_1 \epsilon_{c3}^2 \\
& + 64 \eta_2 \epsilon_{r1} - 32 \eta_2 \epsilon_{r2} + 64 \eta_2 \epsilon_{r3} + 16 \eta_2 \epsilon_{r1}^2 - 32 \eta_2 \epsilon_{r2}^2 + 16 \eta_2 \epsilon_{r3}^2 + 16 \eta_2 \epsilon_{c1}^2 \\
& - 32 \eta_2 \epsilon_{c2}^2 + 16 \eta_2 \epsilon_{c3}^2 + 64 \eta_3 \epsilon_{r1} + 64 \eta_3 \epsilon_{r2} - 32 \eta_3 \epsilon_{r3} + 16 \eta_3 \epsilon_{r1}^2 + 16 \eta_3 \epsilon_{r2}^2 \\
& - 32 \eta_3 \epsilon_{r3}^2 + 16 \eta_3 \epsilon_{c1}^2 + 16 \eta_3 \epsilon_{c2}^2 - 32 \eta_3 \epsilon_{c3}^2 + 16 \eta_1^2 \epsilon_{r1}^2 + 16 \eta_2^2 \epsilon_{r2}^2 + 16 \epsilon_{c2}^2 \eta_2^2 \\
& + 32 \eta_2 \eta_1 + 32 \eta_3 \eta_1 + 32 \eta_2 \eta_3 + 16 \eta_3^2 \epsilon_{r3}^2 - 32 \eta_2^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} + 4 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \\
& + 4 \eta_2^2 \epsilon_{r2}^2 \epsilon_{r1} \epsilon_{c3}^2 + \epsilon_{c3}^2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{c2}^2 + \epsilon_{c1}^2 \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{c2}^2 + 4 \\
& \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1} \epsilon_{r3}^2 + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2} \epsilon_{c3}^2 + \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} + \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \\
& \eta_3^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \epsilon_{c2}^2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r3} + 8 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \eta_2 + 8 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \eta_2 + 4 \eta_3^2 \epsilon_{r3}^2 \epsilon_{r1} \epsilon_{r2} + \\
& \eta_3^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + \eta_3^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \epsilon_{r1}^2 - 4 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \eta_1 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} + \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + \eta_1^2 \\
& \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 16 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} + \eta_1^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} + \eta_1^2 \\
& \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} \eta_3 + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_1 + 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_1 + 2 \eta_2 \eta_3 \epsilon_{r1}^2 \\
& \epsilon_{c2}^2 \epsilon_{c3}^2 + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_3 + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_1 + 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_1 + 2 \eta_3 \epsilon_{c1}^2 \\
& \epsilon_{c2}^2 \epsilon_{r3} \eta_1 + 2 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_1 + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_3 + 2 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} \eta_1 + 2 \eta_3 \epsilon_{r1} \\
& \epsilon_{r2}^2 \epsilon_{r3} \eta_1 + 8 \eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} \eta_1 + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 + 2 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 2 \eta_3 \epsilon_{r1}^2
\end{aligned}$$

$$\begin{aligned}
& \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 8 \eta_1 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 \eta_3 + 2 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_2 + 2 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_3 + 8 \eta_1 \epsilon_{r1} \\
& \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 8 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} \eta_2 + 8 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_3 + 2 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_2 + 2 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \\
& \epsilon_{r3}^2 \eta_3 + 2 \eta_2 \epsilon_{r3}^2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 - 8 \eta_1^2 \epsilon_{r1} \epsilon_{r2}^2 - 32 \eta_1^2 \epsilon_{r1} \epsilon_{r2} + 16 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_1^2 \epsilon_{r1} \\
& \epsilon_{c2}^2 - 8 \eta_1^2 \epsilon_{r1} \epsilon_{r3}^2 - 32 \eta_1^2 \epsilon_{r1} \epsilon_{r3} - 8 \eta_1^2 \epsilon_{r1} \epsilon_{c3}^2 + 8 \eta_2 \epsilon_{r3}^2 \eta_1 + 16 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1} + 8 \eta_3 \\
& \epsilon_{r2}^2 \eta_1 - 16 \eta_2 \epsilon_{r1}^2 \eta_1 + 4 \epsilon_{c2}^2 \eta_2^2 \epsilon_{c1}^2 - 16 \eta_3 \epsilon_{r1}^2 \eta_1 + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r3}^2 + 4 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \\
& \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 + 4 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} + \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 + \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + 16 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 + \\
& \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 64 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} + \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} + \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} \\
& + 4 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 + 4 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} + 16 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 + 4 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} + 4 \epsilon_{r1} \\
& \epsilon_{c2}^2 \epsilon_{c3}^2 + \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 + 16 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} + 4 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 + \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \epsilon_{c1}^2 \\
& \epsilon_{c2}^2 \epsilon_{r3} + \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 - 32 \eta_1 \epsilon_{r1}^2 \epsilon_{r2} - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 - 8 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \\
& - 32 \eta_1 \epsilon_{r1} \epsilon_{r2} - 8 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 - 32 \eta_1 \epsilon_{c1}^2 \epsilon_{r2} - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 - 8 \eta_1 \epsilon_{r1} \\
& \epsilon_{r3}^2 + 2 \eta_2 \epsilon_{r3}^2 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 + 2 \eta_2 \epsilon_{r3}^2 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 + 2 \eta_2 \epsilon_{c3}^2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 + 2 \eta_2 \epsilon_{c3}^2 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 \\
& + 2 \eta_2 \epsilon_{c3}^2 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 + 2 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_2 + 2 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_3 + 8 \eta_2 \epsilon_{r3} \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \\
& + 8 \eta_2 \epsilon_{r3} \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 + 8 \eta_2 \epsilon_{r3} \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 + 2 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_2 + 2 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_3 \\
& + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} \eta_1 + 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} \eta_1 + 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_1 + 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 \eta_1 \\
& + 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} \eta_1 + 2 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_1 + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_1 + 2 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 \eta_1 \\
& + 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} \eta_1 + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 \eta_1 + 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} \eta_1 + 8 \eta_3 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 \\
& + 2 \eta_3 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{r2} \epsilon_{c3}^2 \eta_1 - 32 \eta_1^2 \epsilon_{r1} + 16 \eta_1^2 \epsilon_{r2} + 16 \eta_1^2 \epsilon_{r3} + 4 \eta_1^2 \epsilon_{r2}^2 \\
& + 4 \eta_1^2 \epsilon_{r3}^2 + 4 \eta_1^2 \epsilon_{c2}^2 + 4 \eta_1^2 \epsilon_{c3}^2 + 4 \eta_2^2 \epsilon_{r3}^2 \epsilon_{r2}^2 - 32 \eta_1 \epsilon_{r1} \epsilon_{r3} - 8 \eta_1 \epsilon_{r1} \epsilon_{c3}^2 \\
& + 16 \eta_1 \epsilon_{r2} \epsilon_{r3}^2 + 64 \eta_1 \epsilon_{r2} \epsilon_{r3} + 16 \eta_1 \epsilon_{r2} \epsilon_{c3}^2 - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{r3}^2 - 32 \eta_1 \epsilon_{r1}^2 \epsilon_{r3} - 8 \eta_1 \\
& \epsilon_{r1}^2 \epsilon_{c3}^2 + 4 \eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \eta_1 \epsilon_{r2}^2 \epsilon_{r3} + 4 \eta_1 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{r3}^2 - 32 \eta_1 \epsilon_{c1}^2 \epsilon_{r3} \\
& - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{c3}^2 + 4 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 + 16 \eta_1 \epsilon_{c2}^2 \epsilon_{r3} + 4 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 - 8 \eta_2 \\
& \epsilon_{r1}^2 \epsilon_{r2} - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 - 32 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 - 32 \eta_2 \epsilon_{r1} \epsilon_{r2} - 32 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \\
& - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 + 16 \eta_2 \epsilon_{r1} \epsilon_{r3}^2 + 64 \eta_2 \epsilon_{r1} \epsilon_{r3} + 16 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 \\
& - 8 \eta_2 \epsilon_{r2} \epsilon_{r3}^2 - 32 \eta_2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_2 \epsilon_{r2} \epsilon_{c3}^2 + 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 16 \eta_2 \epsilon_{r1}^2 \epsilon_{r3} + 4 \eta_2 \epsilon_{r1}^2 \\
& \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 32 \eta_2 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \eta_2 \epsilon_{c1}^2 \epsilon_{r3}^2 + 16 \eta_2 \epsilon_{c1}^2 \epsilon_{r3} + 4 \eta_2 \\
& \epsilon_{c1}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 32 \eta_2 \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_2 \epsilon_{c2}^2 \epsilon_{c3}^2 + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 + 16 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}
\end{aligned}$$

$$\begin{aligned}
& + 4\eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 + 16\eta_3 \epsilon_{r1} \epsilon_{r2}^2 + 64\eta_3 \epsilon_{r1} \epsilon_{r2} + 16\eta_3 \epsilon_{r1} \epsilon_{c2}^2 + 4\eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 + 16\eta_3 \\
& \epsilon_{c1}^2 \epsilon_{r2} + 4\eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 - 32\eta_3 \epsilon_{r1} \epsilon_{r3}^2 - 32\eta_3 \epsilon_{r1} \epsilon_{r3} - 32\eta_3 \epsilon_{r1} \epsilon_{c3}^2 - 32\eta_3 \epsilon_{r2} \epsilon_{r3}^2 \\
& - 32\eta_3 \epsilon_{r2} \epsilon_{r3} - 32\eta_3 \epsilon_{r2} \epsilon_{c3}^2 - 8\eta_3 \epsilon_{r1}^2 \epsilon_{r3}^2 - 8\eta_3 \epsilon_{r1}^2 \epsilon_{r3} - 8\eta_3 \epsilon_{r1}^2 \epsilon_{c3}^2 - 8\eta_3 \epsilon_{r2}^2 \\
& \epsilon_{r3}^2 - 8\eta_3 \epsilon_{r2}^2 \epsilon_{r3} - 8\eta_3 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8\eta_3 \epsilon_{c1}^2 \epsilon_{r3}^2 - 8\eta_3 \epsilon_{c1}^2 \epsilon_{r3} - 8\eta_3 \epsilon_{c1}^2 \epsilon_{c3}^2 - 8\eta_3 \epsilon_{c2}^2 \\
& \epsilon_{r3}^2 - 8\eta_3 \epsilon_{c2}^2 \epsilon_{r3} - 8\eta_3 \epsilon_{c2}^2 \epsilon_{c3}^2 + 4\eta_3^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 4\eta_3^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 4\eta_3^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 4\eta_3^2 \epsilon_{c1}^2 \\
& \epsilon_{r3}^2 + 16\eta_3^2 \epsilon_{r1} \epsilon_{r3}^2 + 16\eta_3^2 \epsilon_{r2} \epsilon_{r3}^2 - 16\eta_3 \epsilon_{r3} \eta_1 + 8\eta_3 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 + 2\eta_3 \eta_2 \epsilon_{r1}^2 \\
& \epsilon_{c2}^2 \epsilon_{r3}^2 + 2\eta_3 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 2\eta_3 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 8\eta_3 \epsilon_{r2} \eta_1 \epsilon_{c1}^2 \epsilon_{c3}^2 + 2\eta_3 \epsilon_{r2}^2 \eta_1 \epsilon_{c1}^2 \\
& \epsilon_{c3}^2 + 2\eta_3 \epsilon_{c1}^2 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 + 2\eta_3 \epsilon_{c1}^2 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 + 8\eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 \eta_3 + 2\eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_3 \\
& + 2\eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} \eta_3 + 2\eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_3 + 8\eta_3 \epsilon_{r1}^2 \eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 + 8\eta_3 \epsilon_{c1}^2 \eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 \\
& + 8\eta_3 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 + 2\eta_3 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 2\eta_3 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 2\eta_3 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 \\
& + 8\eta_3 \epsilon_{r2}^2 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 + 2\eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_1 \epsilon_{r3}^2 + 2\eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_1 \epsilon_{c3}^2 + 8\eta_3 \epsilon_{r1}^2 \epsilon_{r2} \eta_1 \epsilon_{c3}^2 \\
& + 2\eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_1 \epsilon_{r3}^2 + 2\eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_1 \epsilon_{c3}^2 + 2\eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \eta_1 \epsilon_{r3}^2 + 8\eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} \eta_2 \\
& + 8\eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 2\eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 - 16\eta_3 \epsilon_{r1} \eta_1 - 16\eta_2 \epsilon_{c1}^2 \eta_1 + 8\eta_2 \epsilon_{c3}^2 \eta_1 \\
& - 16\eta_3 \epsilon_{c1}^2 \eta_1 + 8\eta_3 \epsilon_{c2}^2 \eta_1 - 16\eta_2 \epsilon_{r1} \eta_1 + 4\epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 + 4\epsilon_{c1}^2 \eta_2^2 \epsilon_{c2}^2 + 16\epsilon_{c3}^2 \\
& \eta_3^2 \epsilon_{r1} + 4\epsilon_{c1}^2 \eta_1^2 \epsilon_{c3}^2 + \eta_1^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 4\eta_1^2 \epsilon_{c2}^2 \epsilon_{r3} + \eta_1^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4\eta_1^2 \epsilon_{r2}^2 \epsilon_{r3} \\
& + \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16\eta_1^2 \epsilon_{r2} \epsilon_{r3} + 4\eta_1^2 \epsilon_{r2} \epsilon_{c3}^2 + 4\eta_1^2 \epsilon_{r2} \epsilon_{r3}^2 + 16\epsilon_{c2}^2 \eta_2^2 \epsilon_{r3} + 16\epsilon_{c1}^2 \\
& \eta_1^2 \epsilon_{r3} - 16\eta_2 \epsilon_{r2} \eta_3 - 16\eta_2 \epsilon_{r3} \eta_3 + 8\eta_2 \epsilon_{r1}^2 \eta_3 - 16\eta_2 \epsilon_{r3}^2 \eta_3 + 16\eta_1^2 \epsilon_{r1}^2 \epsilon_{r2} + 4 \\
& \eta_1^2 \epsilon_{r1}^2 \epsilon_{c2}^2 + 4\epsilon_{c3}^2 \eta_3^2 \epsilon_{c2}^2 + 4\eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 - 16\eta_1 \epsilon_{r2}^2 \eta_2 - 16\eta_1 \epsilon_{r3}^2 \eta_3 - 16\eta_1 \epsilon_{c2}^2 \eta_2 \\
& - 16\eta_1 \epsilon_{c3}^2 \eta_3 + 16\epsilon_{c3}^2 \eta_3^2 \epsilon_{r2} + 32\eta_2 \epsilon_{r3} \eta_1 + 4\eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 + 4\eta_3^2 \epsilon_{r1}^2 \epsilon_{r2} + \eta_3^2 \epsilon_{r1}^2 \\
& \epsilon_{c2}^2 + 4\eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 - 4\eta_2 \epsilon_{r1} \epsilon_{c3}^2 \eta_1 - 4\eta_2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_1 - 16\eta_2 \epsilon_{r2} \epsilon_{r3} \eta_1 - 32 \\
& \eta_3^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - 8\eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 8\eta_3^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} - 4\eta_2 \epsilon_{c1}^2 \epsilon_{r3} \eta_3 - 4\eta_2 \epsilon_{c1}^2 \epsilon_{c3}^2 \eta_3 \\
& + 4\epsilon_{c1}^2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3} - 4\eta_2 \epsilon_{r1}^2 \epsilon_{r3}^2 \eta_3 - 4\eta_2 \epsilon_{r1}^2 \epsilon_{r3} \eta_3 - 4\eta_2 \epsilon_{r1}^2 \epsilon_{c3}^2 \eta_3 - 4\eta_2 \epsilon_{c1}^2 \\
& \epsilon_{r3}^2 \eta_3 + 8\eta_2 \epsilon_{r1}^2 \epsilon_{r2} \eta_1 + 8\eta_2 \epsilon_{r1} \epsilon_{r2} \eta_1 + 8\eta_2 \epsilon_{r2} \epsilon_{r3}^2 \eta_3 + 8\eta_2 \epsilon_{r2} \epsilon_{r3} \eta_3 + 8\eta_2 \epsilon_{r2} \\
& \epsilon_{c3}^2 \eta_3 - 8\eta_1^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - 2\eta_1^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 + 16\epsilon_{c1}^2 \eta_1^2 \epsilon_{r2} \epsilon_{r3} - 2\eta_1^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 32 \\
& \eta_1^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - 8\eta_1^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 8\eta_1^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} - 2\eta_1^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 2\eta_1^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 \\
& - 8\eta_1^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 - 4\eta_3 \epsilon_{r2}^2 \epsilon_{r3} \eta_1 + 8\eta_3 \epsilon_{c1}^2 \epsilon_{r3} \eta_1 - 4\eta_3 \epsilon_{c2}^2 \epsilon_{r3} \eta_1 - 16\eta_3 \\
& \epsilon_{c1}^2 \epsilon_{r2} \eta_1 - 4\eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \eta_1 + 8\eta_3 \epsilon_{r1} \epsilon_{r3} \eta_1 - 16\eta_3 \epsilon_{r2} \epsilon_{r3} \eta_1 + 8\eta_3 \epsilon_{r1}^2 \epsilon_{r3} \eta_1
\end{aligned}$$

$$\begin{aligned}
& -4\eta_3 \epsilon_{cl}^2 \epsilon_{r2}^2 \eta_1 + \epsilon_{c2}^2 \eta_2^2 \epsilon_{cl}^2 \epsilon_{r3}^2 - 2\eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8\eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 4\epsilon_{cl}^2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \\
& -16\eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_3 - 16\eta_1 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_3 + 8\eta_3 \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_2 + 8\eta_1 \epsilon_{r1}^2 \epsilon_{r3}^2 \eta_3 + 8\eta_1 \epsilon_{r1}^2 \\
& \epsilon_{c3}^2 \eta_3 + 4\epsilon_{c3}^2 \eta_3^2 \epsilon_{cl}^2 \epsilon_{r2}^2 + 8\eta_2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 + 8\eta_2 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 + \epsilon_{c3}^2 \eta_3^2 \epsilon_{cl}^2 \epsilon_{r2}^2 + 4\epsilon_{c2}^2 \\
& \eta_2^2 \epsilon_{r1}^2 \epsilon_{c3}^2 - 2\eta_2^2 \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 8\eta_2^2 \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 2\eta_2^2 \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4\epsilon_{cl}^2 \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + \epsilon_{cl}^2 \\
& \eta_1^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 4\eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_3 - 16\eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_3 - 2\eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 32\eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \\
& + 16\eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 + 4\eta_3^2 \epsilon_{r1}^2 \epsilon_{c2}^2 + \eta_3^2 \epsilon_{cl}^2 \epsilon_{r2}^2 + 4\eta_3^2 \epsilon_{cl}^2 \epsilon_{r2}^2 + \eta_3^2 \epsilon_{cl}^2 \epsilon_{c2}^2 - 32\eta_3^2 \epsilon_{r1}^2 \epsilon_{r3}^2 \\
& - 32\eta_3^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 4\eta_2^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + \eta_2^2 \epsilon_{r1}^2 \epsilon_{c3}^2 + \eta_2^2 \epsilon_{cl}^2 \epsilon_{r3}^2 + 4\eta_2^2 \epsilon_{cl}^2 \epsilon_{r3}^2 + \\
& \eta_2^2 \epsilon_{cl}^2 \epsilon_{c3}^2 + \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 - 8\eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 - 32\eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 - 8\eta_2^2 \epsilon_{cl}^2 \epsilon_{r2}^2 + 4\eta_2^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 16 \\
& \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 4\eta_2^2 \epsilon_{r1}^2 \epsilon_{c3}^2 - 8\eta_2^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 32\eta_2^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 8\eta_2^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 16\epsilon_{cl}^2 \eta_1^2 \epsilon_{r2}^2 \\
& - 16\eta_3 \epsilon_{r2}^2 \eta_2 - 16\eta_3 \epsilon_{c2}^2 \eta_2 + 4\eta_1^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 4\epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 + 4\eta_1^2 \epsilon_{r1}^2 \epsilon_{c3}^2 + 16\eta_1^2 \\
& \epsilon_{r1}^2 \epsilon_{r3}^2 + 16\eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 + 4\epsilon_{cl}^2 \eta_1^2 \epsilon_{r2}^2 + 16\eta_2^2 \epsilon_{r3}^2 \epsilon_{r2}^2 + 32\eta_2 \epsilon_{r1}^2 \eta_3 - 8\eta_3^2 \epsilon_{c2}^2 \epsilon_{r3}^2 \\
& + 8\eta_2 \epsilon_{cl}^2 \eta_3 - 16\eta_2 \epsilon_{c3}^2 \eta_3 + 32\eta_3 \epsilon_{r2}^2 \eta_1 - 2\eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 8\eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \\
& - 8\eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 2\eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 2\eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 2\eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 8\eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \\
& \epsilon_{c3}^2 - 2\eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 32\eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 2\eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 8\eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 2\eta_1 \epsilon_{cl}^2 \\
& \epsilon_{r2}^2 \epsilon_{r3}^2 - 8\eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 8\eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 2\eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8\eta_1 \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \\
& - 8\eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 8\eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 2\eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 2\eta_1 \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8\eta_1 \epsilon_{cl}^2 \epsilon_{r2}^2 \\
& \epsilon_{r3}^2 - 32\eta_1 \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 8\eta_1 \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 2\eta_1 \epsilon_{cl}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 8\eta_1 \epsilon_{cl}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 2\eta_1 \epsilon_{cl}^2 \\
& \epsilon_{c2}^2 \epsilon_{c3}^2 - 8\eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 8\eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 2\eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8\eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 8\eta_3^2 \\
& \epsilon_{r1}^2 \epsilon_{r3}^2)
\end{aligned}$$

eps_r

$$\begin{aligned}
& -(-64 + 8\epsilon_{c2}^2 \eta_2^2 \epsilon_{c3}^2 + 32\eta_1 + 32\eta_2 + 32\eta_3 - 64\epsilon_{r1} - 64\epsilon_{r2} - 64\epsilon_{r3} - 16\epsilon_{cl}^2 \\
& - 16\epsilon_{c2}^2 - 16\epsilon_{c3}^2 - 16\eta_3^2 \epsilon_{cl}^2 \epsilon_{r3}^2 - 16\eta_3^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 36\epsilon_{cl} \eta_1 \epsilon_{c3} \eta_3 \epsilon_{r2}^2 + 8\epsilon_{c2}^2 \eta_2^2 \epsilon_{r3}^2 \\
& + 8\epsilon_{c3}^2 \eta_3^2 \epsilon_{r2}^2 + 8\eta_2^2 \epsilon_{cl}^2 \epsilon_{r2}^2 - 32\eta_2 \epsilon_{r2}^2 \eta_1 + 16\eta_3 \epsilon_{r1}^2 \eta_1 \epsilon_{r3}^2 - 32\eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_2 \\
& - 32\eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_2 - 8\eta_3 \epsilon_{cl}^2 \epsilon_{r2}^2 \eta_2 - 16\eta_3^2 \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 4\eta_3^2 \epsilon_{cl}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 8\eta_1 \epsilon_{r2}^2 \\
& \epsilon_{r3}^2 \eta_2 - 8\eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_3 - 32\eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_2 + 16\eta_3 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_2 + 8\epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 + 32 \\
& \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 8\epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 32\epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 + 16\eta_2 \eta_1 \epsilon_{cl}^2 \epsilon_{r2}^2 + 16\eta_2 \eta_1 \epsilon_{cl}^2 \\
& \epsilon_{c2}^2 - 8\eta_2 \epsilon_{cl}^2 \epsilon_{r2}^2 \eta_3 - 32\eta_2 \epsilon_{r1}^2 \epsilon_{r3}^2 \eta_3 - 32\eta_2 \epsilon_{r1}^2 \epsilon_{r3}^2 \eta_3 - 32\eta_2 \epsilon_{r1}^2 \epsilon_{c3}^2 \eta_3 + 2\epsilon_{c2}^2
\end{aligned} \tag{2.9}$$

$$\begin{aligned}
& \eta_2^2 \epsilon_{r1}^2 \epsilon_{c3}^2 + 16 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \eta_1 - 8 \eta_2 \epsilon_{r1} \epsilon_{r3}^2 \eta_1 - 32 \eta_2 \epsilon_{r1} \epsilon_{r3} \eta_1 + 8 \epsilon_{c3}^2 \eta_3^2 \epsilon_{c1}^2 \\
& + 16 \eta_3 \epsilon_{c2}^2 \eta_2 \epsilon_{r3}^2 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r3}^2 \eta_1 - 32 \eta_2 \epsilon_{c1}^2 \epsilon_{r3} \eta_1 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{c3}^2 \eta_1 + 16 \eta_3 \\
& \epsilon_{r2}^2 \eta_2 \epsilon_{c3}^2 + 16 \eta_3 \epsilon_{r2}^2 \eta_2 \epsilon_{r3}^2 - 8 \eta_1 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_2 - 8 \eta_1 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_3 - 8 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_2 \\
& - 8 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_3 - 32 \eta_1 \epsilon_{c2}^2 \epsilon_{r3} \eta_2 - 8 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_2 - 8 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_3 - 8 \eta_3 \epsilon_{c1}^2 \\
& \epsilon_{c2}^2 \eta_2 - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r3}^2 \eta_1 - 32 \eta_2 \epsilon_{r1}^2 \epsilon_{r3} \eta_1 - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{c3}^2 \eta_1 + 2 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 2 \epsilon_{c2}^2 \\
& \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 2 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 + 16 \eta_3 \eta_1 \epsilon_{c1}^2 \epsilon_{r3}^2 + 16 \eta_3 \eta_1 \epsilon_{c1}^2 \epsilon_{c3}^2 + 32 \eta_3^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 \\
& + 8 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 + 2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_2 - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_2 - 4 \eta_3^2 \epsilon_{c1}^2 \\
& \epsilon_{r2}^2 \epsilon_{r3} - 4 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 16 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} + 16 \eta_3 \epsilon_{r1}^2 \eta_1 \epsilon_{c3}^2 + 8 \eta_2^2 \epsilon_{c3}^2 \epsilon_{r2}^2 + 8 \eta_1^2 \\
& \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 32 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \eta_1 - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_1 - 8 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \eta_1 - 32 \eta_3 \epsilon_{r1} \epsilon_{r2} \eta_1 \\
& + 144 \epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 + 144 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_1 + 144 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 - 4 \eta_3^2 \\
& \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} + 8 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \eta_3 \epsilon_{c2}^2 \eta_2 \epsilon_{c3}^2 + 32 \epsilon_{c1}^2 \eta_1^2 + 32 \eta_3^2 \epsilon_{r1} + 32 \eta_3^2 \epsilon_{r2} \\
& - 64 \eta_3^2 \epsilon_{r3} + 8 \eta_3^2 \epsilon_{r1}^2 + 8 \eta_3^2 \epsilon_{r2}^2 + 8 \eta_3^2 \epsilon_{c1}^2 + 8 \eta_3^2 \epsilon_{c2}^2 + 32 \eta_2^2 \epsilon_{r1} - 64 \eta_2^2 \epsilon_{r2} \\
& + 32 \eta_2^2 \epsilon_{r3} + 8 \eta_2^2 \epsilon_{r1}^2 + 8 \eta_2^2 \epsilon_{r2}^2 + 8 \eta_2^2 \epsilon_{c1}^2 + 8 \eta_2^2 \epsilon_{c2}^2 + 32 \epsilon_{c3}^2 \eta_3^2 - 4 \epsilon_{r1}^2 \epsilon_{r2}^2 - 16 \\
& \epsilon_{r1}^2 \epsilon_{r2} - 4 \epsilon_{r1}^2 \epsilon_{c2}^2 - 16 \epsilon_{r1} \epsilon_{r2}^2 - 64 \epsilon_{r1} \epsilon_{r2} - 16 \epsilon_{r1} \epsilon_{c2}^2 - 4 \epsilon_{c1}^2 \epsilon_{r2}^2 - 16 \epsilon_{c1}^2 \epsilon_{r2} - 16 \\
& \epsilon_{r1}^2 - 16 \epsilon_{r2}^2 - 16 \epsilon_{r3}^2 + 32 \eta_1^2 + 32 \eta_2^2 + 32 \eta_3^2 - \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 \\
& - \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 - \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 16 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \\
& - 16 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 16 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} - \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 4 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \\
& \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 4 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 4 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 - \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \\
& \epsilon_{c3}^2 - \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 4 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} - \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 - \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 4 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} \\
& - \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} - 4 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 - \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} \\
& - \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 - 16 \eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 + 36 \epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 \epsilon_{c2}^2 \\
& + 144 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \epsilon_{r1} + 36 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \epsilon_{r1}^2 + 36 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \epsilon_{c1}^2 + 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \\
& \epsilon_{r3}^2 \eta_3 + 144 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 \epsilon_{r3} + 36 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 \epsilon_{r3}^2 + 36 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 \epsilon_{c3}^2 \\
& + 144 \epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 \epsilon_{r2} + 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_1 + 16 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 \eta_3 - \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 \\
& - 16 \eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 4 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 4 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} \\
& - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 4 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 16 \eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \\
& - 4 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 4 \eta_3 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 4 \eta_3 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} - 4 \eta_3 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2
\end{aligned}$$

$$\begin{aligned}
& -\eta_3 \epsilon_{cl}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - \eta_3 \epsilon_{cl}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - \eta_3 \epsilon_{cl}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + 2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \\
& + 8 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 2 \eta_2^2 \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 32 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 4 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 16 \\
& \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 8 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 + 2 \epsilon_{c2}^2 \eta_2^2 \epsilon_{cl}^2 \epsilon_{c3}^2 + 8 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{c2}^2 - 16 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \\
& - 4 \epsilon_{cl}^2 \epsilon_{c2}^2 - 16 \epsilon_{r1}^2 \epsilon_{r3}^2 - 64 \epsilon_{r1}^2 \epsilon_{r3}^2 - 16 \epsilon_{r1}^2 \epsilon_{c3}^2 - 16 \epsilon_{r2}^2 \epsilon_{r3}^2 - 64 \epsilon_{r2}^2 \epsilon_{r3}^2 - 16 \epsilon_{r2}^2 \\
& \epsilon_{c3}^2 - 4 \epsilon_{r1}^2 \epsilon_{r3}^2 - 16 \epsilon_{r1}^2 \epsilon_{r3}^2 - 4 \epsilon_{r1}^2 \epsilon_{c3}^2 - 4 \epsilon_{r2}^2 \epsilon_{r3}^2 - 16 \epsilon_{r2}^2 \epsilon_{r3}^2 - 4 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \epsilon_{cl}^2 \epsilon_{r3}^2 \\
& - 16 \epsilon_{cl}^2 \epsilon_{r3}^2 - 4 \epsilon_{cl}^2 \epsilon_{c3}^2 - 4 \epsilon_{c2}^2 \epsilon_{r3}^2 - 16 \epsilon_{c2}^2 \epsilon_{r3}^2 - 4 \epsilon_{c2}^2 \epsilon_{c3}^2 - 16 \eta_1 \epsilon_{r1} + 32 \eta_1 \epsilon_{r2} \\
& + 32 \eta_1 \epsilon_{r3} - 16 \eta_1 \epsilon_{r1}^2 + 8 \eta_1 \epsilon_{r2}^2 + 8 \eta_1 \epsilon_{r3}^2 - 16 \eta_1 \epsilon_{cl}^2 + 8 \eta_1 \epsilon_{c2}^2 + 8 \eta_1 \epsilon_{c3}^2 \\
& + 32 \eta_2 \epsilon_{r1} - 16 \eta_2 \epsilon_{r2} + 32 \eta_2 \epsilon_{r3} + 8 \eta_2 \epsilon_{r1}^2 - 16 \eta_2 \epsilon_{r2}^2 + 8 \eta_2 \epsilon_{r3}^2 + 8 \eta_2 \epsilon_{cl}^2 \\
& - 16 \eta_2 \epsilon_{c2}^2 + 8 \eta_2 \epsilon_{c3}^2 + 32 \eta_3 \epsilon_{r1} + 32 \eta_3 \epsilon_{r2} - 16 \eta_3 \epsilon_{r3} + 8 \eta_3 \epsilon_{r1}^2 + 8 \eta_3 \epsilon_{r2}^2 \\
& - 16 \eta_3 \epsilon_{r3}^2 + 8 \eta_3 \epsilon_{cl}^2 + 8 \eta_3 \epsilon_{c2}^2 - 16 \eta_3 \epsilon_{c3}^2 + 32 \eta_1^2 \epsilon_{r1}^2 + 32 \eta_2^2 \epsilon_{r2}^2 + 32 \epsilon_{c2}^2 \eta_2^2 \\
& + 64 \eta_2 \eta_1 + 64 \eta_3 \eta_1 + 64 \eta_2 \eta_3 + 32 \eta_3^2 \epsilon_{r3}^2 - 64 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 8 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \\
& + 8 \eta_2^2 \epsilon_{r2}^2 \epsilon_{r1}^2 \epsilon_{c3}^2 + 2 \epsilon_{c3}^2 \eta_3^2 \epsilon_{cl}^2 \epsilon_{c2}^2 + 2 \epsilon_{cl}^2 \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 2 \eta_3^2 \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 2 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 \\
& \epsilon_{c2}^2 + 8 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 8 \epsilon_{cl}^2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 2 \epsilon_{cl}^2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 8 \eta_2^2 \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 2 \eta_2^2 \epsilon_{cl}^2 \\
& \epsilon_{r2}^2 \epsilon_{c3}^2 + 8 \eta_3^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 8 \epsilon_{c2}^2 \eta_2^2 \epsilon_{cl}^2 \epsilon_{r3}^2 + 16 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_2 + 16 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_2 + 8 \\
& \eta_3^2 \epsilon_{r3}^2 \epsilon_{r1}^2 \epsilon_{r2}^2 + 2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 2 \eta_3^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \epsilon_{r1}^2 - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_1 + 8 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \\
& + 2 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 2 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 32 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 8 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 2 \eta_1^2 \epsilon_{r1}^2 \\
& \epsilon_{c2}^2 \epsilon_{c3}^2 + 8 \eta_1^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 2 \eta_1^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 16 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_3 + 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_1 \\
& + 16 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_1 + 4 \eta_2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_3 + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_1 \\
& + 16 \eta_3 \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_1 + 4 \eta_3 \epsilon_{cl}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_1 + 4 \eta_3 \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_1 + 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_3 \\
& + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_1 + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_1 + 16 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_1 + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_2 \\
& + 4 \eta_3 \epsilon_{cl}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_2 + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_2 + 16 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_3 + 4 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_2 \\
& + 4 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_3 + 16 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_2 + 16 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_2 + 16 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \\
& \epsilon_{c3}^2 \eta_3 + 4 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_2 + 4 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_3 + 4 \eta_2 \epsilon_{r3}^2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 - 16 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \\
& - 64 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 - 16 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 16 \eta_1^2 \epsilon_{r1}^2 \epsilon_{c2}^2 - 16 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r3}^2 - 64 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r3}^2 - 16 \\
& \eta_1^2 \epsilon_{r1}^2 \epsilon_{c3}^2 + 16 \eta_2 \epsilon_{r3}^2 \eta_1 + 32 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 + 16 \eta_3 \epsilon_{r2}^2 \eta_1 - 32 \eta_2 \epsilon_{r1}^2 \eta_1 + 8 \epsilon_{c2}^2 \eta_2^2 \epsilon_{cl}^2 \\
& - 32 \eta_3 \epsilon_{r1}^2 \eta_1 + 8 \epsilon_{cl}^2 \eta_1^2 \epsilon_{r3}^2 - 4 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 4 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \\
& - 4 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 16 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 64 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2
\end{aligned}$$

$$\begin{aligned}
& -16 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 16 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} - 4 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 4 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \epsilon_{c1}^2 \\
& \epsilon_{r2}^2 \epsilon_{r3} - 16 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - 4 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 4 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 - \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 16 \\
& \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} - 4 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 - \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 4 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} - \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 4 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \\
& - 16 \eta_1 \epsilon_{r1}^2 \epsilon_{r2} - 4 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 - 4 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 - 16 \eta_1 \epsilon_{r1} \epsilon_{r2} - 4 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 - 4 \eta_1 \epsilon_{c1}^2 \\
& \epsilon_{r2}^2 - 16 \eta_1 \epsilon_{c1}^2 \epsilon_{r2} - 4 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 - 4 \eta_1 \epsilon_{r1} \epsilon_{r3}^2 + 4 \eta_2 \epsilon_{r3}^2 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 + 4 \eta_2 \epsilon_{r3}^2 \eta_1 \epsilon_{c1}^2 \\
& \epsilon_{c2}^2 + 4 \eta_2 \epsilon_{c3}^2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 + 4 \eta_2 \epsilon_{c3}^2 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 + 4 \eta_2 \epsilon_{c3}^2 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 + 4 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_2 \\
& + 4 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_3 + 16 \eta_2 \epsilon_{r3} \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 + 16 \eta_2 \epsilon_{r3} \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 + 16 \eta_2 \epsilon_{r3} \eta_1 \epsilon_{c1}^2 \\
& \epsilon_{c2}^2 + 4 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_2 + 4 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_3 + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} \eta_1 + 16 \eta_3 \\
& \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} \eta_1 + 4 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 \eta_1 + 4 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 \eta_1 + 16 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} \eta_1 \\
& + 4 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 \eta_1 + 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 \eta_1 + 4 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 \eta_1 + 16 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} \eta_1 \\
& + 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 \eta_1 + 16 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} \eta_1 + 16 \eta_3 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \eta_3 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \\
& - 8 \eta_2 \epsilon_{r2} \epsilon_{c3}^2 \eta_1 - 64 \eta_1^2 \epsilon_{r1} + 32 \eta_1^2 \epsilon_{r2} + 32 \eta_1^2 \epsilon_{r3} + 8 \eta_1^2 \epsilon_{r2}^2 + 8 \eta_1^2 \epsilon_{r3}^2 + 8 \eta_1^2 \\
& \epsilon_{c2}^2 + 8 \eta_1^2 \epsilon_{c3}^2 + 8 \eta_2^2 \epsilon_{r3}^2 \epsilon_{r2}^2 - 16 \eta_1 \epsilon_{r1} \epsilon_{r3} - 4 \eta_1 \epsilon_{r1} \epsilon_{c3}^2 + 8 \eta_1 \epsilon_{r2} \epsilon_{r3}^2 \\
& + 32 \eta_1 \epsilon_{r2} \epsilon_{r3} + 8 \eta_1 \epsilon_{r2} \epsilon_{c3}^2 - 4 \eta_1 \epsilon_{r1}^2 \epsilon_{r3}^2 - 16 \eta_1 \epsilon_{r1}^2 \epsilon_{r3} - 4 \eta_1 \epsilon_{r1}^2 \epsilon_{c3}^2 + 2 \eta_1 \epsilon_{r2}^2 \\
& \epsilon_{r3}^2 + 8 \eta_1 \epsilon_{r2}^2 \epsilon_{r3} + 2 \eta_1 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \eta_1 \epsilon_{c1}^2 \epsilon_{r3}^2 - 16 \eta_1 \epsilon_{c1}^2 \epsilon_{r3} - 4 \eta_1 \epsilon_{c1}^2 \epsilon_{c3}^2 + 2 \eta_1 \\
& \epsilon_{c2}^2 \epsilon_{r3}^2 + 8 \eta_1 \epsilon_{c2}^2 \epsilon_{r3} + 2 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \\
& - 16 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 - 16 \eta_2 \epsilon_{r1} \epsilon_{r2} - 16 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 - 4 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 - 4 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} - 4 \eta_2 \\
& \epsilon_{c1}^2 \epsilon_{c2}^2 + 8 \eta_2 \epsilon_{r1} \epsilon_{r3}^2 + 32 \eta_2 \epsilon_{r1} \epsilon_{r3} + 8 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{r2} \epsilon_{r3}^2 - 16 \eta_2 \epsilon_{r2} \epsilon_{r3} \\
& - 4 \eta_2 \epsilon_{r2} \epsilon_{c3}^2 + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r3} + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 16 \eta_2 \\
& \epsilon_{r2}^2 \epsilon_{r3} - 4 \eta_2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r3}^2 + 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r3} + 2 \eta_2 \epsilon_{c1}^2 \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{c2}^2 \epsilon_{r3}^2 \\
& - 16 \eta_2 \epsilon_{c2}^2 \epsilon_{r3} - 4 \eta_2 \epsilon_{c2}^2 \epsilon_{c3}^2 + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 + 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 + 8 \eta_3 \epsilon_{r1} \\
& \epsilon_{r2}^2 + 32 \eta_3 \epsilon_{r1} \epsilon_{r2} + 8 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 + 2 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 + 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r2} + 2 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \\
& - 16 \eta_3 \epsilon_{r1} \epsilon_{r3}^2 - 16 \eta_3 \epsilon_{r1} \epsilon_{r3} - 16 \eta_3 \epsilon_{r1} \epsilon_{c3}^2 - 16 \eta_3 \epsilon_{r2} \epsilon_{r3}^2 - 16 \eta_3 \epsilon_{r2} \epsilon_{r3} \\
& - 16 \eta_3 \epsilon_{r2} \epsilon_{c3}^2 - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r3}^2 - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r3} - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{c3}^2 - 4 \eta_3 \epsilon_{r2}^2 \epsilon_{r3}^2 - 4 \eta_3 \\
& \epsilon_{r2}^2 \epsilon_{r3} - 4 \eta_3 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \eta_3 \epsilon_{c1}^2 \epsilon_{r3}^2 - 4 \eta_3 \epsilon_{c1}^2 \epsilon_{r3} - 4 \eta_3 \epsilon_{c1}^2 \epsilon_{c3}^2 - 4 \eta_3 \epsilon_{c2}^2 \epsilon_{r3}^2 - 4 \eta_3 \\
& \epsilon_{c2}^2 \epsilon_{r3} - 4 \eta_3 \epsilon_{c2}^2 \epsilon_{c3}^2 + 8 \eta_3^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 8 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 8 \eta_3^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 8 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r3}^2 + 32 \\
& \eta_3^2 \epsilon_{r1} \epsilon_{r3}^2 + 32 \eta_3^2 \epsilon_{r2} \epsilon_{r3}^2 - 32 \eta_3 \epsilon_{r3} \eta_1 + 16 \eta_3 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 4 \eta_3 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2
\end{aligned}$$

$$\begin{aligned}
& + 4 \eta_3 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 4 \eta_3 \eta_2 \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \eta_3 \epsilon_{r2} \eta_1 \epsilon_{cl}^2 \epsilon_{c3}^2 + 4 \eta_3 \epsilon_{r2}^2 \eta_1 \epsilon_{cl}^2 \epsilon_{c3}^2 \\
& + 4 \eta_3 \epsilon_{cl}^2 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \eta_3 \epsilon_{cl}^2 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 + 16 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 \eta_3 + 4 \eta_2 \epsilon_{cl}^2 \epsilon_{r2} \epsilon_{r3}^2 \eta_3 \\
& + 4 \eta_2 \epsilon_{cl}^2 \epsilon_{r2} \epsilon_{r3} \eta_3 + 4 \eta_2 \epsilon_{cl}^2 \epsilon_{r2} \epsilon_{c3}^2 \eta_3 + 16 \eta_3 \epsilon_{r1}^2 \eta_1 \epsilon_{r2} \epsilon_{r3}^2 + 16 \eta_3 \epsilon_{cl}^2 \eta_1 \epsilon_{r2} \epsilon_{r3}^2 \\
& + 16 \eta_3 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 + 4 \eta_3 \eta_2 \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \eta_3 \eta_2 \epsilon_{cl}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \eta_3 \eta_2 \epsilon_{cl}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 \\
& + 16 \eta_3 \epsilon_{r2}^2 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_1 \epsilon_{r3}^2 + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_1 \epsilon_{c3}^2 + 16 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \eta_1 \epsilon_{c3}^2 \\
& + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_1 \epsilon_{r3}^2 + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_1 \epsilon_{c3}^2 + 4 \eta_3 \epsilon_{cl}^2 \epsilon_{r2}^2 \eta_1 \epsilon_{r3}^2 + 16 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 \\
& + 16 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 4 \eta_3 \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 - 32 \eta_3 \epsilon_{r1} \eta_1 - 32 \eta_2 \epsilon_{cl}^2 \eta_1 + 16 \eta_2 \\
& \epsilon_{c3}^2 \eta_1 - 32 \eta_3 \epsilon_{cl}^2 \eta_1 + 16 \eta_3 \epsilon_{c2}^2 \eta_1 - 32 \eta_2 \epsilon_{r1} \eta_1 + 8 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 + 8 \epsilon_{cl}^2 \eta_1^2 \epsilon_{c2}^2 \\
& + 32 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1} + 8 \epsilon_{cl}^2 \eta_1^2 \epsilon_{c3}^2 + 2 \eta_1^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + 2 \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 8 \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3} + 2 \eta_1^2 \epsilon_{r2}^2 \\
& \epsilon_{c3}^2 + 8 \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3} + 2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 32 \eta_1^2 \epsilon_{r2} \epsilon_{r3} + 8 \eta_1^2 \epsilon_{r2} \epsilon_{c3}^2 + 8 \eta_1^2 \epsilon_{r2} \epsilon_{r3}^2 + 32 \epsilon_{c2}^2 \\
& \eta_2^2 \epsilon_{r3} + 32 \epsilon_{cl}^2 \eta_1^2 \epsilon_{r3} - 32 \eta_2 \epsilon_{r2} \eta_3 - 32 \eta_2 \epsilon_{r3} \eta_3 + 16 \eta_2 \epsilon_{r1}^2 \eta_3 - 32 \eta_2 \epsilon_{r3}^2 \eta_3 \\
& + 32 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2} + 8 \eta_1^2 \epsilon_{r1}^2 \epsilon_{c2}^2 + 8 \epsilon_{c3}^2 \eta_3^2 \epsilon_{c2}^2 + 8 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 - 32 \eta_1 \epsilon_{r2}^2 \eta_2 - 32 \eta_1 \\
& \epsilon_{r3}^2 \eta_3 - 32 \eta_1 \epsilon_{c2}^2 \eta_2 - 32 \eta_1 \epsilon_{c3}^2 \eta_3 + 32 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r2} + 64 \eta_2 \epsilon_{r3} \eta_1 + 8 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \\
& + 8 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2} + 2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{c2}^2 + 8 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 - 8 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 \eta_1 - 8 \eta_2 \epsilon_{r2} \epsilon_{r3}^2 \eta_1 \\
& - 32 \eta_2 \epsilon_{r2} \epsilon_{r3} \eta_1 - 64 \eta_3^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - 16 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 16 \eta_3^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_2 \\
& \epsilon_{cl}^2 \epsilon_{r3} \eta_3 - 8 \eta_2 \epsilon_{cl}^2 \epsilon_{c3}^2 \eta_3 + 8 \epsilon_{cl}^2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r3}^2 \eta_3 - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r3} \eta_3 \\
& - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{c3}^2 \eta_3 - 8 \eta_2 \epsilon_{cl}^2 \epsilon_{r3}^2 \eta_3 + 16 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \eta_1 + 16 \eta_2 \epsilon_{r1} \epsilon_{r2} \eta_1 + 16 \eta_2 \epsilon_{r2} \\
& \epsilon_{r3}^2 \eta_3 + 16 \eta_2 \epsilon_{r2} \epsilon_{r3} \eta_3 + 16 \eta_2 \epsilon_{r2} \epsilon_{c3}^2 \eta_3 - 16 \eta_1^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - 4 \eta_1^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 + 32 \\
& \epsilon_{cl}^2 \eta_1^2 \epsilon_{r2} \epsilon_{r3} - 4 \eta_1^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 64 \eta_1^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - 16 \eta_1^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 16 \eta_1^2 \epsilon_{r1} \\
& \epsilon_{c2}^2 \epsilon_{r3} - 4 \eta_1^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 4 \eta_1^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 - 16 \eta_1^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 - 8 \eta_3 \epsilon_{r2}^2 \epsilon_{r3} \eta_1 \\
& + 16 \eta_3 \epsilon_{cl}^2 \epsilon_{r3} \eta_1 - 8 \eta_3 \epsilon_{c2}^2 \epsilon_{r3} \eta_1 - 32 \eta_3 \epsilon_{cl}^2 \epsilon_{r2} \eta_1 - 8 \eta_3 \epsilon_{cl}^2 \epsilon_{c2}^2 \eta_1 \\
& + 16 \eta_3 \epsilon_{r1} \epsilon_{r3} \eta_1 - 32 \eta_3 \epsilon_{r2} \epsilon_{r3} \eta_1 + 16 \eta_3 \epsilon_{r1}^2 \epsilon_{r3} \eta_1 - 8 \eta_3 \epsilon_{cl}^2 \epsilon_{r2}^2 \eta_1 + 2 \epsilon_{c2}^2 \eta_2^2 \\
& \epsilon_{cl}^2 \epsilon_{r3}^2 - 4 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 16 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} + 8 \epsilon_{cl}^2 \eta_1^2 \epsilon_{r2} \epsilon_{r3}^2 - 32 \eta_1 \epsilon_{r2} \epsilon_{r3}^2 \eta_3 \\
& - 32 \eta_1 \epsilon_{r2} \epsilon_{c3}^2 \eta_3 + 16 \eta_3 \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 16 \eta_1 \epsilon_{r1} \epsilon_{r3}^2 \eta_3 + 16 \eta_1 \epsilon_{r1} \epsilon_{c3}^2 \eta_3 + 8 \epsilon_{c3}^2 \eta_3^2 \\
& \epsilon_{cl}^2 \epsilon_{r2} + 16 \eta_2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 + 16 \eta_2 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 + 2 \epsilon_{c3}^2 \eta_3^2 \epsilon_{cl}^2 \epsilon_{r2}^2 + 8 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1} \epsilon_{c3}^2 - 4 \\
& \eta_2^2 \epsilon_{cl}^2 \epsilon_{r2} \epsilon_{r3}^2 - 16 \eta_2^2 \epsilon_{cl}^2 \epsilon_{r2} \epsilon_{r3} - 4 \eta_2^2 \epsilon_{cl}^2 \epsilon_{r2} \epsilon_{c3}^2 + 8 \epsilon_{cl}^2 \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3} + 2 \epsilon_{cl}^2 \eta_1^2 \epsilon_{c2}^2 \epsilon_{c3}^2 \\
& - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \eta_3 - 32 \eta_2 \epsilon_{r1} \epsilon_{r2} \eta_3 - \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 16 \eta_1 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} + 32 \eta_3^2 \epsilon_{r1} \epsilon_{r2}
\end{aligned}$$

$$\begin{aligned}
& + 8 \eta_3^2 \epsilon_{r1} \epsilon_{c2}^2 + 2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2}^2 + 8 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2} + 2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{c2}^2 - 64 \eta_3^2 \epsilon_{r1} \epsilon_{r3} - 64 \\
& \eta_3^2 \epsilon_{r2} \epsilon_{r3} + 2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 8 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3} + 2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{c3}^2 + 2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r3}^2 + 8 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r3} + 2 \\
& \eta_2^2 \epsilon_{c1}^2 \epsilon_{c3}^2 + 2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 - 16 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2} - 64 \eta_2^2 \epsilon_{r1} \epsilon_{r2} - 16 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2} + 8 \eta_2^2 \epsilon_{r1} \epsilon_{r3}^2 \\
& + 32 \eta_2^2 \epsilon_{r1} \epsilon_{r3} + 8 \eta_2^2 \epsilon_{r1} \epsilon_{c3}^2 - 16 \eta_2^2 \epsilon_{r2} \epsilon_{r3}^2 - 64 \eta_2^2 \epsilon_{r2} \epsilon_{r3} - 16 \eta_2^2 \epsilon_{r2} \epsilon_{c3}^2 + 32 \\
& \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2} - 32 \eta_3 \epsilon_{r2}^2 \eta_2 - 32 \eta_3 \epsilon_{c2}^2 \eta_2 + 8 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 8 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 + 8 \eta_1^2 \epsilon_{r1}^2 \epsilon_{c3}^2 \\
& + 32 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r3} + 32 \eta_2^2 \epsilon_{r1} \epsilon_{r2}^2 + 8 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2}^2 + 32 \eta_2^2 \epsilon_{r3} \epsilon_{r2}^2 + 64 \eta_2 \epsilon_{r1} \eta_3 - 16 \eta_3^2 \\
& \epsilon_{c2}^2 \epsilon_{r3} + 16 \eta_2 \epsilon_{c1}^2 \eta_3 - 32 \eta_2 \epsilon_{c3}^2 \eta_3 + 64 \eta_3 \epsilon_{r2} \eta_1 - \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 4 \eta_1 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 \\
& - 4 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} - \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 - \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 4 \eta_1 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 \\
& - \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 16 \eta_1 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 4 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \\
& - 4 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} - 4 \eta_1 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 - \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 4 \eta_1 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 \\
& - 4 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} - \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 - \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \eta_1 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 16 \eta_1 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} \\
& - 4 \eta_1 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 - \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 4 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} - \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} \\
& - 4 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 4 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} - \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} \\
& - 4 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 16 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r3}^2) / (64 + 4 \epsilon_{c2}^2 \eta_2^2 \\
& \epsilon_{c3}^2 + 64 \eta_1 + 64 \eta_2 + 64 \eta_3 + 64 \epsilon_{r1} + 64 \epsilon_{r2} + 64 \epsilon_{r3} + 16 \epsilon_{c1}^2 + 16 \epsilon_{c2}^2 + 16 \epsilon_{c3}^2 \\
& - 8 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r3} - 8 \eta_3^2 \epsilon_{r2}^2 \epsilon_{r3} + 18 \epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 \epsilon_{r2}^2 + 4 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r3}^2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r2}^2 + 4 \\
& \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2}^2 - 16 \eta_2 \epsilon_{r2} \eta_1 + 8 \eta_3 \epsilon_{r1}^2 \eta_1 \epsilon_{r3}^2 - 16 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \eta_2 - 16 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \eta_2 \\
& - 4 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \eta_2 - 8 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} - 2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 4 \eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_2 - 4 \eta_1 \epsilon_{r2}^2 \\
& \epsilon_{r3}^2 \eta_3 - 16 \eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_2 + 8 \eta_3 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 + 16 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1} \epsilon_{r3} + 4 \\
& \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3} + 16 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1} \epsilon_{r2} + 8 \eta_2 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 + 8 \eta_2 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 - 4 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \eta_3 \\
& - 16 \eta_2 \epsilon_{r1} \epsilon_{r3}^2 \eta_3 - 16 \eta_2 \epsilon_{r1} \epsilon_{r3} \eta_3 - 16 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 \eta_3 + \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{c3}^2 + 8 \eta_2 \\
& \epsilon_{c1}^2 \epsilon_{r2} \eta_1 - 4 \eta_2 \epsilon_{r1} \epsilon_{r3}^2 \eta_1 - 16 \eta_2 \epsilon_{r1} \epsilon_{r3} \eta_1 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{c1}^2 + 8 \eta_3 \epsilon_{c2}^2 \eta_2 \epsilon_{r3}^2 - 4 \eta_2 \\
& \epsilon_{c1}^2 \epsilon_{r3}^2 \eta_1 - 16 \eta_2 \epsilon_{c1}^2 \epsilon_{r3} \eta_1 - 4 \eta_2 \epsilon_{c1}^2 \epsilon_{c3}^2 \eta_1 + 8 \eta_3 \epsilon_{r2}^2 \eta_2 \epsilon_{c3}^2 + 8 \eta_3 \epsilon_{r2}^2 \eta_2 \epsilon_{r3}^2 \\
& - 4 \eta_1 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_2 - 4 \eta_1 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_3 - 4 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_2 - 4 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_3 - 16 \eta_1 \\
& \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_2 - 4 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_2 - 4 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_3 - 4 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \eta_2 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r3}^2 \eta_1 \\
& - 16 \eta_2 \epsilon_{r1}^2 \epsilon_{r3}^2 \eta_1 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{c3}^2 \eta_1 + \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \\
& + 8 \eta_3 \eta_1 \epsilon_{c1}^2 \epsilon_{r3}^2 + 8 \eta_3 \eta_1 \epsilon_{c1}^2 \epsilon_{c3}^2 + 16 \eta_3^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 + 4 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 + \eta_3^2 \epsilon_{c1}^2 \epsilon_{c2}^2 \\
& \epsilon_{r3}^2 - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_2 - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_2 - 2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_3^2
\end{aligned}$$

$$\begin{aligned}
& \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} + 8 \eta_3 \epsilon_{r1}^2 \eta_1 \epsilon_{c3}^2 + 4 \eta_2^2 \epsilon_{c3}^2 \epsilon_{r2}^2 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 16 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \eta_1 - 4 \eta_3 \\
& \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_1 - 4 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \eta_1 - 16 \eta_3 \epsilon_{r1} \epsilon_{r2} \eta_1 + 72 \epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 + 72 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \\
& - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_1 + 72 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 - 2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} + 4 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 + 8 \eta_3 \epsilon_{c2}^2 \eta_2 \\
& \epsilon_{c3}^2 + 16 \epsilon_{c1}^2 \eta_1^2 + 16 \eta_3^2 \epsilon_{r1} + 16 \eta_3^2 \epsilon_{r2} - 32 \eta_3^2 \epsilon_{r3} + 4 \eta_3^2 \epsilon_{r1}^2 + 4 \eta_3^2 \epsilon_{r2}^2 + 4 \eta_3^2 \epsilon_{c1}^2 \\
& + 4 \eta_3^2 \epsilon_{c2}^2 + 16 \eta_2^2 \epsilon_{r1} - 32 \eta_2^2 \epsilon_{r2} + 16 \eta_2^2 \epsilon_{r3} + 4 \eta_2^2 \epsilon_{r1}^2 + 4 \eta_2^2 \epsilon_{r3}^2 + 4 \eta_2^2 \epsilon_{c1}^2 + 4 \\
& \eta_2^2 \epsilon_{c3}^2 + 16 \epsilon_{c3}^2 \eta_3^2 + 4 \epsilon_{r1}^2 \epsilon_{r2}^2 + 16 \epsilon_{r1}^2 \epsilon_{r2} + 4 \epsilon_{r1}^2 \epsilon_{c2}^2 + 16 \epsilon_{r1} \epsilon_{r2}^2 + 64 \epsilon_{r1} \epsilon_{r2} \\
& + 16 \epsilon_{r1} \epsilon_{c2}^2 + 4 \epsilon_{c1}^2 \epsilon_{r2}^2 + 16 \epsilon_{c1}^2 \epsilon_{r2} + 16 \epsilon_{r1}^2 + 16 \epsilon_{r2}^2 + 16 \epsilon_{r3}^2 + 16 \eta_1^2 + 16 \eta_2^2 \\
& + 16 \eta_3^2 - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 \\
& - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 32 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 32 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 2 \eta_2 \epsilon_{c1}^2 \\
& \epsilon_{r2}^2 \epsilon_{r3}^2 - 32 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 8 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} \\
& - 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 - 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \\
& \epsilon_{r3}^2 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} - 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 2 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_2 \epsilon_{c1}^2 \\
& \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_3 \\
& \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 - 32 \eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 + 18 \epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 \epsilon_{c2}^2 \\
& + 72 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \epsilon_{r1} + 18 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \epsilon_{r1}^2 + 18 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \epsilon_{c1}^2 + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \\
& \epsilon_{r3}^2 \eta_3 + 72 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 \epsilon_{r3} + 18 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 \epsilon_{r3}^2 + 18 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 \epsilon_{c3}^2 \\
& + 72 \epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 \epsilon_{r2} + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_1 + 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 \eta_3 - 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 \\
& - 32 \eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 2 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r1} \\
& \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 2 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 32 \eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - 2 \eta_3 \\
& \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 - 2 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} \\
& - 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 2 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 2 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \\
& + \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 4 \eta_2^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 + \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \eta_2^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 \\
& - 8 \eta_2^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2} + \epsilon_{c2}^2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{c3}^2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1} \epsilon_{c2}^2 - 8 \eta_2^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 \\
& + 4 \epsilon_{c1}^2 \epsilon_{c2}^2 + 16 \epsilon_{r1} \epsilon_{r3}^2 + 64 \epsilon_{r1} \epsilon_{r3} + 16 \epsilon_{r1} \epsilon_{c3}^2 + 16 \epsilon_{r2} \epsilon_{r3}^2 + 64 \epsilon_{r2} \epsilon_{r3} + 16 \epsilon_{r2} \\
& \epsilon_{c3}^2 + 4 \epsilon_{r1}^2 \epsilon_{r3}^2 + 16 \epsilon_{r1}^2 \epsilon_{r3} + 4 \epsilon_{r1}^2 \epsilon_{c3}^2 + 4 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \epsilon_{r2}^2 \epsilon_{r3} + 4 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \epsilon_{c1}^2 \epsilon_{r3}^2 \\
& + 16 \epsilon_{c1}^2 \epsilon_{r3} + 4 \epsilon_{c1}^2 \epsilon_{c3}^2 + 4 \epsilon_{c2}^2 \epsilon_{r3}^2 + 16 \epsilon_{c2}^2 \epsilon_{r3} + 4 \epsilon_{c2}^2 \epsilon_{c3}^2 - 32 \eta_1 \epsilon_{r1} + 64 \eta_1 \epsilon_{r2} \\
& + 64 \eta_1 \epsilon_{r3} - 32 \eta_1 \epsilon_{r1}^2 + 16 \eta_1 \epsilon_{r2}^2 + 16 \eta_1 \epsilon_{r3}^2 - 32 \eta_1 \epsilon_{c1}^2 + 16 \eta_1 \epsilon_{c2}^2 + 16 \eta_1 \epsilon_{c3}^2
\end{aligned}$$

$$\begin{aligned}
& + 64 \eta_2 \epsilon_{r1} - 32 \eta_2 \epsilon_{r2} + 64 \eta_2 \epsilon_{r3} + 16 \eta_2 \epsilon_{r1}^2 - 32 \eta_2 \epsilon_{r2}^2 + 16 \eta_2 \epsilon_{r3}^2 + 16 \eta_2 \epsilon_{c1}^2 \\
& - 32 \eta_2 \epsilon_{c2}^2 + 16 \eta_2 \epsilon_{c3}^2 + 64 \eta_3 \epsilon_{r1} + 64 \eta_3 \epsilon_{r2} - 32 \eta_3 \epsilon_{r3} + 16 \eta_3 \epsilon_{r1}^2 + 16 \eta_3 \epsilon_{r2}^2 \\
& - 32 \eta_3 \epsilon_{r3}^2 + 16 \eta_3 \epsilon_{c1}^2 + 16 \eta_3 \epsilon_{c2}^2 - 32 \eta_3 \epsilon_{c3}^2 + 16 \eta_1^2 \epsilon_{r1}^2 + 16 \eta_2^2 \epsilon_{r2}^2 + 16 \epsilon_{c2}^2 \eta_2^2 \\
& + 32 \eta_2 \eta_1 + 32 \eta_3 \eta_1 + 32 \eta_2 \eta_3 + 16 \eta_3^2 \epsilon_{r3}^2 - 32 \eta_2^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} + 4 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \\
& + 4 \eta_2^2 \epsilon_{r2}^2 \epsilon_{r1} \epsilon_{c3}^2 + \epsilon_{c3}^2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{c2}^2 + \epsilon_{c1}^2 \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{c2}^2 + 4 \\
& \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \\
& \eta_3^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \epsilon_{c2}^2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r3}^2 + 8 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \eta_2 + 8 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \eta_2 + 4 \eta_3^2 \epsilon_{r3}^2 \epsilon_{r1}^2 \epsilon_{r2}^2 + \\
& \eta_3^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + \eta_3^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \epsilon_{r1}^2 - 4 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \eta_1 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + \eta_1^2 \\
& \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 16 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + \eta_1^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + \eta_1^2 \\
& \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} \eta_3 + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_1 + 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_1 + 2 \eta_2 \eta_3 \epsilon_{r1}^2 \\
& \epsilon_{c2}^2 \epsilon_{c3}^2 + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_3 + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_1 + 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} \eta_1 + 2 \eta_3 \epsilon_{c1}^2 \\
& \epsilon_{c2}^2 \epsilon_{r3} \eta_1 + 2 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_1 + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} \eta_3 + 2 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} \eta_1 + 2 \eta_3 \epsilon_{r1} \\
& \epsilon_{r2}^2 \epsilon_{r3} \eta_1 + 8 \eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} \eta_1 + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_2 + 2 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_2 + 2 \eta_3 \epsilon_{r1}^2 \\
& \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_2 + 8 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_3 + 2 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_2 + 2 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_3 + 8 \eta_1 \epsilon_{r1} \\
& \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_2 + 8 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_2 + 8 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_3 + 2 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_2 + 2 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \\
& \epsilon_{r3}^2 \eta_3 + 2 \eta_2 \epsilon_{r3}^2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 - 8 \eta_1^2 \epsilon_{r1} \epsilon_{r2}^2 - 32 \eta_1^2 \epsilon_{r1} \epsilon_{r2} + 16 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_1^2 \epsilon_{r1} \\
& \epsilon_{c2}^2 - 8 \eta_1^2 \epsilon_{r1} \epsilon_{r3}^2 - 32 \eta_1^2 \epsilon_{r1} \epsilon_{r3} - 8 \eta_1^2 \epsilon_{r1} \epsilon_{c3}^2 + 8 \eta_2 \epsilon_{r3}^2 \eta_1 + 16 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1} + 8 \eta_3 \\
& \epsilon_{r2}^2 \eta_1 - 16 \eta_2 \epsilon_{r1}^2 \eta_1 + 4 \epsilon_{c2}^2 \eta_2^2 \epsilon_{c1}^2 - 16 \eta_3 \epsilon_{r1}^2 \eta_1 + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r3}^2 + 4 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \\
& \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 + \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + 16 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 + \\
& \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 64 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} + \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} + \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 \\
& + 4 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 4 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 + 4 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 4 \epsilon_{r1} \\
& \epsilon_{c2}^2 \epsilon_{c3}^2 + \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} + 4 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \epsilon_{c1}^2 \\
& \epsilon_{c2}^2 \epsilon_{r3}^2 + \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 - 32 \eta_1 \epsilon_{r1}^2 \epsilon_{r2} - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 - 8 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \\
& - 32 \eta_1 \epsilon_{r1} \epsilon_{r2} - 8 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 - 32 \eta_1 \epsilon_{c1}^2 \epsilon_{r2} - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 - 8 \eta_1 \epsilon_{r1} \\
& \epsilon_{r3}^2 + 2 \eta_2 \epsilon_{r3}^2 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 + 2 \eta_2 \epsilon_{r3}^2 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 + 2 \eta_2 \epsilon_{c3}^2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 + 2 \eta_2 \epsilon_{c3}^2 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 \\
& + 2 \eta_2 \epsilon_{c3}^2 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 + 2 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_2 + 2 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_3 + 8 \eta_2 \epsilon_{r3} \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \\
& + 8 \eta_2 \epsilon_{r3} \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 + 8 \eta_2 \epsilon_{r3} \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 + 2 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_2 + 2 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_3
\end{aligned}$$

$$\begin{aligned}
& + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} \eta_1 + 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} \eta_1 + 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 \eta_1 + 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 \eta_1 \\
& + 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} \eta_1 + 2 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 \eta_1 + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 \eta_1 + 2 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 \eta_1 \\
& + 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} \eta_1 + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 \eta_1 + 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} \eta_1 + 8 \eta_3 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 \\
& + 2 \eta_3 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{r2} \epsilon_{c3}^2 \eta_1 - 32 \eta_1^2 \epsilon_{r1} + 16 \eta_1^2 \epsilon_{r2} + 16 \eta_1^2 \epsilon_{r3} + 4 \eta_1^2 \epsilon_{r2}^2 \\
& + 4 \eta_1^2 \epsilon_{r3}^2 + 4 \eta_1^2 \epsilon_{c2}^2 + 4 \eta_1^2 \epsilon_{c3}^2 + 4 \eta_2^2 \epsilon_{r3}^2 \epsilon_{r2}^2 - 32 \eta_1 \epsilon_{r1} \epsilon_{r3} - 8 \eta_1 \epsilon_{r1} \epsilon_{c3}^2 \\
& + 16 \eta_1 \epsilon_{r2} \epsilon_{r3}^2 + 64 \eta_1 \epsilon_{r2} \epsilon_{r3} + 16 \eta_1 \epsilon_{r2} \epsilon_{c3}^2 - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{r3}^2 - 32 \eta_1 \epsilon_{r1}^2 \epsilon_{r3} - 8 \eta_1 \\
& \epsilon_{r1}^2 \epsilon_{c3}^2 + 4 \eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \eta_1 \epsilon_{r2}^2 \epsilon_{r3} + 4 \eta_1 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{r3}^2 - 32 \eta_1 \epsilon_{c1}^2 \epsilon_{r3} \\
& - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{c3}^2 + 4 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 + 16 \eta_1 \epsilon_{c2}^2 \epsilon_{r3} + 4 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 - 8 \eta_2 \\
& \epsilon_{r1}^2 \epsilon_{r2} - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 - 32 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 - 32 \eta_2 \epsilon_{r1} \epsilon_{r2} - 32 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \\
& - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 + 16 \eta_2 \epsilon_{r1} \epsilon_{r3}^2 + 64 \eta_2 \epsilon_{r1} \epsilon_{r3} + 16 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 \\
& - 8 \eta_2 \epsilon_{r2} \epsilon_{r3}^2 - 32 \eta_2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_2 \epsilon_{r2} \epsilon_{c3}^2 + 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 16 \eta_2 \epsilon_{r1}^2 \epsilon_{r3} + 4 \eta_2 \epsilon_{r1}^2 \\
& \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 32 \eta_2 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \eta_2 \epsilon_{c1}^2 \epsilon_{r3}^2 + 16 \eta_2 \epsilon_{c1}^2 \epsilon_{r3} + 4 \eta_2 \\
& \epsilon_{c1}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 32 \eta_2 \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_2 \epsilon_{c2}^2 \epsilon_{c3}^2 + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 + 16 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \\
& + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 + 16 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 + 64 \eta_3 \epsilon_{r1} \epsilon_{r2} + 16 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 + 4 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 + 16 \eta_3 \\
& \epsilon_{c1}^2 \epsilon_{r2} + 4 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 - 32 \eta_3 \epsilon_{r1} \epsilon_{r3}^2 - 32 \eta_3 \epsilon_{r1} \epsilon_{r3} - 32 \eta_3 \epsilon_{r1} \epsilon_{c3}^2 - 32 \eta_3 \epsilon_{r2} \epsilon_{r3}^2 \\
& - 32 \eta_3 \epsilon_{r2} \epsilon_{r3} - 32 \eta_3 \epsilon_{r2} \epsilon_{c3}^2 - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r3} - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{c3}^2 - 8 \eta_3 \epsilon_{r2}^2 \\
& \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_3 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r3} - 8 \eta_3 \epsilon_{c1}^2 \epsilon_{c3}^2 - 8 \eta_3 \epsilon_{c2}^2 \\
& \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_3 \epsilon_{c2}^2 \epsilon_{c3}^2 + 4 \eta_3^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 4 \eta_3^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 4 \eta_3^2 \epsilon_{c1}^2 \\
& \epsilon_{r3}^2 + 16 \eta_3^2 \epsilon_{r1} \epsilon_{r3}^2 + 16 \eta_3^2 \epsilon_{r2} \epsilon_{r3}^2 - 16 \eta_3 \epsilon_{r3} \eta_1 + 8 \eta_3 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 + 2 \eta_3 \eta_2 \epsilon_{r1}^2 \\
& \epsilon_{c2}^2 \epsilon_{r3}^2 + 2 \eta_3 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 2 \eta_3 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 8 \eta_3 \epsilon_{r2} \eta_1 \epsilon_{c1}^2 \epsilon_{c3}^2 + 2 \eta_3 \epsilon_{r2}^2 \eta_1 \epsilon_{c1}^2 \\
& \epsilon_{c3}^2 + 2 \eta_3 \epsilon_{c1}^2 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 + 2 \eta_3 \epsilon_{c1}^2 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 + 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 \eta_3 + 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_3 \\
& + 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} \eta_3 + 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 \eta_3 + 8 \eta_3 \epsilon_{r1}^2 \eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 + 8 \eta_3 \epsilon_{c1}^2 \eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 \\
& + 8 \eta_3 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 + 2 \eta_3 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 2 \eta_3 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 2 \eta_3 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 \\
& + 8 \eta_3 \epsilon_{r2}^2 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_1 \epsilon_{r3}^2 + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_1 \epsilon_{c3}^2 + 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_1 \epsilon_{c3}^2 \\
& + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_1 \epsilon_{r3}^2 + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_1 \epsilon_{c3}^2 + 2 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \eta_1 \epsilon_{r3}^2 + 8 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} \eta_2 \\
& + 8 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 2 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 - 16 \eta_3 \epsilon_{r1} \eta_1 - 16 \eta_2 \epsilon_{c1}^2 \eta_1 + 8 \eta_2 \epsilon_{c3}^2 \eta_1 \\
& - 16 \eta_3 \epsilon_{c1}^2 \eta_1 + 8 \eta_3 \epsilon_{c2}^2 \eta_1 - 16 \eta_2 \epsilon_{r1} \eta_1 + 4 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{c2}^2 + 16 \epsilon_{c3}^2
\end{aligned}$$

$$\begin{aligned}
& \eta_3^2 \varepsilon_{r1} + 4 \varepsilon_{c1}^2 \eta_1^2 \varepsilon_{c3}^2 + \eta_1^2 \varepsilon_{c2}^2 \varepsilon_{c3}^2 + \eta_1^2 \varepsilon_{c2}^2 \varepsilon_{r3}^2 + 4 \eta_1^2 \varepsilon_{c2}^2 \varepsilon_{r3} + \eta_1^2 \varepsilon_{r2}^2 \varepsilon_{c3}^2 + 4 \eta_1^2 \varepsilon_{r2}^2 \varepsilon_{r3} \\
& + \eta_1^2 \varepsilon_{r2}^2 \varepsilon_{r3}^2 + 16 \eta_1^2 \varepsilon_{r2} \varepsilon_{r3} + 4 \eta_1^2 \varepsilon_{r2} \varepsilon_{c3}^2 + 4 \eta_1^2 \varepsilon_{r2} \varepsilon_{r3}^2 + 16 \varepsilon_{c2}^2 \eta_2^2 \varepsilon_{r3} + 16 \varepsilon_{c1}^2 \\
& \eta_1^2 \varepsilon_{r3} - 16 \eta_2 \varepsilon_{r2} \eta_3 - 16 \eta_2 \varepsilon_{r3} \eta_3 + 8 \eta_2 \varepsilon_{r1}^2 \eta_3 - 16 \eta_2 \varepsilon_{r3}^2 \eta_3 + 16 \eta_1^2 \varepsilon_{r1}^2 \varepsilon_{r2} + 4 \\
& \eta_1^2 \varepsilon_{r1}^2 \varepsilon_{c2}^2 + 4 \varepsilon_{c3}^2 \eta_3^2 \varepsilon_{c2}^2 + 4 \eta_1^2 \varepsilon_{r1}^2 \varepsilon_{r2}^2 - 16 \eta_1 \varepsilon_{r2}^2 \eta_2 - 16 \eta_1 \varepsilon_{r3}^2 \eta_3 - 16 \eta_1 \varepsilon_{c2}^2 \eta_2 \\
& - 16 \eta_1 \varepsilon_{c3}^2 \eta_3 + 16 \varepsilon_{c3}^2 \eta_3^2 \varepsilon_{r2} + 32 \eta_2 \varepsilon_{r3} \eta_1 + 4 \eta_2^2 \varepsilon_{r1}^2 \varepsilon_{r2}^2 + 4 \eta_3^2 \varepsilon_{r1}^2 \varepsilon_{r2} + \eta_3^2 \varepsilon_{r1}^2 \\
& \varepsilon_{c2}^2 + 4 \eta_3^2 \varepsilon_{r1} \varepsilon_{r2}^2 - 4 \eta_2 \varepsilon_{r1} \varepsilon_{c3}^2 \eta_1 - 4 \eta_2 \varepsilon_{r2} \varepsilon_{r3}^2 \eta_1 - 16 \eta_2 \varepsilon_{r2} \varepsilon_{r3} \eta_1 - 32 \\
& \eta_3^2 \varepsilon_{r1} \varepsilon_{r2} \varepsilon_{r3} - 8 \eta_3^2 \varepsilon_{r1} \varepsilon_{r2}^2 \varepsilon_{r3} - 8 \eta_3^2 \varepsilon_{r1} \varepsilon_{c2}^2 \varepsilon_{r3} - 4 \eta_2 \varepsilon_{c1}^2 \varepsilon_{r3} \eta_3 - 4 \eta_2 \varepsilon_{c1}^2 \varepsilon_{c3}^2 \eta_3 \\
& + 4 \varepsilon_{c1}^2 \eta_1^2 \varepsilon_{r2}^2 \varepsilon_{r3} - 4 \eta_2 \varepsilon_{r1}^2 \varepsilon_{r3}^2 \eta_3 - 4 \eta_2 \varepsilon_{r1}^2 \varepsilon_{r3} \eta_3 - 4 \eta_2 \varepsilon_{r1}^2 \varepsilon_{c3}^2 \eta_3 - 4 \eta_2 \varepsilon_{c1}^2 \\
& \varepsilon_{r3}^2 \eta_3 + 8 \eta_2 \varepsilon_{r1}^2 \varepsilon_{r2} \eta_1 + 8 \eta_2 \varepsilon_{r1} \varepsilon_{r2} \eta_1 + 8 \eta_2 \varepsilon_{r2} \varepsilon_{r3}^2 \eta_3 + 8 \eta_2 \varepsilon_{r2} \varepsilon_{r3} \eta_3 + 8 \eta_2 \varepsilon_{r2} \\
& \varepsilon_{c3}^2 \eta_3 - 8 \eta_1^2 \varepsilon_{r1} \varepsilon_{r2} \varepsilon_{r3}^2 - 2 \eta_1^2 \varepsilon_{r1} \varepsilon_{c2}^2 \varepsilon_{c3}^2 + 16 \varepsilon_{c1}^2 \eta_1^2 \varepsilon_{r2} \varepsilon_{r3} - 2 \eta_1^2 \varepsilon_{r1} \varepsilon_{r2}^2 \varepsilon_{c3}^2 - 32 \\
& \eta_1^2 \varepsilon_{r1} \varepsilon_{r2} \varepsilon_{r3} - 8 \eta_1^2 \varepsilon_{r1} \varepsilon_{r2}^2 \varepsilon_{r3} - 8 \eta_1^2 \varepsilon_{r1} \varepsilon_{c2}^2 \varepsilon_{r3} - 2 \eta_1^2 \varepsilon_{r1} \varepsilon_{c2}^2 \varepsilon_{r3}^2 - 2 \eta_1^2 \varepsilon_{r1} \varepsilon_{r2}^2 \varepsilon_{r3}^2 \\
& - 8 \eta_1^2 \varepsilon_{r1} \varepsilon_{r2} \varepsilon_{c3}^2 - 4 \eta_3 \varepsilon_{r2}^2 \varepsilon_{r3} \eta_1 + 8 \eta_3 \varepsilon_{c1}^2 \varepsilon_{r3} \eta_1 - 4 \eta_3 \varepsilon_{c2}^2 \varepsilon_{r3} \eta_1 - 16 \eta_3 \\
& \varepsilon_{c1}^2 \varepsilon_{r2} \eta_1 - 4 \eta_3 \varepsilon_{c1}^2 \varepsilon_{c2}^2 \eta_1 + 8 \eta_3 \varepsilon_{r1} \varepsilon_{r3} \eta_1 - 16 \eta_3 \varepsilon_{r2} \varepsilon_{r3} \eta_1 + 8 \eta_3 \varepsilon_{r1}^2 \varepsilon_{r3} \eta_1 \\
& - 4 \eta_3 \varepsilon_{c1}^2 \varepsilon_{r2}^2 \eta_1 + \varepsilon_{c2}^2 \eta_2^2 \varepsilon_{c1}^2 \varepsilon_{r3}^2 - 2 \eta_2^2 \varepsilon_{r1}^2 \varepsilon_{r2} \varepsilon_{c3}^2 - 8 \eta_2^2 \varepsilon_{r1}^2 \varepsilon_{r2} \varepsilon_{r3} + 4 \varepsilon_{c1}^2 \eta_1^2 \varepsilon_{r2} \varepsilon_{r3}^2 \\
& - 16 \eta_1 \varepsilon_{r2} \varepsilon_{r3}^2 \eta_3 - 16 \eta_1 \varepsilon_{r2} \varepsilon_{c3}^2 \eta_3 + 8 \eta_3 \varepsilon_{c2}^2 \varepsilon_{r3} \eta_2 + 8 \eta_1 \varepsilon_{r1} \varepsilon_{r3}^2 \eta_3 + 8 \eta_1 \varepsilon_{r1} \\
& \varepsilon_{c3}^2 \eta_3 + 4 \varepsilon_{c3}^2 \eta_3^2 \varepsilon_{c1}^2 \varepsilon_{r2} + 8 \eta_2 \eta_1 \varepsilon_{r1}^2 \varepsilon_{c2}^2 + 8 \eta_2 \eta_1 \varepsilon_{r1}^2 \varepsilon_{r2}^2 + \varepsilon_{c3}^2 \eta_3^2 \varepsilon_{c1}^2 \varepsilon_{r2}^2 + 4 \varepsilon_{c2}^2 \\
& \eta_2^2 \varepsilon_{r1} \varepsilon_{c3}^2 - 2 \eta_2^2 \varepsilon_{c1}^2 \varepsilon_{r2} \varepsilon_{r3}^2 - 8 \eta_2^2 \varepsilon_{c1}^2 \varepsilon_{r2} \varepsilon_{r3} - 2 \eta_2^2 \varepsilon_{c1}^2 \varepsilon_{r2} \varepsilon_{c3}^2 + 4 \varepsilon_{c1}^2 \eta_1^2 \varepsilon_{c2}^2 \varepsilon_{r3} + \varepsilon_{c1}^2 \\
& \eta_1^2 \varepsilon_{c2}^2 \varepsilon_{c3}^2 - 4 \eta_2 \varepsilon_{r1}^2 \varepsilon_{r2} \eta_3 - 16 \eta_2 \varepsilon_{r1} \varepsilon_{r2} \eta_3 - 2 \eta_3 \varepsilon_{r1}^2 \varepsilon_{c2}^2 \varepsilon_{c3}^2 - 32 \eta_1 \varepsilon_{r1}^2 \varepsilon_{r2} \varepsilon_{r3} \\
& + 16 \eta_3^2 \varepsilon_{r1} \varepsilon_{r2} + 4 \eta_3^2 \varepsilon_{r1} \varepsilon_{c2}^2 + \eta_3^2 \varepsilon_{c1}^2 \varepsilon_{r2}^2 + 4 \eta_3^2 \varepsilon_{c1}^2 \varepsilon_{r2} + \eta_3^2 \varepsilon_{c1}^2 \varepsilon_{c2}^2 - 32 \eta_3^2 \varepsilon_{r1} \varepsilon_{r3} \\
& - 32 \eta_3^2 \varepsilon_{r2} \varepsilon_{r3} + \eta_2^2 \varepsilon_{r1}^2 \varepsilon_{r3}^2 + 4 \eta_2^2 \varepsilon_{r1}^2 \varepsilon_{r3} + \eta_2^2 \varepsilon_{r1}^2 \varepsilon_{c3}^2 + \eta_2^2 \varepsilon_{c1}^2 \varepsilon_{r3}^2 + 4 \eta_2^2 \varepsilon_{c1}^2 \varepsilon_{r3} + \\
& \eta_2^2 \varepsilon_{c1}^2 \varepsilon_{c3}^2 + \eta_3^2 \varepsilon_{r1}^2 \varepsilon_{r2}^2 - 8 \eta_2^2 \varepsilon_{r1}^2 \varepsilon_{r2} - 32 \eta_2^2 \varepsilon_{r1} \varepsilon_{r2} - 8 \eta_2^2 \varepsilon_{c1}^2 \varepsilon_{r2} + 4 \eta_2^2 \varepsilon_{r1} \varepsilon_{r3}^2 + 16 \\
& \eta_2^2 \varepsilon_{r1} \varepsilon_{r3} + 4 \eta_2^2 \varepsilon_{r1} \varepsilon_{c3}^2 - 8 \eta_2^2 \varepsilon_{r2} \varepsilon_{r3}^2 - 32 \eta_2^2 \varepsilon_{r2} \varepsilon_{r3} - 8 \eta_2^2 \varepsilon_{r2} \varepsilon_{c3}^2 + 16 \varepsilon_{c1}^2 \eta_1^2 \varepsilon_{r2} \\
& - 16 \eta_3 \varepsilon_{r2}^2 \eta_2 - 16 \eta_3 \varepsilon_{c2}^2 \eta_2 + 4 \eta_1^2 \varepsilon_{r1}^2 \varepsilon_{r3}^2 + 4 \varepsilon_{c3}^2 \eta_3^2 \varepsilon_{r1}^2 + 4 \eta_1^2 \varepsilon_{r1}^2 \varepsilon_{c3}^2 + 16 \eta_1^2 \\
& \varepsilon_{r1}^2 \varepsilon_{r3} + 16 \eta_2^2 \varepsilon_{r1} \varepsilon_{r2}^2 + 4 \varepsilon_{c1}^2 \eta_1^2 \varepsilon_{r2}^2 + 16 \eta_2^2 \varepsilon_{r3} \varepsilon_{r2}^2 + 32 \eta_2 \varepsilon_{r1} \eta_3 - 8 \eta_3^2 \varepsilon_{c2}^2 \varepsilon_{r3} \\
& + 8 \eta_2 \varepsilon_{c1}^2 \eta_3 - 16 \eta_2 \varepsilon_{c3}^2 \eta_3 + 32 \eta_3 \varepsilon_{r2} \eta_1 - 2 \eta_1 \varepsilon_{r1} \varepsilon_{c2}^2 \varepsilon_{r3}^2 - 8 \eta_1 \varepsilon_{r1}^2 \varepsilon_{r2} \varepsilon_{c3}^2 \\
& - 8 \eta_1 \varepsilon_{r1}^2 \varepsilon_{c2}^2 \varepsilon_{r3} - 2 \eta_1 \varepsilon_{r1}^2 \varepsilon_{r2}^2 \varepsilon_{c3}^2 - 2 \eta_1 \varepsilon_{r1} \varepsilon_{r2}^2 \varepsilon_{r3}^2 - 2 \eta_1 \varepsilon_{r1}^2 \varepsilon_{c2}^2 \varepsilon_{c3}^2 - 8 \eta_1 \varepsilon_{r1} \varepsilon_{r2} \\
& \varepsilon_{c3}^2 - 2 \eta_1 \varepsilon_{r1}^2 \varepsilon_{c2}^2 \varepsilon_{r3}^2 - 32 \eta_1 \varepsilon_{r1} \varepsilon_{r2} \varepsilon_{r3} - 2 \eta_1 \varepsilon_{r1}^2 \varepsilon_{r2}^2 \varepsilon_{r3}^2 - 8 \eta_1 \varepsilon_{r1} \varepsilon_{r2}^2 \varepsilon_{r3} - 2 \eta_1 \varepsilon_{c1}^2
\end{aligned}$$

$$\begin{aligned}
& \epsilon_{r2}^2 \epsilon_{r3}^2 - 8 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 2 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} \\
& - 8 \eta_1 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 2 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 - 2 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{r2} \\
& \epsilon_{r3}^2 - 32 \eta_1 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 2 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_1 \epsilon_{c1}^2 \\
& \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_2^2 \\
& \epsilon_{r1}^2 \epsilon_{r3}^2)
\end{aligned}$$

to string
→

$$\begin{aligned}
& "- (-64C8*\epsilon[c2]^2*\eta[2]^2*\epsilon[c3]^2C32*\eta[1]C32*\eta[2]C32*\eta[3]-64*\epsilon[r1]-64*\epsilon[r2]-64*\epsilon[r3]-16*\epsilon[c1]^2-16* \\
& \epsilon[c2]^2-16*\epsilon[c3]^2-16*\eta[3]^2*\epsilon[c1]^2*\epsilon[r3] \\
& -16*\eta[3]^2*\epsilon[r2]^2*\epsilon[r3]C36*\epsilon[c1]*\eta[1]*\epsilon[c3] \\
& *\eta[3]*\epsilon[r2]^2C8*\epsilon[c2]^2*\eta[2]^2*\epsilon[r3]^2C8*\epsilon[c3]^2*\eta[3]^2*\epsilon[r2]^2C8*\eta[2]^2*\epsilon[c1]^2*\epsilon[r2]^2 \\
& -32*\eta[2]*\epsilon[r2]*\eta[1]C16*\eta[3]*\epsilon[r1]^2*\eta[1]*\epsilon[r3] \\
& ^2-32*\eta[3]*\epsilon[r1]*\epsilon[r2]^2*\eta[2]-32*\eta[3]*\epsilon[r1]* \\
& \epsilon[c2]^2*\eta[2]-8*\eta[3]*\epsilon[c1]^2*\epsilon[r2]^2*\eta[2]-16*\eta[3]^2*\epsilon[c1]^2*\epsilon[r2]*\epsilon[r3]-4*\eta[3]^2*\epsilon[c1]^2* \\
& \epsilon[c2]^2*\epsilon[r3]-8*\eta[1]*\epsilon[r2]^2*\epsilon[r3]^2*\eta[2]-8* \\
& \eta[1]*\epsilon[r2]^2*\epsilon[r3]^2*\eta[3]-32*\eta[1]*\epsilon[r2]^2*\epsilon[r3] \\
& *\eta[2]C16*\eta[3]*\epsilon[r2]^2*\epsilon[r3]*\eta[2]C8*\epsilon[c3]^2* \\
& \eta[3]^2*\epsilon[r1]*\epsilon[r2]^2C32*\epsilon[c2]^2*\eta[2]^2*\epsilon[r1] \\
& *\epsilon[r3]C8*\epsilon[c2]^2*\eta[2]^2*\epsilon[r1]^2*\epsilon[r3]C32* \\
& \epsilon[c3]^2*\eta[3]^2*\epsilon[r1]*\epsilon[r2]C16*\eta[2]*\eta[1]*\epsilon[c1]^2*\epsilon[c2]^2-8*\eta[2]*\epsilon[c1]^2*\epsilon[r2]*\eta[3]-32*\eta[2]*\epsilon[r1]*\epsilon[r3]^2* \\
& \eta[3]-32*\eta[2]*\epsilon[r1]*\epsilon[r3]*\eta[3]-32*\eta[2]*\epsilon[r1]* \\
& \epsilon[c3]^2*\eta[3]C2*\epsilon[c2]^2*\eta[2]^2*\epsilon[r1]^2*\epsilon[c3] \\
& ^2C16*\eta[2]*\epsilon[c1]^2*\epsilon[r2]*\eta[1]-8*\eta[2]*\epsilon[r1]* \\
& \epsilon[r3]^2*\eta[1]-32*\eta[2]*\epsilon[r1]*\epsilon[r3]*\eta[1]C8*\epsilon[c3]^2*\eta[3]^2*\epsilon[c1]^2C16*\eta[3]*\epsilon[c2]^2*\eta[2]*\epsilon[r3] \\
& ^2-8*\eta[2]*\epsilon[c1]^2*\epsilon[r3]^2*\eta[1]-32*\eta[2]*\epsilon[c1]^2* \\
& \epsilon[r3]*\eta[1]-8*\eta[2]*\epsilon[c1]^2*\epsilon[c3]^2*\eta[1]C16*\eta[3]* \\
& \epsilon[r2]^2*\eta[2]*\epsilon[c3]^2C16*\eta[3]*\epsilon[r2]^2*\eta[2]*\epsilon[r3] \\
& ^2-8*\eta[1]*\epsilon[r2]^2*\epsilon[c3]^2*\eta[2]-8*\eta[1]*\epsilon[r2] \\
& ^2*\epsilon[c3]^2*\eta[3]-8*\eta[1]*\epsilon[c2]^2*\epsilon[r3]^2*\eta[2]-8*\eta[1]*\epsilon[c2]^2*\epsilon[r3]^2*\eta[3]-32*\eta[1]*\epsilon[c2]^2*\epsilon[r3] \\
& *\eta[2]-8*\eta[1]*\epsilon[c2]^2*\epsilon[c3]^2*\eta[2]-8*\eta[1]*\epsilon[c2]
\end{aligned}$$

(2.10)

$$\begin{aligned}
&^2*\epsilon[c3]^2*\eta[3]-8*\eta[3]*\epsilon[c1]^2*\epsilon[c2]^2*\eta[2]-8* \\
&\eta[2]*\epsilon[r1]^2*\epsilon[r3]^2*\eta[1]-32*\eta[2]*\epsilon[r1]^2*\epsilon[r3]^2*\eta[1]-8*\eta[2]*\epsilon[r1]^2*\epsilon[c3]^2*\eta[1]C^2*\epsilon[c1]^2* \\
&\eta[1]^2*\epsilon[r2]^2*\epsilon[r3]^2C^2*\epsilon[c2]^2*\eta[2]^2*\epsilon[r1]^2*\epsilon[r3]^2C^2*\epsilon[c3]^2*\eta[3]^2*\epsilon[r1]^2*\epsilon[r2]^ \\
&^2C^{16}*\eta[3]*\eta[1]*\epsilon[c1]^2*\epsilon[r3]^2C^{16}*\eta[3]*\eta[1]* \\
&\epsilon[c1]^2*\epsilon[c3]^2C^{32}*\eta[3]^2*\epsilon[r1]*\epsilon[r2]*\epsilon[r3]^2C^8*\eta[3]^2*\epsilon[c1]^2*\epsilon[r2]*\epsilon[r3]^2C^2*\eta[3]^2* \\
&\epsilon[c1]^2*\epsilon[c2]^2*\epsilon[r3]^2-8*\eta[3]*\epsilon[r1]^2*\epsilon[r2]^2*\eta[2]-8*\eta[3]*\epsilon[r1]^2*\epsilon[c2]^2*\eta[2]-4*\eta[3]^2* \\
&\epsilon[c1]^2*\epsilon[r2]^2*\epsilon[r3]-4*\eta[3]^2*\epsilon[r1]^2*\epsilon[r2]^2*\epsilon[r3]-16*\eta[3]^2*\epsilon[r1]^2*\epsilon[r2]*\epsilon[r3]C^{16}* \\
&\eta[3]*\epsilon[r1]^2*\eta[1]*\epsilon[c3]^2C^8*\eta[2]^2*\epsilon[c3]^2* \\
&\epsilon[r2]^2C^8*\eta[1]^2*\epsilon[r1]^2*\epsilon[r2]*\epsilon[c3]^2-32*\eta[3]*\epsilon[r1]^2*\epsilon[r2]*\eta[1]-8*\eta[3]*\epsilon[r1]^2*\epsilon[c2]^2* \\
&\eta[1]-8*\eta[3]*\epsilon[r1]*\epsilon[r2]^2*\eta[1]-32*\eta[3]*\epsilon[r1]* \\
&\epsilon[r2]*\eta[1]C^{144}*\epsilon[c1]*\eta[1]*\epsilon[c3]*\eta[3]C^{144}*\epsilon[c2]*\eta[2]*\epsilon[c3]*\eta[3]-8*\eta[3]*\epsilon[r1]^2*\epsilon[r2]^2*\eta[1] \\
&C^{144}*\epsilon[c1]*\eta[1]*\epsilon[c2]*\eta[2]-4*\eta[3]^2*\epsilon[r1]^2* \\
&\epsilon[c2]^2*\epsilon[r3]C^8*\eta[3]^2*\epsilon[r1]*\epsilon[r2]^2*\epsilon[r3]^2C^{16}*\eta[3]*\epsilon[c2]^2*\eta[2]*\epsilon[c3]^2C^{32}*\epsilon[c1]^2* \\
&\eta[1]^2C^{32}*\eta[3]^2*\epsilon[r1]C^{32}*\eta[3]^2*\epsilon[r2]-64*\eta[3]^2* \\
&\epsilon[r3]C^8*\eta[3]^2*\epsilon[r1]^2C^8*\eta[3]^2*\epsilon[r2]^2C^8*\eta[3]^2*\epsilon[c1]^2C^8*\eta[3]^2*\epsilon[c2]^2C^{32}*\eta[2]^2*\epsilon[r1]-64* \\
&\eta[2]^2*\epsilon[r2]C^{32}*\eta[2]^2*\epsilon[r3]C^8*\eta[2]^2*\epsilon[r1]^2C^8*\eta[2]^2*\epsilon[r3]^2C^8*\eta[2]^2*\epsilon[c1]^2C^8*\eta[2]^2* \\
&\epsilon[c3]^2C^{32}*\epsilon[c3]^2*\eta[3]^2-4*\epsilon[r1]^2*\epsilon[r2]^2 \\
&-16*\epsilon[r1]^2*\epsilon[r2]-4*\epsilon[r1]^2*\epsilon[c2]^2-16*\epsilon[r1]*\epsilon[r2]^2-64*\epsilon[r1]*\epsilon[r2]-16*\epsilon[r1]*\epsilon[c2]^2-4* \\
&\epsilon[c1]^2*\epsilon[r2]^2-16*\epsilon[c1]^2*\epsilon[r2]-16* \\
&\epsilon[r1]^2-16*\epsilon[r2]^2-16*\epsilon[r3]^2C^{32}*\eta[1]^2C^{32}*\eta[2]^2C^{32}*\eta[3]^2-\eta[2]*\epsilon[r1]^2*\epsilon[r2]^2*\epsilon[c3]^2-4*\eta[2]* \\
&\epsilon[r1]*\epsilon[r2]^2*\epsilon[r3]^2-\eta[2]*\epsilon[r1]^2*\epsilon[c2]^2*\epsilon[c3]^2-4*\eta[2]*\epsilon[r1]*\epsilon[r2]*\epsilon[c3]^2-\eta[2]* \\
&\epsilon[r1]^2*\epsilon[c2]^2*\epsilon[r3]^2-16*\eta[2]*\epsilon[r1]*\epsilon[r2]*\epsilon[r3]-\eta[2]*\epsilon[r1]^2*\epsilon[r2]^2*\epsilon[r3]^2-16*\eta[2]* \\
&\epsilon[r1]*\epsilon[r2]^2*\epsilon[r3]-\eta[2]*\epsilon[c1]^2*\epsilon[r2]^2*\epsilon[r3]^2-16*\eta[2]*\epsilon[r1]*\epsilon[c2]^2*\epsilon[r3]-\eta[2]*
\end{aligned}$$

[illegible]

$$\begin{aligned}
& -4*\eta[2]^2*\epsilon[r1]^2*\epsilon[r2]*\epsilon[r3]^2-16*\eta[2]^2*\epsilon[r1]*\epsilon[r2]*\epsilon[r3]^2C8*\epsilon[c3]^2*\eta[3]^2*\epsilon[r1]^2* \\
& \epsilon[r2]C2*\epsilon[c2]^2*\eta[2]^2*\epsilon[c1]^2*\epsilon[c3]^2C8* \\
& \epsilon[c3]^2*\eta[3]^2*\epsilon[r1]*\epsilon[c2]^2-16*\eta[2]^2*\epsilon[r1]* \\
& \epsilon[r2]*\epsilon[c3]^2-4*\epsilon[c1]^2*\epsilon[c2]^2-16*\epsilon[r1]* \\
& \epsilon[r3]^2-64*\epsilon[r1]*\epsilon[r3]-16*\epsilon[r1]*\epsilon[c3]^2 \\
& -16*\epsilon[r2]*\epsilon[r3]^2-64*\epsilon[r2]*\epsilon[r3]-16*\epsilon[r2]* \\
& \epsilon[c3]^2-4*\epsilon[r1]^2*\epsilon[r3]^2-16*\epsilon[r1]^2*\epsilon[r3] \\
& -4*\epsilon[r1]^2*\epsilon[c3]^2-4*\epsilon[r2]^2*\epsilon[r3]^2-16*\epsilon[r2]^2*\epsilon[r3]-4*\epsilon[r2]^2*\epsilon[c3]^2-4*\epsilon[c1]^2*\epsilon[r3]^2-16*\epsilon[c1]^2*\epsilon[r3]-4*\epsilon[c1]^2*\epsilon[c3]^2-4* \\
& \epsilon[c2]^2*\epsilon[r3]^2-16*\epsilon[c2]^2*\epsilon[r3]-4*\epsilon[c2]^2*\epsilon[c3]^2-16*\eta[1]*\epsilon[r1]C32*\eta[1]*\epsilon[r2]C32*\eta[1]* \\
& \epsilon[r3]-16*\eta[1]*\epsilon[r1]^2C8*\eta[1]*\epsilon[r2]^2C8*\eta[1]* \\
& \epsilon[r3]^2-16*\eta[1]*\epsilon[c1]^2C8*\eta[1]*\epsilon[c2]^2C8*\eta[1]* \\
& \epsilon[c3]^2C32*\eta[2]*\epsilon[r1]-16*\eta[2]*\epsilon[r2]C32*\eta[2]* \\
& \epsilon[r3]C8*\eta[2]*\epsilon[r1]^2-16*\eta[2]*\epsilon[r2]^2C8*\eta[2]* \\
& \epsilon[r3]^2C8*\eta[2]*\epsilon[c1]^2-16*\eta[2]*\epsilon[c2]^2C8*\eta[2]* \\
& \epsilon[c3]^2C32*\eta[3]*\epsilon[r1]C32*\eta[3]*\epsilon[r2]-16*\eta[3]* \\
& \epsilon[r3]C8*\eta[3]*\epsilon[r1]^2C8*\eta[3]*\epsilon[r2]^2-16*\eta[3]* \\
& \epsilon[r3]^2C8*\eta[3]*\epsilon[c1]^2C8*\eta[3]*\epsilon[c2]^2-16*\eta[3]* \\
& \epsilon[c3]^2C32*\eta[1]^2*\epsilon[r1]^2C32*\eta[2]^2*\epsilon[r2]^2C32* \\
& \epsilon[c2]^2*\eta[2]^2C64*\eta[2]*\eta[1]C64*\eta[3]*\eta[1]C64*\eta[2]*\eta[3]C32*\eta[3]^2*\epsilon[r3]^2-64*\eta[2]^2*\epsilon[r1]*\epsilon[r2]* \\
& \epsilon[r3]C8*\eta[2]^2*\epsilon[r1]^2*\epsilon[r2]^2*\epsilon[r3]C8*\eta[2]^2*\epsilon[r2]^2*\epsilon[r1]*\epsilon[c3]^2C2*\epsilon[c3]^2*\eta[3]^2* \\
& \epsilon[c1]^2*\epsilon[c2]^2C2*\epsilon[c1]^2*\eta[1]^2*\epsilon[c2]^2* \\
& \epsilon[r3]^2C2*\eta[3]^2*\epsilon[c1]^2*\epsilon[r2]^2*\epsilon[r3]^2C2* \\
& \epsilon[c3]^2*\eta[3]^2*\epsilon[r1]^2*\epsilon[c2]^2C8*\epsilon[c2]^2*\eta[2]^2*\epsilon[r1]*\epsilon[r3]^2C8*\epsilon[c1]^2*\eta[1]^2*\epsilon[r2]* \\
& \epsilon[c3]^2C2*\epsilon[c1]^2*\eta[1]^2*\epsilon[r2]^2*\epsilon[c3]^2C8* \\
& \eta[2]^2*\epsilon[c1]^2*\epsilon[r2]^2*\epsilon[r3]C2*\eta[2]^2*\epsilon[c1]^2*\epsilon[r2]^2*\epsilon[c3]^2C8*\eta[3]^2*\epsilon[r1]*\epsilon[c2]^2* \\
& \epsilon[r3]^2C8*\epsilon[c2]^2*\eta[2]^2*\epsilon[c1]^2*\epsilon[r3]C16* \\
& \eta[1]*\epsilon[r1]*\epsilon[r2]^2*\eta[2]C16*\eta[1]*\epsilon[r1]*\epsilon[c2]^2*\eta[2]C8*\eta[3]^2*\epsilon[r3]^2*\epsilon[r1]^2*\epsilon[r2]C2*\eta[3]^2*\epsilon[r1]^2*\epsilon[c2]^2*\epsilon[r3]^2C2*\eta[3]^2*\epsilon[r2]^2* \\
& \epsilon[r3]^2*\epsilon[r1]^2-8*\eta[3]*\epsilon[r1]*\epsilon[c2]^2*\eta[1]C8*
\end{aligned}$$

[illegible]

$$\begin{aligned}
& \epsilon[r_3] - \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[r_3]^2 - 16 \epsilon[r_1] * \\
& \epsilon[r_2]^2 \epsilon[r_3] - \epsilon[c_1]^2 \epsilon[r_2]^2 \epsilon[r_3]^2 - 16 * \\
& \epsilon[r_1] \epsilon[c_2]^2 \epsilon[r_3] - 4 \epsilon[r_1]^2 \epsilon[r_2] \epsilon[r_3]^2 - 4 * \\
& \epsilon[r_1] \epsilon[r_2]^2 \epsilon[c_3]^2 - 4 \epsilon[c_1]^2 \epsilon[r_2]^2 \epsilon[r_3] - 16 \epsilon[r_1] \epsilon[r_2] \epsilon[r_3]^2 - 4 \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[r_3] - 4 \epsilon[r_1] \epsilon[c_2]^2 \epsilon[c_3]^2 - \\
& \epsilon[c_1]^2 \epsilon[r_2]^2 \epsilon[c_3]^2 - 4 \epsilon[c_1]^2 \epsilon[r_2] * \\
& \epsilon[r_3]^2 - 16 \epsilon[c_1]^2 \epsilon[r_2] \epsilon[r_3] - 4 \epsilon[c_1]^2 * \\
& \epsilon[r_2] \epsilon[c_3]^2 - \epsilon[c_1]^2 \epsilon[c_2]^2 \epsilon[r_3]^2 - 4 * \\
& \epsilon[c_1]^2 \epsilon[c_2]^2 \epsilon[r_3] - \epsilon[c_1]^2 \epsilon[c_2]^2 * \\
& \epsilon[c_3]^2 - 4 \eta[1] \epsilon[r_1]^2 \epsilon[r_2]^2 - 16 \eta[1] \epsilon[r_1]^2 \epsilon[r_2] - 4 \eta[1] \epsilon[r_1]^2 \epsilon[c_2]^2 - 4 \eta[1] \epsilon[r_1] * \\
& \epsilon[r_2]^2 - 16 \eta[1] \epsilon[r_1] \epsilon[r_2] - 4 \eta[1] \epsilon[r_1] * \\
& \epsilon[c_2]^2 - 4 \eta[1] \epsilon[c_1]^2 \epsilon[r_2]^2 - 16 \eta[1] \epsilon[c_1]^2 \epsilon[r_2] - 4 \eta[1] \epsilon[c_1]^2 \epsilon[c_2]^2 - 4 \eta[1] \epsilon[r_1] * \\
& \epsilon[r_3]^2 C 4 \eta[2] \epsilon[r_3]^2 \eta[1] \epsilon[c_1]^2 \epsilon[r_2]^2 C 4 \eta[2] * \\
& \epsilon[c_3]^2 \eta[1] \epsilon[r_1]^2 \epsilon[c_2]^2 C 4 \eta[2] \epsilon[c_3]^2 * \\
& \eta[1] \epsilon[c_1]^2 \epsilon[r_2]^2 C 4 \eta[2] \epsilon[c_3]^2 \eta[1] \epsilon[c_1]^2 \epsilon[c_2]^2 C 4 \eta[1] \epsilon[r_1] \epsilon[c_2]^2 \epsilon[r_3]^2 * \\
& \eta[2] C 4 \eta[1] \epsilon[r_1] \epsilon[c_2]^2 \epsilon[r_3]^2 \eta[3] C 16 \eta[2] * \\
& \epsilon[r_3] \eta[1] \epsilon[r_1]^2 \epsilon[c_2]^2 C 16 \eta[2] \epsilon[r_3] \eta[1] \epsilon[c_1]^2 \epsilon[r_2]^2 C 16 \eta[2] \epsilon[r_3] \eta[1] \epsilon[c_1]^2 \epsilon[c_2]^2 C 4 \eta[1] \epsilon[r_1] \epsilon[c_2]^2 \epsilon[c_3]^2 \eta[2] C 4 \eta[1] \epsilon[r_1] \epsilon[c_2]^2 \epsilon[c_3]^2 \eta[3] C 4 \eta[3] * \\
& \epsilon[r_1]^2 \epsilon[c_2]^2 \epsilon[r_3] \eta[1] C 16 \eta[3] \epsilon[r_1]^2 * \\
& \epsilon[r_2] \epsilon[r_3] \eta[1] C 4 \eta[2] \epsilon[c_1]^2 \epsilon[r_2] \epsilon[r_3]^2 \eta[1] C 4 \eta[2] \epsilon[c_1]^2 \epsilon[r_2] \epsilon[r_3]^2 \eta[1] C 16 \eta[2] \epsilon[c_1]^2 \epsilon[r_2] \epsilon[r_3] \eta[1] C 4 \eta[2] \epsilon[r_1] \epsilon[r_2] \epsilon[r_3]^2 \eta[1] C 4 \eta[2] \epsilon[r_1]^2 \epsilon[r_2] * \\
& \epsilon[r_3]^2 \eta[1] C 4 \eta[2] \epsilon[r_1] \epsilon[r_2] \epsilon[c_3]^2 \eta[1] C 16 \eta[2] \epsilon[r_1] \epsilon[r_2] \epsilon[r_3] \eta[1] C 4 \eta[2] \epsilon[r_1]^2 \epsilon[r_2] * \\
& \epsilon[r_3] \eta[1] C 16 \eta[3] \eta[2] \epsilon[r_1] \epsilon[c_2]^2 \epsilon[r_3]^2 C 4 \eta[3] \eta[2] \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[c_3]^2 - 8 \eta[2] * \\
& \epsilon[r_2] \epsilon[c_3]^2 \eta[1] - 64 \eta[1]^2 \epsilon[r_1] C 32 \eta[1]^2 * \\
& \epsilon[r_2] C 32 \eta[1]^2 \epsilon[r_3] C 8 \eta[1]^2 \epsilon[r_2]^2 C 8 \eta[1]^2 * \\
& \epsilon[r_3]^2 C 8 \eta[1]^2 \epsilon[c_2]^2 C 8 \eta[1]^2 \epsilon[c_3]^2 C 8 *
\end{aligned}$$

$$\begin{aligned}
& \eta[2]^2 \epsilon[r3]^2 \epsilon[r2]^2 - 16 \eta[1] \epsilon[r1] \epsilon[r3] - 4 \eta[1] \epsilon[r1] \epsilon[c3]^2 C 8 \eta[1] \epsilon[r2] \epsilon[r3]^2 C 32 \\
& \eta[1] \epsilon[r2] \epsilon[r3] C 8 \eta[1] \epsilon[r2] \epsilon[c3]^2 - 4 \eta[1] \epsilon[r1]^2 \epsilon[r3]^2 - 16 \eta[1] \epsilon[r1]^2 \epsilon[r3] - 4 \eta[1] \\
& \epsilon[r1]^2 \epsilon[c3]^2 C 2 \eta[1] \epsilon[r2]^2 \epsilon[r3]^2 C 8 \eta[1] \epsilon[r2]^2 \epsilon[r3] C 2 \eta[1] \epsilon[r2]^2 \epsilon[c3]^2 - 4 \eta[1] \\
& \epsilon[c1]^2 \epsilon[r3]^2 - 16 \eta[1] \epsilon[c1]^2 \epsilon[r3] - 4 \eta[1] \epsilon[c1]^2 \epsilon[c3]^2 C 2 \eta[1] \epsilon[c2]^2 \epsilon[r3]^2 C 8 \\
& \eta[1] \epsilon[c2]^2 \epsilon[r3] C 2 \eta[1] \epsilon[c2]^2 \epsilon[c3]^2 - 4 \eta[2] \epsilon[r1]^2 \epsilon[r2]^2 - 4 \eta[2] \epsilon[r1]^2 \epsilon[r2] - 4 \eta[2] \\
& \epsilon[r1]^2 \epsilon[c2]^2 - 16 \eta[2] \epsilon[r1] \epsilon[r2]^2 - 16 \eta[2] \epsilon[r1] \epsilon[r2] - 16 \eta[2] \epsilon[r1] \epsilon[c2]^2 - 4 \eta[2] \\
& \epsilon[c1]^2 \epsilon[r2]^2 - 4 \eta[2] \epsilon[c1]^2 \epsilon[r2] - 4 \eta[2] \epsilon[c1]^2 \epsilon[c2]^2 C 8 \eta[2] \epsilon[r1] \epsilon[r3]^2 C 32 \\
& \eta[2] \epsilon[r1] \epsilon[r3] C 8 \eta[2] \epsilon[r1] \epsilon[c3]^2 - 4 \eta[2] \epsilon[r2] \epsilon[r3]^2 - 16 \eta[2] \epsilon[r2] \epsilon[r3] - 4 \eta[2] \\
& \epsilon[r2] \epsilon[c3]^2 C 2 \eta[2] \epsilon[r1]^2 \epsilon[r3]^2 C 8 \eta[2] \epsilon[r1]^2 \epsilon[r3] C 2 \eta[2] \epsilon[r1]^2 \epsilon[c3]^2 - 4 \eta[2] \\
& \epsilon[r2]^2 \epsilon[r3]^2 - 16 \eta[2] \epsilon[r2]^2 \epsilon[r3] - 4 \eta[2] \epsilon[r2]^2 \epsilon[c3]^2 C 2 \eta[2] \epsilon[c1]^2 \epsilon[r3]^2 C 8 \\
& \eta[2] \epsilon[c1]^2 \epsilon[r3] C 2 \eta[2] \epsilon[c1]^2 \epsilon[c3]^2 - 4 \eta[2] \epsilon[c2]^2 \epsilon[r3]^2 - 16 \eta[2] \epsilon[c2]^2 \epsilon[r3] - 4 \eta[2] \\
& \epsilon[c2]^2 \epsilon[c3]^2 C 2 \eta[3] \epsilon[r1]^2 \epsilon[r2]^2 C 8 \eta[3] \epsilon[r1]^2 \epsilon[r2] C 2 \eta[3] \epsilon[r1]^2 \epsilon[c2]^2 C 8 \\
& \eta[3] \epsilon[r1] \epsilon[r2]^2 C 32 \eta[3] \epsilon[r1] \epsilon[r2] C 8 \eta[3] \epsilon[r1] \epsilon[c2]^2 C 2 \eta[3] \epsilon[c1]^2 \epsilon[r2]^2 C 8 \\
& \eta[3] \epsilon[c1]^2 \epsilon[r2] C 2 \eta[3] \epsilon[c1]^2 \epsilon[c2]^2 - 16 \eta[3] \epsilon[r1] \epsilon[r3]^2 - 16 \eta[3] \epsilon[r1] \epsilon[r3] - 16 \eta[3] \epsilon[r1] \epsilon[c3]^2 \\
& - 16 \eta[3] \epsilon[r2] \epsilon[r3]^2 - 16 \eta[3] \epsilon[r2] \epsilon[r3] - 16 \eta[3] \epsilon[r2] \epsilon[c3]^2 - 4 \eta[3] \epsilon[r1]^2 \epsilon[r3]^2 - 4 \eta[3] \epsilon[r1]^2 \epsilon[c3]^2 \\
& - 4 \eta[3] \epsilon[r2]^2 \epsilon[r3]^2 - 4 \eta[3] \epsilon[r2]^2 \epsilon[c3]^2 - 4 \eta[3] \epsilon[c1]^2 \epsilon[r3]^2 - 4 \eta[3] \epsilon[c1]^2 \epsilon[c3]^2 \\
& - 4 \eta[3] \epsilon[c2]^2 \epsilon[r3]^2 - 4 \eta[3] \epsilon[c2]^2 \epsilon[c3]^2 C 8 \eta[3]^2 \epsilon[c2]^2 \epsilon[r3]^2 C 8 \eta[3]^2 \epsilon[r1]^2 \epsilon[r2]^2 \epsilon[r3]
\end{aligned}$$

[illegible]

$$\begin{aligned}
&^2\epsilon[r_1]^2\epsilon[c_2]^2C^8\epsilon[c_3]^2\eta[3]^2\epsilon[c_2] \\
&^2C^8\eta[1]^2\epsilon[r_1]^2\epsilon[r_2]^2-32\eta[1]\epsilon[r_2]^2\eta[2]-32\eta[1]\epsilon[r_3]^2\eta[3]-32\eta[1]\epsilon[c_2]^2\eta[2]-32\eta[1]\epsilon[c_3]^2\eta[3]C^32\epsilon[c_3]^2\eta[3]^2\epsilon[r_2]C^64\eta[2] \\
&^2\epsilon[r_3]\eta[1]C^8\eta[2]^2\epsilon[r_1]^2\epsilon[r_2]^2C^8\eta[3]^2\epsilon[r_1]^2\epsilon[r_2]C^2\eta[3]^2\epsilon[r_1]^2\epsilon[c_2]^2C^8\eta[3]^2\epsilon[r_1]\epsilon[r_2]^2-8\eta[2]\epsilon[r_1]\epsilon[c_3]^2\eta[1] \\
&-8\eta[2]\epsilon[r_2]\epsilon[r_3]^2\eta[1]-32\eta[2]\epsilon[r_2]\epsilon[r_3]\eta[1]-64\eta[3]^2\epsilon[r_1]\epsilon[r_2]\epsilon[r_3]-16\eta[3]^2\epsilon[r_1]\epsilon[r_2]^2\epsilon[r_3]-16\eta[3]^2\epsilon[r_1]\epsilon[c_2] \\
&^2\epsilon[r_3]-8\eta[2]\epsilon[c_1]^2\epsilon[r_3]\eta[3]-8\eta[2]\epsilon[c_1]^2\epsilon[c_3]^2\eta[3]C^8\epsilon[c_1]^2\eta[1]^2\epsilon[r_2]^2\epsilon[r_3]-8\eta[2]\epsilon[r_1]^2\epsilon[r_3]^2\eta[3]-8\eta[2]\epsilon[r_1]^2\epsilon[c_3]^2\eta[3]-8\eta[2]\epsilon[c_1]^2\epsilon[r_3]^2\eta[3]C^16\eta[2]\epsilon[r_1]^2\epsilon[r_2]\eta[1]C^16\eta[2]\epsilon[r_1]\epsilon[r_2]\eta[1]C^16\eta[2]\epsilon[r_2] \\
&^2\epsilon[r_3]^2\eta[3]C^16\eta[2]\epsilon[r_2]\epsilon[r_3]\eta[3]C^16\eta[2]^2\epsilon[r_2]\epsilon[c_3]^2\eta[3]-16\eta[1]^2\epsilon[r_1]\epsilon[r_2]\epsilon[r_3]^2-4\eta[1]^2\epsilon[r_1]\epsilon[c_2]^2\epsilon[c_3]^2C^32\epsilon[c_1]^2\eta[1]^2\epsilon[r_2]\epsilon[r_3]-4\eta[1]^2\epsilon[r_1]\epsilon[r_2]^2\epsilon[c_3]^2-64\eta[1]^2\epsilon[r_1]\epsilon[r_2]\epsilon[r_3]-16\eta[1]^2\epsilon[r_1]\epsilon[r_2]^2\epsilon[r_3]-16\eta[1]^2\epsilon[r_1]\epsilon[c_2]^2\epsilon[r_3]-4\eta[1]^2\epsilon[r_1]\epsilon[c_2]^2\epsilon[r_3]^2-4\eta[1]^2\epsilon[r_1]\epsilon[r_2]^2\epsilon[r_3]^2-16\eta[1]^2\epsilon[r_1]\epsilon[r_2]\epsilon[c_3]^2-8\eta[3]\epsilon[r_2]^2\epsilon[r_3]\eta[1]C^16\eta[3]\epsilon[c_1]^2\epsilon[r_3]\eta[1]-8\eta[3]\epsilon[c_2]^2\epsilon[r_3]\eta[1]-32\eta[3]\epsilon[c_1]^2\epsilon[r_2]\eta[1]-8\eta[3]\epsilon[c_1]^2\epsilon[c_2]^2\eta[1]C^16\eta[3]\epsilon[r_1]\epsilon[r_3]\eta[1]-32\eta[3]\epsilon[r_2]\epsilon[r_3]\eta[1]C^16\eta[3]\epsilon[r_1]^2\epsilon[r_3]\eta[1]-8\eta[3]\epsilon[c_1]^2\epsilon[r_2]^2\eta[1]C^2\epsilon[c_2]^2\eta[2]^2\epsilon[c_1]^2\epsilon[r_3]^2-4\eta[2]^2\epsilon[r_1]^2\epsilon[r_2]\epsilon[c_3]^2-16\eta[2]^2\epsilon[r_1]^2\epsilon[r_2]\epsilon[r_3]C^8\epsilon[c_1]^2\eta[1]^2\epsilon[r_2]\epsilon[r_3]^2-32\eta[1]\epsilon[r_2]\epsilon[r_3]^2\eta[3]-32\eta[1]\epsilon[r_2]\epsilon[c_3]^2\eta[3]C^16\eta[3]\epsilon[c_2]^2\epsilon[r_3]\eta[2]C^16\eta[1]\epsilon[r_1]\epsilon[r_3]^2\eta[3]C^16\eta[1]\epsilon[r_1]\epsilon[c_3]^2\eta[3]C^8\epsilon[c_3]^2\eta[3]^2\epsilon[c_1]^2\epsilon[r_2]C^16\eta[2]\eta[1]\epsilon[r_1]^2\epsilon[c_2]^2C^16\eta[2]\eta[1]\epsilon[r_1]^2\epsilon[r_2]\epsilon[r_3]
\end{aligned}$$

$$\begin{aligned} & [r_2]^2 c^2 \epsilon [c_3]^2 \eta [c_1]^2 \epsilon [r_2]^2 c^8 \epsilon \\ & [c_2]^2 \eta [2]^2 \epsilon [r_1] \epsilon [c_3]^2 - 4 \eta [2]^2 \epsilon [c_1]^2 * \\ & \epsilon [r_2] \epsilon [r_3]^2 - 16 \eta [2]^2 \epsilon [c_1]^2 \epsilon [r_2] \epsilon \\ & [r_3] - 4 \eta [2]^2 \epsilon [c_1]^2 \epsilon [r_2] \epsilon [c_3]^2 c^8 \epsilon [c_1]^2 * \\ & \eta [1]^2 \epsilon [c_2]^2 \epsilon [r_3] c^2 \epsilon [c_1]^2 \eta [1]^2 \epsilon [c_2] \\ & ^2 \epsilon [c_3]^2 - 8 \eta [2] \epsilon [r_1]^2 \epsilon [r_2] \eta [3] - 32 \eta [2] * \\ & \epsilon [r_1] \epsilon [r_2] \eta [3] - \eta [3] \epsilon [r_1]^2 \epsilon [c_2]^2 \epsilon \\ & [c_3]^2 - 16 \eta [1] \epsilon [r_1]^2 \epsilon [r_2] \epsilon [r_3] c^{32} \eta [3]^2 * \\ & \epsilon [r_1] \epsilon [r_2] c^8 \eta [3]^2 \epsilon [r_1] \epsilon [c_2]^2 c^2 \eta [3]^2 * \\ & \epsilon [c_1]^2 \epsilon [r_2]^2 c^8 \eta [3]^2 \epsilon [c_1]^2 \epsilon [r_2] c^2 \eta \\ & [3]^2 \epsilon [c_1]^2 \epsilon [c_2]^2 - 64 \eta [3]^2 \epsilon [r_1] \epsilon [r_3] \\ & - 64 \eta [3]^2 \epsilon [r_2] \epsilon [r_3] c^2 \eta [2]^2 \epsilon [r_1]^2 \epsilon \\ & [r_3]^2 c^8 \eta [2]^2 \epsilon [r_1]^2 \epsilon [r_3] c^2 \eta [2]^2 \epsilon [r_1]^2 * \\ & \epsilon [c_3]^2 c^2 \eta [2]^2 \epsilon [c_1]^2 \epsilon [r_3]^2 c^8 \eta [2]^2 * \\ & \epsilon [c_1]^2 \epsilon [r_3] c^2 \eta [2]^2 \epsilon [c_1]^2 \epsilon [c_3]^2 c^2 \eta \\ & [3]^2 \epsilon [r_1]^2 \epsilon [r_2]^2 - 16 \eta [2]^2 \epsilon [r_1]^2 \epsilon [r_2] \\ & - 64 \eta [2]^2 \epsilon [r_1] \epsilon [r_2] - 16 \eta [2]^2 \epsilon [c_1]^2 \epsilon \\ & [r_2] c^8 \eta [2]^2 \epsilon [r_1] \epsilon [r_3]^2 c^{32} \eta [2]^2 \epsilon [r_1] * \\ & \epsilon [r_3] c^8 \eta [2]^2 \epsilon [r_1] \epsilon [c_3]^2 - 16 \eta [2]^2 \epsilon [r_2] \\ & * \epsilon [r_3]^2 - 64 \eta [2]^2 \epsilon [r_2] \epsilon [r_3] - 16 \eta [2]^2 \epsilon \\ & [r_2] \epsilon [c_3]^2 c^{32} \epsilon [c_1]^2 \eta [1]^2 \epsilon [r_2] - 32 \eta [3] * \\ & \epsilon [r_2]^2 \eta [2] - 32 \eta [3] \epsilon [c_2]^2 \eta [2] c^8 \eta [1]^2 \epsilon \\ & [r_1]^2 \epsilon [r_3]^2 c^8 \epsilon [c_3]^2 \eta [3]^2 \epsilon [r_1]^2 c^8 \eta [1]^2 * \\ & \epsilon [r_1]^2 \epsilon [c_3]^2 c^{32} \eta [1]^2 \epsilon [r_1]^2 \epsilon [r_3] c^{32} * \\ & \eta [2]^2 \epsilon [r_1] \epsilon [r_2]^2 c^8 \epsilon [c_1]^2 \eta [1]^2 \epsilon [r_2] \\ & ^2 c^{32} \eta [2]^2 \epsilon [r_3] \epsilon [r_2]^2 c^{64} \eta [2] \epsilon [r_1] \eta [3] \\ & - 16 \eta [3]^2 \epsilon [c_2]^2 \epsilon [r_3] c^{16} \eta [2] \epsilon [c_1]^2 \eta [3] \\ & - 32 \eta [2] \epsilon [c_3]^2 \eta [3] c^{64} \eta [3] \epsilon [r_2] \eta [1] - \eta [1] * \\ & \epsilon [r_1] \epsilon [c_2]^2 \epsilon [r_3]^2 - 4 \eta [1] \epsilon [r_1]^2 \epsilon \\ & [r_2] \epsilon [c_3]^2 - 4 \eta [1] \epsilon [r_1]^2 \epsilon [c_2]^2 \epsilon [r_3] - \eta [1] \\ & * \epsilon [r_1]^2 \epsilon [r_2]^2 \epsilon [c_3]^2 - \eta [1] \epsilon [r_1] \epsilon [r_2] \\ & ^2 \epsilon [r_3]^2 - \eta [1] \epsilon [r_1]^2 \epsilon [c_2]^2 \epsilon [c_3]^2 - 4 \eta \\ & [1] \epsilon [r_1] \epsilon [r_2] \epsilon [c_3]^2 - \eta [1] \epsilon [r_1]^2 \epsilon [c_2] \\ & ^2 \epsilon [r_3]^2 - 16 \eta [1] \epsilon [r_1] \epsilon [r_2] \epsilon [r_3] - \eta [1] * \\ & \epsilon [r_1]^2 \epsilon [r_2]^2 \epsilon [r_3]^2 - 4 \eta [1] \epsilon [r_1] \epsilon \\ & [r_2]^2 \epsilon [r_3] - \eta [1] \epsilon [c_1]^2 \epsilon [r_2]^2 \epsilon [r_3]^2 - 4 \eta \\ & [1] \epsilon [r_1] \epsilon [c_2]^2 \epsilon [r_3] - 4 \eta [1] \epsilon [r_1]^2 \epsilon \\ & [r_2] \epsilon [r_3]^2 - \eta [1] \epsilon [r_1] \epsilon [r_2]^2 \epsilon [c_3]^2 - 4 \eta [1] \end{aligned}$$

$$\begin{aligned}
& * \epsilon[c_1]^2 \epsilon[r_2]^2 \epsilon[r_3] - 4 \eta[1] \epsilon[r_1] \epsilon[r_2] * \\
& \epsilon[r_3]^2 - 4 \eta[1] \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[r_3] - \eta[1] * \\
& \epsilon[r_1] \epsilon[c_2]^2 \epsilon[c_3]^2 - \eta[1] \epsilon[c_1]^2 \epsilon[r_2] \\
& ^2 \epsilon[c_3]^2 - 4 \eta[1] \epsilon[c_1]^2 \epsilon[r_2] \epsilon[r_3]^2 - 16 \eta \\
& [1] \epsilon[c_1]^2 \epsilon[r_2] \epsilon[r_3] - 4 \eta[1] \epsilon[c_1]^2 \epsilon[r_2] \\
& \epsilon[c_3]^2 - \eta[1] \epsilon[c_1]^2 \epsilon[c_2]^2 \epsilon[r_3]^2 - 4 \eta \\
& [1] \epsilon[c_1]^2 \epsilon[c_2]^2 \epsilon[r_3] - \eta[1] \epsilon[c_1]^2 \epsilon[r_2] \\
& \epsilon[c_2]^2 \epsilon[c_3]^2 - 4 \eta[2] \epsilon[r_1]^2 \epsilon[r_2] \epsilon[r_3] - 4 \eta \\
& [2] \epsilon[r_1] \epsilon[c_2]^2 \epsilon[r_3]^2 - \eta[2] \epsilon[r_1]^2 \epsilon[r_2] \\
& \epsilon[c_3]^2 - 4 \eta[2] \epsilon[r_1]^2 \epsilon[c_2]^2 \epsilon[r_3] - 16 * \\
& \eta[3]^2 \epsilon[r_1]^2 \epsilon[r_3]) / (64 C 4 \epsilon[c_2]^2 \eta[2]^2 \epsilon[r_3] \\
& \epsilon[c_3]^2 C 64 \eta[1] C 64 \eta[2] C 64 \eta[3] C 64 \epsilon[r_1] C 64 \epsilon[r_2] \\
& C 64 \epsilon[r_3] C 16 \epsilon[c_1]^2 C 16 \epsilon[c_2]^2 C 16 \epsilon[c_3]^2 \\
& - 8 \eta[3]^2 \epsilon[c_1]^2 \epsilon[r_3] - 8 \eta[3]^2 \epsilon[r_2]^2 \epsilon[r_3] C 18 \epsilon[c_1] * \eta[1] \epsilon[c_3] * \eta[3] \epsilon[r_2]^2 C 4 \epsilon[c_2] \\
& ^2 \eta[2]^2 \epsilon[r_3]^2 C 4 \epsilon[c_3]^2 \eta[3]^2 \epsilon[r_2]^2 C 4 \eta \\
& [2]^2 \epsilon[c_1]^2 \epsilon[r_2]^2 - 16 \eta[2] \epsilon[r_2] \eta[1] C 8 \eta[3] * \\
& \epsilon[r_1]^2 \eta[1] \epsilon[r_3]^2 - 16 \eta[3] \epsilon[r_1] \epsilon[r_2]^2 * \\
& \eta[2] - 16 \eta[3] \epsilon[r_1] \epsilon[c_2]^2 \eta[2] - 4 \eta[3] \epsilon[c_1]^2 * \\
& \epsilon[r_2]^2 \eta[2] - 8 \eta[3]^2 \epsilon[c_1]^2 \epsilon[r_2] \epsilon[r_3] - 2 * \\
& \eta[3]^2 \epsilon[c_1]^2 \epsilon[c_2]^2 \epsilon[r_3] - 4 \eta[1] \epsilon[r_2]^2 * \\
& \epsilon[r_3]^2 \eta[2] - 4 \eta[1] \epsilon[r_2]^2 \epsilon[r_3]^2 \eta[3] - 16 \eta[1] \\
& * \epsilon[r_2]^2 \epsilon[r_3] \eta[2] C 8 \eta[3] \epsilon[r_2]^2 \epsilon[r_3] \eta[2] \\
& C 4 \epsilon[c_3]^2 \eta[3]^2 \epsilon[r_1] \epsilon[r_2]^2 C 16 \epsilon[c_2] \\
& ^2 \eta[2]^2 \epsilon[r_1] \epsilon[r_3] C 4 \epsilon[c_2]^2 \eta[2]^2 \epsilon[r_1] \\
& ^2 \epsilon[r_3] C 16 \epsilon[c_3]^2 \eta[3]^2 \epsilon[r_1] \epsilon[r_2] C 8 \eta[2] \\
& \eta[1] \epsilon[c_1]^2 \epsilon[r_2]^2 C 8 \eta[2] \eta[1] \epsilon[c_1]^2 * \\
& \epsilon[c_2]^2 - 4 \eta[2] \epsilon[c_1]^2 \epsilon[r_2] \eta[3] - 16 \eta[2] \epsilon[r_1] \\
& \epsilon[r_3]^2 \eta[3] - 16 \eta[2] \epsilon[r_1] \epsilon[r_3] \eta[3] - 16 \eta[2] \\
& \epsilon[r_1] \epsilon[c_3]^2 \eta[3] C \epsilon[c_2]^2 \eta[2]^2 \epsilon[r_1] \\
& ^2 \epsilon[c_3]^2 C 8 \eta[2] \epsilon[c_1]^2 \epsilon[r_2] \eta[1] - 4 \eta[2] * \\
& \epsilon[r_1] \epsilon[r_3]^2 \eta[1] - 16 \eta[2] \epsilon[r_1] \epsilon[r_3] \eta[1] \\
& C 4 \epsilon[c_3]^2 \eta[3]^2 \epsilon[c_1]^2 C 8 \eta[3] \epsilon[c_2]^2 \eta[2] * \\
& \epsilon[r_3]^2 - 4 \eta[2] \epsilon[c_1]^2 \epsilon[r_3]^2 \eta[1] - 16 \eta[2] * \\
& \epsilon[c_1]^2 \epsilon[r_3] \eta[1] - 4 \eta[2] \epsilon[c_1]^2 \epsilon[c_3]^2 \eta[1] \\
& C 8 \eta[3] \epsilon[r_2]^2 \eta[2] \epsilon[c_3]^2 C 8 \eta[3] \epsilon[r_2]^2 * \\
& \eta[2] \epsilon[r_3]^2 - 4 \eta[1] \epsilon[r_2]^2 \epsilon[c_3]^2 \eta[2] - 4 \eta[1] * \\
& \epsilon[r_2]^2 \epsilon[c_3]^2 \eta[3] - 4 \eta[1] \epsilon[c_2]^2 \epsilon[r_3]^2 *
\end{aligned}$$

$$\begin{aligned}
& \text{eta}[2]-4*\text{eta}[1]*\text{epsilon}[\text{c}2]^2*\text{epsilon}[\text{r}3]^2*\text{eta}[3]-16*\text{eta}[1]*\text{epsilon}[\text{c}2] \\
& ^2*\text{epsilon}[\text{r}3]*\text{eta}[2]-4*\text{eta}[1]*\text{epsilon}[\text{c}2]^2*\text{epsilon}[\text{c}3]^2*\text{eta}[2]-4*\text{eta}[1] \\
& *\text{epsilon}[\text{c}2]^2*\text{epsilon}[\text{c}3]^2*\text{eta}[3]-4*\text{eta}[3]*\text{epsilon}[\text{c}1]^2*\text{epsilon}[\text{c}2]^2* \\
& \text{eta}[2]-4*\text{eta}[2]*\text{epsilon}[\text{r}1]^2*\text{epsilon}[\text{r}3]^2*\text{eta}[1]-16*\text{eta}[2]*\text{epsilon}[\text{r}1] \\
& ^2*\text{epsilon}[\text{r}3]*\text{eta}[1]-4*\text{eta}[2]*\text{epsilon}[\text{r}1]^2*\text{epsilon}[\text{c}3]^2*\text{eta}[1]\text{C epsilon} \\
& [\text{c}1]^2*\text{eta}[1]^2*\text{epsilon}[\text{r}2]^2*\text{epsilon}[\text{r}3]^2\text{C epsilon}[\text{c}2]^2*\text{eta}[2]^2* \\
& \text{epsilon}[\text{r}1]^2*\text{epsilon}[\text{r}3]^2\text{C epsilon}[\text{c}3]^2*\text{eta}[3]^2*\text{epsilon}[\text{r}1]^2* \\
& \text{epsilon}[\text{r}2]^2\text{C }8*\text{eta}[3]*\text{eta}[1]*\text{epsilon}[\text{c}1]^2*\text{epsilon}[\text{r}3]^2\text{C }8*\text{eta}[3]*\text{eta} \\
& [1]*\text{epsilon}[\text{c}1]^2*\text{epsilon}[\text{c}3]^2\text{C }16*\text{eta}[3]^2*\text{epsilon}[\text{r}1]*\text{epsilon}[\text{r}2]* \\
& \text{epsilon}[\text{r}3]^2\text{C }4*\text{eta}[3]^2*\text{epsilon}[\text{c}1]^2*\text{epsilon}[\text{r}2]*\text{epsilon}[\text{r}3]^2\text{C eta}[3] \\
& ^2*\text{epsilon}[\text{c}1]^2*\text{epsilon}[\text{c}2]^2*\text{epsilon}[\text{r}3]^2-4*\text{eta}[3]*\text{epsilon}[\text{r}1]^2* \\
& \text{epsilon}[\text{r}2]^2*\text{eta}[2]-4*\text{eta}[3]*\text{epsilon}[\text{r}1]^2*\text{epsilon}[\text{c}2]^2*\text{eta}[2]-2*\text{eta}[3] \\
& ^2*\text{epsilon}[\text{c}1]^2*\text{epsilon}[\text{r}2]^2*\text{epsilon}[\text{r}3]-2*\text{eta}[3]^2*\text{epsilon}[\text{r}1]^2* \\
& \text{epsilon}[\text{r}2]^2*\text{epsilon}[\text{r}3]-8*\text{eta}[3]^2*\text{epsilon}[\text{r}1]^2*\text{epsilon}[\text{r}2]*\text{epsilon} \\
& [\text{r}3]\text{C }8*\text{eta}[3]*\text{epsilon}[\text{r}1]^2*\text{eta}[1]*\text{epsilon}[\text{c}3]^2\text{C }4*\text{eta}[2]^2*\text{epsilon}[\text{c}3] \\
& ^2*\text{epsilon}[\text{r}2]^2\text{C }4*\text{eta}[1]^2*\text{epsilon}[\text{r}1]^2*\text{epsilon}[\text{r}2]*\text{epsilon}[\text{c}3]^2-16* \\
& \text{eta}[3]*\text{epsilon}[\text{r}1]^2*\text{epsilon}[\text{r}2]*\text{eta}[1]-4*\text{eta}[3]*\text{epsilon}[\text{r}1]^2*\text{epsilon}[\text{c}2] \\
& ^2*\text{eta}[1]-4*\text{eta}[3]*\text{epsilon}[\text{r}1]*\text{epsilon}[\text{r}2]^2*\text{eta}[1]-16*\text{eta}[3]*\text{epsilon}[\text{r}1]* \\
& \text{epsilon}[\text{r}2]*\text{eta}[1]\text{C }72*\text{epsilon}[\text{c}1]*\text{eta}[1]*\text{epsilon}[\text{c}3]*\text{eta}[3]\text{C }72*\text{epsilon} \\
& [\text{c}2]*\text{eta}[2]*\text{epsilon}[\text{c}3]*\text{eta}[3]-4*\text{eta}[3]*\text{epsilon}[\text{r}1]^2*\text{epsilon}[\text{r}2]^2*\text{eta}[1] \\
& \text{C }72*\text{epsilon}[\text{c}1]*\text{eta}[1]*\text{epsilon}[\text{c}2]*\text{eta}[2]-2*\text{eta}[3]^2*\text{epsilon}[\text{r}1]^2* \\
& \text{epsilon}[\text{c}2]^2*\text{epsilon}[\text{r}3]\text{C }4*\text{eta}[3]^2*\text{epsilon}[\text{r}1]*\text{epsilon}[\text{r}2]^2*\text{epsilon} \\
& [\text{r}3]^2\text{C }8*\text{eta}[3]*\text{epsilon}[\text{c}2]^2*\text{eta}[2]*\text{epsilon}[\text{c}3]^2\text{C }16*\text{epsilon}[\text{c}1]^2*\text{eta} \\
& [1]^2\text{C }16*\text{eta}[3]^2*\text{epsilon}[\text{r}1]\text{C }16*\text{eta}[3]^2*\text{epsilon}[\text{r}2]-32*\text{eta}[3]^2* \\
& \text{epsilon}[\text{r}3]\text{C }4*\text{eta}[3]^2*\text{epsilon}[\text{r}1]^2\text{C }4*\text{eta}[3]^2*\text{epsilon}[\text{r}2]^2\text{C }4*\text{eta}[3] \\
& ^2*\text{epsilon}[\text{c}1]^2\text{C }4*\text{eta}[3]^2*\text{epsilon}[\text{c}2]^2\text{C }16*\text{eta}[2]^2*\text{epsilon}[\text{r}1]-32* \\
& \text{eta}[2]^2*\text{epsilon}[\text{r}2]\text{C }16*\text{eta}[2]^2*\text{epsilon}[\text{r}3]\text{C }4*\text{eta}[2]^2*\text{epsilon}[\text{r}1] \\
& ^2\text{C }4*\text{eta}[2]^2*\text{epsilon}[\text{r}3]^2\text{C }4*\text{eta}[2]^2*\text{epsilon}[\text{c}1]^2\text{C }4*\text{eta}[2]^2* \\
& \text{epsilon}[\text{c}3]^2\text{C }16*\text{epsilon}[\text{c}3]^2*\text{eta}[3]^2\text{C }4*\text{epsilon}[\text{r}1]^2*\text{epsilon}[\text{r}2] \\
& ^2\text{C }16*\text{epsilon}[\text{r}1]^2*\text{epsilon}[\text{r}2]\text{C }4*\text{epsilon}[\text{r}1]^2*\text{epsilon}[\text{c}2]^2\text{C }16* \\
& \text{epsilon}[\text{r}1]*\text{epsilon}[\text{r}2]^2\text{C }64*\text{epsilon}[\text{r}1]*\text{epsilon}[\text{r}2]\text{C }16*\text{epsilon}[\text{r}1]* \\
& \text{epsilon}[\text{c}2]^2\text{C }4*\text{epsilon}[\text{c}1]^2*\text{epsilon}[\text{r}2]^2\text{C }16*\text{epsilon}[\text{c}1]^2*\text{epsilon} \\
& [\text{r}2]\text{C }16*\text{epsilon}[\text{r}1]^2\text{C }16*\text{epsilon}[\text{r}2]^2\text{C }16*\text{epsilon}[\text{r}3]^2\text{C }16*\text{eta}[1] \\
& ^2\text{C }16*\text{eta}[2]^2\text{C }16*\text{eta}[3]^2-2*\text{eta}[2]*\text{epsilon}[\text{r}1]^2*\text{epsilon}[\text{r}2]^2* \\
& \text{epsilon}[\text{c}3]^2-8*\text{eta}[2]*\text{epsilon}[\text{r}1]*\text{epsilon}[\text{r}2]^2*\text{epsilon}[\text{r}3]^2-2*\text{eta}[2]* \\
& \text{epsilon}[\text{r}1]^2*\text{epsilon}[\text{c}2]^2*\text{epsilon}[\text{c}3]^2-8*\text{eta}[2]*\text{epsilon}[\text{r}1]*\text{epsilon} \\
& [\text{r}2]*\text{epsilon}[\text{c}3]^2-2*\text{eta}[2]*\text{epsilon}[\text{r}1]^2*\text{epsilon}[\text{c}2]^2*\text{epsilon}[\text{r}3]^2 \\
& -32*\text{eta}[2]*\text{epsilon}[\text{r}1]*\text{epsilon}[\text{r}2]*\text{epsilon}[\text{r}3]-2*\text{eta}[2]*\text{epsilon}[\text{r}1]^2*
\end{aligned}$$

$$\begin{aligned}
& \epsilon[r_2]^2 \epsilon[r_3]^2 - 32 \eta[2] \epsilon[r_1] \epsilon[r_2]^2 \epsilon[r_3] - 2 \eta[2] \epsilon[c_1]^2 \epsilon[r_2]^2 \epsilon[r_3]^2 - 32 \eta[2] \epsilon[r_1] \epsilon[c_2]^2 \epsilon[r_3] - 2 \eta[2] \epsilon[r_1]^2 \epsilon[r_2] \epsilon[r_3]^2 - 8 \eta[2] \epsilon[r_1] \epsilon[r_2]^2 \epsilon[c_3]^2 - 8 \eta[2] \epsilon[c_1]^2 \epsilon[r_2]^2 \epsilon[r_3] - 8 \eta[2] \epsilon[r_1] \epsilon[r_2] \epsilon[r_3]^2 - 8 \eta[2] \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[r_3] - 8 \eta[2] \epsilon[r_1] \epsilon[c_2]^2 \epsilon[c_3]^2 - 2 \eta[2] \epsilon[c_1]^2 \epsilon[r_2]^2 \epsilon[c_3]^2 - 2 \eta[2] \epsilon[c_1]^2 \epsilon[r_2] \epsilon[r_3]^2 - 8 \eta[2] \epsilon[c_1]^2 \epsilon[r_2] \epsilon[r_3] - 2 \eta[2] \epsilon[c_1]^2 \epsilon[r_2]^2 \epsilon[c_3]^2 - 2 \eta[2] \epsilon[c_1]^2 \epsilon[c_2]^2 \epsilon[r_3]^2 - 8 \eta[2] \epsilon[c_1]^2 \epsilon[c_2]^2 \epsilon[r_3] - 2 \eta[2] \epsilon[c_1]^2 \epsilon[c_2]^2 \epsilon[r_3]^2 - 8 \eta[3] \epsilon[r_1]^2 \epsilon[r_2] \epsilon[r_3] - 8 \eta[3] \epsilon[r_1] \epsilon[c_2]^2 \epsilon[r_3]^2 - 8 \eta[3] \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[c_3]^2 - 2 \eta[3] \epsilon[r_1]^2 \epsilon[c_2]^2 \epsilon[r_3] - 2 \eta[3] \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[c_3]^2 - 8 \eta[3] \epsilon[r_1] \epsilon[r_2]^2 \epsilon[r_3]^2 - 32 \eta[3] \epsilon[r_1] \epsilon[r_2]^2 \epsilon[c_3]^2 \\
& + 18 \epsilon[c_1] \eta[1] \epsilon[c_3] \eta[3] \epsilon[c_2] \\
& + 72 \epsilon[c_2] \eta[2] \epsilon[c_3] \eta[3] \epsilon[r_1] + 18 \epsilon[c_2] \eta[2] \epsilon[c_3] \eta[3] \epsilon[r_1]^2 + 18 \epsilon[c_2] \eta[2] \epsilon[c_3] \eta[3] \epsilon[r_1]^2 \epsilon[r_2] \epsilon[r_3]^2 + \eta[3] \epsilon[c_1]^2 + 2 \eta[2] \epsilon[r_1]^2 \epsilon[r_2] \epsilon[r_3]^2 + \eta[3] \epsilon[c_1] \eta[1] \epsilon[c_2] \eta[2] \epsilon[r_3]^2 + 18 \epsilon[c_1] \eta[1] \epsilon[c_2] \eta[2] \epsilon[r_3]^2 + 18 \epsilon[c_1] \eta[1] \epsilon[c_2] \eta[2] \epsilon[r_3]^2 + 72 \epsilon[c_1] \eta[1] \epsilon[c_3] \eta[3] \epsilon[r_2] + 2 \eta[2] \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[r_3]^2 + \eta[1] \epsilon[r_1] \epsilon[r_2] \epsilon[c_3]^2 + 2 \eta[3] \epsilon[r_1]^2 \epsilon[c_2]^2 \epsilon[r_3] - 2 \eta[3] \epsilon[r_1]^2 \epsilon[c_2]^2 \epsilon[r_3]^2 - 32 \eta[3] \epsilon[r_1] \epsilon[r_2] \epsilon[r_3] - 2 \eta[3] \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[r_3]^2 - 8 \eta[3] \epsilon[r_1] \epsilon[r_2]^2 \epsilon[r_3]^2 - 2 \eta[3] \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[r_3]^2 - 8 \eta[3] \epsilon[r_1] \epsilon[c_2]^2 \epsilon[r_3] - 8 \eta[3] \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[r_3]^2 - 2 \eta[3] \epsilon[c_1]^2 \epsilon[r_2]^2 \epsilon[r_3] - 32 \eta[3] \epsilon[r_1] \epsilon[r_2] \epsilon[r_3]^2 - 2 \eta[3] \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[r_3]^2 - 8 \eta[3] \epsilon[r_1] \epsilon[c_2]^2 \epsilon[c_3]^2 - 2 \eta[3] \epsilon[c_1]^2 \epsilon[r_2]^2 \epsilon[c_3]^2 - 8 \eta[3] \epsilon[c_1]^2 \epsilon[r_2] \epsilon[r_3]^2 - 8 \eta[3] \epsilon[c_1]^2 \epsilon[r_2] \epsilon[r_3] - 8 \eta[3] \epsilon[c_1]^2 \epsilon[r_2] \epsilon[c_3]^2 - 2 \eta[3] \epsilon[c_1]^2 \epsilon[r_2]^2 \epsilon[c_3]^2 - 2 \eta[3] \epsilon[c_1]^2 \epsilon[c_2]^2 \epsilon[r_3] - 2 \eta[3] \epsilon[c_1]^2 \epsilon[c_2]^2 \epsilon[r_3]^2 + 2 \eta[2]^2 \epsilon[r_1]^2
\end{aligned}$$

$$\begin{aligned}
& \epsilon[r_2]^2 \epsilon[c_3]^2 \eta[2]^2 \epsilon[r_1]^2 \epsilon[r_2]^2 \\
& \epsilon[r_3]^2 C^4 \eta[2]^2 \epsilon[r_1] \epsilon[r_2]^2 \epsilon[r_3]^2 C \eta[2]^2 \epsilon[c_1]^2 \epsilon[r_2]^2 \epsilon[r_3]^2 C^{16} \eta[2]^2 \epsilon[r_1] \\
& \epsilon[r_2]^2 \epsilon[r_3] - 2 \eta[2]^2 \epsilon[r_1]^2 \epsilon[r_2] \epsilon[r_3]^2 - 8 \eta[2]^2 \epsilon[r_1] \epsilon[r_2] \epsilon[r_3]^2 C^4 \epsilon[c_3]^2 \\
& \eta[3]^2 \epsilon[r_1]^2 \epsilon[r_2] C \epsilon[c_2]^2 \eta[2]^2 \epsilon[c_1]^2 \\
& \epsilon[c_3]^2 C^4 \epsilon[c_3]^2 \eta[3]^2 \epsilon[r_1] \epsilon[c_2]^2 - 8 \eta[2]^2 \epsilon[r_1] \epsilon[r_2] \epsilon[c_3]^2 C^4 \epsilon[c_1]^2 \epsilon[c_2]^2 \\
& C^{16} \epsilon[r_1] \epsilon[r_3]^2 C^{64} \epsilon[r_1] \epsilon[r_3] C^{16} \epsilon[r_1] \epsilon[c_3]^2 C^{16} \epsilon[r_2] \epsilon[r_3]^2 C^{64} \epsilon[r_2] \epsilon[r_3] C^{16} \epsilon[r_2] \epsilon[c_3]^2 C^4 \epsilon[r_1]^2 \epsilon[r_3]^2 C^{16} \\
& \epsilon[r_1]^2 \epsilon[r_3] C^4 \epsilon[r_1]^2 \epsilon[c_3]^2 C^4 \epsilon[r_2]^2 \epsilon[r_3]^2 C^{16} \epsilon[r_2]^2 \epsilon[r_3] C^4 \epsilon[r_2]^2 \epsilon[c_3]^2 C^4 \epsilon[c_1]^2 \epsilon[r_3]^2 C^{16} \epsilon[c_1]^2 \epsilon[r_3] C^4 \\
& \epsilon[c_1]^2 \epsilon[c_3]^2 C^4 \epsilon[c_2]^2 \epsilon[r_3]^2 C^{16} \epsilon[c_2]^2 \epsilon[r_3] C^4 \epsilon[c_2]^2 \epsilon[c_3]^2 - 32 \eta[1] \epsilon[r_1] C^{64} \eta[1] \epsilon[r_2] C^{64} \eta[1] \epsilon[r_3] - 32 \eta[1] \epsilon[r_1] \\
& ^2 C^{16} \eta[1] \epsilon[r_2]^2 C^{16} \eta[1] \epsilon[r_3]^2 - 32 \eta[1] \epsilon[c_1]^2 C^{16} \eta[1] \epsilon[c_2]^2 C^{16} \eta[1] \epsilon[c_3]^2 C^{64} \eta[2]^2 \epsilon[r_1] - 32 \eta[2]^2 \epsilon[r_2] C^{64} \eta[2]^2 \epsilon[r_3] C^{16} \eta[2]^2 \\
& \epsilon[r_1]^2 - 32 \eta[2]^2 \epsilon[r_2]^2 C^{16} \eta[2]^2 \epsilon[r_3]^2 C^{16} \eta[2]^2 \epsilon[c_1]^2 - 32 \eta[2]^2 \epsilon[c_2]^2 C^{16} \eta[2]^2 \epsilon[c_3]^2 C^{64} \eta[3]^2 \epsilon[r_1] C^{64} \eta[3]^2 \epsilon[r_2] - 32 \eta[3]^2 \epsilon[r_3] C^{16} \eta[3]^2 \\
& \epsilon[r_1]^2 C^{16} \eta[3]^2 \epsilon[r_2]^2 - 32 \eta[3]^2 \epsilon[r_3]^2 C^{16} \eta[3]^2 \epsilon[c_1]^2 C^{16} \eta[3]^2 \epsilon[c_2]^2 - 32 \eta[3]^2 \epsilon[c_3]^2 C^{16} \eta[1]^2 \epsilon[r_1]^2 C^{16} \eta[2]^2 \epsilon[r_2]^2 C^{16} \epsilon[c_2]^2 \eta[2]^2 \\
& ^2 C^{32} \eta[2]^2 \eta[1] C^{32} \eta[3]^2 \eta[1] C^{32} \eta[2]^2 \eta[3] C^{16} \eta[3]^2 \epsilon[r_3]^2 - 32 \eta[2]^2 \epsilon[r_1] \epsilon[r_2] \epsilon[r_3] C^4 \eta[2]^2 \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[r_3] C^4 \eta[2]^2 \epsilon[r_2]^2 \\
& \epsilon[r_1] \epsilon[c_3]^2 C \epsilon[c_3]^2 \eta[3]^2 \epsilon[c_1]^2 \epsilon[c_2]^2 C \epsilon[c_1]^2 \eta[1]^2 \epsilon[c_2]^2 \epsilon[r_3]^2 C \eta[3]^2 \epsilon[c_1]^2 \epsilon[r_2]^2 \epsilon[r_3]^2 C \epsilon[c_3]^2 \eta[3]^2 \\
& \epsilon[r_1]^2 \epsilon[c_2]^2 C^4 \epsilon[c_2]^2 \eta[2]^2 \epsilon[r_1] \epsilon[r_3]^2 C \epsilon[c_1]^2 \eta[1]^2 \epsilon[r_2]^2 \epsilon[c_3]^2 C \epsilon[c_1]^2 \eta[1]^2 \epsilon[r_2]^2 \epsilon[c_3]^2 C^4 \eta[2]^2 \epsilon[c_1]^2 \\
& \epsilon[r_2]^2 \epsilon[r_3] C \eta[2]^2 \epsilon[c_1]^2 \epsilon[r_2]^2 \epsilon[c_3]^2 C^4 \eta[3]^2 \epsilon[r_1] \epsilon[c_2]^2 \epsilon[r_3]^2 C^4 \epsilon[c_2]^2 \eta[2]^2 \epsilon[c_1]^2 \epsilon[r_3] C^8 \eta[1] \epsilon[r_1] \epsilon[r_2]
\end{aligned}$$

$$\begin{aligned}
&^2\eta[2]C^8\eta[1]\epsilon[r1]\epsilon[c2]^2\eta[2]C^4\eta[3]^2\epsilon[r3]^2\epsilon[r1]^2\epsilon[r2]C\eta[3]^2\epsilon[r1]^2\epsilon[c2]^2\epsilon[r3]^2C\eta[3]^2\epsilon[r2]^2\epsilon[r3]^2\epsilon[r1]^2-4\eta[3]\epsilon[r1]\epsilon[c2]^2\eta[1]C^4\eta[1]^2\epsilon[r1]^2\epsilon[r2]^2\epsilon[r3]C\eta[1]^2\epsilon[r1]^2\epsilon[r2]^2\epsilon[r3]^2C\eta[1]^2\epsilon[r1]^2\epsilon[r2]^2\epsilon[c3]^2C^{16}\eta[1]^2\epsilon[r1]^2\epsilon[r2]\epsilon[r3]C^4\eta[1]^2\epsilon[r1]^2\epsilon[r2]\epsilon[r3]^2C\eta[1]^2\epsilon[r1]^2\epsilon[c2]^2\epsilon[c3]^2C^4\eta[1]^2\epsilon[r1]^2\epsilon[c2]^2\epsilon[r3]C\eta[1]^2\epsilon[r1]^2\epsilon[c2]^2\epsilon[r3]^2C^8\eta[2]\epsilon[r1]\epsilon[r2]\epsilon[r3]\eta[3]C^2\eta[2]\epsilon[r1]^2\epsilon[r2]^2\epsilon[c3]^2\eta[1]C^8\eta[2]\epsilon[r1]^2\epsilon[r2]^2\epsilon[r3]\eta[1]C^2\eta[2]\eta[3]\epsilon[r1]^2\epsilon[c2]^2\epsilon[c3]^2C^2\eta[2]\epsilon[r1]^2\epsilon[r2]\epsilon[c3]^2\eta[3]C^2\eta[3]\epsilon[r1]^2\epsilon[r2]^2\epsilon[r3]\eta[1]C^8\eta[3]\epsilon[c1]^2\epsilon[r2]\epsilon[r3]\eta[1]C^2\eta[3]\epsilon[c1]^2\epsilon[c2]^2\epsilon[r3]\eta[1]C^2\eta[3]\epsilon[c1]^2\epsilon[r2]^2\epsilon[r3]\eta[1]C^2\eta[2]\epsilon[r1]^2\epsilon[r2]\epsilon[r3]\eta[3]C^2\eta[3]\epsilon[r1]\epsilon[c2]^2\epsilon[r3]\eta[1]C^2\eta[3]\epsilon[r1]\epsilon[r2]^2\epsilon[r3]\eta[1]C^8\eta[3]\epsilon[r1]\epsilon[r2]\epsilon[r3]\eta[1]C^2\eta[3]\epsilon[r1]^2\epsilon[r2]^2\epsilon[r3]\eta[2]C^2\eta[3]\epsilon[c1]^2\epsilon[c2]^2\epsilon[r3]\eta[2]C^2\eta[3]\epsilon[r1]^2\epsilon[c2]^2\epsilon[r3]\eta[2]C^8\eta[1]\epsilon[r1]\epsilon[r2]\epsilon[r3]^2\eta[3]C^2\eta[1]\epsilon[r1]\epsilon[r2]^2\epsilon[c3]^2\eta[3]C^8\eta[1]\epsilon[r1]\epsilon[r2]^2\epsilon[r3]\eta[2]C^8\eta[1]\epsilon[r1]\epsilon[r2]\epsilon[c3]^2\eta[3]C^2\eta[1]\epsilon[r1]\epsilon[r2]^2\epsilon[r3]^2\eta[2]C^2\eta[1]\epsilon[r1]\epsilon[r2]^2\epsilon[r3]^2\eta[3]C^2\eta[2]\epsilon[r3]^2\eta[1]\epsilon[r1]^2\epsilon[c2]^2-8\eta[1]^2\epsilon[r1]\epsilon[r2]^2-32\eta[1]^2\epsilon[r1]\epsilon[r2]C^{16}\epsilon[r1]^2\epsilon[r2]\epsilon[r3]-8\eta[1]^2\epsilon[r1]\epsilon[c2]^2-8\eta[1]^2\epsilon[r1]\epsilon[r3]^2-32\eta[1]^2\epsilon[r1]\epsilon[r3]-8\eta[1]^2\epsilon[r1]\epsilon[c3]^2C^8\eta[2]\epsilon[r3]^2\eta[1]C^{16}\epsilon[c2]^2\eta[2]^2\epsilon[r1]C^8\eta[3]\epsilon[r2]^2\eta[1]-16\eta[2]\epsilon[r1]^2\eta[1]C^4\epsilon[c2]^2\eta[2]^2\epsilon[c1]^2-16\eta[3]\epsilon[r1]^2\eta[1]C^4\epsilon[c1]^2\eta[1]^2\epsilon[r3]^2C^4\epsilon[r1]\epsilon[c2]^2\epsilon[r3]^2C^4\epsilon[r1]^2\epsilon[r2]\epsilon[c3]^2C^4\epsilon[r1]^2\epsilon[c2]^2\epsilon[r3]C\epsilon[r1]^2
\end{aligned}$$

$$\begin{aligned} & \epsilon[r_2]^2 \epsilon[c_3]^{2C} 4^* \epsilon[r_1] \epsilon[r_2]^2 \epsilon[r_3]^{2C} \\ & \epsilon[r_1]^2 \epsilon[c_2]^2 \epsilon[c_3]^{2C} 16^* \epsilon[r_1] \epsilon[r_2]^* \\ & \epsilon[c_3]^{2C} \epsilon[r_1]^2 \epsilon[c_2]^2 \epsilon[r_3]^{2C} 64^* \epsilon[r_1]^* \\ & \epsilon[r_2]^* \epsilon[r_3] C \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[r_3]^{2C} 16^* \\ & \epsilon[r_1]^* \epsilon[r_2]^2 \epsilon[r_3] C \epsilon[c_1]^2 \epsilon[r_2]^2^* \\ & \epsilon[r_3]^{2C} 16^* \epsilon[r_1]^* \epsilon[c_2]^2 \epsilon[r_3] C 4^* \epsilon[r_1]^2^* \\ & \epsilon[r_2]^* \epsilon[r_3]^{2C} 4^* \epsilon[r_1]^* \epsilon[r_2]^2 \epsilon[c_3]^{2C} 4^* \\ & \epsilon[c_1]^2 \epsilon[r_2]^2 \epsilon[r_3] C 16^* \epsilon[r_1]^* \epsilon[r_2]^* \\ & \epsilon[r_3]^{2C} 4^* \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[r_3] C 4^* \epsilon[r_1]^* \\ & \epsilon[c_2]^2 \epsilon[c_3]^{2C} \epsilon[c_1]^2 \epsilon[r_2]^2 \epsilon[c_3] \\ & ^{2C} 4^* \epsilon[c_1]^2 \epsilon[r_2]^* \epsilon[r_3]^{2C} 16^* \epsilon[c_1]^2 \epsilon[r_2]^* \epsilon[r_3] C 4^* \epsilon[c_1]^2 \epsilon[r_2]^* \epsilon[c_3]^{2C} \epsilon[c_1]^2 \epsilon[c_2]^2 \epsilon[r_3]^{2C} 4^* \epsilon[c_1]^2 \epsilon[c_2]^2 \epsilon[r_3] C \epsilon[c_1]^2 \epsilon[c_2]^2 \epsilon[c_3]^{2-8*\eta[1]} \epsilon[r_1]^2^* \epsilon[r_2]^{2-32*\eta[1]} \epsilon[r_1]^2 \epsilon[r_2]-8*\eta[1] \epsilon[r_1]^2^* \epsilon[c_2]^{2-8*\eta[1]} \epsilon[r_1]^* \epsilon[r_2]^{2-32*\eta[1]} \epsilon[r_1]^* \epsilon[r_2]-8*\eta[1] \epsilon[r_1]^* \epsilon[c_2]^{2-8*\eta[1]} \epsilon[c_1]^2^* \epsilon[r_2]^{2-32*\eta[1]} \epsilon[c_1]^2 \epsilon[r_2]-8*\eta[1] \epsilon[c_1]^2 \epsilon[c_2]^{2-8*\eta[1]} \epsilon[r_1]^* \epsilon[r_3]^{2C} 2^* \eta[2]^* \epsilon[r_3]^{2^*} \eta[1]^* \epsilon[c_1]^2 \epsilon[c_2]^{2C} 2^* \eta[2]^* \epsilon[c_3]^{2^*} \eta[1]^* \epsilon[r_1]^2 \epsilon[c_2]^{2C} 2^* \eta[2]^* \epsilon[c_3]^{2^*} \eta[1]^* \epsilon[c_1]^2 \epsilon[c_2]^{2C} 2^* \eta[1]^* \epsilon[r_1]^* \epsilon[c_2]^{2^*} \epsilon[r_3]^{2^*} \eta[2] C 2^* \eta[1]^* \epsilon[r_1]^* \epsilon[c_2]^{2^*} \epsilon[r_3]^{2^*} \eta[3] C 8^* \eta[2]^* \epsilon[r_3]^* \eta[1]^* \epsilon[r_1]^2 \epsilon[c_2]^{2C} 8^* \eta[2]^* \epsilon[r_3]^* \eta[1]^* \epsilon[c_1]^2 \epsilon[r_2]^{2C} 8^* \eta[2]^* \epsilon[r_3]^* \eta[1]^* \epsilon[c_1]^2 \epsilon[c_2]^{2C} 2^* \eta[1]^* \epsilon[r_1]^* \epsilon[c_2]^{2^*} \epsilon[c_3]^{2^*} \eta[2] C 2^* \eta[1]^* \epsilon[r_1]^* \epsilon[c_2]^{2^*} \epsilon[c_3]^{2^*} \eta[3] C 2^* \eta[3]^* \epsilon[r_1]^2 \epsilon[c_2]^{2^*} \epsilon[r_3]^* \eta[1] C 8^* \eta[3]^* \epsilon[r_1]^2 \epsilon[r_2]^* \epsilon[r_3]^* \eta[1] C 2^* \eta[2]^* \epsilon[c_1]^2 \epsilon[r_2]^* \epsilon[r_3]^{2^*} \eta[1] C 2^* \eta[2]^* \epsilon[r_1]^2 \epsilon[r_2]^* \epsilon[r_3]^{2^*} \eta[1] C 2^* \eta[2]^* \epsilon[r_1]^* \epsilon[r_2]^* \epsilon[c_3]^{2^*} \eta[1] C 8^* \eta[2]^* \epsilon[r_1]^* \epsilon[r_2]^* \epsilon[r_3]^* \eta[1] C 8^* \eta[3]^* \eta[2]^* \epsilon[r_1]^* \epsilon[c_2]^{2^*} \epsilon[r_3]^{2C} 2^* \eta[3]^* \eta[2]^* \epsilon[r_1]\end{aligned}$$

$$\begin{aligned}
&^2\epsilon[r_2]^2\epsilon[c_3]^2-4\eta[2]\epsilon[r_2]\epsilon[c_3]^2\eta[1] \\
&-32\eta[1]^2\epsilon[r_1]C_{16}\eta[1]^2\epsilon[r_2]C_{16}\eta[1]^2\epsilon[r_3]C_4\eta[1]^2\epsilon[r_2]^2C_4\eta[1]^2\epsilon[r_3]^2C_4\eta[1]^2\epsilon[c_2]^2C_4\eta[1]^2\epsilon[c_3]^2C_4\eta[2]^2\epsilon[r_3]^2\epsilon[r_2]^2-32\eta[1]\epsilon[r_1]\epsilon[r_3]-8\eta[1]\epsilon[r_1]\epsilon[c_3]^2C_{16}\eta[1]\epsilon[r_2]\epsilon[r_3]^2C_{64}\eta[1]\epsilon[r_2]\epsilon[r_3]C_{16}\eta[1]\epsilon[r_2]\epsilon[c_3]^2-8\eta[1]\epsilon[r_1]^2\epsilon[r_3]^2-32\eta[1]\epsilon[r_1]^2\epsilon[r_3]-8\eta[1]\epsilon[r_1]^2\epsilon[c_3]^2C_4\eta[1]\epsilon[r_2]^2\epsilon[r_3]^2C_{16}\eta[1]\epsilon[r_2]^2\epsilon[r_3]C_4\eta[1]\epsilon[r_2]^2\epsilon[c_3]^2-8\eta[1]\epsilon[c_1]^2\epsilon[r_3]^2-32\eta[1]\epsilon[c_1]^2\epsilon[r_3]-8\eta[1]\epsilon[c_1]^2\epsilon[c_3]^2C_4\eta[1]\epsilon[c_2]^2\epsilon[r_3]^2C_{16}\eta[1]\epsilon[c_2]^2\epsilon[r_3]C_4\eta[1]\epsilon[c_2]^2\epsilon[c_3]^2-8\eta[2]\epsilon[r_1]^2\epsilon[r_2]^2-8\eta[2]\epsilon[r_1]^2\epsilon[r_2]-8\eta[2]\epsilon[r_1]^2\epsilon[c_2]^2-32\eta[2]\epsilon[r_1]\epsilon[r_2]^2-32\eta[2]\epsilon[r_1]\epsilon[r_2]-32\eta[2]\epsilon[r_1]\epsilon[c_2]^2-8\eta[2]\epsilon[c_1]^2\epsilon[r_2]^2-8\eta[2]\epsilon[c_1]^2\epsilon[r_2]-8\eta[2]\epsilon[c_1]^2\epsilon[c_2]^2C_{16}\eta[2]\epsilon[r_1]\epsilon[r_3]^2C_{64}\eta[2]\epsilon[r_1]\epsilon[r_3]C_{16}\eta[2]\epsilon[r_1]\epsilon[c_3]^2-8\eta[2]\epsilon[r_2]\epsilon[r_3]^2-32\eta[2]\epsilon[r_2]\epsilon[r_3]-8\eta[2]\epsilon[r_2]\epsilon[c_3]^2C_4\eta[2]\epsilon[r_1]^2\epsilon[r_3]^2C_{16}\eta[2]\epsilon[r_1]^2\epsilon[r_3]C_4\eta[2]\epsilon[r_1]^2\epsilon[c_3]^2-8\eta[2]\epsilon[r_2]^2\epsilon[r_3]^2-32\eta[2]\epsilon[r_2]^2\epsilon[r_3]-8\eta[2]\epsilon[r_2]^2\epsilon[c_3]^2C_4\eta[2]\epsilon[c_1]^2\epsilon[r_3]^2C_{16}\eta[2]\epsilon[c_1]^2\epsilon[r_3]C_4\eta[2]\epsilon[c_1]^2\epsilon[c_3]^2-8\eta[2]\epsilon[c_2]^2\epsilon[r_3]^2-32\eta[2]\epsilon[c_2]^2\epsilon[r_3]-8\eta[2]\epsilon[c_2]^2\epsilon[c_3]^2C_4\eta[3]\epsilon[r_1]^2\epsilon[r_2]^2C_{16}\eta[3]\epsilon[r_1]^2\epsilon[r_2]C_{16}\eta[3]\epsilon[r_1]\epsilon[r_2]^2C_{64}\eta[3]\epsilon[r_1]\epsilon[r_2]C_{16}\eta[3]\epsilon[r_1]\epsilon[c_2]^2C_4\eta[3]\epsilon[c_1]^2\epsilon[r_2]^2C_{16}\eta[3]\epsilon[c_1]^2\epsilon[r_2]C_4\eta[3]\epsilon[c_1]^2\epsilon[c_2]^2-32\eta[3]\epsilon[r_1]\epsilon[r_3]^2-32\eta[3]\epsilon[r_1]\epsilon[r_3]-32\eta[3]\epsilon[r_1]\epsilon[c_3]^2-32\eta[3]\epsilon[r_2]\epsilon[r_3]^2-32\eta[3]\epsilon[r_2]\epsilon[r_3]-32\eta[3]\epsilon[r_2]\epsilon[c_3]^2-8\eta[3]\epsilon[r_1]^2\epsilon[r_3]^2-8\eta[3]\epsilon[r_1]^2\epsilon[c_3]^2-8\eta[3]\epsilon[r_2]^2\epsilon[r_3]^2-8\eta[3]\epsilon[r_2]^2\epsilon[r_3]-8\eta[3]\epsilon[r_2]^2\epsilon[c_3]^2-8\eta[3]\epsilon[c_1]^2\epsilon[r_3]^2-8\eta[3]\epsilon[c_1]^2\epsilon[r_3]
\end{aligned}$$

$$\begin{aligned}
& [c1]^2 \epsilon[r3] - 8 \eta[3] \epsilon[c1]^2 \epsilon[c3]^2 - 8 \eta[3] \epsilon[c2]^2 \epsilon[r3]^2 - 8 \eta[3] \epsilon[c2]^2 \epsilon[c3]^2 C^4 \eta[3]^2 \epsilon[c2]^2 \epsilon[r3]^2 C^4 \eta[3]^2 \epsilon[r1]^2 \epsilon[r3]^2 C^4 \eta[3]^2 \epsilon[r2]^2 \epsilon[r3]^2 C^4 \eta[3]^2 \epsilon[c1]^2 \epsilon[r3]^2 C^{16} \eta[3]^2 \epsilon[r1]^* \epsilon[r3]^2 C^{16} \eta[3]^2 \epsilon[r2]^* \epsilon[r3]^2 - 16 \eta[3] \epsilon[r3]^* \eta[1] C^8 \eta[3]^* \eta[2]^* \epsilon[r1]^* \epsilon[r2]^2 \epsilon[r3]^2 C^2 \eta[3]^* \eta[2]^* \epsilon[r1]^2 \epsilon[c2]^2 \epsilon[r3]^2 C^2 \eta[3]^* \eta[2]^* \epsilon[r1]^2 \epsilon[r2]^2 \epsilon[r3]^2 C^2 \eta[3]^* \eta[2]^* \epsilon[c1]^2 \epsilon[r2]^2 \epsilon[r3]^2 C^8 \eta[3]^* \epsilon[r2]^* \eta[1]^* \epsilon[c1]^2 \epsilon[c3]^2 C^2 \eta[3]^* \epsilon[r2]^2 \eta[1]^* \epsilon[c1]^2 \epsilon[c3]^2 C^2 \eta[3]^* \epsilon[c1]^2 \eta[1]^* \epsilon[c2]^2 \epsilon[r3]^2 C^2 \eta[3]^* \epsilon[c1]^2 \eta[1]^* \epsilon[c2]^2 \epsilon[c3]^2 C^8 \eta[2]^* \epsilon[r1]^* \epsilon[r2]^* \epsilon[r3]^2 \eta[3] C^2 \eta[2]^* \epsilon[c1]^2 \epsilon[r2]^* \epsilon[r3]^* \eta[3] C^2 \eta[2]^* \epsilon[c1]^2 \epsilon[r2]^* \epsilon[c3]^2 \eta[3] C^8 \eta[3]^* \epsilon[r1]^2 \eta[1]^* \epsilon[r2]^* \epsilon[r3]^2 C^8 \eta[3]^* \epsilon[c1]^2 \eta[1]^* \epsilon[r2]^* \epsilon[r3]^2 C^8 \eta[3]^* \eta[2]^* \epsilon[r1]^* \epsilon[c2]^2 \epsilon[c3]^2 C^2 \eta[3]^* \eta[2]^* \epsilon[c1]^2 \epsilon[r2]^2 \epsilon[c3]^2 C^2 \eta[3]^* \eta[2]^* \epsilon[c1]^2 \epsilon[c2]^2 \epsilon[c3]^2 C^8 \eta[3]^* \epsilon[r2]^2 \eta[2]^* \epsilon[r1]^* \epsilon[c3]^2 C^2 \eta[3]^* \epsilon[r1]^2 \epsilon[r2]^2 \eta[1]^* \epsilon[c3]^2 C^8 \eta[3]^* \epsilon[r1]^2 \epsilon[r2]^* \eta[1]^* \epsilon[c3]^2 C^2 \eta[3]^* \epsilon[r1]^2 \epsilon[c2]^2 \eta[1]^* \epsilon[r3]^2 C^2 \eta[3]^* \epsilon[c2]^2 \eta[1]^* \epsilon[r3]^2 C^8 \eta[3]^* \epsilon[r1]^* \epsilon[r2]^2 \epsilon[r3]^* \eta[2] C^8 \eta[3]^* \epsilon[r1]^* \epsilon[c2]^2 \epsilon[r3]^* \eta[2] C^2 \eta[3]^* \epsilon[c1]^2 \epsilon[r2]^2 \epsilon[r3]^* \eta[2] - 16 \eta[3]^* \epsilon[r1]^* \eta[1] - 16 \eta[3]^* \epsilon[c1]^2 \eta[1] C^8 \eta[3]^* \epsilon[c2]^2 \eta[1] - 16 \eta[2]^* \epsilon[r1]^* \eta[1] C^4 \epsilon[c2]^2 \eta[2]^2 \epsilon[r1]^2 C^4 \epsilon[c1]^2 \eta[1]^2 \epsilon[c2]^2 C^{16} \epsilon[c3]^2 \eta[3]^2 \epsilon[r1] C^4 \epsilon[c1]^2 \eta[1]^2 \epsilon[c3]^2 C^2 \eta[1]^2 \epsilon[c2]^2 \epsilon[c3]^2 C^2 \eta[1]^2 \epsilon[c2]^2 \epsilon[r3]^2 C^4 \eta[1]^2 \epsilon[c2]^2 \epsilon[r3] C^2 \eta[1]^2 \epsilon[r2]^2 \epsilon[r3]^2 C^{16} \eta[1]^2 \epsilon[r2]^* \epsilon[r3] C^4 \eta[1]^2 \epsilon[r2]^* \epsilon[c3]^2 C^4 \eta[1]^2 \epsilon[r2]^* \epsilon[r3]
\end{aligned}$$

[illegible]

$$\begin{aligned}
& \epsilon[r_3]^2 \eta[3] C 8 \eta[1] \epsilon[r_1] \epsilon[c_3]^2 \eta[3] C 4 \epsilon[c_3]^2 \eta[3]^2 \epsilon[c_1]^2 \epsilon[r_2] C 8 \eta[2] \eta[1] \epsilon[r_1]^2 \epsilon[r_2]^2 C \epsilon[c_3]^2 \eta[3]^2 \epsilon[c_1]^2 \epsilon[r_2]^2 C 4 \epsilon[c_2]^2 \eta[2]^2 \epsilon[r_1] \epsilon[c_3]^2 - 2 \eta[2]^2 \epsilon[c_1]^2 \epsilon[r_2] \epsilon[r_3]^2 - 8 \eta[2]^2 \epsilon[c_1]^2 \epsilon[r_2] \epsilon[r_3] - 2 \eta[2]^2 \epsilon[c_1]^2 \epsilon[r_2] \epsilon[c_3]^2 C 4 \epsilon[c_1]^2 \eta[1]^2 \epsilon[c_2]^2 \epsilon[c_3]^2 - 4 \eta[2] \epsilon[r_1]^2 \epsilon[r_2] \eta[3] - 16 \eta[2] \epsilon[r_1] \epsilon[r_2] \eta[3] - 2 \eta[3] \epsilon[r_1]^2 \epsilon[c_2]^2 \epsilon[c_3]^2 - 32 \eta[1] \epsilon[r_1]^2 \epsilon[r_2] \epsilon[r_3] C 16 \eta[3]^2 \epsilon[r_1] \epsilon[r_2] C 4 \eta[3]^2 \epsilon[r_1] \epsilon[c_2]^2 C \eta[3]^2 \epsilon[c_1]^2 \epsilon[r_2]^2 C 4 \eta[3]^2 \epsilon[c_1]^2 \epsilon[r_2] C \eta[3]^2 \epsilon[c_1]^2 \epsilon[c_2]^2 - 32 \eta[3]^2 \epsilon[r_1] \epsilon[r_3] - 32 \eta[3]^2 \epsilon[r_2] \epsilon[r_3] C \eta[2]^2 \epsilon[r_1]^2 \epsilon[r_3]^2 C 4 \eta[2]^2 \epsilon[r_1]^2 \epsilon[r_3] C \eta[2]^2 \epsilon[c_1]^2 \epsilon[r_3]^2 C 4 \eta[2]^2 \epsilon[c_1]^2 \epsilon[r_3] C \eta[2]^2 \epsilon[c_1]^2 \epsilon[c_3]^2 C \eta[3]^2 \epsilon[r_1]^2 \epsilon[r_2]^2 - 8 \eta[2]^2 \epsilon[r_1]^2 \epsilon[r_2] - 32 \eta[2]^2 \epsilon[r_1] \epsilon[r_2] - 8 \eta[2]^2 \epsilon[c_1]^2 \epsilon[r_2] C 4 \eta[2]^2 \epsilon[r_1] \epsilon[r_3]^2 C 16 \eta[2]^2 \epsilon[r_1] \epsilon[r_3] C 4 \eta[2]^2 \epsilon[r_1] \epsilon[c_3]^2 - 8 \eta[2]^2 \epsilon[r_2] \epsilon[r_3]^2 - 32 \eta[2]^2 \epsilon[r_2] \epsilon[r_3] - 8 \eta[2]^2 \epsilon[r_2] \epsilon[c_3]^2 C 16 \epsilon[c_1]^2 \eta[1]^2 \epsilon[r_2] - 16 \eta[3] \epsilon[r_2]^2 \eta[2] - 16 \eta[3] \epsilon[c_2]^2 \eta[2] C 4 \eta[1]^2 \epsilon[r_1]^2 \epsilon[r_3]^2 C 4 \epsilon[c_3]^2 \eta[3]^2 \epsilon[r_1]^2 C 4 \eta[1]^2 \epsilon[r_1]^2 \epsilon[c_3]^2 C 16 \eta[1]^2 \epsilon[r_1]^2 \epsilon[r_3] C 16 \eta[2]^2 \epsilon[r_1] \epsilon[r_2]^2 C 4 \epsilon[c_1]^2 \eta[1]^2 \epsilon[r_2]^2 C 16 \eta[2]^2 \epsilon[r_3] \epsilon[r_2]^2 C 32 \eta[2] \epsilon[r_1] \eta[3] - 8 \eta[3]^2 \epsilon[c_2]^2 \epsilon[r_3] C 8 \eta[2] \epsilon[c_1]^2 \eta[3] - 16 \eta[2] \epsilon[c_3]^2 \eta[3] C 32 \eta[3] \epsilon[r_2] \eta[1] - 2 \eta[1] \epsilon[r_1] \epsilon[c_2]^2 \epsilon[r_3]^2 - 8 \eta[1] \epsilon[r_1]^2 \epsilon[r_2] \epsilon[c_3]^2 - 8 \eta[1] \epsilon[r_1]^2 \epsilon[c_2]^2 \epsilon[r_3] - 2 \eta[1] \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[c_3]^2 - 2 \eta[1] \epsilon[r_1] \epsilon[r_2]^2 \epsilon[r_3]^2 - 2 \eta[1] \epsilon[r_1]^2 \epsilon[c_2]^2 \epsilon[r_3]^2 - 8 \eta[1] \epsilon[r_1] \epsilon[r_2] \epsilon[c_3]^2 - 2 \eta[1] \epsilon[r_1]^2 \epsilon[r_2] \epsilon[r_3]^2 - 8 \eta[1] \epsilon[r_1] \epsilon[r_2]^2 \epsilon[r_3] - 2 \eta[1]^2 \epsilon[r_3]^2 - 8 \eta[1] \epsilon[r_1] \epsilon[r_2]^2 \epsilon[r_3] - 2 \eta[1]^2 \epsilon[r_3]^2
\end{aligned}$$

$$\begin{aligned}
& \epsilon[c1]^2 \epsilon[r2]^2 \epsilon[r3]^2 - 8 \eta[1] \epsilon[r1] \epsilon[c2]^2 \epsilon[r3] - 8 \eta[1] \epsilon[r1]^2 \epsilon[r2] \epsilon[r3]^2 - 2 \eta[1] \epsilon[r1] \epsilon[r2]^2 \epsilon[c3]^2 - 8 \eta[1] \epsilon[c1]^2 \epsilon[r2]^2 \epsilon[r3] - 8 \eta[1] \epsilon[r1]^2 \epsilon[r2]^2 \epsilon[r3]^2 - 2 \eta[1] \epsilon[r1] \epsilon[c2]^2 \epsilon[c3]^2 - 2 \eta[1] \epsilon[c1]^2 \epsilon[r2]^2 \epsilon[r3]^2 - 8 \eta[1] \epsilon[c1]^2 \epsilon[r2] \epsilon[r3]^2 - 32 \eta[1] \epsilon[c1]^2 \epsilon[r2] \epsilon[r3] - 8 \eta[1] \epsilon[c1]^2 \epsilon[r2] \epsilon[c3]^2 - 2 \eta[1] \epsilon[c1]^2 \epsilon[c2]^2 \epsilon[r3]^2 - 8 \eta[1] \epsilon[c1]^2 \epsilon[c2]^2 \epsilon[r3] - 2 \eta[1] \epsilon[c1]^2 \epsilon[c2]^2 \epsilon[r3]^2 - 8 \eta[2] \epsilon[r1]^2 \epsilon[r2] \epsilon[r3] - 8 \eta[2] \epsilon[r1] \epsilon[c2]^2 \epsilon[r3]^2 - 2 \eta[2] \epsilon[r1]^2 \epsilon[r2] \epsilon[c3]^2 - 8 \eta[2] \epsilon[r1]^2 \epsilon[c2]^2 \epsilon[r3] - 8 \eta[3] \epsilon[r1]^2 \epsilon[r3]^2 \\
& \epsilon[r3] - 8 \eta[3] \epsilon[r1]^2 \epsilon[r3]^2)
\end{aligned}$$

eps_c

$$\begin{aligned}
& \left(9 \left(16 \epsilon_{cl} \eta_1 + 16 \epsilon_{c2} \eta_2 + 16 \epsilon_{c3} \eta_3 + 16 \epsilon_{cl} \eta_1 \epsilon_{r2} + 16 \epsilon_{cl} \eta_1 \epsilon_{r3} + 4 \epsilon_{cl} \eta_1 \epsilon_{r2}^2 \right. \right. \\
& + 4 \epsilon_{cl} \eta_1 \epsilon_{r3}^2 + 4 \epsilon_{cl} \eta_1 \epsilon_{c2}^2 + 4 \epsilon_{cl} \eta_1 \epsilon_{c3}^2 + 16 \epsilon_{c2} \eta_2 \epsilon_{r1} + 16 \epsilon_{c2} \eta_2 \epsilon_{r3} + 4 \epsilon_{c2} \eta_2 \epsilon_{r1}^2 \\
& + 4 \epsilon_{c2} \eta_2 \epsilon_{r3}^2 + 4 \epsilon_{c2} \eta_2 \epsilon_{cl}^2 + 4 \epsilon_{c2} \eta_2 \epsilon_{c3}^2 + 16 \epsilon_{c3} \eta_3 \epsilon_{r1} + 16 \epsilon_{c3} \eta_3 \epsilon_{r2} \\
& + 4 \epsilon_{c3} \eta_3 \epsilon_{r1}^2 + 4 \epsilon_{c3} \eta_3 \epsilon_{r2}^2 + 4 \epsilon_{c3} \eta_3 \epsilon_{cl}^2 + 4 \epsilon_{c3} \eta_3 \epsilon_{c2}^2 + 4 \epsilon_{cl} \eta_1 \epsilon_{r2} \epsilon_{r3}^2 \\
& + 16 \epsilon_{cl} \eta_1 \epsilon_{r2} \epsilon_{r3} + 4 \epsilon_{cl} \eta_1 \epsilon_{r2} \epsilon_{c3}^2 + \epsilon_{cl} \eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 + 4 \epsilon_{cl} \eta_1 \epsilon_{r2}^2 \epsilon_{r3} + \epsilon_{cl} \eta_1 \epsilon_{r2}^2 \epsilon_{c3}^2 \\
& + \epsilon_{cl} \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \epsilon_{cl} \eta_1 \epsilon_{c2}^2 \epsilon_{r3} + \epsilon_{cl} \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 + 4 \epsilon_{c2} \eta_2 \epsilon_{r1} \epsilon_{r3}^2 + 16 \epsilon_{c2} \eta_2 \epsilon_{r1} \epsilon_{r3} \\
& + 4 \epsilon_{c2} \eta_2 \epsilon_{r1} \epsilon_{c3}^2 + \epsilon_{c2} \eta_2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 4 \epsilon_{c2} \eta_2 \epsilon_{r1}^2 \epsilon_{r3} + \epsilon_{c2} \eta_2 \epsilon_{r1}^2 \epsilon_{c3}^2 + \epsilon_{c2} \eta_2 \epsilon_{cl}^2 \epsilon_{r3}^2 \\
& + 4 \epsilon_{c2} \eta_2 \epsilon_{cl}^2 \epsilon_{r3} + \epsilon_{c2} \eta_2 \epsilon_{cl}^2 \epsilon_{c3}^2 + \epsilon_{c3} \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 + 4 \epsilon_{c3} \eta_3 \epsilon_{r1}^2 \epsilon_{r2} + \epsilon_{c3} \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 + 4 \epsilon_{c3} \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \\
& + 4 \epsilon_{c3} \eta_3 \epsilon_{r1} \epsilon_{r2}^2 + 16 \epsilon_{c3} \eta_3 \epsilon_{r1} \epsilon_{r2} + 4 \epsilon_{c3} \eta_3 \epsilon_{r1} \epsilon_{c2}^2 + \epsilon_{c3} \eta_3 \epsilon_{cl}^2 \epsilon_{r2}^2 + 4 \epsilon_{c3} \eta_3 \epsilon_{cl}^2 \epsilon_{r2} + \epsilon_{c3} \eta_3 \epsilon_{cl}^2 \epsilon_{c2}^2 \Big) \Big/ \Big(64 + 4 \epsilon_{c2}^2 \eta_2^2 \epsilon_{c3}^2 + 64 \eta_1 + 64 \eta_2 + 64 \eta_3 + 64 \epsilon_{r1} \\
& + 64 \epsilon_{r2} + 64 \epsilon_{r3} + 16 \epsilon_{cl}^2 + 16 \epsilon_{c2}^2 + 16 \epsilon_{c3}^2 - 8 \eta_3^2 \epsilon_{cl}^2 \epsilon_{r3} - 8 \eta_3^2 \epsilon_{r2}^2 \epsilon_{r3} \\
& + 18 \epsilon_{cl} \eta_1 \epsilon_{c3} \eta_3 \epsilon_{r2}^2 + 4 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r3}^2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r2}^2 + 4 \eta_2^2 \epsilon_{cl}^2 \epsilon_{r2}^2 - 16 \eta_2 \epsilon_{r2} \eta_1 \\
& + 8 \eta_3 \epsilon_{r1}^2 \eta_1 \epsilon_{r3}^2 - 16 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \eta_2 - 16 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \eta_2 - 4 \eta_3 \epsilon_{cl}^2 \epsilon_{r2}^2 \eta_2 - 8 \eta_3^2 \epsilon_{cl}^2 \epsilon_{r2} \epsilon_{r3} - 2 \eta_3^2 \epsilon_{cl}^2 \epsilon_{c2}^2 \epsilon_{r3} - 4 \eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_2 - 4 \eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_3 - 16 \eta_1 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 \\
& + 8 \eta_3 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 + 16 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1} \epsilon_{r3} + 4 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3} + 16 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1} \epsilon_{r2} + 8 \eta_2 \eta_1 \epsilon_{cl}^2 \epsilon_{r2}^2 + 8 \eta_2 \eta_1 \epsilon_{cl}^2 \epsilon_{c2}^2 - 4 \eta_2 \epsilon_{cl}^2 \epsilon_{r2} \eta_3 - 16 \eta_2 \epsilon_{r1} \epsilon_{r3}^2 \eta_3 \Big)
\end{aligned} \tag{2.11}$$

$$\begin{aligned}
& -16 \eta_2 \epsilon_{r1} \epsilon_{r3} \eta_3 - 16 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 \eta_3 + \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{c3}^2 + 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \eta_1 - 4 \eta_2 \epsilon_{r1} \\
& \epsilon_{r3}^2 \eta_1 - 16 \eta_2 \epsilon_{r1} \epsilon_{r3} \eta_1 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{c1}^2 + 8 \eta_3 \epsilon_{c2}^2 \eta_2 \epsilon_{r3}^2 - 4 \eta_2 \epsilon_{c1}^2 \epsilon_{r3}^2 \eta_1 - 16 \eta_2 \\
& \epsilon_{c1}^2 \epsilon_{r3} \eta_1 - 4 \eta_2 \epsilon_{c1}^2 \epsilon_{c3}^2 \eta_1 + 8 \eta_3 \epsilon_{r2}^2 \eta_2 \epsilon_{c3}^2 + 8 \eta_3 \epsilon_{r2}^2 \eta_2 \epsilon_{r3}^2 - 4 \eta_1 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_2 - 4 \eta_1 \\
& \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_3 - 4 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_2 - 4 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_3 - 16 \eta_1 \epsilon_{c2}^2 \epsilon_{r3} \eta_2 - 4 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_2 \\
& - 4 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_3 - 4 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \eta_2 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r3}^2 \eta_1 - 16 \eta_2 \epsilon_{r1}^2 \epsilon_{r3} \eta_1 - 4 \eta_2 \epsilon_{r1}^2 \\
& \epsilon_{c3}^2 \eta_1 + \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 + 8 \eta_3 \eta_1 \epsilon_{c1}^2 \epsilon_{r3}^2 + 8 \eta_3 \eta_1 \epsilon_{c1}^2 \\
& \epsilon_{c3}^2 + 16 \eta_3^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 + 4 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 + \eta_3^2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_2 - 4 \eta_3 \epsilon_{r1}^2 \\
& \epsilon_{c2}^2 \eta_2 - 2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 8 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} + 8 \eta_3 \epsilon_{r1}^2 \eta_1 \epsilon_{c3}^2 + 4 \eta_2^2 \\
& \epsilon_{c3}^2 \epsilon_{r2}^2 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 16 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \eta_1 - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_1 - 4 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \eta_1 \\
& - 16 \eta_3 \epsilon_{r1} \epsilon_{r2} \eta_1 + 72 \epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 + 72 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_1 \\
& + 72 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 - 2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 + 8 \eta_3 \epsilon_{c2}^2 \eta_2 \epsilon_{c3}^2 + 16 \epsilon_{c1}^2 \eta_1^2 \\
& + 16 \eta_3^2 \epsilon_{r1} + 16 \eta_3^2 \epsilon_{r2} - 32 \eta_3^2 \epsilon_{r3} + 4 \eta_3^2 \epsilon_{r1}^2 + 4 \eta_3^2 \epsilon_{r2}^2 + 4 \eta_3^2 \epsilon_{c1}^2 + 4 \eta_3^2 \epsilon_{c2}^2 \\
& + 16 \eta_2^2 \epsilon_{r1} - 32 \eta_2^2 \epsilon_{r2} + 16 \eta_2^2 \epsilon_{r3} + 4 \eta_2^2 \epsilon_{r1}^2 + 4 \eta_2^2 \epsilon_{r2}^2 + 4 \eta_2^2 \epsilon_{c1}^2 + 4 \eta_2^2 \epsilon_{c3}^2 \\
& + 16 \epsilon_{c3}^2 \eta_3^2 + 4 \epsilon_{r1}^2 \epsilon_{r2}^2 + 16 \epsilon_{r1}^2 \epsilon_{r2} + 4 \epsilon_{r1}^2 \epsilon_{c2}^2 + 16 \epsilon_{r1} \epsilon_{r2}^2 + 64 \epsilon_{r1} \epsilon_{r2} + 16 \epsilon_{r1} \epsilon_{c2}^2 \\
& + 4 \epsilon_{c1}^2 \epsilon_{r2}^2 + 16 \epsilon_{c1}^2 \epsilon_{r2} + 16 \epsilon_{r1}^2 + 16 \epsilon_{r2}^2 + 16 \epsilon_{r3}^2 + 16 \eta_1^2 + 16 \eta_2^2 + 16 \eta_3^2 - 2 \eta_2 \\
& \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 \\
& - 32 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 32 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 32 \eta_2 \epsilon_{r1} \\
& \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 8 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - 8 \eta_2 \\
& \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 - 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} \\
& - 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 2 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_3 \\
& \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \\
& - 8 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 32 \eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 + 18 \epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 \epsilon_{c2}^2 + 72 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \epsilon_{r1} \\
& + 18 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \epsilon_{r1}^2 + 18 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \epsilon_{c1}^2 + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 \eta_3 \\
& + 72 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 \epsilon_{r3} + 18 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 \epsilon_{r3}^2 + 18 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 \epsilon_{c3}^2 \\
& + 72 \epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 \epsilon_{r2} + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_1 + 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 \eta_3 - 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 \\
& - 32 \eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 2 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r1} \\
& \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 2 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 32 \eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - 2 \eta_3
\end{aligned}$$

$$\begin{aligned}
& \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 - 2 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} \\
& - 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 2 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 2 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \\
& + \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 4 \eta_2^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 + \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \eta_2^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 \\
& - 8 \eta_2^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2} + \epsilon_{c2}^2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{c3}^2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1} \epsilon_{c2}^2 - 8 \eta_2^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 \\
& + 4 \epsilon_{c1}^2 \epsilon_{c2}^2 + 16 \epsilon_{r1} \epsilon_{r3}^2 + 64 \epsilon_{r1} \epsilon_{r3} + 16 \epsilon_{r1} \epsilon_{c3}^2 + 16 \epsilon_{r2} \epsilon_{r3}^2 + 64 \epsilon_{r2} \epsilon_{r3} + 16 \epsilon_{r2} \\
& \epsilon_{c3}^2 + 4 \epsilon_{r1}^2 \epsilon_{r3}^2 + 16 \epsilon_{r1}^2 \epsilon_{r3} + 4 \epsilon_{r1}^2 \epsilon_{c3}^2 + 4 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \epsilon_{r2}^2 \epsilon_{r3} + 4 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \epsilon_{c1}^2 \epsilon_{r3}^2 \\
& + 16 \epsilon_{c1}^2 \epsilon_{r3} + 4 \epsilon_{c1}^2 \epsilon_{c3}^2 + 4 \epsilon_{c2}^2 \epsilon_{r3}^2 + 16 \epsilon_{c2}^2 \epsilon_{r3} + 4 \epsilon_{c2}^2 \epsilon_{c3}^2 - 32 \eta_1 \epsilon_{r1} + 64 \eta_1 \epsilon_{r2} \\
& + 64 \eta_1 \epsilon_{r3} - 32 \eta_1 \epsilon_{r1}^2 + 16 \eta_1 \epsilon_{r2}^2 + 16 \eta_1 \epsilon_{r3}^2 - 32 \eta_1 \epsilon_{c1}^2 + 16 \eta_1 \epsilon_{c2}^2 + 16 \eta_1 \epsilon_{c3}^2 \\
& + 64 \eta_2 \epsilon_{r1} - 32 \eta_2 \epsilon_{r2} + 64 \eta_2 \epsilon_{r3} + 16 \eta_2 \epsilon_{r1}^2 - 32 \eta_2 \epsilon_{r2}^2 + 16 \eta_2 \epsilon_{r3}^2 + 16 \eta_2 \epsilon_{c1}^2 \\
& - 32 \eta_2 \epsilon_{c2}^2 + 16 \eta_2 \epsilon_{c3}^2 + 64 \eta_3 \epsilon_{r1} + 64 \eta_3 \epsilon_{r2} - 32 \eta_3 \epsilon_{r3} + 16 \eta_3 \epsilon_{r1}^2 + 16 \eta_3 \epsilon_{r2}^2 \\
& - 32 \eta_3 \epsilon_{r3}^2 + 16 \eta_3 \epsilon_{c1}^2 + 16 \eta_3 \epsilon_{c2}^2 - 32 \eta_3 \epsilon_{c3}^2 + 16 \eta_1^2 \epsilon_{r1}^2 + 16 \eta_2^2 \epsilon_{r2}^2 + 16 \epsilon_{c2}^2 \eta_2^2 \\
& + 32 \eta_2 \eta_1 + 32 \eta_3 \eta_1 + 32 \eta_2 \eta_3 + 16 \eta_3^2 \epsilon_{r3}^2 - 32 \eta_2^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} + 4 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \\
& + 4 \eta_2^2 \epsilon_{r2}^2 \epsilon_{r1} \epsilon_{c3}^2 + \epsilon_{c3}^2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{c2}^2 + \epsilon_{c1}^2 \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{c2}^2 + 4 \\
& \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} + \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \\
& \eta_3^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \epsilon_{c2}^2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r3} + 8 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \eta_2 + 8 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \eta_2 + 4 \eta_3^2 \epsilon_{r3}^2 \epsilon_{r1}^2 \epsilon_{r2}^2 + \\
& \eta_3^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + \eta_3^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \epsilon_{r1}^2 - 4 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \eta_1 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} + \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + \eta_1^2 \\
& \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 16 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 + \eta_1^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} + \eta_1^2 \\
& \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} \eta_3 + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_1 + 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_1 + 2 \eta_2 \eta_3 \epsilon_{r1}^2 \\
& \epsilon_{c2}^2 \epsilon_{c3}^2 + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 \eta_3 + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_1 + 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} \eta_1 + 2 \eta_3 \epsilon_{c1}^2 \\
& \epsilon_{c2}^2 \epsilon_{r3} \eta_1 + 2 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_1 + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} \eta_3 + 2 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} \eta_1 + 2 \eta_3 \epsilon_{r1} \\
& \epsilon_{r2}^2 \epsilon_{r3} \eta_1 + 8 \eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} \eta_1 + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 + 2 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 2 \eta_3 \epsilon_{r1}^2 \\
& \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 8 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_3 + 2 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_2 + 2 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_3 + 8 \eta_1 \epsilon_{r1} \\
& \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 8 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} \eta_2 + 8 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_3 + 2 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_2 + 2 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \\
& \epsilon_{r3}^2 \eta_3 + 2 \eta_2 \epsilon_{r3}^2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 - 8 \eta_1^2 \epsilon_{r1} \epsilon_{r2}^2 - 32 \eta_1^2 \epsilon_{r1} \epsilon_{r2} + 16 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_1^2 \epsilon_{r1} \\
& \epsilon_{c2}^2 - 8 \eta_1^2 \epsilon_{r1} \epsilon_{r3}^2 - 32 \eta_1^2 \epsilon_{r1} \epsilon_{r3} - 8 \eta_1^2 \epsilon_{r1} \epsilon_{c3}^2 + 8 \eta_2 \epsilon_{r3}^2 \eta_1 + 16 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1} + 8 \eta_3 \\
& \epsilon_{r2}^2 \eta_1 - 16 \eta_2 \epsilon_{r1}^2 \eta_1 + 4 \epsilon_{c2}^2 \eta_2^2 \epsilon_{c1}^2 - 16 \eta_3 \epsilon_{r1}^2 \eta_1 + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r3}^2 + 4 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \\
& \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 + 4 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} + \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} + \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + 16 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 +
\end{aligned}$$

$$\begin{aligned}
& \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 64 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} + \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} + \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} \\
& + 4 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 + 4 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} + 16 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 + 4 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} + 4 \epsilon_{r1} \\
& \epsilon_{c2}^2 \epsilon_{c3}^2 + \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 + 16 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} + 4 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 + \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \epsilon_{c1}^2 \\
& \epsilon_{c2}^2 \epsilon_{r3} + \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 - 32 \eta_1 \epsilon_{r1}^2 \epsilon_{r2} - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 - 8 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \\
& - 32 \eta_1 \epsilon_{r1} \epsilon_{r2} - 8 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 - 32 \eta_1 \epsilon_{c1}^2 \epsilon_{r2} - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 - 8 \eta_1 \epsilon_{r1} \\
& \epsilon_{r3}^2 + 2 \eta_2 \epsilon_{r3}^2 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 + 2 \eta_2 \epsilon_{r3}^2 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 + 2 \eta_2 \epsilon_{c3}^2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 + 2 \eta_2 \epsilon_{c3}^2 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 \\
& + 2 \eta_2 \epsilon_{c3}^2 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 + 2 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_2 + 2 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_3 + 8 \eta_2 \epsilon_{r3} \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \\
& + 8 \eta_2 \epsilon_{r3} \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 + 8 \eta_2 \epsilon_{r3} \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 + 2 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_2 + 2 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_3 \\
& + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} \eta_1 + 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} \eta_1 + 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_1 + 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_1 \\
& + 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} \eta_1 + 2 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_1 + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_1 + 2 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_1 \\
& + 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} \eta_1 + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_1 + 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} \eta_1 + 8 \eta_3 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 \\
& + 2 \eta_3 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{r2} \epsilon_{c3}^2 \eta_1 - 32 \eta_1^2 \epsilon_{r1} + 16 \eta_1^2 \epsilon_{r2} + 16 \eta_1^2 \epsilon_{r3} + 4 \eta_1^2 \epsilon_{r2}^2 \\
& + 4 \eta_1^2 \epsilon_{r3}^2 + 4 \eta_1^2 \epsilon_{c2}^2 + 4 \eta_1^2 \epsilon_{c3}^2 + 4 \eta_2^2 \epsilon_{r3}^2 \epsilon_{r2}^2 - 32 \eta_1 \epsilon_{r1} \epsilon_{r3} - 8 \eta_1 \epsilon_{r1} \epsilon_{c3}^2 \\
& + 16 \eta_1 \epsilon_{r2} \epsilon_{r3}^2 + 64 \eta_1 \epsilon_{r2} \epsilon_{r3} + 16 \eta_1 \epsilon_{r2} \epsilon_{c3}^2 - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{r3}^2 - 32 \eta_1 \epsilon_{r1}^2 \epsilon_{r3} - 8 \eta_1 \\
& \epsilon_{r1}^2 \epsilon_{c3}^2 + 4 \eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \eta_1 \epsilon_{r2}^2 \epsilon_{r3} + 4 \eta_1 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{r3}^2 - 32 \eta_1 \epsilon_{c1}^2 \epsilon_{r3} \\
& - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{c3}^2 + 4 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 + 16 \eta_1 \epsilon_{c2}^2 \epsilon_{r3} + 4 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 - 8 \eta_2 \\
& \epsilon_{r1}^2 \epsilon_{r2} - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 - 32 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 - 32 \eta_2 \epsilon_{r1} \epsilon_{r2} - 32 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \\
& - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 + 16 \eta_2 \epsilon_{r1} \epsilon_{r3}^2 + 64 \eta_2 \epsilon_{r1} \epsilon_{r3} + 16 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 \\
& - 8 \eta_2 \epsilon_{r2} \epsilon_{r3}^2 - 32 \eta_2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_2 \epsilon_{r2} \epsilon_{c3}^2 + 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 16 \eta_2 \epsilon_{r1}^2 \epsilon_{r3} + 4 \eta_2 \epsilon_{r1}^2 \\
& \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 32 \eta_2 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \eta_2 \epsilon_{c1}^2 \epsilon_{r3}^2 + 16 \eta_2 \epsilon_{c1}^2 \epsilon_{r3} + 4 \eta_2 \\
& \epsilon_{c1}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 32 \eta_2 \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_2 \epsilon_{c2}^2 \epsilon_{c3}^2 + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 + 16 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \\
& + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 + 16 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 + 64 \eta_3 \epsilon_{r1} \epsilon_{r2} + 16 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 + 4 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 + 16 \eta_3 \\
& \epsilon_{c1}^2 \epsilon_{r2} + 4 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 - 32 \eta_3 \epsilon_{r1} \epsilon_{r3}^2 - 32 \eta_3 \epsilon_{r1} \epsilon_{r3} - 32 \eta_3 \epsilon_{r1} \epsilon_{c3}^2 - 32 \eta_3 \epsilon_{r2} \epsilon_{r3}^2 \\
& - 32 \eta_3 \epsilon_{r2} \epsilon_{r3} - 32 \eta_3 \epsilon_{r2} \epsilon_{c3}^2 - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r3} - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{c3}^2 - 8 \eta_3 \epsilon_{r2}^2 \\
& \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_3 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r3} - 8 \eta_3 \epsilon_{c1}^2 \epsilon_{c3}^2 - 8 \eta_3 \epsilon_{c2}^2 \\
& \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_3 \epsilon_{c2}^2 \epsilon_{c3}^2 + 4 \eta_3^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \eta_3^2 \epsilon_{c2}^2 \epsilon_{r3} + 4 \eta_3^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 4 \eta_3^2 \epsilon_{c1}^2 \\
& \epsilon_{r3}^2 + 16 \eta_3^2 \epsilon_{r1} \epsilon_{r3}^2 + 16 \eta_3^2 \epsilon_{r2} \epsilon_{r3}^2 - 16 \eta_3 \epsilon_{r3} \eta_1 + 8 \eta_3 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 + 2 \eta_3 \eta_2 \epsilon_{r1}^2
\end{aligned}$$

$$\begin{aligned}
& \epsilon_{c2}^2 \epsilon_{r3}^2 + 2 \eta_3 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 2 \eta_3 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 8 \eta_3 \epsilon_{r2} \eta_1 \epsilon_{c1}^2 \epsilon_{c3}^2 + 2 \eta_3 \epsilon_{r2}^2 \eta_1 \epsilon_{c1}^2 \\
& \epsilon_{c3}^2 + 2 \eta_3 \epsilon_{c1}^2 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 + 2 \eta_3 \epsilon_{c1}^2 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 + 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 \eta_3 + 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_3 \\
& + 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} \eta_3 + 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_3 + 8 \eta_3 \epsilon_{r1}^2 \eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 + 8 \eta_3 \epsilon_{c1}^2 \eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 \\
& + 8 \eta_3 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + 2 \eta_3 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 2 \eta_3 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 2 \eta_3 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 \\
& + 8 \eta_3 \epsilon_{r2}^2 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_1 \epsilon_{r3}^2 + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_1 \epsilon_{c3}^2 + 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \eta_1 \epsilon_{c3}^2 \\
& + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_1 \epsilon_{r3}^2 + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_1 \epsilon_{c3}^2 + 2 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \eta_1 \epsilon_{r3}^2 + 8 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} \eta_2 \\
& + 8 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 2 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 - 16 \eta_3 \epsilon_{r1} \eta_1 - 16 \eta_2 \epsilon_{c1}^2 \eta_1 + 8 \eta_2 \epsilon_{c3}^2 \eta_1 \\
& - 16 \eta_3 \epsilon_{c1}^2 \eta_1 + 8 \eta_3 \epsilon_{c2}^2 \eta_1 - 16 \eta_2 \epsilon_{r1} \eta_1 + 4 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{c2}^2 + 16 \epsilon_{c3}^2 \\
& \eta_3^2 \epsilon_{r1} + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{c3}^2 + \eta_1^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3} + \eta_1^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3} \\
& + \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \eta_1^2 \epsilon_{r2} \epsilon_{r3} + 4 \eta_1^2 \epsilon_{r2} \epsilon_{c3}^2 + 4 \eta_1^2 \epsilon_{r2} \epsilon_{r3}^2 + 16 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r3} + 16 \epsilon_{c1}^2 \\
& \eta_1^2 \epsilon_{r3} - 16 \eta_2 \epsilon_{r2} \eta_3 - 16 \eta_2 \epsilon_{r3} \eta_3 + 8 \eta_2 \epsilon_{r1}^2 \eta_3 - 16 \eta_2 \epsilon_{r3}^2 \eta_3 + 16 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2} + 4 \\
& \eta_1^2 \epsilon_{r1}^2 \epsilon_{c2}^2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{c2}^2 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 - 16 \eta_1 \epsilon_{r2}^2 \eta_2 - 16 \eta_1 \epsilon_{r3}^2 \eta_3 - 16 \eta_1 \epsilon_{c2}^2 \eta_2 \\
& - 16 \eta_1 \epsilon_{c3}^2 \eta_3 + 16 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r2} + 32 \eta_2 \epsilon_{r3} \eta_1 + 4 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 + 4 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2} + \eta_3^2 \epsilon_{r1}^2 \\
& \epsilon_{c2}^2 + 4 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 - 4 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 \eta_1 - 4 \eta_2 \epsilon_{r2} \epsilon_{r3}^2 \eta_1 - 16 \eta_2 \epsilon_{r2} \epsilon_{r3} \eta_1 - 32 \\
& \eta_3^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - 8 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_3^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} - 4 \eta_2 \epsilon_{c1}^2 \epsilon_{r3} \eta_3 - 4 \eta_2 \epsilon_{c1}^2 \epsilon_{c3}^2 \eta_3 \\
& + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3} - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r3}^2 \eta_3 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r3} \eta_3 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{c3}^2 \eta_3 - 4 \eta_2 \epsilon_{c1}^2 \\
& \epsilon_{r3}^2 \eta_3 + 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \eta_1 + 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \eta_1 + 8 \eta_2 \epsilon_{r2} \epsilon_{r3}^2 \eta_3 + 8 \eta_2 \epsilon_{r2} \epsilon_{r3} \eta_3 + 8 \eta_2 \epsilon_{r2} \\
& \epsilon_{c3}^2 \eta_3 - 8 \eta_1^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - 2 \eta_1^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 + 16 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2} \epsilon_{r3} - 2 \eta_1^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 32 \\
& \eta_1^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - 8 \eta_1^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_1^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_1^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 2 \eta_1^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 \\
& - 8 \eta_1^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 - 4 \eta_3 \epsilon_{r2}^2 \epsilon_{r3} \eta_1 + 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r3} \eta_1 - 4 \eta_3 \epsilon_{c2}^2 \epsilon_{r3} \eta_1 - 16 \eta_3 \\
& \epsilon_{c1}^2 \epsilon_{r2} \eta_1 - 4 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \eta_1 + 8 \eta_3 \epsilon_{r1} \epsilon_{r3} \eta_1 - 16 \eta_3 \epsilon_{r2} \epsilon_{r3} \eta_1 + 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r3} \eta_1 \\
& - 4 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \eta_1 + \epsilon_{c2}^2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r3}^2 - 2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2} \epsilon_{r3}^2 \\
& - 16 \eta_1 \epsilon_{r2} \epsilon_{r3}^2 \eta_3 - 16 \eta_1 \epsilon_{r2} \epsilon_{c3}^2 \eta_3 + 8 \eta_3 \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 8 \eta_1 \epsilon_{r1} \epsilon_{r3}^2 \eta_3 + 8 \eta_1 \epsilon_{r1} \\
& \epsilon_{c3}^2 \eta_3 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2} + 8 \eta_2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 + 8 \eta_2 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 + \epsilon_{c3}^2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2}^2 + 4 \epsilon_{c2}^2 \\
& \eta_2^2 \epsilon_{r1} \epsilon_{c3}^2 - 2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 8 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} - 2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3} + \epsilon_{c1}^2 \\
& \eta_1^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \eta_3 - 16 \eta_2 \epsilon_{r1} \epsilon_{r2} \eta_3 - 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 32 \eta_1 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} \\
& + 16 \eta_3^2 \epsilon_{r1} \epsilon_{r2} + 4 \eta_3^2 \epsilon_{r1} \epsilon_{c2}^2 + \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2}^2 + 4 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2} + \eta_3^2 \epsilon_{c1}^2 \epsilon_{c2}^2 - 32 \eta_3^2 \epsilon_{r1} \epsilon_{r3}
\end{aligned}$$

$$\begin{aligned}
& -32 \eta_3^2 \epsilon_{r2} \epsilon_{r3} + \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 4 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3} + \eta_2^2 \epsilon_{r1}^2 \epsilon_{c3}^2 + \eta_2^2 \epsilon_{c1}^2 \epsilon_{r3}^2 + 4 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r3} + \\
& \eta_2^2 \epsilon_{c1}^2 \epsilon_{c3}^2 + \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 - 8 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2} - 32 \eta_2^2 \epsilon_{r1} \epsilon_{r2} - 8 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2} + 4 \eta_2^2 \epsilon_{r1} \epsilon_{r3}^2 + 16 \\
& \eta_2^2 \epsilon_{r1} \epsilon_{r3} + 4 \eta_2^2 \epsilon_{r1} \epsilon_{c3}^2 - 8 \eta_2^2 \epsilon_{r2} \epsilon_{r3}^2 - 32 \eta_2^2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_2^2 \epsilon_{r2} \epsilon_{c3}^2 + 16 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2} \\
& - 16 \eta_3 \epsilon_{r2}^2 \eta_2 - 16 \eta_3 \epsilon_{c2}^2 \eta_2 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{c3}^2 + 16 \eta_1^2 \\
& \epsilon_{r1}^2 \epsilon_{r3} + 16 \eta_2^2 \epsilon_{r1} \epsilon_{r2}^2 + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2}^2 + 16 \eta_2^2 \epsilon_{r3} \epsilon_{r2}^2 + 32 \eta_2 \epsilon_{r1} \eta_3 - 8 \eta_3^2 \epsilon_{c2}^2 \epsilon_{r3} \\
& + 8 \eta_2 \epsilon_{c1}^2 \eta_3 - 16 \eta_2 \epsilon_{c3}^2 \eta_3 + 32 \eta_3 \epsilon_{r2} \eta_1 - 2 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 \\
& - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 2 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 - 2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_1 \epsilon_{r1} \epsilon_{r2} \\
& \epsilon_{c3}^2 - 2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 32 \eta_1 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - 2 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 8 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 2 \eta_1 \epsilon_{c1}^2 \\
& \epsilon_{r2}^2 \epsilon_{r3}^2 - 8 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 2 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} \\
& - 8 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 2 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 - 2 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{r2} \\
& \epsilon_{r3}^2 - 32 \eta_1 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 2 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_1 \epsilon_{c1}^2 \\
& \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_3^2 \\
& \epsilon_{r1}^2 \epsilon_{r3}^2)
\end{aligned}$$

to string
→

$$\begin{aligned}
& "9*(16*\epsilon[c1]*\eta[1]C\ 16*\epsilon[c2]*\eta[2]C\ 16*\epsilon[c3]*\eta[3]C\ 16* \quad (2.12) \\
& \epsilon[c1]*\eta[1]*\epsilon[r2]C\ 16*\epsilon[c1]*\eta[1]*\epsilon[r3]C\ 4* \\
& \epsilon[c1]*\eta[1]*\epsilon[r2]^2C\ 4*\epsilon[c1]*\eta[1]*\epsilon[r3]^2C\ 4* \\
& \epsilon[c1]*\eta[1]*\epsilon[c2]^2C\ 4*\epsilon[c1]*\eta[1]*\epsilon[c3]^2C\ 16* \\
& \epsilon[c2]*\eta[2]*\epsilon[r1]C\ 16*\epsilon[c2]*\eta[2]*\epsilon[r3]C\ 4* \\
& \epsilon[c2]*\eta[2]*\epsilon[r1]^2C\ 4*\epsilon[c2]*\eta[2]*\epsilon[r3]^2C\ 4* \\
& \epsilon[c2]*\eta[2]*\epsilon[c1]^2C\ 4*\epsilon[c2]*\eta[2]*\epsilon[c3]^2C\ 16* \\
& \epsilon[c3]*\eta[3]*\epsilon[r1]C\ 16*\epsilon[c3]*\eta[3]*\epsilon[r2]C\ 4* \\
& \epsilon[c3]*\eta[3]*\epsilon[r1]^2C\ 4*\epsilon[c3]*\eta[3]*\epsilon[r2]^2C\ 4* \\
& \epsilon[c3]*\eta[3]*\epsilon[c1]^2C\ 4*\epsilon[c3]*\eta[3]*\epsilon[c2]^2C\ 4* \\
& \epsilon[c1]*\eta[1]*\epsilon[r2]*\epsilon[r3]^2C\ 16*\epsilon[c1]*\eta[1]* \\
& \epsilon[r2]*\epsilon[r3]C\ 4*\epsilon[c1]*\eta[1]*\epsilon[r2]*\epsilon[r3]^2C \\
& \epsilon[c1]*\eta[1]*\epsilon[r2]^2*\epsilon[r3]^2C\ 4*\epsilon[c1]*\eta[1]* \\
& \epsilon[r2]^2*\epsilon[r3]C\ \epsilon[c1]*\eta[1]*\epsilon[r2]^2*\epsilon[r3] \\
& ^2C\ \epsilon[c1]*\eta[1]*\epsilon[c2]^2*\epsilon[r3]^2C\ 4*\epsilon[c1]*\eta[1]* \\
& \epsilon[c2]^2*\epsilon[r3]C\ \epsilon[c1]*\eta[1]*\epsilon[c2]^2*\epsilon[r3] \\
& ^2C\ 4*\epsilon[c2]*\eta[2]*\epsilon[r1]*\epsilon[r3]^2C\ 16*\epsilon[c2]*\eta[2]* \\
& \epsilon[r1]*\epsilon[r3]C\ 4*\epsilon[c2]*\eta[2]*\epsilon[r1]*\epsilon[r3]^2C \\
& \epsilon[c2]*\eta[2]*\epsilon[r1]^2*\epsilon[r3]^2C\ 4*\epsilon[c2]*\eta[2]*
\end{aligned}$$

$$\begin{aligned}
& \epsilon[r_1]^2 \epsilon[r_3] C \epsilon[c_2] \eta[2] \epsilon[r_1]^2 \epsilon[c_3] \\
& ^2 C \epsilon[c_2] \eta[2] \epsilon[c_1]^2 \epsilon[r_3]^2 C^4 \epsilon[c_2] \eta[2] * \\
& \epsilon[c_1]^2 \epsilon[r_3] C \epsilon[c_2] \eta[2] \epsilon[c_1]^2 \epsilon[c_3] \\
& ^2 C \epsilon[c_3] \eta[3] \epsilon[r_1]^2 \epsilon[r_2]^2 C^4 \epsilon[c_3] \eta[3] * \\
& \epsilon[r_1]^2 \epsilon[r_2] C \epsilon[c_3] \eta[3] \epsilon[r_1]^2 \epsilon[c_2] \\
& ^2 C^4 \epsilon[c_3] \eta[3] \epsilon[r_1] \epsilon[r_2]^2 C^{16} \epsilon[c_3] \eta[3] * \\
& \epsilon[r_1] \epsilon[r_2] C^4 \epsilon[c_3] \eta[3] \epsilon[r_1] \epsilon[c_2]^2 C \\
& \epsilon[c_3] \eta[3] \epsilon[c_1]^2 \epsilon[r_2]^2 C^4 \epsilon[c_3] \eta[3] * \\
& \epsilon[c_1]^2 \epsilon[r_2] C \epsilon[c_3] \eta[3] \epsilon[c_1]^2 \epsilon[c_2]^2) \\
& / (64 C^4 \epsilon[c_2]^2 \eta[2]^2 \epsilon[c_3]^2 C^{64} \eta[1] C^{64} \eta[2] C^{64} * \\
& \eta[3] C^{64} \epsilon[r_1] C^{64} \epsilon[r_2] C^{64} \epsilon[r_3] C^{16} \epsilon[c_1] \\
& ^2 C^{16} \epsilon[c_2]^2 C^{16} \epsilon[c_3]^2 - 8 \eta[3]^2 \epsilon[c_1]^2 \epsilon[r_3] - 8 \eta[3]^2 \epsilon[r_2]^2 \epsilon[r_3] C^{18} \epsilon[c_1] \eta[1] \epsilon[c_3] \eta[3] \epsilon[r_2]^2 C^4 \epsilon[c_2]^2 \eta[2]^2 \epsilon[r_3]^2 C^4 * \\
& \epsilon[c_3]^2 \eta[3]^2 \epsilon[r_2]^2 C^4 \eta[2]^2 \epsilon[c_1]^2 \epsilon[r_2]^2 - 16 \eta[2] \epsilon[r_2] \eta[1] C^8 \eta[3] \epsilon[r_1]^2 \eta[1] \epsilon[r_3]^2 - 16 \eta[3] \epsilon[r_1] \epsilon[r_2]^2 \eta[2] - 16 \eta[3] \epsilon[r_1] * \\
& \epsilon[c_2]^2 \eta[2] - 4 \eta[3] \epsilon[c_1]^2 \epsilon[r_2]^2 \eta[2] - 8 \eta[3] \\
& ^2 \epsilon[c_1]^2 \epsilon[r_2] \epsilon[r_3] - 2 \eta[3]^2 \epsilon[c_1]^2 * \\
& \epsilon[c_2]^2 \epsilon[r_3] - 4 \eta[1] \epsilon[r_2]^2 \epsilon[r_3]^2 \eta[2] - 4 * \\
& \eta[1] \epsilon[r_2]^2 \epsilon[r_3]^2 \eta[3] - 16 \eta[1] \epsilon[r_2]^2 \epsilon[r_3] * \eta[2] C^8 \eta[3] \epsilon[r_2]^2 \epsilon[r_3] \eta[2] C^4 \epsilon[c_3]^2 \eta[3] \\
& ^2 \epsilon[r_1] \epsilon[r_2]^2 C^{16} \epsilon[c_2]^2 \eta[2]^2 \epsilon[r_1] * \\
& \epsilon[r_3] C^4 \epsilon[c_2]^2 \eta[2]^2 \epsilon[r_1]^2 \epsilon[r_3] C^{16} * \\
& \epsilon[c_3]^2 \eta[3]^2 \epsilon[r_1] \epsilon[r_2] C^8 \eta[2] \eta[1] \epsilon[c_1]^2 \epsilon[c_2]^2 - 4 \eta[2] * \\
& \epsilon[c_1]^2 \epsilon[r_2] \eta[3] - 16 \eta[2] \epsilon[r_1] \epsilon[r_3]^2 \eta[3] \\
& - 16 \eta[2] \epsilon[r_1] \epsilon[r_3] \eta[3] - 16 \eta[2] \epsilon[r_1] \epsilon[c_3]^2 \eta[3] C \epsilon[c_2]^2 \eta[2]^2 \epsilon[r_1]^2 \epsilon[c_3]^2 C^8 \eta[2] \\
& ^2 \epsilon[c_1]^2 \epsilon[r_2] \eta[1] - 4 \eta[2] \epsilon[r_1] \epsilon[r_3]^2 * \\
& \eta[1] - 16 \eta[2] \epsilon[r_1] \epsilon[r_3] \eta[1] C^4 \epsilon[c_3]^2 \eta[3]^2 * \\
& \epsilon[c_1]^2 C^8 \eta[3] \epsilon[c_2]^2 \eta[2] \epsilon[r_3]^2 - 4 \eta[2] * \\
& \epsilon[c_1]^2 \epsilon[r_3]^2 \eta[1] - 16 \eta[2] \epsilon[c_1]^2 \epsilon[r_3] * \\
& \eta[1] - 4 \eta[2] \epsilon[c_1]^2 \epsilon[c_3]^2 \eta[1] C^8 \eta[3] \epsilon[r_2] \\
& ^2 \eta[2] \epsilon[c_3]^2 C^8 \eta[3] \epsilon[r_2]^2 \eta[2] \epsilon[r_3]^2 - 4 * \\
& \eta[1] \epsilon[r_2]^2 \epsilon[c_3]^2 \eta[2] - 4 \eta[1] \epsilon[r_2]^2 \epsilon[c_3]^2 \eta[3] - 4 \eta[1] \epsilon[c_2]^2 \epsilon[r_3]^2 \eta[2] - 4 \eta[1] \epsilon[c_2]^2 \epsilon[r_3]^2 \eta[3] - 16 \eta[1] \epsilon[c_2]^2 \epsilon[r_3] \eta[2] - 4 *
\end{aligned}$$

$$\begin{aligned}
& \eta[1] \epsilon[c_2]^2 \epsilon[c_3]^2 \eta[2] - 4 \eta[1] \epsilon[c_2]^2 \epsilon[c_3]^2 \eta[3] - 4 \eta[3] \epsilon[c_1]^2 \epsilon[c_2]^2 \eta[2] - 4 \eta[2]^* \\
& \epsilon[r_1]^2 \epsilon[r_3]^2 \eta[1] - 16 \eta[2]^* \epsilon[r_1]^2 \epsilon[r_3]^* \eta[1] - 4 \eta[2]^* \epsilon[r_1]^2 \epsilon[c_3]^2 \eta[1] C \epsilon[c_1]^2 \eta[1]^2 \\
& \epsilon[r_2]^2 \epsilon[r_3]^2 C \epsilon[c_2]^2 \eta[2]^2 \epsilon[r_1]^2^* \epsilon[r_3]^2 C \epsilon[c_3]^2 \eta[3]^2 \epsilon[r_1]^2 \epsilon[r_2]^2 C 8^* \eta[3]^* \eta[1]^* \epsilon[c_1]^2^* \\
& \epsilon[c_3]^2 C 16^* \eta[3]^2 \epsilon[r_1]^* \epsilon[r_2]^* \epsilon[r_3]^2 C 4^* \eta[3]^2 \epsilon[c_1]^2^* \epsilon[r_2]^* \epsilon[r_3]^2 C \eta[3]^2 \epsilon[c_1]^2^* \\
& \epsilon[c_2]^2 \epsilon[r_3]^2 - 4 \eta[3]^* \epsilon[r_1]^2 \epsilon[r_2]^2 \eta[2] - 4 \eta[3]^* \epsilon[r_1]^2 \epsilon[c_2]^2 \eta[2] - 2 \eta[3]^2 \epsilon[c_1]^2^* \\
& \epsilon[r_2]^2 \epsilon[r_3] - 2 \eta[3]^2 \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[r_3] - 8 \eta[3]^2 \epsilon[r_1]^2 \epsilon[r_2]^* \epsilon[r_3] C 8^* \eta[3]^* \epsilon[r_1]^2 \eta[1]^* \epsilon[c_3]^2 C 4^* \eta[2]^2 \epsilon[c_3]^2 \epsilon[r_2]^2 C 4^* \eta[1]^2 \epsilon[r_1]^2 \epsilon[r_2]^* \epsilon[c_3]^2 - 16 \eta[3]^* \epsilon[r_1]^2^* \epsilon[r_2]^* \eta[1] - 4 \eta[3]^* \epsilon[r_1]^2 \epsilon[c_2]^2 \eta[1] - 4 \eta[3]^* \epsilon[r_1]^* \epsilon[r_2]^2 \eta[1] - 16 \eta[3]^* \epsilon[r_1]^* \epsilon[r_2]^* \eta[1] C 72^* \epsilon[c_1]^* \eta[1]^* \epsilon[c_3]^* \eta[3] C 72^* \epsilon[c_2]^* \eta[2]^* \epsilon[c_3]^* \eta[3] - 4 \eta[3]^* \epsilon[r_1]^2 \epsilon[r_2]^2 \eta[1] C 72^* \epsilon[c_1]^* \eta[1]^* \epsilon[c_2]^* \eta[2] - 2 \eta[3]^2 \epsilon[r_1]^2 \epsilon[c_2]^2 \epsilon[r_3] C 4^* \eta[3]^2 \epsilon[r_1]^* \epsilon[r_2]^2 \epsilon[r_3]^2 C 8^* \eta[3]^* \epsilon[c_2]^2 \eta[2]^* \epsilon[c_3]^2 C 16^* \epsilon[c_1]^2 \eta[1]^2 C 16^* \eta[3]^2^* \epsilon[r_1] C 16^* \eta[3]^2^* \epsilon[r_2] - 32 \eta[3]^2^* \epsilon[r_3] C 4^* \eta[3]^2^* \epsilon[r_1]^2 C 4^* \eta[3]^2^* \epsilon[r_2]^2 C 4^* \eta[3]^2^* \epsilon[c_1]^2 C 4^* \eta[3]^2^* \epsilon[c_2]^2 C 16^* \eta[2]^2^* \epsilon[r_1] - 32 \eta[2]^2^* \epsilon[r_2] C 16^* \eta[2]^2^* \epsilon[r_3] C 4^* \eta[2]^2^* \epsilon[r_1]^2 C 4^* \eta[2]^2^* \epsilon[r_3]^2 C 4^* \eta[2]^2^* \epsilon[c_1]^2 C 4^* \eta[2]^2^* \epsilon[c_3]^2 C 16^* \epsilon[c_3]^2 \eta[3]^2 C 4^* \epsilon[r_1]^2 \epsilon[r_2]^2 C 16^* \epsilon[r_1]^2 \epsilon[r_2] C 4^* \epsilon[r_1]^2 \epsilon[c_2]^2 C 16^* \epsilon[r_1]^* \epsilon[r_2]^2 C 64^* \epsilon[r_1]^* \epsilon[r_2] C 16^* \epsilon[c_1]^2 \epsilon[r_2]^2 C 16^* \epsilon[c_1]^2 \epsilon[r_2] C 16^* \epsilon[r_1]^2 C 16^* \epsilon[r_2]^2 C 16^* \epsilon[r_3]^2 C 16^* \eta[1]^2 C 16^* \eta[2]^2 C 16^* \eta[3]^2 - 2 \eta[2]^* \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[c_3]^2 - 8 \eta[2]^* \epsilon[r_1]^* \epsilon[r_2]^2 \epsilon[r_3]^2 - 2 \eta[2]^* \epsilon[r_1]^2 \epsilon[c_2]^2 \epsilon[c_3]^2 - 8 \eta[2]^* \epsilon[r_1]^* \epsilon[r_2]^* \epsilon[c_3]^2 - 2 \eta[2]^* \epsilon[r_1]^2 \epsilon[c_2]^2 \epsilon[r_3]^2 - 32 \eta[2]^* \epsilon[r_1]^* \epsilon[r_2]^* \epsilon[r_3] - 2 \eta[2]^* \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[r_3]^2 - 32 \eta[2]^* \epsilon[r_1]^* \epsilon[r_2]^2 \epsilon[r_3] - 2 \eta[2]^* \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[r_3]^2 - 2 \eta[2]^* \epsilon[c_1]^2 \epsilon[r_2]^2^*
\end{aligned}$$

$$\begin{aligned}
& \epsilon[r_3]^2 - 32 \eta[2] \epsilon[r_1] \epsilon[c_2]^2 \epsilon[r_3] - 2 \eta[2] \epsilon[r_1]^2 \epsilon[r_2] \epsilon[r_3]^2 - 8 \eta[2] \epsilon[r_1] \epsilon[r_2]^2 \epsilon[c_3]^2 - 8 \eta[2] \epsilon[c_1]^2 \epsilon[r_2]^2 \epsilon[r_3] - 8 \eta[2] \epsilon[r_1] \epsilon[r_2] \epsilon[r_3]^2 - 8 \eta[2] \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[r_3] - 8 \eta[2] \epsilon[r_1] \epsilon[c_2]^2 \epsilon[c_3]^2 - 2 \eta[2] \epsilon[c_1]^2 \epsilon[r_2]^2 \epsilon[c_3]^2 - 2 \eta[2] \epsilon[c_1]^2 \epsilon[r_2] \epsilon[r_3]^2 - 8 \eta[2] \epsilon[c_1]^2 \epsilon[r_2] \epsilon[r_3] - 2 \eta[2] \epsilon[c_1]^2 \epsilon[r_2] \epsilon[c_3]^2 - 2 \eta[2] \epsilon[c_1]^2 \epsilon[r_2] \epsilon[r_3] - 2 \eta[2] \epsilon[c_2]^2 \epsilon[r_3]^2 - 8 \eta[2] \epsilon[c_1]^2 \epsilon[c_2]^2 \epsilon[r_3] - 2 \eta[2] \epsilon[c_1]^2 \epsilon[c_2]^2 \epsilon[c_3]^2 - 8 \eta[3] \epsilon[r_1]^2 \epsilon[r_2] \epsilon[r_3] - 8 \eta[3] \epsilon[r_1] \epsilon[c_2]^2 \epsilon[r_3]^2 - 8 \eta[3] \epsilon[r_1]^2 \epsilon[r_2] \epsilon[c_3]^2 - 2 \eta[3] \epsilon[r_1]^2 \epsilon[c_2]^2 \epsilon[r_3] - 2 \eta[3] \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[c_3]^2 - 8 \eta[3] \epsilon[r_1] \epsilon[r_2]^2 \epsilon[r_3]^2 - 32 \eta[3] \epsilon[r_1] \epsilon[r_2] \epsilon[c_3]^2 C_{18} \epsilon[c_1] \eta[1] \epsilon[c_3] \eta[3] \epsilon[c_2]^2 C_{72} \epsilon[c_2] \eta[2] \epsilon[c_3] \eta[3] \epsilon[r_1] C_{18} \epsilon[c_2] \eta[2] \epsilon[c_3] \eta[3] \epsilon[r_1]^2 C_{18} \epsilon[c_2] \eta[2] \epsilon[c_3] \eta[3] \epsilon[c_1]^2 C_2 \eta[2] \epsilon[r_1]^2 \epsilon[r_2] \epsilon[r_3]^2 \eta[3] C_{72} \epsilon[c_1] \eta[1] \epsilon[c_2] \eta[2] \epsilon[r_3] C_{18} \epsilon[c_1] \eta[1] \epsilon[c_2] \eta[2] \epsilon[r_3]^2 C_{18} \epsilon[c_1] \eta[1] \epsilon[c_2] \eta[2] \epsilon[c_3]^2 C_{72} \epsilon[c_1] \eta[1] \epsilon[c_3] \eta[3] \epsilon[r_2] C_2 \eta[2] \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[r_3]^2 \eta[1] C_8 \eta[2] \epsilon[r_1] \epsilon[r_2] \epsilon[c_3]^2 \eta[3] - 2 \eta[3] \epsilon[r_1]^2 \epsilon[c_2]^2 \epsilon[r_3]^2 - 32 \eta[3] \epsilon[r_1] \epsilon[r_2] \epsilon[r_3] - 2 \eta[3] \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[r_3]^2 - 8 \eta[3] \epsilon[r_1] \epsilon[r_2]^2 \epsilon[r_3] - 2 \eta[3] \epsilon[c_1]^2 \epsilon[r_2]^2 \epsilon[r_3] - 8 \eta[3] \epsilon[r_1] \epsilon[c_2]^2 \epsilon[r_3] - 8 \eta[3] \epsilon[r_1]^2 \epsilon[r_2] \epsilon[c_3]^2 - 2 \eta[3] \epsilon[r_1]^2 \epsilon[r_2] \epsilon[c_3]^2 - 2 \eta[3] \epsilon[c_1]^2 \epsilon[r_2]^2 \epsilon[r_3] - 32 \eta[3] \epsilon[r_1] \epsilon[r_2] \epsilon[r_3]^2 - 2 \eta[3] \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[r_3] - 8 \eta[3] \epsilon[r_1] \epsilon[c_2]^2 \epsilon[c_3]^2 - 2 \eta[3] \epsilon[c_1]^2 \epsilon[r_2]^2 \epsilon[c_3]^2 - 8 \eta[3] \epsilon[c_1]^2 \epsilon[r_2] \epsilon[r_3]^2 - 8 \eta[3] \epsilon[c_1]^2 \epsilon[r_2] \epsilon[c_3]^2 - 2 \eta[3] \epsilon[c_1]^2 \epsilon[r_2] \epsilon[r_3] - 8 \eta[3] \epsilon[c_2]^2 \epsilon[r_3]^2 - 2 \eta[3] \epsilon[c_1]^2 \epsilon[c_2]^2 \epsilon[r_3] - 2 \eta[3] \epsilon[c_1]^2 \epsilon[c_2]^2 \epsilon[c_3]^2 C_{\eta[2]^2} \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[c_3]^2 C_{\eta[2]^2} \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[r_3]^2 C_4 \eta[2]^2 \epsilon[r_1] \epsilon[r_2]^2 \epsilon[r_3]^2 C_{\eta[2]}
\end{aligned}$$

$$\begin{aligned}
& \epsilon^2[c_1]\epsilon^2[r_2]\epsilon^2[r_3]2C16\epsilon^2[2]\epsilon^2[r_1]* \\
& \epsilon[r_2]\epsilon^2[r_3]-2\epsilon^2[2]\epsilon^2[r_1]\epsilon^2[r_2]\epsilon[r_3]^2-8\epsilon^2[2]\epsilon^2[r_1]\epsilon[r_2]\epsilon[r_3]^2C4\epsilon^2[c_3]^2* \\
& \epsilon^2[3]\epsilon^2[r_1]\epsilon^2[r_2]C\epsilon^2[c_2]\epsilon^2[2]\epsilon^2[c_1]^2* \\
& \epsilon[c_3]^2C4\epsilon^2[c_3]^2\epsilon^2[3]\epsilon^2[r_1]\epsilon^2[c_2]^2-8\epsilon^2[2]\epsilon^2[r_1]\epsilon[r_2]\epsilon[c_3]^2C4\epsilon^2[c_1]^2\epsilon^2[c_2] \\
& ^2C16\epsilon^2[r_1]\epsilon^2[r_3]^2C64\epsilon^2[r_1]\epsilon^2[r_3]C16\epsilon^2[r_1]\epsilon^2[c_3]^2C16\epsilon^2[r_2]\epsilon^2[r_3]^2C64\epsilon^2[r_2]\epsilon^2[r_3]C16\epsilon^2[r_2]\epsilon^2[c_3]^2C4\epsilon^2[r_1]^2\epsilon^2[r_3]^2C16* \\
& \epsilon[r_1]^2\epsilon^2[r_3]C4\epsilon^2[r_1]^2\epsilon^2[c_3]^2C4\epsilon^2[r_2] \\
& ^2\epsilon^2[r_3]^2C16\epsilon^2[r_2]^2\epsilon^2[r_3]C4\epsilon^2[r_2]^2\epsilon^2[c_3]^2C4\epsilon^2[c_1]^2\epsilon^2[r_3]^2C16\epsilon^2[c_1]^2\epsilon^2[c_3]^2C4\epsilon^2[c_2]^2\epsilon^2[r_3]C4\epsilon^2[c_2]^2\epsilon^2[c_3]^2-32\epsilon^2[1]\epsilon^2[r_1] \\
& C64\epsilon^2[1]\epsilon^2[r_2]C64\epsilon^2[1]\epsilon^2[r_3]-32\epsilon^2[1]\epsilon^2[r_1] \\
& ^2C16\epsilon^2[1]\epsilon^2[r_2]^2C16\epsilon^2[1]\epsilon^2[r_3]^2-32\epsilon^2[1]\epsilon^2[r_1] \\
& [c_1]^2C16\epsilon^2[1]\epsilon^2[c_2]^2C16\epsilon^2[1]\epsilon^2[c_3]^2C64\epsilon^2[2]* \\
& \epsilon[r_1]-32\epsilon^2[2]\epsilon^2[r_2]C64\epsilon^2[2]\epsilon^2[r_3]C16\epsilon^2[2]* \\
& \epsilon[r_1]^2-32\epsilon^2[2]\epsilon^2[r_2]^2C16\epsilon^2[2]\epsilon^2[r_3]^2C16\epsilon^2[2] \\
& *\epsilon^2[c_1]^2-32\epsilon^2[2]\epsilon^2[c_2]^2C16\epsilon^2[2]\epsilon^2[c_3]^2C64\epsilon^2[3]* \\
& \epsilon[r_1]C64\epsilon^2[3]\epsilon^2[r_2]-32\epsilon^2[3]\epsilon^2[r_3]C16\epsilon^2[3]* \\
& \epsilon[r_1]^2C16\epsilon^2[3]\epsilon^2[r_2]^2-32\epsilon^2[3]\epsilon^2[r_3]^2C16\epsilon^2[3] \\
& *\epsilon^2[c_1]^2C16\epsilon^2[3]\epsilon^2[c_2]^2-32\epsilon^2[3]\epsilon^2[c_3]^2C16\epsilon^2[1]^2\epsilon^2[r_1]^2C16\epsilon^2[2]^2\epsilon^2[r_2]^2C16\epsilon^2[c_2]^2\epsilon^2[2] \\
& ^2C32\epsilon^2[2]\epsilon^2[1]C32\epsilon^2[3]\epsilon^2[1]C32\epsilon^2[2]\epsilon^2[3]C16\epsilon^2[3]^2* \\
& \epsilon[r_3]^2-32\epsilon^2[2]^2\epsilon^2[r_1]\epsilon^2[r_2]\epsilon^2[r_3]C4\epsilon^2[2] \\
& ^2\epsilon^2[r_1]^2\epsilon^2[r_2]^2\epsilon^2[r_3]C4\epsilon^2[2]^2\epsilon^2[r_2]^2* \\
& \epsilon[r_1]\epsilon^2[c_3]^2C\epsilon^2[c_3]^2\epsilon^2[3]^2\epsilon^2[c_1]^2\epsilon^2[c_2]^2C\epsilon^2[c_1]^2\epsilon^2[1]^2\epsilon^2[c_2]^2\epsilon^2[r_3]^2C\epsilon^2[3]^2* \\
& \epsilon[c_1]^2\epsilon^2[r_2]^2\epsilon^2[r_3]^2C\epsilon^2[c_3]^2\epsilon^2[3]^2* \\
& \epsilon[r_1]^2\epsilon^2[c_2]^2C4\epsilon^2[c_2]^2\epsilon^2[2]^2\epsilon^2[r_1]\epsilon^2[r_3]^2C4\epsilon^2[c_1]^2\epsilon^2[1]^2\epsilon^2[r_2]\epsilon^2[c_3]^2C\epsilon^2[c_1] \\
& ^2\epsilon^2[1]^2\epsilon^2[r_2]^2\epsilon^2[c_3]^2C4\epsilon^2[2]^2\epsilon^2[c_1]^2* \\
& \epsilon[r_2]^2\epsilon^2[r_3]C\epsilon^2[2]^2\epsilon^2[c_1]^2\epsilon^2[r_2]^2\epsilon^2[c_3]^2C4\epsilon^2[3]^2\epsilon^2[r_1]\epsilon^2[c_2]^2\epsilon^2[r_3]^2C4\epsilon^2[c_2] \\
& ^2\epsilon^2[2]^2\epsilon^2[c_1]^2\epsilon^2[r_3]C8\epsilon^2[1]\epsilon^2[r_1]\epsilon^2[r_2] \\
& ^2\epsilon^2[2]C8\epsilon^2[1]\epsilon^2[r_1]\epsilon^2[c_2]^2\epsilon^2[2]C4\epsilon^2[3]^2\epsilon^2[r_3]^2\epsilon^2[r_1]^2\epsilon^2[r_2]C\epsilon^2[3]^2\epsilon^2[r_1]^2\epsilon^2[c_2]^2*
\end{aligned}$$

$$\begin{aligned}
& \epsilon[r_3]^2 C \eta[3]^2 \epsilon[r_2]^2 \epsilon[r_3]^2 \epsilon[r_1]^2 - 4 \eta[3] \epsilon[r_1] \epsilon[c_2]^2 \eta[1] C 4 \eta[1]^2 \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[r_3]^2 C \eta[1]^2 \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[c_3]^2 C 16 \eta[1]^2 \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[r_3]^2 C 4 \eta[1]^2 \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[c_3]^2 C 4 \eta[1]^2 \epsilon[r_1]^2 \epsilon[c_2]^2 \epsilon[r_3]^2 C \eta[1]^2 \epsilon[r_1]^2 \epsilon[c_2]^2 \epsilon[r_3]^2 C 8 \eta[2] \epsilon[r_1] \epsilon[r_2] \epsilon[r_3] \eta[3] C 2 \eta[2] \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[c_3]^2 \eta[1] C 8 \eta[2] \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[r_3] \eta[1] C 2 \eta[2] \eta[3] \epsilon[r_1]^2 \epsilon[c_2]^2 \epsilon[c_3]^2 C 2 \eta[2] \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[c_3]^2 \eta[3] C 2 \eta[3] \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[r_3] \eta[1] C 8 \eta[3] \epsilon[c_1]^2 \epsilon[r_2] \epsilon[r_3] \eta[1] C 2 \eta[3] \epsilon[c_1]^2 \epsilon[c_2]^2 \epsilon[r_3] \eta[1] C 2 \eta[3] \epsilon[c_1]^2 \epsilon[r_2]^2 \epsilon[r_3] \eta[1] C 2 \eta[2] \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[r_3] \eta[3] C 2 \eta[3] \epsilon[r_1] \epsilon[c_2]^2 \epsilon[r_3] \eta[1] C 2 \eta[3] \epsilon[r_1] \epsilon[r_2]^2 \epsilon[r_3] \eta[1] C 8 \eta[3] \epsilon[r_1] \epsilon[r_2] \epsilon[r_3] \eta[1] C 2 \eta[3] \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[r_3] \eta[2] C 2 \eta[3] \epsilon[c_1]^2 \epsilon[c_2]^2 \epsilon[r_3] \eta[2] C 2 \eta[3] \epsilon[r_1]^2 \epsilon[c_2]^2 \epsilon[r_3] \eta[2] C 8 \eta[1] \epsilon[r_1] \epsilon[r_2] \epsilon[r_3]^2 \eta[3] C 2 \eta[1] \epsilon[r_1] \epsilon[r_2]^2 \epsilon[c_3]^2 \eta[3] C 8 \eta[1] \epsilon[r_1] \epsilon[r_2]^2 \epsilon[r_3] \eta[2] C 8 \eta[1] \epsilon[r_1] \epsilon[r_2] \epsilon[c_3]^2 \eta[3] C 2 \eta[1] \epsilon[r_1] \epsilon[r_2]^2 \epsilon[r_3]^2 \eta[2] C 2 \eta[2] \epsilon[r_3]^2 \eta[1] \epsilon[r_1]^2 \epsilon[c_2]^2 - 8 \eta[1]^2 \epsilon[r_1] \epsilon[r_2]^2 - 32 \eta[1]^2 \epsilon[r_1] \epsilon[r_2] C 16 \epsilon[r_1]^2 \epsilon[r_2] \epsilon[r_3] - 8 \eta[1]^2 \epsilon[r_1] \epsilon[c_2]^2 - 8 \eta[1]^2 \epsilon[r_1] \epsilon[r_3]^2 - 32 \eta[1]^2 \epsilon[r_1] \epsilon[r_3] - 8 \eta[1]^2 \epsilon[r_1] \epsilon[c_3]^2 C 8 \eta[2] \epsilon[r_3]^2 \eta[1] C 16 \epsilon[c_2]^2 \eta[2]^2 \epsilon[r_1] C 8 \eta[3] \epsilon[r_2]^2 \eta[1] - 16 \eta[2] \epsilon[r_1]^2 \eta[1] C 4 \epsilon[c_2]^2 \eta[2]^2 \epsilon[c_1]^2 - 16 \eta[3] \epsilon[r_1]^2 \eta[1] C 4 \epsilon[c_1]^2 \eta[1]^2 \epsilon[r_3]^2 C 4 \epsilon[r_1] \epsilon[c_2]^2 \epsilon[r_3]^2 C 4 \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[c_3]^2 C 4 \epsilon[r_1]^2 \epsilon[c_2]^2 \epsilon[r_3]^2 C \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[c_3]^2 C 4 \epsilon[r_1] \epsilon[r_2]^2 \epsilon[r_3]^2 C \epsilon[r_1]^2 \epsilon[c_2]^2 \epsilon[c_3]^2 C 16 \epsilon[r_1] \epsilon[r_2]^2
\end{aligned}$$

$$\begin{aligned}
& \epsilon[c_3]^2 C \epsilon[r_1]^2 \epsilon[c_2]^2 \epsilon[r_3]^2 C 64 \epsilon[r_1]^* \\
& \epsilon[r_2]^* \epsilon[r_3] C \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[r_3]^2 C 16^* \\
& \epsilon[r_1]^* \epsilon[r_2]^2 \epsilon[r_3] C \epsilon[c_1]^2 \epsilon[r_2]^2^* \\
& \epsilon[r_3]^2 C 16^* \epsilon[r_1]^* \epsilon[c_2]^2 \epsilon[r_3] C 4^* \epsilon[r_1]^2^* \\
& \epsilon[r_2]^* \epsilon[r_3]^2 C 4^* \epsilon[r_1]^* \epsilon[r_2]^2 \epsilon[c_3]^2 C 4^* \\
& \epsilon[c_1]^2 \epsilon[r_2]^2 \epsilon[r_3] C 16^* \epsilon[r_1]^* \epsilon[r_2]^* \\
& \epsilon[r_3]^2 C 4^* \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[r_3] C 4^* \epsilon[r_1]^* \\
& \epsilon[c_2]^2 \epsilon[c_3]^2 C \epsilon[c_1]^2 \epsilon[r_2]^2 \epsilon[c_3] \\
& ^2 C 4^* \epsilon[c_1]^2 \epsilon[r_2]^* \epsilon[r_3]^2 C 16^* \epsilon[c_1]^2 \epsilon[r_2]^* \\
& \epsilon[r_3] C 4^* \epsilon[c_1]^2 \epsilon[r_2]^* \epsilon[c_3]^2 C \epsilon[c_1] \\
& ^2 \epsilon[c_2]^2 \epsilon[r_3]^2 C 4^* \epsilon[c_1]^2 \epsilon[c_2]^2 \epsilon[r_3] C \\
& \epsilon[c_1]^2 \epsilon[c_2]^2 \epsilon[c_3]^2 - 8^* \eta[1]^* \epsilon[r_1]^2^* \\
& \epsilon[r_2]^2 - 32^* \eta[1]^* \epsilon[r_1]^2 \epsilon[r_2] - 8^* \eta[1]^* \epsilon[r_1]^2^* \\
& \epsilon[c_2]^2 - 8^* \eta[1]^* \epsilon[r_1]^* \epsilon[r_2]^2 - 32^* \eta[1]^* \epsilon[r_1]^* \\
& \epsilon[r_2] - 8^* \eta[1]^* \epsilon[r_1]^* \epsilon[c_2]^2 - 8^* \eta[1]^* \epsilon[c_1]^2^* \\
& \epsilon[r_2]^2 - 32^* \eta[1]^* \epsilon[c_1]^2 \epsilon[r_2] - 8^* \eta[1]^* \epsilon[c_1]^2^* \\
& \epsilon[c_2]^2 - 8^* \eta[1]^* \epsilon[r_1]^* \epsilon[r_3]^2 C 2^* \eta[2]^* \epsilon[r_3]^2^* \\
& \eta[1]^* \epsilon[c_1]^2 \epsilon[r_2]^2 C 2^* \eta[2]^* \epsilon[r_3]^2 \eta[1]^* \epsilon[r_2]^* \\
& \epsilon[c_1]^2 \epsilon[c_2]^2 C 2^* \eta[2]^* \epsilon[c_3]^2 \eta[1]^* \epsilon[r_1]^2 \epsilon[r_2]^* \\
& \epsilon[c_2]^2 C 2^* \eta[2]^* \epsilon[c_3]^2 \eta[1]^* \epsilon[c_1]^2 \epsilon[r_2]^2 C 2^* \eta[2]^* \\
& \epsilon[c_3]^2 \eta[1]^* \epsilon[c_1]^2 \epsilon[c_2]^2 C 2^* \eta[1]^* \epsilon[r_1]^* \\
& ^* \epsilon[c_2]^2 \epsilon[r_3]^2 \eta[2] C 2^* \eta[1]^* \epsilon[r_1]^* \epsilon[c_2]^2^* \\
& \epsilon[r_3]^2 \eta[3] C 8^* \eta[2]^* \epsilon[r_3]^* \eta[1]^* \epsilon[r_1]^2 \epsilon[c_2] \\
& ^2 C 8^* \eta[2]^* \epsilon[r_3]^* \eta[1]^* \epsilon[c_1]^2 \epsilon[r_2]^2 C 8^* \eta[2]^* \\
& \epsilon[r_3]^* \eta[1]^* \epsilon[c_1]^2 \epsilon[c_2]^2 C 2^* \eta[1]^* \epsilon[r_1]^* \\
& \epsilon[c_2]^2 \epsilon[c_3]^2 \eta[2] C 2^* \eta[1]^* \epsilon[r_1]^* \epsilon[c_2]^2^* \\
& \epsilon[c_3]^2 \eta[3] C 2^* \eta[3]^* \epsilon[r_1]^2 \epsilon[c_2]^2 \epsilon[r_3]^* \eta[1] \\
& C 8^* \eta[3]^* \epsilon[r_1]^2 \epsilon[r_2]^* \epsilon[r_3]^* \eta[1] C 2^* \eta[2]^* \\
& \epsilon[c_1]^2 \epsilon[r_2]^* \epsilon[c_3]^2 \eta[1] C 2^* \eta[2]^* \epsilon[c_1]^2^* \\
& \epsilon[r_2]^* \epsilon[r_3]^2 \eta[1] C 8^* \eta[2]^* \epsilon[c_1]^2 \epsilon[r_2]^* \\
& \epsilon[r_3]^* \eta[1] C 2^* \eta[2]^* \epsilon[r_1]^* \epsilon[r_2]^* \epsilon[r_3]^2 \eta[1] \\
& C 2^* \eta[2]^* \epsilon[r_1]^2 \epsilon[r_2]^* \epsilon[r_3]^2 \eta[1] C 2^* \eta[2]^* \\
& \epsilon[r_1]^* \epsilon[r_2]^* \epsilon[c_3]^2 \eta[1] C 8^* \eta[2]^* \epsilon[r_1]^* \epsilon[r_2]^* \\
& \epsilon[r_3]^* \eta[1] C 2^* \eta[2]^* \epsilon[r_1]^2 \epsilon[r_2]^* \epsilon[c_3]^2^* \\
& \eta[1] C 8^* \eta[2]^* \epsilon[r_1]^2 \epsilon[r_2]^* \epsilon[r_3]^* \eta[1] C 8^* \eta[3]^* \eta[2]^* \\
& \epsilon[r_1]^* \epsilon[c_2]^2 \epsilon[r_3]^2 C 2^* \eta[3]^* \eta[2]^* \epsilon[r_1] \\
& ^2 \epsilon[r_2]^2 \epsilon[c_3]^2 - 4^* \eta[2]^* \epsilon[r_2]^* \epsilon[c_3]^2 \eta[1] \\
& - 32^* \eta[1]^2 \epsilon[r_1] C 16^* \eta[1]^2 \epsilon[r_2] C 16^* \eta[1]^2 \epsilon[r_3]
\end{aligned}$$

$$\begin{aligned}
& [r_3]C^4\eta[1]^2\epsilon[r_2]^2C^4\eta[1]^2\epsilon[r_3]^2C^4\eta[1]^2\epsilon[r_2]^2C^4\eta[1]^2\epsilon[r_3]^2C^4\eta[2]^2\epsilon[r_3]^2\epsilon[r_2]^2-32\eta[1]\epsilon[r_1]\epsilon[r_3]-8\eta[1]\epsilon[r_1]\epsilon[r_3]^2C^{16}\eta[1]\epsilon[r_2]\epsilon[r_3]^2C^{64}\eta[1]\epsilon[r_2]\epsilon[r_3]C^{16}\eta[1]\epsilon[r_2]\epsilon[r_3]^2-8\eta[1]\epsilon[r_1]^2\epsilon[r_3]^2-32\eta[1]\epsilon[r_1]^2\epsilon[r_3]-8\eta[1]\epsilon[r_1]^2\epsilon[r_3]^2C^4\eta[1]\epsilon[r_2]^2\epsilon[r_3]^2C^{16}\eta[1]\epsilon[r_2]^2\epsilon[r_3]C^4\eta[1]\epsilon[r_2]^2\epsilon[r_3]^2-8\eta[1]\epsilon[r_1]^2\epsilon[r_3]^2-32\eta[1]\epsilon[r_1]^2\epsilon[r_3]-8\eta[1]\epsilon[r_1]^2\epsilon[r_3]^2C^4\eta[1]\epsilon[r_2]^2\epsilon[r_3]^2C^{16}\eta[1]\epsilon[r_2]^2\epsilon[r_3]C^4\eta[1]\epsilon[r_2]^2\epsilon[r_3]^2-8\eta[2]\epsilon[r_1]^2\epsilon[r_2]^2-8\eta[2]\epsilon[r_1]^2\epsilon[r_2]-8\eta[2]\epsilon[r_1]^2\epsilon[r_2]-32\eta[2]\epsilon[r_1]\epsilon[r_2]^2-32\eta[2]\epsilon[r_1]\epsilon[r_2]-32\eta[2]\epsilon[r_1]\epsilon[r_2]^2-8\eta[2]\epsilon[r_1]\epsilon[r_2]-8\eta[2]\epsilon[r_1]\epsilon[r_2]^2C^{16}\eta[2]\epsilon[r_1]\epsilon[r_3]^2C^{64}\eta[2]\epsilon[r_1]\epsilon[r_3]C^{16}\eta[2]\epsilon[r_1]\epsilon[r_3]^2-8\eta[2]\epsilon[r_2]\epsilon[r_3]^2-32\eta[2]\epsilon[r_2]\epsilon[r_3]-8\eta[2]\epsilon[r_2]\epsilon[r_3]^2C^4\eta[2]\epsilon[r_1]^2\epsilon[r_3]^2C^{16}\eta[2]\epsilon[r_1]^2\epsilon[r_3]C^4\eta[2]\epsilon[r_1]^2\epsilon[r_3]^2-8\eta[2]\epsilon[r_2]^2\epsilon[r_3]^2-32\eta[2]\epsilon[r_2]^2\epsilon[r_3]-8\eta[2]\epsilon[r_2]^2\epsilon[r_3]^2C^4\eta[2]\epsilon[r_1]^2\epsilon[r_3]^2C^{16}\eta[2]\epsilon[r_1]^2\epsilon[r_3]C^4\eta[2]\epsilon[r_1]^2\epsilon[r_3]^2-8\eta[2]\epsilon[r_2]^2\epsilon[r_3]^2-32\eta[2]\epsilon[r_2]^2\epsilon[r_3]-8\eta[2]\epsilon[r_2]^2\epsilon[r_3]^2C^4\eta[3]\epsilon[r_1]^2\epsilon[r_2]^2C^{16}\eta[3]\epsilon[r_1]\epsilon[r_2]^2C^{64}\eta[3]\epsilon[r_1]\epsilon[r_2]C^{16}\eta[3]\epsilon[r_1]\epsilon[r_2]^2C^4\eta[3]\epsilon[r_1]^2\epsilon[r_2]^2C^{16}\eta[3]\epsilon[r_1]^2\epsilon[r_2]C^4\eta[3]\epsilon[r_1]^2\epsilon[r_2]^2-32\eta[3]\epsilon[r_1]\epsilon[r_3]^2-32\eta[3]\epsilon[r_1]\epsilon[r_3]^2-32\eta[3]\epsilon[r_2]\epsilon[r_3]^2-32\eta[3]\epsilon[r_2]\epsilon[r_3]-8\eta[3]\epsilon[r_1]^2\epsilon[r_3]^2-8\eta[3]\epsilon[r_1]^2\epsilon[r_3]-8\eta[3]\epsilon[r_1]^2\epsilon[r_3]^2-8\eta[3]\epsilon[r_2]^2\epsilon[r_3]^2-8\eta[3]\epsilon[r_2]^2\epsilon[r_3]-8\eta[3]\epsilon[r_2]^2\epsilon[r_3]^2C^4\eta[3]\epsilon[r_1]^2\epsilon[r_3]^2-8\eta[3]\epsilon[r_1]^2\epsilon[r_3]-8\eta[3]\epsilon[r_1]^2\epsilon[r_3]^2C^4\eta[3]\epsilon[r_2]^2\epsilon[r_3]^2-8\eta[3]\epsilon[r_2]^2\epsilon[r_3]-8\eta[3]\epsilon[r_2]^2\epsilon[r_3]^2
\end{aligned}$$

$$\begin{aligned}
& [c2]^2 \epsilon [c3]^2 C^4 \eta [3]^2 \epsilon [c2]^2 \epsilon [r3]^2 C^4 \eta [3] \\
& ^2 \epsilon [r1]^2 \epsilon [r3]^2 C^4 \eta [3]^2 \epsilon [r2]^2 \epsilon [r3] \\
& ^2 C^4 \eta [3]^2 \epsilon [c1]^2 \epsilon [r3]^2 C^{16} \eta [3]^2 \epsilon [r1]^* \\
& \epsilon [r3]^2 C^{16} \eta [3]^2 \epsilon [r2]^* \epsilon [r3]^2 - 16 \eta [3]^* \epsilon [r3]^* \eta [1] C^8 \eta [3]^* \eta [2]^* \epsilon [r1]^* \epsilon [r2]^2 \epsilon [r3]^2 C^2 \eta [3]^* \eta [2]^* \epsilon [r1]^2 \epsilon [c2]^2 \epsilon [r3]^2 C^2 \eta [3]^* \eta [2]^* \\
& \epsilon [r1]^2 \epsilon [r2]^2 \epsilon [r3]^2 C^2 \eta [3]^* \eta [2]^* \epsilon [c1]^2 * \\
& \epsilon [r2]^2 \epsilon [r3]^2 C^8 \eta [3]^* \epsilon [r2]^* \eta [1]^* \epsilon [c1]^2 * \\
& \epsilon [c3]^2 C^2 \eta [3]^* \epsilon [r2]^2 \eta [1]^* \epsilon [c1]^2 \epsilon [c3] \\
& ^2 C^2 \eta [3]^* \epsilon [c1]^2 \eta [1]^* \epsilon [c2]^2 \epsilon [r3]^2 C^2 \eta [3]^* \\
& \epsilon [c1]^2 \eta [1]^* \epsilon [c2]^2 \epsilon [c3]^2 C^8 \eta [2]^* \epsilon [r1]^* \\
& \epsilon [r2]^* \epsilon [r3]^2 \eta [3] C^2 \eta [2]^* \epsilon [c1]^2 \epsilon [r2]^* \epsilon [r3]^* \eta [3] \\
& C^2 \eta [2]^* \epsilon [c1]^2 \epsilon [r2]^* \epsilon [c3]^2 \eta [3] C^8 \eta [3]^* \\
& \epsilon [r1]^2 \eta [1]^* \epsilon [r2]^* \epsilon [r3]^2 C^8 \eta [3]^* \epsilon [c1]^2 \eta [1]^* \epsilon [r2]^* \epsilon [r3]^2 C^8 \eta [3]^* \eta [2]^* \epsilon [r1]^* \epsilon [c2]^2 * \\
& \epsilon [c3]^2 C^2 \eta [3]^* \eta [2]^* \epsilon [c1]^2 \epsilon [r2]^2 \epsilon [c3] \\
& ^2 C^2 \eta [3]^* \eta [2]^* \epsilon [c1]^2 \epsilon [c2]^2 \epsilon [r3]^2 C^2 \eta [3]^* \\
& \eta [2]^* \epsilon [c1]^2 \epsilon [c2]^2 \epsilon [c3]^2 C^8 \eta [3]^* \epsilon [r2]^2 * \\
& \eta [2]^* \epsilon [r1]^* \epsilon [c3]^2 C^2 \eta [3]^* \epsilon [r1]^2 \epsilon [r2]^2 \eta [1]^* \epsilon [r3]^2 C^2 \eta [3]^* \epsilon [r1]^2 \epsilon [r2]^2 \eta [1]^* \epsilon [c3] \\
& ^2 C^8 \eta [3]^* \epsilon [r1]^2 \epsilon [r2]^* \eta [1]^* \epsilon [c3]^2 C^2 \eta [3]^* \\
& \epsilon [r1]^2 \epsilon [c2]^2 \eta [1]^* \epsilon [r3]^2 C^2 \eta [3]^* \epsilon [r1]^2 * \\
& \epsilon [c2]^2 \eta [1]^* \epsilon [c3]^2 C^2 \eta [3]^* \epsilon [c1]^2 \epsilon [r2]^2 * \\
& \eta [1]^* \epsilon [r3]^2 C^8 \eta [3]^* \epsilon [r1]^* \epsilon [r2]^2 \epsilon [r3]^* \eta [2] \\
& C^8 \eta [3]^* \epsilon [r1]^* \epsilon [c2]^2 \epsilon [r3]^* \eta [2] C^2 \eta [3]^* \epsilon [c1]^2 \epsilon [r2]^2 \epsilon [r3]^* \eta [2] - 16 \eta [3]^* \epsilon [r1]^* \eta [1] - 16 * \\
& \eta [2]^* \epsilon [c1]^2 \eta [1] C^8 \eta [2]^* \epsilon [c3]^2 \eta [1] - 16 \eta [3]^* \\
& \epsilon [c1]^2 \eta [1] C^8 \eta [3]^* \epsilon [c2]^2 \eta [1] - 16 \eta [2]^* \epsilon [r1]^* \\
& \eta [1] C^4 \epsilon [c2]^2 \eta [2]^2 \epsilon [r1]^2 C^4 \epsilon [c1]^2 \eta [1]^2 * \\
& \epsilon [c2]^2 C^{16} \epsilon [c3]^2 \eta [3]^2 \epsilon [r1] C^4 \epsilon [c1]^2 * \\
& \eta [1]^2 \epsilon [c3]^2 C^4 \eta [1]^2 \epsilon [c2]^2 \epsilon [c3]^2 C^4 \eta [1]^2 * \\
& \epsilon [c2]^2 \epsilon [r3]^2 C^4 \eta [1]^2 \epsilon [c2]^2 \epsilon [r3] C^4 \eta [1]^2 * \\
& \epsilon [r2]^2 \epsilon [c3]^2 C^4 \eta [1]^2 \epsilon [r2]^2 \epsilon [r3] C^4 \eta [1]^2 * \\
& \epsilon [r2]^2 \epsilon [r3]^2 C^{16} \eta [1]^2 \epsilon [r2]^* \epsilon [r3] \\
& C^4 \eta [1]^2 \epsilon [r2]^* \epsilon [c3]^2 C^4 \eta [1]^2 \epsilon [r2]^* \epsilon [r3] \\
& ^2 C^{16} \epsilon [c2]^2 \eta [2]^2 \epsilon [r3] C^{16} \epsilon [c1]^2 \eta [1]^2 * \\
& \epsilon [r3] - 16 \eta [2]^* \epsilon [r2]^* \eta [3] - 16 \eta [2]^* \epsilon [r3]^* \eta [3] C^8 *
\end{aligned}$$

$$\begin{aligned}
& \epsilon[2] \epsilon[r_1]^2 \epsilon[3] - 16 \epsilon[2] \epsilon[r_3]^2 \epsilon[3] C_{16} \epsilon[1]^2 \epsilon[r_1]^2 \epsilon[r_2] C_4 \epsilon[1]^2 \epsilon[r_1]^2 \epsilon[c_2]^2 C_4 \epsilon[c_3]^2 \epsilon[3]^2 \epsilon[c_2]^2 C_4 \epsilon[1]^2 \epsilon[r_1]^2 \epsilon[r_2]^2 - 16 \epsilon[1] \epsilon[r_2]^2 \epsilon[2] - 16 \epsilon[1] \epsilon[r_3]^2 \epsilon[3] - 16 \epsilon[1] \epsilon[c_2]^2 \epsilon[2] - 16 \epsilon[1] \epsilon[c_3]^2 \epsilon[3] C_{16} \epsilon[c_3]^2 \epsilon[3]^2 \epsilon[r_2] C_{32} \epsilon[2] \epsilon[r_3] \epsilon[1] C_4 \epsilon[2]^2 \epsilon[r_1]^2 \epsilon[r_2]^2 C_4 \epsilon[3]^2 \epsilon[r_1]^2 \epsilon[r_2] C \epsilon[3]^2 \epsilon[r_1]^2 \epsilon[c_2]^2 C_4 \epsilon[3]^2 \epsilon[r_1] \epsilon[r_2]^2 - 4 \epsilon[2] \epsilon[r_1] \epsilon[c_3]^2 \epsilon[1] - 4 \epsilon[2] \epsilon[r_2] \epsilon[r_3]^2 \epsilon[1] - 16 \epsilon[2] \epsilon[r_2] \epsilon[r_3] \epsilon[1] - 32 \epsilon[3]^2 \epsilon[r_1] \epsilon[r_2] \epsilon[r_3] - 8 \epsilon[3]^2 \epsilon[r_1] \epsilon[r_2]^2 \epsilon[r_3] - 8 \epsilon[3]^2 \epsilon[r_1] \epsilon[c_2]^2 \epsilon[r_3] - 4 \epsilon[2] \epsilon[c_1]^2 \epsilon[r_3] \epsilon[3] - 4 \epsilon[2] \epsilon[c_1]^2 \epsilon[c_3]^2 \epsilon[3] C_4 \epsilon[c_1]^2 \epsilon[1]^2 \epsilon[r_2]^2 \epsilon[r_3] - 4 \epsilon[2] \epsilon[r_1]^2 \epsilon[r_3]^2 \epsilon[3] - 4 \epsilon[2] \epsilon[r_1]^2 \epsilon[r_3] \epsilon[3] - 4 \epsilon[2] \epsilon[r_1]^2 \epsilon[c_3]^2 \epsilon[3] - 4 \epsilon[2] \epsilon[c_1]^2 \epsilon[r_3]^2 \epsilon[3] C_8 \epsilon[2] \epsilon[r_1]^2 \epsilon[r_2] \epsilon[1] C_8 \epsilon[2] \epsilon[r_1] \epsilon[r_2] \epsilon[1] C_8 \epsilon[2] \epsilon[r_2] \epsilon[r_3]^2 \epsilon[3] C_8 \epsilon[2] \epsilon[r_2] \epsilon[r_3] \epsilon[3] C_8 \epsilon[2] \epsilon[r_2] \epsilon[c_3]^2 \epsilon[3] - 8 \epsilon[1]^2 \epsilon[r_1] \epsilon[r_2] \epsilon[r_3]^2 - 2 \epsilon[1]^2 \epsilon[r_1] \epsilon[c_2]^2 \epsilon[c_3]^2 C_{16} \epsilon[c_1]^2 \epsilon[1]^2 \epsilon[r_2] \epsilon[r_3] - 2 \epsilon[1]^2 \epsilon[r_1] \epsilon[r_2]^2 \epsilon[c_3]^2 - 32 \epsilon[1]^2 \epsilon[r_1] \epsilon[r_2] \epsilon[r_3] - 8 \epsilon[1]^2 \epsilon[r_1] \epsilon[r_2]^2 \epsilon[r_3] - 8 \epsilon[1]^2 \epsilon[r_1] \epsilon[c_2]^2 \epsilon[r_3] - 2 \epsilon[1]^2 \epsilon[r_1] \epsilon[r_2]^2 \epsilon[r_3]^2 - 8 \epsilon[1]^2 \epsilon[r_1] \epsilon[r_2] \epsilon[c_3]^2 - 4 \epsilon[3] \epsilon[r_2]^2 \epsilon[r_3] \epsilon[1] C_8 \epsilon[3] \epsilon[c_1]^2 \epsilon[r_3] \epsilon[1] - 4 \epsilon[3] \epsilon[c_2]^2 \epsilon[r_3] \epsilon[1] - 16 \epsilon[3] \epsilon[c_1]^2 \epsilon[c_2]^2 \epsilon[1] C_8 \epsilon[3] \epsilon[r_1] \epsilon[r_3] \epsilon[1] - 16 \epsilon[3] \epsilon[r_2] \epsilon[r_3] \epsilon[1] C_8 \epsilon[3] \epsilon[r_1]^2 \epsilon[r_3] \epsilon[1] - 4 \epsilon[3] \epsilon[c_1]^2 \epsilon[r_2]^2 \epsilon[1] C \epsilon[c_2]^2 \epsilon[2]^2 \epsilon[c_1]^2 \epsilon[r_3]^2 - 2 \epsilon[2]^2 \epsilon[r_1]^2 \epsilon[r_2] \epsilon[c_3]^2 - 8 \epsilon[2]^2 \epsilon[r_1]^2 \epsilon[r_2] \epsilon[r_3] C_4 \epsilon[c_1]^2 \epsilon[1]^2 \epsilon[r_2] \epsilon[r_3]^2 - 16 \epsilon[1] \epsilon[r_2] \epsilon[r_3]^2 \epsilon[3] - 16 \epsilon[1] \epsilon[r_2] \epsilon[c_3]^2 \epsilon[3] C_8 \epsilon[3] \epsilon[c_2]^2 \epsilon[r_3] \epsilon[2] C_8 \epsilon[1] \epsilon[r_1] \epsilon[r_3]^2 \epsilon[3] C_8 \epsilon[1] \epsilon[r_1] \epsilon[c_3]^2 \epsilon[3] C_4 \epsilon[c_3]^2 \epsilon[3]^2 \epsilon[c_1]^2 \epsilon[r_2] C_8 \epsilon[2] \epsilon[1] \epsilon[r_1]^2
\end{aligned}$$

$$\begin{aligned}
& \epsilon[c_2]^2 C 8 \eta[2] \eta[1] \epsilon[r_1]^2 \epsilon[r_2]^2 C \epsilon[c_3] \\
& \wedge^2 \eta[3]^2 \epsilon[c_1]^2 \epsilon[r_2]^2 C 4 \epsilon[c_2]^2 \eta[2]^2 * \\
& \epsilon[r_1] \epsilon[c_3]^2 - 2 \eta[2]^2 \epsilon[c_1]^2 \epsilon[r_2] \epsilon[r_3]^2 - 8 \eta[2]^2 \epsilon[c_1]^2 \epsilon[r_2] \epsilon[r_3] - 2 \eta[2]^2 * \\
& \epsilon[c_1]^2 \epsilon[r_2] \epsilon[c_3]^2 C 4 \epsilon[c_1]^2 \eta[1]^2 * \\
& \epsilon[c_2]^2 \epsilon[r_3] C \epsilon[c_1]^2 \eta[1]^2 \epsilon[c_2]^2 \epsilon[c_3]^2 - 4 \eta[2]^2 \epsilon[r_1]^2 \epsilon[r_2] \eta[3] - 16 \eta[2]^2 \epsilon[r_1] * \\
& \epsilon[r_2] \eta[3] - 2 \eta[3]^2 \epsilon[r_1]^2 \epsilon[c_2]^2 \epsilon[c_3]^2 - 32 * \\
& \eta[1]^2 \epsilon[r_1]^2 \epsilon[r_2] \epsilon[r_3] C 16 \eta[3]^2 \epsilon[r_1] * \\
& \epsilon[r_2] C 4 \eta[3]^2 \epsilon[r_1] \epsilon[c_2]^2 C \eta[3]^2 \epsilon[c_1] \\
& \wedge^2 \epsilon[r_2]^2 C 4 \eta[3]^2 \epsilon[c_1]^2 \epsilon[r_2] C \eta[3]^2 \epsilon[c_1] \\
& \wedge^2 \epsilon[c_2]^2 - 32 \eta[3]^2 \epsilon[r_1] \epsilon[r_3] - 32 \eta[3]^2 * \\
& \epsilon[r_2] \epsilon[r_3] C \eta[2]^2 \epsilon[r_1]^2 \epsilon[r_3]^2 C 4 \eta[2]^2 * \\
& \epsilon[r_1]^2 \epsilon[r_3] C \eta[2]^2 \epsilon[r_1]^2 \epsilon[c_3]^2 C \eta[2] \\
& \wedge^2 \epsilon[c_1]^2 \epsilon[r_3]^2 C 4 \eta[2]^2 \epsilon[c_1]^2 \epsilon[r_3] C \eta[2] \\
& \wedge^2 \epsilon[c_1]^2 \epsilon[c_3]^2 C \eta[3]^2 \epsilon[r_1]^2 \epsilon[r_2]^2 \\
& - 8 \eta[2]^2 \epsilon[r_1]^2 \epsilon[r_2] - 32 \eta[2]^2 \epsilon[r_1] \epsilon[r_2] \\
& - 8 \eta[2]^2 \epsilon[c_1]^2 \epsilon[r_2] C 4 \eta[2]^2 \epsilon[r_1] \epsilon[r_3] \\
& \wedge^2 C 16 \eta[2]^2 \epsilon[r_1] \epsilon[r_3] C 4 \eta[2]^2 \epsilon[r_1] \epsilon[c_3]^2 - 8 \eta[2]^2 \epsilon[r_2] \epsilon[r_3]^2 - 32 \eta[2]^2 \epsilon[r_2] * \\
& \epsilon[r_3] - 8 \eta[2]^2 \epsilon[r_2] \epsilon[c_3]^2 C 16 \epsilon[c_1]^2 \eta[1]^2 * \\
& \epsilon[r_2] - 16 \eta[3]^2 \epsilon[r_2]^2 \eta[2] - 16 \eta[3]^2 \epsilon[c_2]^2 \eta[2] \\
& C 4 \eta[1]^2 \epsilon[r_1]^2 \epsilon[r_3]^2 C 4 \epsilon[c_3]^2 \eta[3]^2 * \\
& \epsilon[r_1]^2 C 4 \eta[1]^2 \epsilon[r_1]^2 \epsilon[c_3]^2 C 16 \eta[1]^2 * \\
& \epsilon[r_1]^2 \epsilon[r_3] C 16 \eta[2]^2 \epsilon[r_1] \epsilon[r_2]^2 C 4 * \\
& \epsilon[c_1]^2 \eta[1]^2 \epsilon[r_2]^2 C 16 \eta[2]^2 \epsilon[r_3] \epsilon[r_2] \\
& \wedge^2 C 32 \eta[2]^2 \epsilon[r_1] \eta[3] - 8 \eta[3]^2 \epsilon[c_2]^2 \epsilon[r_3] C 8 * \\
& \eta[2]^2 \epsilon[c_1]^2 \eta[3] - 16 \eta[2]^2 \epsilon[c_3]^2 \eta[3] C 32 \eta[3]^2 * \\
& \epsilon[r_2] \eta[1] - 2 \eta[1]^2 \epsilon[r_1] \epsilon[c_2]^2 \epsilon[r_3]^2 - 8 \eta[1]^2 * \\
& \epsilon[r_1]^2 \epsilon[r_2] \epsilon[c_3]^2 - 8 \eta[1]^2 \epsilon[r_1]^2 * \\
& \epsilon[c_2]^2 \epsilon[r_3] - 2 \eta[1]^2 \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[c_3]^2 - 2 \eta[1]^2 \epsilon[r_1] \epsilon[r_2]^2 \epsilon[r_3]^2 - 2 \eta[1]^2 \epsilon[r_1]^2 * \\
& \epsilon[c_2]^2 \epsilon[c_3]^2 - 8 \eta[1]^2 \epsilon[r_1] \epsilon[r_2]^2 * \\
& \epsilon[c_3]^2 - 2 \eta[1]^2 \epsilon[r_1]^2 \epsilon[c_2]^2 \epsilon[r_3]^2 - 32 \eta[1]^2 * \\
& \epsilon[r_1] \epsilon[r_2] \epsilon[r_3] - 2 \eta[1]^2 \epsilon[r_1]^2 \epsilon[r_2]^2 * \\
& \epsilon[r_3]^2 - 8 \eta[1]^2 \epsilon[r_1] \epsilon[r_2]^2 \epsilon[r_3] - 2 \eta[1]^2 * \\
& \epsilon[c_1]^2 \epsilon[r_2]^2 \epsilon[r_3]^2 - 8 \eta[1]^2 \epsilon[r_1] \epsilon[c_2]^2 * \\
& \epsilon[r_3] - 8 \eta[1]^2 \epsilon[r_1]^2 \epsilon[r_2] \epsilon[r_3]^2 - 2 \eta[1]^2 *
\end{aligned}$$

```
[1]*epsilon[r1]*epsilon[r2]^2*epsilon[c3]^2-8*eta[1]*epsilon[c1]^2*
epsilon[r2]^2*epsilon[r3]-8*eta[1]*epsilon[r1]*epsilon[r2]*epsilon[r3]^2
-8*eta[1]*epsilon[r1]^2*epsilon[r2]^2*epsilon[r3]-2*eta[1]*epsilon[r1]*
epsilon[c2]^2*epsilon[c3]^2-2*eta[1]*epsilon[c1]^2*epsilon[r2]^2*epsilon
[c3]^2-8*eta[1]*epsilon[c1]^2*epsilon[r2]*epsilon[r3]^2-32*eta[1]*epsilon
[c1]^2*epsilon[r2]*epsilon[r3]-8*eta[1]*epsilon[c1]^2*epsilon[r2]*epsilon
[c3]^2-2*eta[1]*epsilon[c1]^2*epsilon[c2]^2*epsilon[r3]^2-8*eta[1]*
epsilon[c1]^2*epsilon[c2]^2*epsilon[r3]-2*eta[1]*epsilon[c1]^2*epsilon
[c2]^2*epsilon[c3]^2-8*eta[2]*epsilon[r1]^2*epsilon[r2]*epsilon[r3]-8*eta
[2]*epsilon[r1]*epsilon[c2]^2*epsilon[r3]^2-2*eta[2]*epsilon[r1]^2*
epsilon[r2]*epsilon[c3]^2-8*eta[2]*epsilon[r1]^2*epsilon[c2]^2*epsilon
[r3]-8*eta[3]^2*epsilon[r1]^2*epsilon[r3])"
```

Let's code the expression in Python now.

Numerical application

$$\begin{aligned} \varepsilon_1 &:= -9.04639807977 + I \cdot 28.8437268491; \eta_1 := (1e0 - \eta_2 - \eta_3); \\ &\quad -9.04639807977 + 28.8437268491 I \\ &\quad 0.7308 \end{aligned} \tag{2.13}$$

$$\begin{aligned} \varepsilon_2 &:= -0.62993 + I \cdot 24.915; \eta_2 := 16.87e-2; \#Cr \\ &\quad -0.62993 + 24.915 I \\ &\quad 0.1687 \end{aligned} \tag{2.14}$$

$$\begin{aligned} \varepsilon_3 &:= -17.8077402901e0 + I \cdot 28.97412511e0; \eta_3 := 10.05e-2; \#Ni \\ &\quad -17.8077402901 + 28.97412511 I \\ &\quad 0.1005 \end{aligned} \tag{2.15}$$

$$\begin{aligned} \varepsilon_{r1} &:= \text{Re}(\varepsilon_1) : \varepsilon_{c1} := \text{Im}(\varepsilon_1) : \varepsilon_{r2} := \text{Re}(\varepsilon_2) : \varepsilon_{c2} := \text{Im}(\varepsilon_2) : \varepsilon_{r3} := \text{Re}(\varepsilon_3) : \varepsilon_{c3} \\ &:= \text{Im}(\varepsilon_3) : \\ eps_r & \\ &\quad -7.977157246 \end{aligned} \tag{2.16}$$

$$\begin{aligned} eps_c & \\ &\quad 28.75781330 \end{aligned} \tag{2.17}$$

restart

Formula used by Inam Mirza.

$$\varepsilon_{eff} := \varepsilon_m \cdot \left(1 + \frac{3 \cdot fI \cdot \text{beta}}{1 - fI \cdot \text{beta}} \right); \text{beta} := \frac{(\varepsilon_{np} - \varepsilon_m)}{\varepsilon_{np} - 2 \cdot \varepsilon_m}$$

$$1.625 + \frac{(3.475945863 + 1.205338949 I) fI}{1 + (-0.7130145361 - 0.2472490151 I) fI}$$

$$0.7130145361 + 0.2472490151 I \quad (1)$$

$$\varepsilon_{np} := 0 - I \cdot 2.8; fI :=$$

$$-2.8 I \quad (2)$$

$$\varepsilon_m := 1.625 - 0 \cdot I$$

$$1.625 \quad (3)$$

$$\text{evalc}(\varepsilon_{eff})$$

$$1.625 + \frac{3.475945863 fI (1 - 0.7130145361 fI)}{(1 - 0.7130145361 fI)^2 + 0.06113207547 fI^2}$$

$$- \frac{0.2980188680 fI^2}{(1 - 0.7130145361 fI)^2 + 0.06113207547 fI^2}$$

$$+ I \left(\frac{1.205338949 fI (1 - 0.7130145361 fI)}{(1 - 0.7130145361 fI)^2 + 0.06113207547 fI^2} \right.$$

$$\left. + \frac{0.8594241912 fI^2}{(1 - 0.7130145361 fI)^2 + 0.06113207547 fI^2} \right) \quad (4)$$